Estimation and Probabilistic Learning

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Farzad Farnoud

Chapter 0

Review of Probability

In this chapter, we will review some concepts from probability theory and linear algebra that will be useful in the rest of the course.

This review is not comprehensive. You can refer to the course webpage for more resources.

0.1 What is probability?

Intuitively, probability is a way of systematically studying events whose outcomes are uncertain (e.g., the probability of rolling a 6 is $1/6$, or the probability of rain at 10 am tomorrow is 20%). It enables us to quantify information and uncertainty and provides ways to transfer our knowledge about one random quantity to another. Mathematically, probability deals with sets, and functions that assign real values to those sets, in a way that certain axioms are satisfied; like geometry or number theory, it can be used to model the real world, but it can also be studied as an abstract subject.

0.1.1 Definitions

Assuming an experiment with different possible outcomes, consider the following definitions.[1]

- Ω: the sample space, the set of all possibilities (*outcomes*)
- $E \subseteq \Omega$: an event, i.e., a set of outcomes
- $\Pr$: A function from subsets of Ω to $\mathbb{R}$. $\Pr(E)$ is the probability of the event E.

0.1.2 Axioms

- $\Pr(E) \geq 0$ for all $E \subseteq \Omega$.
- $\Pr(\Omega) = 1$
- $\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2)$ if $E_1 \cap E_2 = \varnothing$.

Based on these axioms, many theorems and other results can be proven. For $A, B \subseteq \Omega$:

- If $A \subseteq B$, then $\Pr(A) \leq \Pr(B)$.
- $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$.
- $\Pr(A^c) = 1 - \Pr(A)$, where A^c is the complement of A, i.e., the event that A does not occur.
- $\Pr(A) \leq 1$, which is the first result with $B = \Omega$.

[1] These definitions and the following axioms are simplified. We cannot always assign probability to all subsets of Ω. Also, for the third axiom, for any *countable* sequence of mutually exclusive events $E_1, E_2, \ldots$, we require that $\Pr(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \Pr(E_i)$.

What these theorems and definitions 'mean' depends on what we think probability means.

0.1.3 Interpretations of probability

> *"Probability is the most important concept in modern science, especially as nobody has the slightest notion what it means."*
>
> Bertrand Russell

How do we assign probability to events? What does it mean, for example, to say that $\Pr(E) = 1/3$?

Classical: If there are K possible outcomes and no reason for some to be more likely than others, each outcome gets probability $1/K$: rolling a 3 has probability 1/6.

Frequentist: In N repetitions of a "random" experiment, with N very large, E occurs approximately $N \Pr(E)$ times: a coin with $\Pr(H) = 1/3$ tossed 3000 times shows heads around 1000 times.

Bayesian: Probability is a degree of belief, consistent with the axioms; this covers events that are, strictly speaking, not random, such as the stock market hitting a threshold this year, a nuclear war this century, or a defendant being guilty.

Each interpretation can be criticized: the classical for circularity (a coin is fair if the outcomes are equally likely, which is what we set out to define); the frequentist for vagueness (how large is "large"?); and the Bayesian for subjectivity and for assigning probabilities to one-time events. From a purely mathematical perspective, none of this matters: we only need to assign probabilities in a way that satisfies the axioms. But the interpretation shapes how problems are approached and can lead to different real-world decisions.

0.2 Sets and their sizes

Finding the probability of an event is easiest when all outcomes are equally likely. In such cases, if we can measure the size of the set A of desirable outcomes, dividing it by the size of the sample space yields the probability,

$$\Pr(A) = \frac{|A|}{|\Omega|},$$

where $|A|$ denotes the size of the set A.

Definition 0.1 (Finite, countable, uncountable)**.** A set A is **finite** if there is a natural number n such that the number of elements in A is less than n. Otherwise, it is **infinite**. If the elements of A can be counted, i.e., there is a one-to-one function from A to natural numbers, then A is **countable**. Otherwise, it is **uncountable**. A countable set may be finite (e.g., $\{1, 5, 6\}$) or infinite (e.g., integers, prime numbers, rational numbers).

If A is finite, we define its size (aka, cardinality) as the number of elements. Counting the elements of a finite set usually rests on the **product principle**: if each element is specified by a sequence of n choices, with k_1 options for the first choice, k_2 for the second, and so on, then the set has $k_1 k_2 \cdots k_n$ elements. In particular, there are k^n strings of length n over an alphabet of k symbols, since each of the n positions can be filled in k ways. The same count applied to membership shows that a set of n elements has 2^n subsets, since each element is either in or out; in particular, a sample space with n outcomes has 2^n events. As an example with unequal k_i, arranging n objects in a row means choosing which object comes first (n options), then which of the rest comes second ($n - 1$ options), and so on, so the number of arrangements is

$$n! = n \times (n-1) \times \cdots \times 2 \times 1.$$

Example 0.2 (Bits for a die roll)**.** A die is rolled, showing X. How many bits are needed to represent the value of X?

Solution. A string of m bits takes 2^m values, so it can represent X only if $2^m \geq 6$, i.e., $m \geq \log_2 6 \simeq 2.58$. As m is an integer, $m = 3$ bits are needed. In general, a quantity with k possible values needs $\lceil \log_2 k \rceil$ bits.

△

The number of ways to choose k objects from a set of n objects, when the order of the choice does not matter, is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Exercise 0.3 (Bytes with three ones)**.** How many 8-bit bytes are there? How many of these have exactly 3 ones? If we pick a random byte, what is the probability that it has exactly 3 ones (binomial distribution)? △

If the sample space has an infinite, even uncountable, number of outcomes, we may still be able to think of the outcomes as equally likely. For example, if we pick a random number between 0 and 1, we may assume all outcomes are equally likely. In such cases, the size of the set can be measured via length, area, volume, etc.

Exercise 0.4 (Point in a square)**.** A random point is chosen in a square of unit side. What is the probability that it is inside the circle of diameter one inscribed in the square? What is the probability that it is on the circle? △

0.3 Random variables and distributions

A **random variable** (RV) is a function that assigns real values to outcomes in Ω. In most cases, there is a very natural mapping. For example, let X denote the number showing on a die. Now X is a random variable, mapping each outcome of the form "the die shows i" to the real number i. For this reason, the fact that random variables are really functions is often overlooked. Information about the probabilities of different outcomes is given by the **distribution** of the random variable.

A random variable is **discrete** if there are a countable number of possibilities (could be infinite but countable, like natural numbers). It can also be **continuous** (uncountable number of outcomes, defined over the real line or a subset of a Euclidean space).

For example, a random variable that is 1 if heads shows when a given coin is flipped and is 0 otherwise is discrete and finite; the number of phone calls made in a given hour is discrete and infinite; the arrival time of a plane from midnight is continuous.

0.3.1 Discrete distributions

The distribution of a discrete random variable X is given by its **probability mass function** (pmf) denoted by $p_X(x)$, where

$$p_X(x) = \Pr(X = x).$$

Clearly, $p_X(x) \geq 0$ for all x and

$$\sum_x p_X(x) = 1. \tag{0.1}$$

If clear from the context, we drop the X in the subscript.

Example 0.5 (Poisson distribution)**.** An RV X has the Poisson distribution with parameter λ if

$$p(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \in \{0, 1, \dots\}.$$

The number of times an event, e.g., phone calls or car accidents, occurs in a given interval of time is often assumed to have a Poisson distribution (with good reason). △

Exercise 0.6 (Red die, blue die)**.** A red die and a blue die are rolled. Let X denote the number showing on the red die and Y denote the sum of the two dice. Find the pmf of X and the pmf of Y. △

Exercise 0.7 (Drawing aces)**.** Two cards are drawn at random from a standard deck of 52 cards and let Z denote the number of Aces drawn. Find the pmf of Z. △

0.3.2 Continuous distributions

The distribution of a continuous random variable X is given by its **probability density function** (pdf) $p_X(x)$, also sometimes denoted $f_X(x)$. Roughly speaking,

$$\Pr\left(x - \frac{dx}{2} \le X \le x + \frac{dx}{2}\right) = p_X(x)dx.$$

For two real numbers a, b,

$$\Pr(a \le X \le b) = \int_a^b p_X(x)dx.$$

For any pdf, we have $p_X(x) \ge 0$ and

$$\int_{-\infty}^{\infty} p_X(x)dx = 1.$$

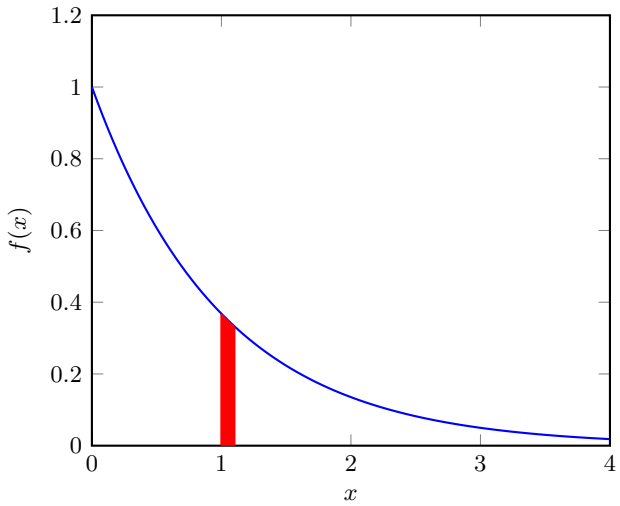
A pdf is a probability per unit length rather than a probability, so unlike a pmf it may exceed 1; the exponential density below equals λ at $x = 0$. What it assigns to a single point is zero: for a continuous RV, $\Pr(X = x) = 0$ for every x, and only intervals carry probability.

Example 0.8 (Exponential distribution)**.** An exponential random variable X with parameter λ has distribution

$$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0.$$

For $\lambda = 1$, approximate the probability that X is between 1 and 1.1.

Solution. Over a short interval, the probability is approximately the density times the length of the interval: $f(1)\times 0.1 = e^{-1}\times 0.1 = 0.37\times 0.1 = 0.037$. In the figure below, the area colored red represents this probability.



△

0.3.3 Cumulative distribution functions

Cumulative distribution functions (CDFs) are defined for both discrete and continuous RVs as $F_X(x) = \Pr(X \leq x)$ and can be found via summation or integration:

$$F_X(x) = \sum_{k \leq x} p_X(k)$$
$$F_X(x) = \int_{-\infty}^{x} p_X(t)dt$$

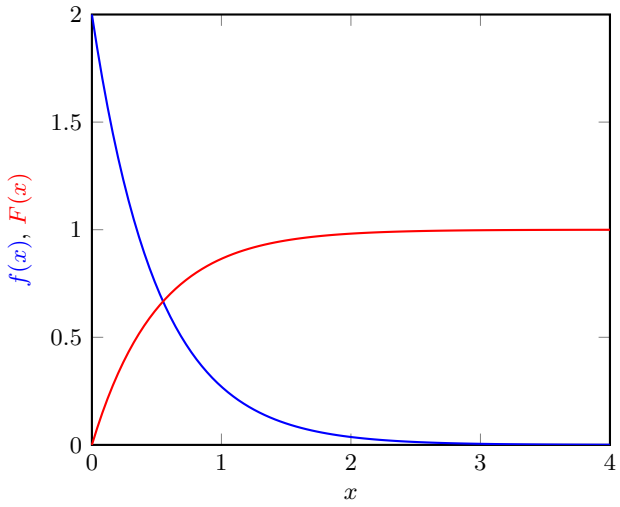
Each relation can also be run backwards, recovering the distribution from the CDF: for an integer-valued RV, $p_X(k) = F_X(k) - F_X(k-1)$, the height of the jump the staircase makes at k; for a continuous RV, the density is the derivative of the CDF, $p_X(x) = \frac{d}{dx}F_X(x)$.

Example 0.9 (Exponential CDF)**.** Find the CDF of the exponential RV in Example 0.8.

Solution. For $x \geq 0$,

$$F_X(x) = \int_0^x \lambda e^{-\lambda t}dt = 1 - e^{-\lambda x},$$

shown below for $\lambda = 2$, along with the pdf.



△

0.3.4 Expected value

The **expected value** or the **mean** $\mathbb{E}[X]$ of a random variable X with distribution $p(x)$ is given by

$$\mathbb{E}[X] = \sum_x xp(x),$$
$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xp(x)dx.$$

One way to think about the expected value is as the average of a large number of experiments. For example, if a game pays out $\$X$ each time you play with probability distribution $p(x)$, if you play the game many times, on average you will win $\$\,\mathbb{E}[X]$ per game. That is, if you play n times, winning $\$x_i$ in game i, and n is large, then

$$\frac{1}{n}(x_1 + x_2 + \cdots + x_n) \simeq \mathbb{E}[X].$$

Exercise 0.10 (Means from earlier examples)**.** Find the expected value of the discrete and continuous RVs in the examples above. △

Exercise 0.11 (Expectation of a constant)**.** Find $\mathbb{E}[1]$. △

0.3.4.1 Indicator random variables

For an event A, the **indicator** random variable $\mathbb{1}(A)$ equals 1 if A occurs and 0 otherwise. Its expectation is the probability of the event,

$$\mathbb{E}[\mathbb{1}(A)] = 1 \cdot \Pr(A) + 0 \cdot \Pr(A^c) = \Pr(A),$$

which lets us move between probabilities and expectations.

Example 0.12 (Indicator of a die roll)**.** A die is rolled, showing X. What is $\mathbb{E}[\mathbb{1}(X \geq 5)]$?

Solution. $\mathbb{E}[\mathbb{1}(X \geq 5)] = \Pr(X \geq 5) = 1/3$.

△

0.3.4.2 Expectation of functions of random variables

For an RV X and a function $f(x)$ it follows from the definition that

$$\begin{aligned} \mathbb{E}[f(X)] &= \sum_x f(x)p(x), \\ \mathbb{E}[f(X)] &= \int_{-\infty}^{\infty} f(x)p(x)dx. \end{aligned} \tag{0.2}$$

Exercise 0.13 (A four-value pmf)**.** A random variable X has distribution

$$p_X(-1) = 0.1,\ p_X(0) = 0.2,\ p_X(1) = 0.3,\ p_X(2) = 0.4.$$

Find $\mathbb{E}\,X$. Let $Y = X^2$. Find $\mathbb{E}\,Y$, both by finding the distribution of Y and by using (0.2). △

0.3.4.3 Linearity of expectation

For a RV X, functions $f(x)$ and $g(x)$, and real numbers a and b,

$$\mathbb{E}[af(X) + bg(X)] = a\,\mathbb{E}[f(X)] + b\,\mathbb{E}[g(X)],$$

which can be proven easily from the definition of expectation.

Example 0.14 (Expanding a square)**.** For an RV X and a constant a, express $\mathbb{E}[(X-a)^2]$ in terms of $\mathbb{E}[X^2]$ and $\mathbb{E}\,X$.

Solution. $\mathbb{E}[(X-a)^2] = \mathbb{E}[X^2 - 2aX + a^2] = \mathbb{E}[X^2] - 2a\,\mathbb{E}\,X + a^2$.

△

Consider a collection of random variables $X_1, X_2, \ldots, X_n$. By the linearity of expectation

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}\,X_i. \tag{0.3}$$

If all variables are identically distributed, then

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = n\,\mathbb{E}\,X_1. \tag{0.4}$$

Example 0.15 (Shared birthdays)**.** In a class of n students, what is the expected number of pairs of students who have the same birthday?

Solution. For two students i and j, let X_{ij} be equal to 1 if they share a birthday and 0 otherwise (the indicator of that event) and let $X = \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} X_{ij}$. Now,

$$\mathbb{E}\,X = \binom{n}{2}\mathbb{E}\,X_{12} = \binom{n}{2}\Pr(X_{12}=1) = \binom{n}{2}\frac{1}{365} \simeq \frac{n^2}{730}. \tag{0.5}$$

In particular, having $n = \sqrt{730} \simeq 27$ students in a class is enough to have on average one pair with the same birthday. With $n = 60$ and $n = 85$ students, there should be around 5 and 10 such pairs, respectively.

△

Example 0.16 (The forgetful waiter)**.** In a restaurant, n diners each order a different dish, and the forgetful waiter places the dishes in front of the diners at random. How many diners do we expect to receive the dish they ordered?

Solution. For any given diner, the probability of receiving the correct dish is $\frac{1}{n}$. Letting X_i be the indicator of diner i being served correctly, the number of correctly served diners is $X = \sum_{i=1}^{n} X_i$, so

$$\mathbb{E}\,X = \sum_{i=1}^{n}\mathbb{E}\,X_i = n\cdot\frac{1}{n} = 1,$$

regardless of n. Note that the X_i are far from independent. For example, if $n-1$ diners got the right dish, so did the remaining one. Linearity of expectation, however, does not require independence.

△

Exercise 0.17 (Heads as indicators)**.** A coin with probability of heads p is tossed n times and X is the number of heads. Write X as a sum of n indicator random variables and use linearity to show that $\mathbb{E}\,X = np$. △

0.3.4.4 Variance

Suppose someone offers you a game in which your expected winning is \$100. Will you accept? Which game would you play?

Sure thing: You always win exactly \$100.

Coin flip: You win \$0 with probability 1/2 and \$200 with probability 1/2.

High stakes: You win \$1200 with probability 1/2 and lose \$1000 with probability 1/2.

All three have the same mean. So what's different between them?

The mean helps us represent a distribution with one value, which describes the average behavior of the RV. But as this example shows, the behavior around the mean is also important. Denoting the mean of X by μ_X, the variability around the mean is captured to a degree by the variance $\mathrm{Var}[X]$,

$$\mathrm{Var}[X] = \mathbb{E}[(X-\mu_X)^2].$$

The variance gives a sense of *how far X is from its mean μ_X, on average.* The **standard deviation**, σ_X, is defined as

$$\sigma_X = \sqrt{\text{Var}[X]},$$

and the variance is usually denoted as σ_X^2.

Exercise 0.18 (Variance identity)**.** Prove that

$$\text{Var}[X] = \mathbb{E}\, X^2 - (\mathbb{E}\, X)^2.$$

△

Exercise 0.19 (Scaling and shifting)**.** Find the mean and variance of each of the following RVs [1]:

- $X + c$
- aX
- $aX + c$
- $\frac{X-\mu_X}{\sigma_X}$ (called the **standardized version** of X)

△

0.3.4.5 Mode and median

Two other single-value summaries of a distribution are used alongside the mean. The **mode** is a value at which the pmf or pdf is maximized, and a **median** is a value m such that $\Pr(X \leq m) \geq 1/2$ and $\Pr(X \geq m) \geq 1/2$ (for a continuous RV, a solution of $F(m) = 1/2$). The three generally differ; as we will see when discussing Bayesian point estimates, each is the best single-value summary under a different measure of error. The median is less sensitive to outliers than the mean: if a billionaire walks into a bar, the mean income of the patrons jumps by millions while the median hardly moves.

Example 0.20 (Summaries of the exponential)**.** Find the mode, the median, and the mean of the exponential RV of Example 0.8 with parameter λ.

Solution. The pdf $\lambda e^{-\lambda x}$ is maximized at $x = 0$, the mode. The median solves $1 - e^{-\lambda m} = 1/2$, giving $m = (\ln 2)/\lambda \simeq 0.69/\lambda$. The mean is $1/\lambda$.

△

Exercise 0.21 (Discrete mode and median)**.** Find the mode and the median of the RV of Exercise 0.13. △

0.3.5 Common distributions

We denote X having distribution 'Dist' by $X \sim \text{Dist}(a, b, \dots)$, where $a, b, \dots$, are the parameters of the distribution.

0.3.5.1 Discrete distributions

- $X \sim \text{Ber}(p)$: $\Pr(X = 1) = p$, $\Pr(X = 0) = 1 - p$, $\mathbb{E}[X] = p$, $\text{Var}[X] = p(1-p)$.
- $X \sim \text{Cat}(\theta_1, \dots, \theta_k)$: $\Pr(X = i) = \theta_i$, $1 \leq i \leq k$, $\sum_{i=1}^k \theta_i = 1$. The **categorical** distribution describes a single trial with k possible outcomes; $\text{Ber}(p)$ is the special case $k = 2$. The vector of counts of the k outcomes in n independent trials has the *multinomial* distribution, which we will meet in a later chapter.
- $X \sim \text{Bin}(n, p)$:[2] $p(x) = \binom{n}{x} p^x (1-p)^{n-x}$, $0 \leq x \leq n$, $\mathbb{E}[X] = np$, $\text{Var}[X] = np(1-p)$.

[2] Note that sometimes p is used both as a parameter and as the distribution. The meaning should be clear from the context.

- $X \sim \text{Geo}(p)$: $p(x) = (1-p)^{x-1}p,\ x \geq 1, \quad \mathbb{E}[X] = 1/p, \quad \text{Var}[X] = \frac{1-p}{p^2}$.
- $X \sim \text{Poi}(\lambda)$: $p(x) = \frac{\lambda^x e^{-\lambda}}{x!},\ x \geq 0, \quad \mathbb{E}[X] = \lambda, \quad \text{Var}[X] = \lambda$.
- $X \sim \text{Uni}[a,b]$: $p(x) = \frac{1}{b-a+1},\ x \in \mathbb{Z}, a \leq x \leq b, \quad \mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var}[X] = \frac{(b-a+1)^2-1}{12}$.

Example 0.22 (A character-level language model)**.** A simple model of English text treats the next character as a categorical random variable: with an alphabet consisting of the letters a–z and the space character, we have $k = 27$, and $\theta_1, \ldots, \theta_{27}$ can be set to the letter frequencies of English. Generating text then amounts to repeatedly sampling from this distribution (Section 0.3.6 shows how). Modern language models produce text the same way: at each step, the next token is drawn from a categorical distribution over a large vocabulary, whose parameters are computed from the text so far. △

0.3.5.2 Continuous distributions

- $X \sim \text{Uni}(a,b)$: $p(x) = \frac{1}{b-a},\ x \in (a,b), \quad \mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var}[X] = \frac{(b-a)^2}{12}$.
- $X \sim \mathcal{N}(\mu, \sigma^2)$: $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-\mu)^2}{2\sigma^2}),\ x \in \mathbb{R}, \quad \mathbb{E}[X] = \mu, \quad \text{Var}[X] = \sigma^2$.
- $X \sim \text{Exp}(\lambda)$: $p(x) = \lambda e^{-\lambda x},\ x \geq 0, \quad \mathbb{E}[X] = 1/\lambda, \quad \text{Var}[X] = 1/\lambda^2$.

A useful property of the normal distribution is closure under linear transformations: if $X \sim \mathcal{N}(\mu, \sigma^2)$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$, and a sum of *independent* normal RVs is again normal. We will study the multivariate version of this fact in a later chapter.

Sometimes, we drop the normalization constant, that is, the constant by which we divide to ensure that the distribution integrates to 1. This could be because the constant is not important (e.g., in Bayesian inference) or because it is hard to determine. In such cases, we use $\propto$ to show proportionality rather than equality. We should be careful which of the entities appearing is the *variable*. For example, viewed as a function of x, we have $f(x) = \frac{\lambda^x e^{-\lambda}}{x!} \propto \frac{\lambda^x}{x!}$ and as a function of λ, we have $g(\lambda) = \frac{\lambda^x e^{-\lambda}}{x!} \propto \lambda^x e^{-\lambda}$.

- $X \sim \text{Beta}(\alpha, \beta)$: $p(x) \propto x^{\alpha-1}(1-x)^{\beta-1},\ 0 \leq x \leq 1, \quad \mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}, \quad \text{Var}[X] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.
- $X \sim \text{Gamma}(\alpha, \beta)$: $p(x) \propto x^{\alpha-1}e^{-\beta x},\ x > 0, \quad \mathbb{E}[X] = \frac{\alpha}{\beta}, \quad \text{Var}[X] = \frac{\alpha}{\beta^2}$.

Example 0.23 (Swapping variable and parameter)**.** For the distributions given in this section, try changing what the variable is and what the parameters are and check whether another distribution from the list can be obtained with appropriate normalization.

Solution. For example, $\text{Bin}(n,p)$ viewed as a distribution in p turns into $\text{Beta}(x+1, n-x+1)$.

△

0.3.6 Functions of a random variable and sampling

If X is a random variable and g a function, then $Y = g(X)$ is also a random variable, whose distribution is determined by that of X. For a discrete X,

$$p_Y(y) = \sum_{x:\, g(x)=y} p_X(x).$$

Example 0.24 (Distribution of $|X|$)**.** Let X be uniform on $\{-2, -1, 0, 1, 2\}$ and $Y = |X|$. Find the pmf of Y.

Solution. $p_Y(0) = 1/5$ and $p_Y(1) = p_Y(2) = 2/5$.

△

For a continuous X and a strictly increasing g, the CDF of $Y = g(X)$ is

$$F_Y(y) = \Pr(g(X) \le y) = \Pr\big(X \le g^{-1}(y)\big) = F_X\big(g^{-1}(y)\big),$$

and differentiating gives the **change-of-variables formula**

$$p_Y(y) = p_X\big(g^{-1}(y)\big)\frac{dg^{-1}(y)}{dy},$$

where for a strictly decreasing g the derivative is replaced by its absolute value.

Example 0.25 (From uniform to exponential)**.** Let $X \sim \text{Uni}(0,1)$ and $Y = -\ln X$. Find the distribution of Y.

Solution. For $y \ge 0$,

$$F_Y(y) = \Pr(-\ln X \le y) = \Pr\big(X \ge e^{-y}\big) = 1 - e^{-y},$$

so $Y \sim \text{Exp}(1)$: transforming a uniform random variable has produced an exponential one.

△

0.3.6.1 Sampling and the inverse-CDF transform

Computers readily produce pseudo-random samples from $\text{Uni}(0,1)$. Samples from other distributions, needed whenever we simulate a model or generate data from it, are obtained by transforming such uniform samples, as in Example 0.25. The general recipe inverts the CDF:

Definition 0.26 (Generalized inverse CDF)**.** For a CDF F and $0 < u < 1$, the **generalized inverse** of F, also called its **quantile function**, is

$$F^{-1}(u) = \min\{x : F(x) \ge u\},$$

the smallest value at which the CDF has reached u.[3]

This definition applies to any random variable, discrete or continuous. If F is continuous and strictly increasing, $F^{-1}(u)$ is the ordinary inverse, the unique x with $F(x) = u$. If X is discrete, F is a staircase and $F^{-1}(u)$ is the first value of X whose cumulative probability reaches u.

Theorem 0.27 (Inverse-CDF transform)**.** *Let F be a CDF and $U \sim \text{Uni}(0,1)$. Then $X = F^{-1}(U)$ has CDF F.*

Proof. For any c, we have $F^{-1}(u) \le c$ if and only if $u \le F(c)$. If $u \le F(c)$, then c belongs to the set $\{x : F(x) \ge u\}$, so the minimum of that set is at most c. Conversely, if $F^{-1}(u) \le c$, then $F(c) \ge F(F^{-1}(u)) \ge u$ because F is non-decreasing. Hence $\Pr(X \le c) = \Pr\big(F^{-1}(U) \le c\big) = \Pr(U \le F(c)) = F(c)$. □

Figure 1 shows a CDF and its inverse side by side, drawn to the same scales, so that the inverse is the CDF reflected with its two axes exchanged. Two intervals of the same width are shaded on the F^{-1} axis. Reading each across to the curve and down gives the values of u that are mapped into it, and those two sets of u have very different lengths: $-1 \le X \le 0$ has probability 0.23 and $2 \le X \le 3$ has probability 0.07, so a uniform U produces a value in the first interval more than three times as often. An interval of u-values of length ℓ is always mapped to a set of values of probability ℓ.

Example 0.28 (Sampling an exponential)**.** Find a function $G : (0,1) \to \mathbb{R}$ such that $G(U)$ is exponentially distributed with parameter λ, where $U \sim \text{Uni}(0,1)$.

[3] The minimum is attained for every CDF, since F is non-decreasing and right-continuous.

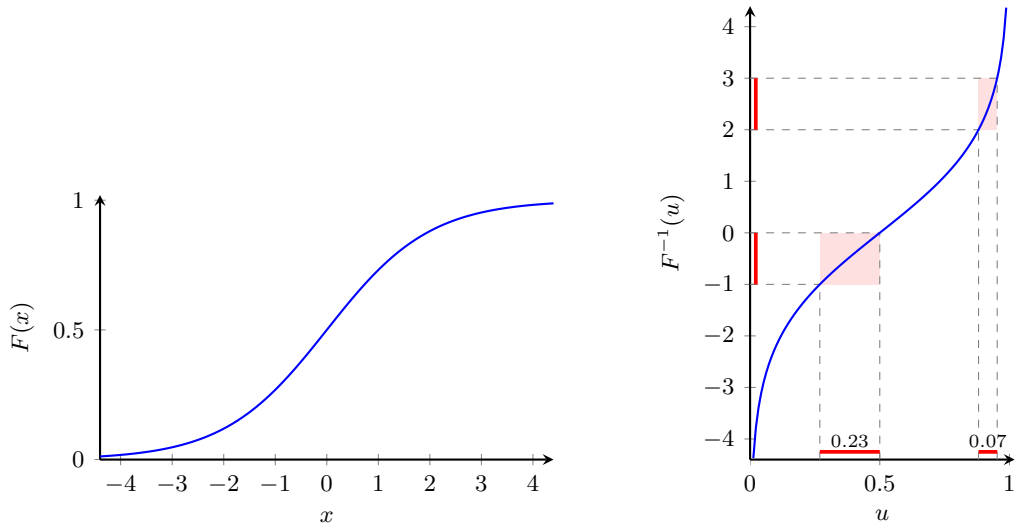


Figure 1: A CDF (left) and its inverse (right), drawn to the same scales: the u axis carries the same scale as the F axis, and the F^{-1} axis the same as the x axis. Two intervals of equal width are shaded on the F^{-1} axis. The values of u mapped into them, marked on the u axis, occupy lengths 0.23 and 0.07, which are the probabilities that X falls in the two intervals.

Solution. We have $F(x) = 1 - e^{-\lambda x}$. Set $G(u) = F^{-1}(u) = -\ln(1-u)/\lambda$. So $-\ln(1-U)/\lambda$, with $U \sim \text{Uni}(0,1)$, is a sample from $\text{Exp}(\lambda)$. This is consistent with Example 0.25, since $1-U$ is also uniform on $(0,1)$.

△

For a discrete distribution, the CDF is a staircase and so is its inverse: the jump of size θ_i that the CDF makes at i becomes an interval of u of length θ_i on which F^{-1} equals i (Figure 2). Sampling therefore amounts to partitioning $[0,1)$ into consecutive intervals of lengths $\theta_1, \theta_2, \ldots$ and outputting i if U falls in the ith interval, which happens with probability θ_i, as required.

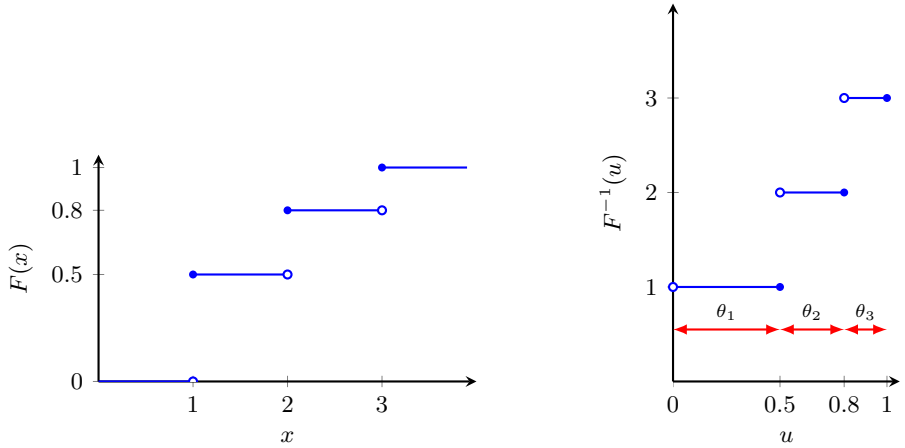


Figure 2: The CDF of $\text{Cat}(0.5, 0.3, 0.2)$ (left) and its inverse (right), drawn to the same scales. The jump of height θ_i that the CDF makes at i becomes the step of width θ_i on which F^{-1} equals i, so a uniform U produces i with probability θ_i. At a jump, F^{-1} takes the lower of the two values, since it is the first value at which the CDF reaches u.

★ **Sampling general 1-D random variables**
To sample from a distribution, take $U \sim \text{Uni}(0,1)$ and apply to it the inverse CDF.

Example 0.29 (Sampling a categorical)**.** How can we sample from $\text{Cat}(0.5, 0.3, 0.2)$?

Solution. Draw $U \sim \text{Uni}(0,1)$; output 1 if $U < 0.5$, output 2 if $0.5 \leq U < 0.8$, and output 3 otherwise. Repeating this with the letter frequencies of English generates (zero-order) random text.

△

Exercise 0.30 (Sampling a geometric)**.** Give an algorithm that turns a $\text{Uni}(0,1)$ sample into a sample from $\text{Geo}(p)$. △

0.4 Joint probability distributions

Joint probability distributions allow us to encode information about relationships between quantities.

For random variables X and Y, the CDF and the pmf/pdf give their joint distribution, depending on their type,

$$F_{X,Y}(x,y) = \Pr(X \leq x, Y \leq y), \qquad \text{CDF for continuous and discrete}$$
$$p_{X,Y}(x,y) = \Pr(X = x, Y = y), \qquad \text{pmf for discrete}$$
$$p_{X,Y}(x,y)dxdy \simeq \Pr\left(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}, y - \frac{dy}{2} \leq Y \leq y + \frac{dy}{2}\right), \qquad \text{pdf for continuous}$$

We can find the distribution for each random variable (in this context these are called the **marginals**) by integration/summation,

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x,y)dy.$$

Example 0.31 (A joint pmf table)**.** A fair coin is flipped twice. Let X be 1 if the first flip is heads and 0 otherwise, and let Y be the total number of heads. Find the joint pmf of X and Y and the two marginals.

Solution. Each of the four outcomes has probability $1/4$, giving the joint pmf below. The pmfs of X and Y are obtained by summing each row and column, and are written, as is customary, in the margins of the table, which is where the name *marginal* comes from.

$p_{X,Y}(x,y)$	$y=0$	$y=1$	$y=2$	$p_X(x)$
$x=0$	$1/4$	$1/4$	0	$1/2$
$x=1$	0	$1/4$	$1/4$	$1/2$
$p_Y(y)$	$1/4$	$1/2$	$1/4$	

For instance, $p_Y(1) = p_{X,Y}(0,1) + p_{X,Y}(1,1) = 1/4 + 1/4 = 1/2$.

△

0.4.1 Expectation, correlation, and covariance

Given two or more RVs, we may be interested in finding the expected value of a function of these RVs, e.g., $\mathbb{E}[XY]$. In such cases, similar to (0.2), we have

$$\mathbb{E}[f(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)p(x,y)dxdy, \tag{0.6}$$

and similarly for discrete variables.

The **correlation** between X and Y is $\mathbb{E}[XY] = \int\int xyp(x,y)dxdy$. The **covariance** $\mathrm{Cov}(X,Y)$ and the **correlation coefficient** $\rho_{X,Y}$ are defined as

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mu_X)(Y-\mu_Y)]$$
$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}.$$

It can be shown that $-1 \le \rho_{X,Y} \le 1$. If $\rho = 0$, then the random variables are **uncorrelated**.

What does the correlation coefficient mean? Let X and Y be random variables, for example, weight and height of a person chosen at random. Suppose that we want to predict the value of Y given X but we are restricted to linear functions of X. Then, in a certain sense,[4] the best predictor $\hat{Y}$ of Y is

$$\hat{Y} = \mathbb{E}\,Y + \rho\frac{\sigma_Y}{\sigma_X}(X - \mathbb{E}\,X),$$

with the "error" being

$$\sigma_Y^2(1-\rho^2).$$

In particular, if X and Y are standardized, $\hat{Y} = \rho X$ with error $1-\rho^2$.

Exercise 0.32 (Strong correlation)**.** If $|\rho|$ is close to 1, the RVs are said to be **strongly correlated**. Why? △

Exercise 0.33 (Covariance identity)**.** Show that $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}\,X\,\mathbb{E}\,Y$. △

Example 0.34 (Bivariate normal)**.** The bivariate jointly Gaussian distribution for X,Y with means μ_X and μ_Y, standard deviations σ_X and σ_Y, and correlation coefficient ρ is given as

$$p(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2}+\frac{(y-\mu_Y)^2}{\sigma_Y^2}-\frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right]}.$$

Examples of this pdf are given in Figure 3. △

Exercise 0.35 (Covariance properties)**.** For random variables X, Y, Z and constants a, b, c, d, e, prove that

- $\mathrm{Var}(X) = \mathrm{Cov}(X,X)$
- $\mathrm{Cov}(X+Y,Z) = \mathrm{Cov}(X,Z) + \mathrm{Cov}(Y,Z)$
- $\mathrm{Cov}(aX,Y) = a\,\mathrm{Cov}(X,Y)$
- $\mathrm{Cov}(X,b) = 0$
- $\mathrm{Cov}(aX+bY+c, dZ+e) = ad\,\mathrm{Cov}(X,Z) + bd\,\mathrm{Cov}(Y,Z)$
- $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$

△

[4] Minimizing the Mean Square Error

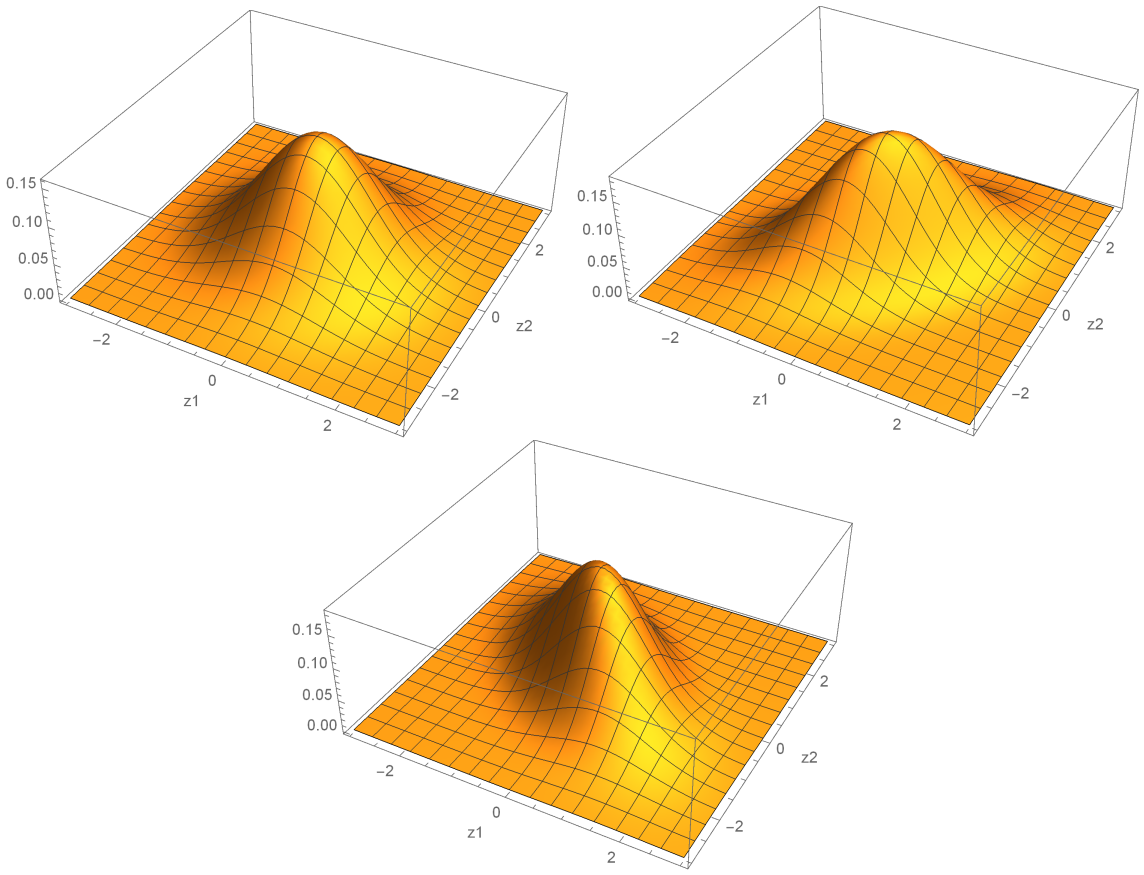


Figure 3: Bivariate Normal pdfs with $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$, with $\rho = 0$ (uncorrelated), $\rho = .5$ (positively correlated), and $\rho = -.5$ (negatively correlated), respectively.

Exercise 0.36 (Dice covariance)**.** Find the expected values and variances of X and Y from Exercise 0.6. Find $\text{Cov}(X, Y)$. △

0.4.2 Independence

Recall that two events A and B are independent, denoted $A \perp\!\!\!\perp B$, iff (if and only if) $\Pr(A \cap B) = \Pr(A)\Pr(B)$. Two random variables X and Y are independent if $\{X \in S_1\}$ and $\{Y \in S_2\}$ are independent for all sets S_1 and S_2. This implies that

$$p(x, y) = p(x)p(y). \tag{0.7}$$

Example 0.37 (Hearts and aces)**.** A card is drawn at random from a standard 52-card deck. Determine whether the following pairs of events are independent: 'heart' and 'ace'; 'red' and 'heart'; 'black' and 'heart'.

Solution. The events 'heart' and 'ace' are independent:

$$\Pr(\text{heart}) = \frac{1}{4}, \quad \Pr(\text{ace}) = \frac{1}{13}, \quad \Pr(\text{heart and ace}) = \frac{1}{52} = \frac{1}{4} \cdot \frac{1}{13}.$$

The events 'red' and 'heart' are not independent: $\Pr(\text{red and heart}) = \frac{1}{4} \neq \frac{1}{2} \cdot \frac{1}{4}$. Neither are 'black' and 'heart', which are disjoint: $\Pr(\text{black and heart}) = 0 \neq \frac{1}{2} \cdot \frac{1}{4}$. In fact, disjoint events with nonzero probabilities are never independent: each rules the other out, which is as far from 'no information' as possible.

△

For two independent random variables, we have

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] \tag{0.8}$$

and $\text{Cov}(X, Y) = 0$.

Exercise 0.38 (Product of independent RVs)**.** Prove (0.8) using (0.7). △

Exercise 0.39 (Independent sums)**.** For two independent RVs X and Y, find $\text{Var}[X+Y]$ and $\mathbb{E}[(X-Y)^2 + 3XY + 5]$ in terms of means and variances of X and Y. △

A collection $X_1, \ldots, X_n$ of random variables that are independent from each other but have the same distribution are called **independent and identically distributed** (iid). We have

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i). \tag{0.9}$$

Exercise 0.40 (Variance of a sum)**.** For iid RVs $X_1, \ldots, X_n$, let $S_n = \sum_{i=1}^n X_i$. Show that

$$\text{Var}(S_n) = \sum_{i=1}^{n} \text{Var}(X_i). \tag{0.10}$$

△

Exercise 0.41 (The sample mean)**.** For iid RVs $X_1, \ldots, X_n$, suppose $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, and let $\bar{X}$ be their average. Show that

$$\mathbb{E}[\bar{X}] = \mu, \qquad \text{Var}[\bar{X}] = \frac{\sigma^2}{n}. \tag{0.11}$$

△

0.4.3 Conditional probability and conditional distributions

For events A and B with $\Pr(B) \neq 0$, the **conditional probability** of A given B is defined as

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

For two discrete variables X and Y, the conditional probability distribution of Y given X is given by

$$p_{Y|X}(y|x) = \Pr(Y = y|X = x) = \frac{\Pr(Y = y, X = x)}{\Pr(X = x)} = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

For continuous RVs, we also have $p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$. In this case, however, we interpret the conditional density as

$$p_{Y|X}(y|x) \simeq \frac{\Pr(y - \epsilon/2 \leq Y \leq y + \epsilon/2 | x - \epsilon/2 \leq X \leq x + \epsilon/2)}{\epsilon},$$

for small positive ϵ. This essentially says to find $p_{Y|X}(y|x)$, we first assume that X is in a narrow strip around x and then find the density for Y given this assumption.

Law of total probability. Let $A_1, A_2, \ldots, A_n$ be a partition of the sample space. That is, $\cup_{i=1}^n A_i = \Omega$ and for all $i \neq j$, we have $A_i \cap A_j = \varnothing$. For an event B, we have

$$\Pr(B) = \sum_{i=1}^{n} \Pr(B \cap A_i) = \sum_{i=1}^{n} \Pr(B|A_i) \Pr(A_i).$$

In particular, if X can take on $\{1, 2, \ldots, n\}$, then for another RV Y,

$$p_Y(y) = \sum_{x=1}^{n} p_{Y|X}(y|x) p_X(x).$$

†

Exercise 0.42 (Monty Hall)**.** On a game show, a prize is hidden behind one of three doors. You pick a door; the host, who knows where the prize is, opens one of the other two doors, revealing no prize, and offers you the chance to switch to the remaining unopened door. Using the law of total probability (conditioning on which door hides the prize), show that switching wins with probability $\frac{2}{3}$, while staying with your original choice wins with probability $\frac{1}{3}$.[5] △

Chain rule of probability. For events $A_1, \ldots, A_n$, we have

$$\Pr(A_1 \cap A_2 \cap \cdots \cap A_n) = \Pr(A_1) \Pr(A_2|A_1) \Pr(A_3|A_1, A_2) \cdots \Pr(A_n|A_1, \ldots, A_{n-1}),$$

which can be easily proven by induction. A similar rule holds for random variables $X_1, \ldots, X_n$:

$$p(x_1, \ldots, x_n) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2) \ldots p(x_n|x_1, \ldots, x_{n-1}).$$

0.4.3.1 Conditional independence

Two events A and B are **conditionally independent** given an event C with $\Pr(C) > 0$ if

$$\Pr(A \cap B|C) = \Pr(A|C) \Pr(B|C).$$

Similarly, random variables X and Z are conditionally independent given Y, denoted $X \perp\!\!\!\perp Z \mid Y$, if

$$p(x, z|y) = p(x|y)\, p(z|y) \qquad \text{for all } x, y, z,$$

[5] You can play the Monty Hall game at https://fhfarnoud.github.io/moi/conditional.html.

or, equivalently, $p(z|x,y) = p(z|y)$: once Y is known, also learning X provides no further information about Z.

Conditional independence neither implies nor is implied by (unconditional) independence, as the following two examples show.

Example 0.43 (Alice, Bob, and traffic)**.** Alice and Bob take the same route to work. Let T indicate whether traffic is heavy, with $\Pr(T=1) = 1/2$, and let X and Z indicate whether Alice and Bob are late, respectively. Given the state of traffic, they are late independently: conditioned on $T=1$, X and Z are independent Ber(0.8) RVs, and conditioned on $T=0$, they are independent Ber(0.2) RVs. So $X \perp\!\!\!\perp Z \mid T$ by construction. Are X and Z independent?

Solution. They are not:

$$\Pr(X=1) = \Pr(Z=1) = \frac{1}{2}(0.8) + \frac{1}{2}(0.2) = 0.5,$$

while

$$\Pr(X=1, Z=1) = \frac{1}{2}(0.8)^2 + \frac{1}{2}(0.2)^2 = 0.34 \neq 0.25 = \Pr(X=1)\Pr(Z=1).$$

Intuitively, learning that Alice is late makes heavy traffic more likely, which in turn makes it more likely that Bob is late.

△

Example 0.44 (Agreement of two coins)**.** Two fair coins are flipped independently, giving X and Z (each 1 for heads), and let $Y = \mathbb{1}(X=Z)$ indicate whether they agree. Then X and Z are independent. Are they conditionally independent given Y?

Solution. No. Given $Y=1$, knowing X determines Z exactly: $p(z|x,Y=1)$ equals 1 for $z=x$ and 0 otherwise, while $p(z|Y=1) = 1/2$. So $X \not\perp\!\!\!\perp Z \mid Y$.

△

Exercise 0.45 (Learning from Alice)**.** In Example 0.43, find $\Pr(Z=1|X=1)$ and compare it with $\Pr(Z=1)$. △

Conditional expectations are defined based on conditional distributions, e.g.,

$$\mathbb{E}[X|Y=y] = \sum_x x p_{X|Y}(x|y).$$

Exercise 0.46 (A conditional pmf table)**.** Suppose the joint pmf is given as

$p_{X,Y}(x,y)$	$x=0$	$x=1$
$y=0$	0.25	0
$y=1$	0.5	0.25

Find $p(y|x)$, $p(x|y)$, $\mathbb{E}[Y|X=0]$, $\mathbb{E}[Y|X=1]$, $\mathbb{E}[X|Y=0]$, $\mathbb{E}[X|Y=1]$. △

Exercise 0.47 (Point in a triangle)**.** A point is chosen uniformly at random in a triangle with vertices on $(0,0),(1,0),(1,1)$. Let X and Y determine the x and y coordinates of the chosen point. Find $p(x|y)$, $p(y|x)$, $\mathbb{E}[X|Y=y]$, $\mathbb{E}[Y|X=x]$. △

0.4.3.2 Law of iterated expectations

Consider a random variable X and a function $g(x)$. We can now obtain $g(X)$ by replacing the deterministic value for x with a random one. Note that $g(X)$ is a random variable. For example, if $X \sim \text{Uni}(-1, 1)$ and $g(x) = |x|$, then $g(X)$ is a random variable with distribution $\text{Uni}(0, 1)$.

Now let $g(x) = \mathbb{E}[Y|X = x]$. This is, of course, a well-defined function. We define $\mathbb{E}[Y|X] = g(X)$, which is as discussed a random variable. Now that we have a random variable, we can compute its expectation, i.e., $\mathbb{E}[\mathbb{E}[Y|X]]$.

Exercise 0.48 (Die then coins)**.** A die is rolled, showing X. A coin is then flipped X times resulting in Y heads. Find the conditional pmf $p_{Y|X}(y|3)$, $\mathbb{E}[Y]$, $\mathbb{E}[Y|X = x]$, the pmf of $\mathbb{E}[Y|X]$, and $\mathbb{E}[\mathbb{E}[Y|X]]$. △

It can be shown that

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y], \qquad \mathbb{E}[\mathbb{E}[Y|X, Z]|Z] = \mathbb{E}[Y|Z]. \tag{0.12}$$

0.4.4 Bayes' rule

In Exercise 0.48, the conditional distribution $p(y|x)$ is readily available as

$$p(y|x) = \binom{x}{y} 2^{-x}.$$

But what if we are interested in $p(x|y)$? Since $p(x|y) = \frac{p(x,y)}{p(y)}$ and $p(x, y) = p(y|x)p(x)$, we have

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)} = \frac{p(y|x)p(x)}{\sum_{x'} p(y|x')p(x')},$$

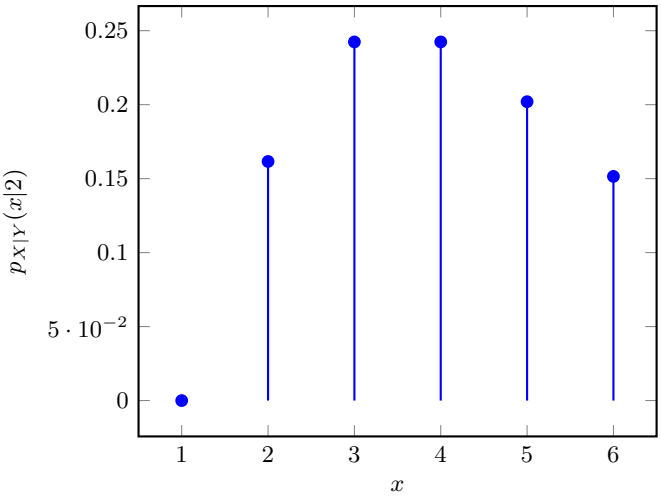
which is called **Bayes' rule**.

Example 0.49 (Inferring the die roll)**.** In Exercise 0.48, find $p(x|y)$. What are the likeliest values for X if $Y = 2$?

Solution. By Bayes' rule,

$$p(x|y) = \frac{\binom{x}{y} 2^{-x}(1/6)}{\sum_{x'=y}^{6} \binom{x'}{y} 2^{-x'}(1/6)} = \frac{\binom{x}{y} 2^{-x}}{\sum_{x'=y}^{6} \binom{x'}{y} 2^{-x'}}.$$

The plot below shows $p_{X|Y}(x|2)$, the conditional distribution of X given $Y = 2$. We can see that the likeliest values for X are $3, 4$.



△

Bayes' rule is used in *evidential reasoning*, examples of which we will see in the next chapter. In this setting, the goal is to find the probabilities of different causes based on the evidence.

Bayesian inference takes its name from Bayes' rule. In this setting, it is often the case that we know the distribution of data given the parameters. But what we actually have is data and need to find the distribution of the parameters. Bayes' rule allows us to find this conditional distribution, a topic we will discuss in detail later.

0.5 Limit theorems

Limits in probability provide a way to understand what happens when the number of experiments grows or many random effects accumulate. Limit theorems are beneficial given that we often deal with large volumes of data. The following limit theorems will be helpful to us later in the course.

0.5.1 Law of large numbers

Let $X_1, \ldots, X_n$ be random variables with mean μ and variance $\leq \sigma^2$ and suppose that for each $i \neq j$, X_i and X_j are uncorrelated (as is the case, in particular, if they are independent). Also, let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then, for any $\epsilon > 0$,

$$\Pr\big(|\bar{X}_n - \mu| \geq \epsilon\big) \leq \frac{\sigma^2}{n\epsilon^2}. \tag{0.13}$$

As n becomes large the right side becomes smaller and smaller. So for large n the probability of $\bar{X}_n$ being too far from the mean is very small. This is referred to as the **Law of Large Numbers** (LLN). In other words, if we take n independent samples from a random variable X, then the average of those samples will be close to the mean $\mathbb{E}\,X$,

$$\frac{1}{n}(x_1 + x_2 + \cdots + x_n) \simeq \mathbb{E}[X],$$

which is what we used to motivate expected value. This mode of convergence, $\Pr(|\bar{X}_n - \mu| \geq \epsilon) \to 0$ for every $\epsilon > 0$, is called **convergence in probability**: $\bar{X}_n$ converges in probability to μ. We will encounter it again when defining what it means for an estimator to be consistent. †

Exercise 0.50 (Proving the LLN)**.** Use the Chebyshev inequality (Section 0.7) to prove LLN when random variables are independent and all have the same variance σ^2. △

Example 0.51 (LLN in simulation)**.** Suppose $X_i \sim \text{Poi}(2)$, $1 \le i \le 500$, and let $\bar{X}_n$ be the average of the first n X_is. Figure 4 shows the plot for $\bar{X}_n$ for a realization of X_is obtained via computer simulation. It is observed that for large values of n, $\bar{X}_n$ is close to 2, the mean of the Poisson distribution. △

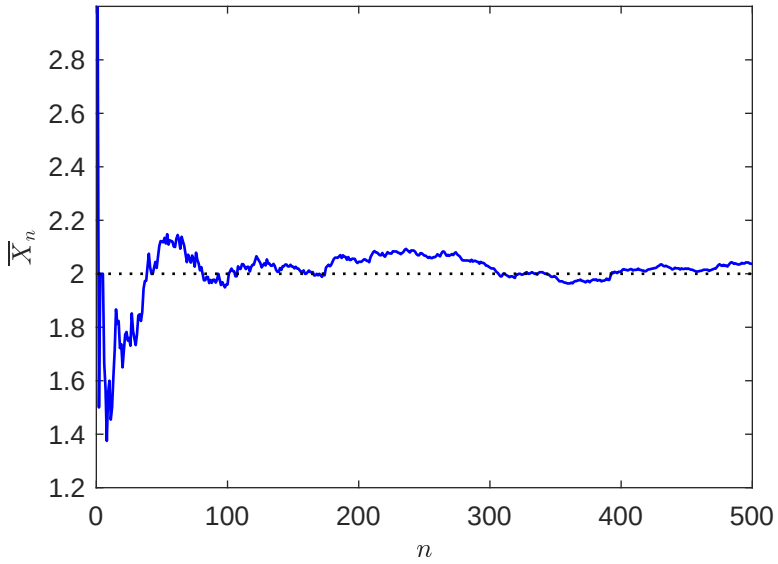


Figure 4: $\bar{X}_n$ based on $X_i \sim \text{Poi}(2)$ as a function of n.

0.5.2 Central limit theorem

Let $X_1, X_2, \ldots$ be iid random variables with mean μ and variance σ^2 and let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$. The **Central Limit Theorem** (CLT) states that, as $n \to \infty$,

$$\text{distribution of } \sqrt{n}(\bar{X}_n - \mu) \quad \to \quad \mathcal{N}(0, \sigma^2). \tag{0.14}$$

That is, the distribution of $\sqrt{n}(\bar{X}_n - \mu)$ approaches the distribution of a normal random variable with mean 0 and variance σ^2. This mode of convergence, in which the CDF of a sequence of random variables approaches the CDF of a limiting distribution, is called **convergence in distribution**.

Loosely speaking, the CLT also means $S_n = \sum_{i=1}^n X_i$ has distribution $\mathcal{N}(n\mu, n\sigma^2)$.

> ★ **Long-term averages approach the mean, deviating normally**
> The average of many independent samples concentrates around $\mathbb{E}\,X$, and by the CLT its fluctuations around the mean are approximately normal.

Example 0.52 (CLT in simulation)**.** Let $X_i \sim \text{Uni}(0,1), 1 \le i \le n = 10$. We produce $50,000$ samples of $\bar{X}_n$ (and S_n), and plot the normalized histograms for $\sqrt{n}(\bar{X}_n - \mu)$ and the pdf of $\mathcal{N}(0, \sigma^2)$ and the normalized histogram for S_n and the pdf of $\mathcal{N}(n\mu, n\sigma^2)$ in Figure 5. △

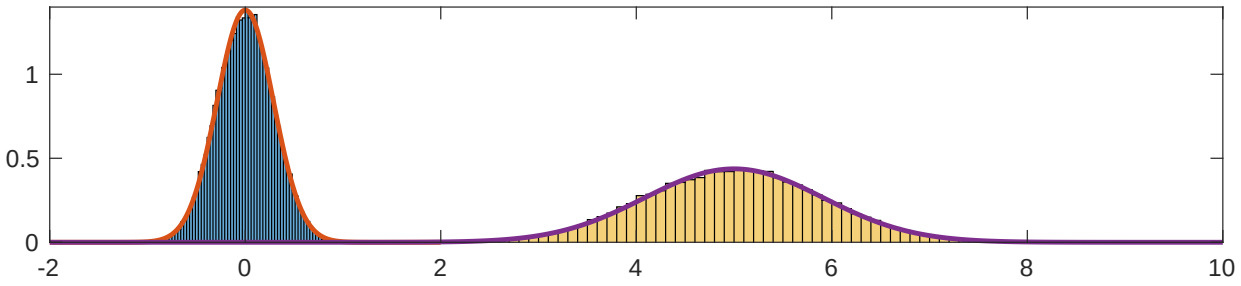


Figure 5: The normalized histograms for $\sqrt{n}(\bar{X}_n - \mu)$ and the pdf of $\mathcal{N}(0, \sigma^2)$ (on the left) and the normalized histogram for S_n and the pdf of $\mathcal{N}(n\mu, n\sigma^2)$ (on the right) for uniform X_i with $\mu = 1/2$ and $\sigma^2 = 1/12$ and with $n = 10$.

0.6 Random vectors

A **random vector** is a vector of random variables.[6] Consider the random vectors $\boldsymbol{X}$ and $\boldsymbol{Y}$

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_m \end{pmatrix}, \quad \boldsymbol{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}. \tag{0.15}$$

The **expected value** of $\boldsymbol{X}$ is

$$\mathbb{E}\,\boldsymbol{X} = \begin{pmatrix} \mathbb{E}\,X_1 \\ \vdots \\ \mathbb{E}\,X_m \end{pmatrix}. \tag{0.16}$$

The **correlation matrix** of $\boldsymbol{X}$ and $\boldsymbol{Y}$ is the $m \times n$ matrix $\mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T]$, whose i, jth element is $\mathbb{E}[X_i Y_j]$. The **cross-covariance matrix** $\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y})$ of $\boldsymbol{X}$ and $\boldsymbol{Y}$ is the matrix $\mathbb{E}[(\boldsymbol{X} - \mathbb{E}\,\boldsymbol{X})(\boldsymbol{Y} - \mathbb{E}\,\boldsymbol{Y})^T]$, whose i, jth element is $\mathrm{Cov}(X_i, Y_j)$. The covariance of a vector $\boldsymbol{X}$ is $\mathrm{Cov}(\boldsymbol{X}) = \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{X})$. The **conditional expectation** $\mathbb{E}[\boldsymbol{X}|\boldsymbol{Y}]$ of $\boldsymbol{X}$ given $\boldsymbol{Y}$ is a vector whose ith element is $\mathbb{E}[X_i|\boldsymbol{Y}]$.

If the elements of $\boldsymbol{X}$ are uncorrelated, then $\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$ and the covariance matrix becomes diagonal. If, in addition, $\mathrm{Cov}(X_i, X_i) = \mathrm{Var}(X_i) = \sigma^2$, i.e., all elements of $\boldsymbol{X}$ have the same variance σ^2, then $\mathrm{Cov}(\boldsymbol{X}) = \sigma^2 I$.

0.6.1 Properties of expectation and covariance

For deterministic matrices A, B, deterministic vectors $\boldsymbol{a}, \boldsymbol{b}$, and random vectors $\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{W}, \boldsymbol{Z}$, we have [1]

1. $\mathbb{E}[\mathsf{A}\boldsymbol{X} + \boldsymbol{a}] = \mathsf{A}\,\mathbb{E}\,\boldsymbol{X} + \boldsymbol{a}$
2. $\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y}) = \mathbb{E}[\boldsymbol{X}(\boldsymbol{Y} - \mathbb{E}\,\boldsymbol{Y})^T] = \mathbb{E}[(\boldsymbol{X} - \mathbb{E}\,\boldsymbol{X})\boldsymbol{Y}^T] = \mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T] - \mathbb{E}\,\boldsymbol{X}\,\mathbb{E}\,\boldsymbol{Y}^T$
3. $\mathbb{E}[(\mathsf{A}\boldsymbol{X})(\mathsf{B}\boldsymbol{Y})^T] = \mathsf{A}\,\mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T]\mathsf{B}^T$
4. $\mathrm{Cov}(\mathsf{A}\boldsymbol{X} + \boldsymbol{a}, \mathsf{B}\boldsymbol{Y} + \boldsymbol{b}) = \mathsf{A}\,\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y})\mathsf{B}^T$
5. $\mathrm{Cov}(\mathsf{A}\boldsymbol{X} + \boldsymbol{a}) = \mathsf{A}\,\mathrm{Cov}(\boldsymbol{X})\mathsf{A}^T$
6. $\mathrm{Cov}(\boldsymbol{W} + \boldsymbol{X}, \boldsymbol{Y} + \boldsymbol{Z}) = \mathrm{Cov}(\boldsymbol{W}, \boldsymbol{Y}) + \mathrm{Cov}(\boldsymbol{W}, \boldsymbol{Z}) + \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y}) + \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Z})$

Example 0.53 (Covariance under scaling)**.** For a random vector $\boldsymbol{X}$ and constants $a, \boldsymbol{b}$, property 5 gives $\mathrm{Cov}(a\boldsymbol{X} + \boldsymbol{b}) = a^2\,\mathrm{Cov}(\boldsymbol{X})$. Prove this using the other properties.

[6] We use lowercase bold letters to denote deterministic vectors, uppercase bold letters to denote random vectors, and uppercase sans serif letters, such as A, to denote matrices.

Solution. The relevant properties are given in each step.

$$\begin{aligned}\operatorname{Cov}(a\boldsymbol{X}+\boldsymbol{b}) &= \operatorname{Cov}(a\boldsymbol{X}+\boldsymbol{b}, a\boldsymbol{X}+\boldsymbol{b}) && (0.17)\\ &\overset{6}{=} \operatorname{Cov}(a\boldsymbol{X}, a\boldsymbol{X}) + \operatorname{Cov}(a\boldsymbol{X}, \boldsymbol{b}) + \operatorname{Cov}(\boldsymbol{b}, a\boldsymbol{X}) + \operatorname{Cov}(\boldsymbol{b}, \boldsymbol{b}) && (0.18)\\ &\overset{4}{=} a^2 \operatorname{Cov}(\boldsymbol{X}, \boldsymbol{X}) + 0 + 0 + 0 && (0.19)\end{aligned}$$

△ †

0.7 Concentration inequalities

The following inequalities bound the probability that a random variable is far from its mean, using only the mean, or the mean and the variance. Chebyshev's inequality is the tool behind the proof of the law of large numbers.

0.7.1 Markov inequality

Suppose the average length of a blue whale is 22 meters and we do not know anything else about the distribution of the lengths of blue whales. Can we say anything about the probability that the length of a randomly chosen blue whale is at least 30 meters? For example, is it possible that this probability is 0.8 or larger? No, since in that case, the average would be at least $0.8 \times 30 = 24$ meters. So only knowing the mean enables us to say something about the extremes of the probability distribution.

This observation is formalized via the **Markov inequality**. For a *non-negative* random variable X, we have

$$\Pr(X \geq a) \leq \frac{\mathbb{E}\, X}{a}.$$

Exercise 0.54 (Proving Markov's inequality)**.** Prove the Markov inequality. △

A special case of this occurs when X counts something, i.e., it only takes non-negative integer values. Then,

$$\Pr(X \geq 1) = \Pr(X > 0) \leq \mathbb{E}\, X, \qquad \Pr(X = 0) \geq 1 - \mathbb{E}\, X.$$

In particular, if the mean $\mathbb{E}\, X$ is small, then there is a large probability that $X = 0$.

0.7.2 Chebyshev inequality

If in addition to the mean, we also have the variance, we can use the Chebyshev bound. For a random variable X with mean μ and variance σ^2,

$$\Pr\left(\left|\frac{X-\mu}{\sigma}\right| \geq a\right) \leq \frac{1}{a^2}.$$

That is, being a standard deviations away from the mean has probability at most $1/a^2$; for example, being 10 standard deviations away from the mean has probability at most 1%.

Exercise 0.55 (Proving Chebyshev's inequality)**.** Prove the Chebyshev bound using the Markov bound. △

References

[1] Bruce Hajek. *Random Processes for Engineers*. Illinois, 2014. URL: http://hajek.ece.illinois.edu/Papers/randomprocJuly14.pdf (visited on 01/30/2017).

Chapter 1

Probability in Reasoning, Inference, and Learning

1.1 Probability as the language of uncertainty

How do we represent information and uncertainty mathematically? When you type "good" into your phone, the keyboard offers a few likely continuations: perhaps "morning", "luck", and "night". These suggestions reflect a state of knowledge: "morning" may follow with probability 0.45, "luck" with probability 0.25, "night" with probability 0.15, and other words with the remaining 0.15 (the numbers are illustrative). A probability distribution over the possible next words captures both what the options are and how strongly each is expected. It also encodes uncertainty: your phone does not know for sure what you are going to write next.

Probability theory is our main tool for representing knowledge and uncertainty. Additionally, it provides a framework for formulating assumptions. In this chapter, we will study the role of probability in reasoning, inference, and machine learning. We first describe these tasks, bearing in mind that there is no universally accepted definition for them.

Probabilistic reasoning: Given probability distributions for a set of random variables, we can use probabilistic reasoning to draw conclusions about unobserved variables based on observed variables.

Inference: Drawing conclusions about the real world or a given system based on evidence or observations. Our models typically aim to reflect reality. For example, determining whether a gene is linked to a disease.

(Supervised) machine learning: Learning from examples of (input, output) pairs and generalizing to predict the output for new inputs. For example, predicting the price of a house based on its features. Machine learning models and algorithms are not necessarily focused on representing reality and are more concerned with the accuracy of their predictions.

When considering these problems, we deal with uncertainty. Sources of uncertainty, for example in machine learning, include:

- **Noise**: aggregate contribution of factors that we do not (wish to) consider (models focus on the most important quantities). For example, your height can be predicted well from your parents' heights and general patterns of nutrition and sleep, which we can include in our model. But it may also depend on every meal you ate and how much you slept each night throughout your youth, which we cannot include.
- Finite sample size: finite size of data makes it impossible to fully determine relationships (i.e., probability distributions) as some configuration may never happen or happen a few times in finite data.

1.1.1 Relationships, distributions, and probabilistic reasoning

Is there any relationship between the arrival times of two people working at a business (opening at 9:00 am), both living in the same area? If so, how can we represent this relationship? How can we make predictions about one being late given the other is late (e.g., if we need at least one person to be present)?

In the same way that we can encode our information about a random quantity as a distribution, we can encode information about random quantities, as well as their relationships, as joint distributions.

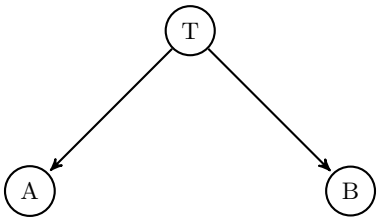
In our example, there's obviously a relationship, that is, the arrival times are not independent. For example, both are affected by traffic. Let

$$\begin{aligned} T_O &: \text{normal traffic} \\ T_H &: \text{heavy traffic} \\ A_O &: \text{Alice is on time} \\ A_L &: \text{Alice is late} \\ B_O &: \text{Bob is on time} \\ B_L &: \text{Bob is late} \end{aligned}$$

and assume

$$\begin{aligned} \Pr(T_O) &= 0.75, \\ \Pr(A_O|T_O) &= 0.8, \\ \Pr(B_O|T_O) &= 0.6, \\ \Pr(A_O|T_H) &= 0.6, \\ \Pr(B_O|T_H) &= 0.2. \end{aligned}$$

Finally, conditioned on the traffic situation, Alice and Bob's arrival times are independent. This information completely determines all probabilities. As we will see in greater depth later, the fact that Alice and Bob's arrival times are only related through traffic can be shown *graphically* as



Causal reasoning:

$$\begin{aligned} \Pr(A_O) &= \Pr(T_O)\Pr(A_O|T_O) + \Pr(T_H)\Pr(A_O|T_H) \\ &= (0.75 \times 0.8) + (0.25 \times 0.6) = 0.75, \\ \Pr(B_O) &= \Pr(T_O)\Pr(B_O|T_O) + \Pr(T_H)\Pr(B_O|T_H) \\ &= (0.75 \times 0.6) + (0.25 \times 0.2) = 0.5. \end{aligned}$$

Evidential reasoning (inverse probabilities, uses Bayes rule):

$$\begin{aligned} \Pr(T_O|A_O) &= \Pr(A_O|T_O)\Pr(T_O)/\Pr(A_O) = 0.8 \times 0.75/0.75 = 0.8 \\ \Pr(T_O|B_O) &= \Pr(B_O|T_O)\Pr(T_O)/\Pr(B_O) = 0.6 \times 0.75/0.5 = 0.9 \end{aligned}$$

The common cause makes the events A_i and B_i dependent. Recall that two events E_1 and E_2 are independent,

denoted $E_1 \perp\!\!\!\perp E_2$ if $\Pr(E_1E_2) = \Pr(E_1)\Pr(E_2)$, or, if $\Pr(E_2) \neq 0$, $\Pr(E_1|E_2) = \Pr(E_1)$. We have

$$\begin{aligned}
&\Pr(A_O|B_O) = \Pr(A_OB_O)/\Pr(B_O)\\
&\Pr(A_OB_O) = (0.75 \times 0.8 \times 0.6) + (0.25 \times 0.6 \times 0.2) = 0.39\\
&\Pr(A_O|B_O) = 0.39/0.5 = 0.78 \neq \Pr(A_O)\\
&\Pr(B_O|A_O) = 0.39/0.75 = 0.52 \neq \Pr(B_O)
\end{aligned}$$

So $A_O \not\perp\!\!\!\perp B_O$.

However, they are *conditionally independent*, by assumption

$$\Pr(A_OB_O|T_O) = \Pr(A_O|T_O)\Pr(B_O|T_O),$$

which is denoted as $A_O \perp\!\!\!\perp B_O|T_O$.

What is the source of uncertainty in this problem? Since we have assumed the distribution is known, sample size is not an issue. The source is noise. For example, if we had information about other factors affecting Bob, e.g., how reliable his car is, if he needs to drop off his kids, etc., we could reduce the amount of noise and make better predictions.

1.2 Inference and machine learning

The reasoning in the previous section relied on distributions that were fully known: five numbers specified the model, and probability answered every question we asked. The other two problem types, inference and machine learning, begin instead from data, because the distributions they need are unknown. Each is illustrated here briefly, together with its sources of uncertainty. The full treatments come in later chapters.

1.2.1 Inference

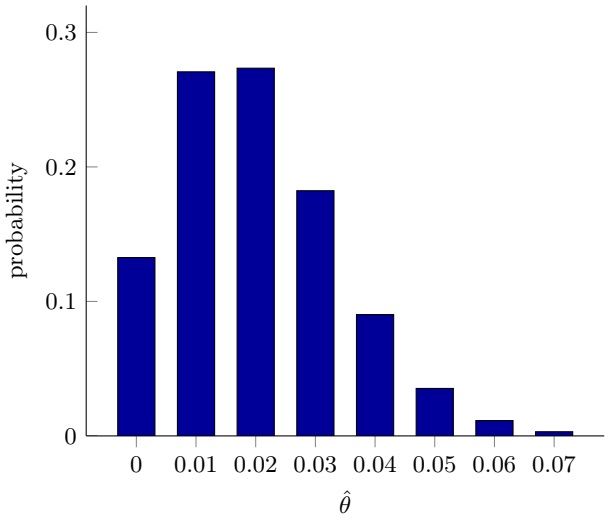
Inference is the process of using data to draw conclusions about the real world; for example, estimating an unknown quantity. Suppose that the probability that someone with a given allele of a gene will develop a certain disease is θ. We are interested in determining θ; an answer is a number for θ together with a statement of how sure we are about it. To find one, we collect data: among a sample of 100 people with this allele, 2 have the disease. How we approach this problem depends on whether we view θ as deterministic or random: the same data and the same question, but two different notions of what "unknown" means.

1.2.1.1 The frequentist view

In the **frequentist view**, θ is fixed and deterministic, but unknown. The frequentist framework does not prescribe a single way to estimate θ from data; many **estimators** are possible. Suppose we take the **sample mean** $\hat{\theta} = 2/100 = 0.02$ as our estimate. How accurate is this estimate? *We cannot know: judging it would require knowing the true value of* θ. What we can ask is how accurate the *estimator* is, and that depends on two things:

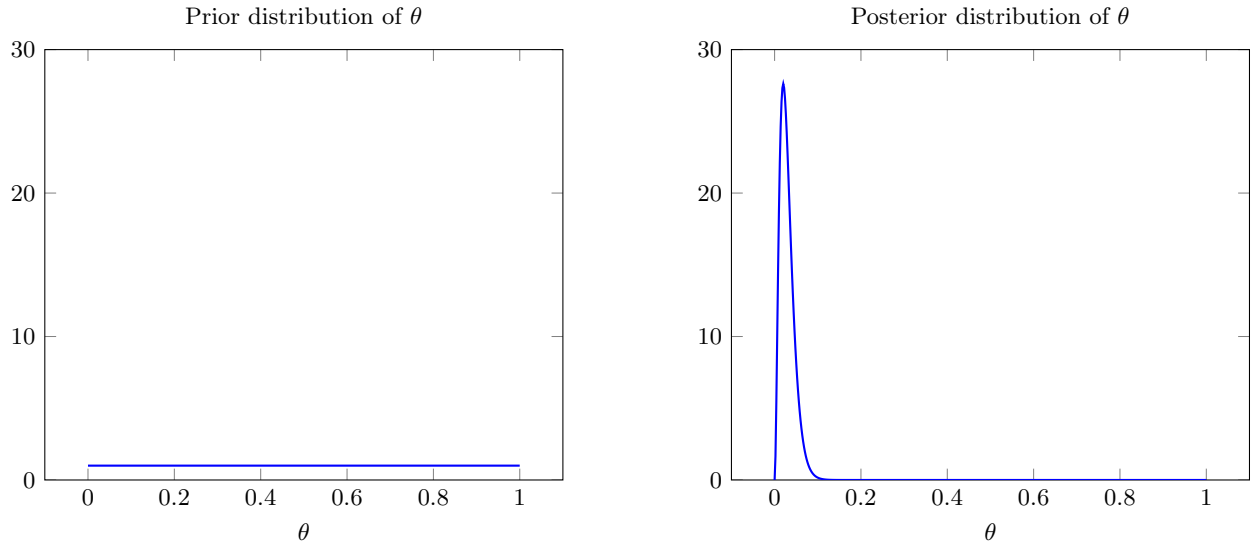
- how well a random sample represents the true value, and
- how well the method (here, the sample mean) extracts information from the sample.

Accuracy in the frequentist view is instead a property of the procedure, evaluated over all the samples we could have gotten. If θ were really 0.02, the number of diseased people in a sample of 100 would have the distribution $\text{Bin}(100, 0.02)$ and $\hat{\theta}$ would be that number divided by 100; the resulting distribution of $\hat{\theta}$ is shown below. The estimate equals 0.02 only about 27% of the time. Quantifying this spread, and resolving the apparent contradiction of assuming we know θ in order to study the estimator, are the subjects of the chapter on frequentist parameter estimation.



1.2.1.2 The Bayesian view

In the **Bayesian view**, θ is a random variable: we could be living in a universe where θ is 0.01, or 0.05, or any other value, and in many cases we will never find out which. Treating θ as a fixed number would then be a contradiction. We describe our initial belief about θ using a **prior distribution**, say a uniform distribution over $[0, 1]$, and update it using the two observed cases to obtain the **posterior distribution** $p(\theta \mid \text{data})$. How to calculate the posterior from the prior and the data is the subject of the chapter on Bayesian parameter estimation; for our data it is shown below. The posterior is peaked at 0.02, its mode, with mean 0.029, but still allows other values. It is the whole answer: a number for θ and how sure we are about it, in one distribution.



1.2.1.3 Source of uncertainty

In both views, the source of uncertainty is **finite sample size**. If we knew the disease status of 20 out of 1000 people with the allele, the same rate with ten times the data, both the distribution of the estimate $\hat{\theta}$ and the posterior distribution of θ would be more concentrated, with lower variance. Estimating θ from the sample and quantifying the remaining uncertainty are the subjects of the chapters on frequentist and Bayesian parameter estimation.

1.2.2 Supervised machine learning

Consider the house-price problem from the start of the chapter: predict the sale price of a house from features such as location, lot size, square footage, and number of bedrooms. More generally, supervised machine learning problems have the following components:

- **Data**: $\mathcal{D} = \{(x_1, y_1), \ldots, (x_n, y_n)\}, x_i \in \mathcal{X}, y_i \in \mathcal{Y}$. $\mathcal{X}$ is called the feature space, and $\mathcal{Y}$ is called the label space. As an example, each x_i could be a vector providing information about a house, e.g., (location, lot size, square footage, number of bedrooms, ...), and y_i can be the sale price of the house.
- **Goal**: Find the "best" function f to predict y corresponding to a given x. In other words, the function f produces an estimate $\hat{y} = f(x)$ of y given features x. Continuing our example, y would be the true but unknown price of the house with features x, and $f(x)$ would be a prediction (similar to what Zillow does).
- **Evaluation**: How do we define "*best*"? For a given data point (x, y), evaluate the success of f using a loss function $L(y, f(x))$, e.g., $L(y, f(x)) = |y - f(x)|$. Ideally, we would like to minimize the expected loss over all possible outcomes weighted by their probabilities, so we define

$$\mathcal{L}(f) = \mathbb{E}[L(Y, f(X))], \tag{1.1}$$

  also known as the **population risk**, where the expectation is over the distribution $p(x, y)$ of (X, Y).
- **Learning Algorithm**: The algorithm that tries to find

$$f^* = \arg\min_f \mathcal{L}(f) = \arg\min_f \mathbb{E}[L(Y, f(X))]. \tag{1.2}$$

If we have the joint distribution $p(x, y)$, we can solve this problem exactly, at least in principle (see Section 1.2.2.2). But what we actually have is the data set $\mathcal{D}$. We will discuss how to address this mismatch through empirical risk minimization or by estimating the unknown distribution $p(x, y)$ from $\mathcal{D}$ and then predicting with the estimate.

1.2.2.1 Source of uncertainty

Both sources of uncertainty appear in the house-price problem. The features do not determine the price: two houses identical on paper can sell for different amounts. There is **noise** from factors left out of the model. The relationship between features and prices must also be learned from a finite record of past sales, so finite sample size introduces uncertainty about that relationship. Section 1.3 makes this limitation concrete by showing how quickly finite data runs out when a language model is learned by counting.

1.2.2.2 What would knowing the distribution give us?

If the joint distribution $p(x, y)$ of the features X and the target Y were known exactly, the supervised-learning problem would be solved: the expected loss could be evaluated, and the optimal predictor f^* could be found without learning from a finite dataset. Two common problems in supervised learning illustrate this:

- **Regression**: $\mathcal{Y}$ consists of *scalars or vectors of reals*. For example, predicting stock price based on financial information, or determining the score someone will assign a movie based on previous scores. A common loss function is the **quadratic** or **squared error** loss function:

$$L(y, f(x)) = (y - f(x))^2. \tag{1.3}$$

  For this loss, *if the distribution is known*, the best predictor is

$$f^*(x) = \mathbb{E}[Y|X = x]. \tag{1.4}$$

  For a fixed x, $\mathbb{E}[(Y - c)^2 \mid X = x]$ is smallest at $c = \mathbb{E}[Y \mid X = x]$, so predicting the conditional mean minimizes the expected loss at every x, and hence overall.

- **Classification**: $\mathcal{Y}$ consists of *classes or categories*. For example, speech recognition, handwriting recognition, the presence or absence of a disease. A common loss function is the **0-1 loss**:

$$L(y, f(x)) = \begin{cases} 1, & \text{if } y \neq f(x). \\ 0, & \text{if } y = f(x). \end{cases} \tag{1.5}$$

  In this case, *if the distribution is known*, then the best classifier is

$$f^*(x) = \arg\max_{y \in \mathcal{Y}} p(y|x). \tag{1.6}$$

  Given x, the probability of error is $1 - p(f(x)|x)$, which is smallest when $f(x)$ is the most likely class.

We emphasize again that to solve the problem optimally as in (1.4) and (1.6), we need to know the joint distribution of (X, Y) or the conditional distribution of Y given X.

Knowing the joint buys one more thing: **generation**. We can sample a fresh pair (X, Y) from $p(x, y)$, or, given x, sample Y from $p(y|x)$, using the sampling methods of Chapter 0. In the context of **large language models**, discussed later, the task may be to generate a response y given a prompt x.

1.3 From Coin Flips to Language Models

A Bernoulli random variable is about as simple as a model gets: one number, the probability of heads, says everything there is to say about a biased coin. A month of daily weather asks for more, because a model of the whole record has to say how each day depends on the days before it. A model of English asks for much more again, since it must give a probability to every sentence anyone might write, and with a vocabulary of ten thousand words there are 10^{80} strings of twenty words to account for. All three are joint distributions over the quantities we care about. What separates them is how complex they are and how much structure we impose to keep the model small enough to learn from data.

The tools for imposing that structure are already in Chapter 0: the chain rule, which factors a joint distribution into conditional distributions, and conditional independence, which lets each factor forget most of what it conditions on. This section applies them to sequences and follows one construction the whole way, from a sequence of Bernoulli variables recording rain or no rain to the next-token models that large language models are built from.

1.3.1 Why the full joint distribution grows so quickly

Suppose we record the weather at one location for d days. Let $X_t = 1$ mean that it rains on day t, and let $X_t = 0$ mean that it does not. Our goal is to model the complete sequence

$$X_{1:d} = (X_1, \ldots, X_d),$$

not just the weather on one particular day. With such a model, we could assign a probability to an event such as a week of continuous rain, or we could generate possible weather sequences.

For three days, there are eight possible sequences:

$$000,\ 001,\ 010,\ 011,\ 100,\ 101,\ 110,\ 111.$$

In general, d binary observations have 2^d possible sequences. A table that gives a probability to every sequence therefore needs

$$2^d - 1 \tag{1.7}$$

free parameters. The subtraction by one appears because all the probabilities must sum to one. For a 30-day record, the number is

$$2^{30} - 1 = 1{,}073{,}741{,}823.$$

We would have trouble even filling such a table with useful estimates: an ordinary weather dataset would contain little or no information about most of the possible 30-day sequences.

1.3.2 The joint distribution in autoregressive form

The rain record showed how fast a distribution over a whole sequence grows. The chain rule rewrites such a distribution as a chain of next-step predictions.

That form matches how sequences are used. Words arrive one after another, and what we usually want from a model is the distribution of the next word given what has been written so far, whether we are generating text, scoring a sentence, or filling in a blank. A model of those next-step distributions is already a model of the whole sequence, with nothing lost.

It is also the form in which approximations can be made. A table over all sequences has one number for each of them and no structure to take advantage of. Once the distribution is written as a chain, every factor conditions on a history, and the history is where an assumption can be applied: keep the last few tokens and drop the rest, or summarize the history with a function that has a fixed number of parameters. Either one turns an unusable table into a model that can be estimated from a finite corpus, and both are developed in the rest of this section.

Text will be our running example, though the construction applies to any ordered data, including music, DNA, and a sequence of measurements. We write the vocabulary of possible tokens as $\mathcal{V}$ and its size as $V = |\mathcal{V}|$.

Let $W_1, W_2, \ldots, W_T$ be the random tokens in a sequence, and let $w_1, w_2, \ldots, w_T$ denote a particular sequence of values. The chain rule gives

$$p(w_1, \ldots, w_T) = p(w_1) \prod_{t=2}^{T} p(w_t \mid w_1, \ldots, w_{t-1}). \tag{1.8}$$

It is convenient to abbreviate $(w_1, \ldots, w_{t-1})$ as $w_{1:t-1}$, so the same factorization can be written as

$$p(w_{1:T}) = \prod_{t=1}^{T} p(w_t \mid w_{1:t-1}),$$

where the factor for $t = 1$ is $p(w_1)$. It says that we can describe the first token, then the second given the first, then the third given the first two, and so on. Equation (1.8) is an identity: it makes no independence assumption and discards no part of the history. For the same reason it does not by itself make the model any smaller. If every next-token distribution may depend on the entire history, the conditional tables together hold as many free parameters as the joint distribution they came from, which for the 30-day weather record is the same $2^{30} - 1$ numbers as before.

> ★ **Chain rule versus assumptions**
> The chain rule turns a joint distribution into a sequence of next-step distributions. We need additional assumptions before the number of parameters becomes smaller.

Definition 1.1 (Autoregressive sequence model)**.** An **autoregressive sequence model** specifies the joint distribution of a sequence through a left-to-right collection of conditional distributions, with one distribution $p(w_t \mid w_{1:t-1})$ for each next token. For a discrete vocabulary, each next-token distribution is categorical.

Thus a model that predicts only the next token is also a model of the entire sequence. The product of all of its next-token probabilities gives the joint probability of that sequence. Conversely, sampling once from each conditional distribution, from left to right, produces a sample from the joint distribution.

Sentences do not all have the same length, so it is useful to add a special end-of-sequence token EOS. If $w_{T+1} = \text{EOS}$, then

$$p(w_{1:T}, \text{EOS}) = \prod_{t=1}^{T+1} p(w_t \mid w_{1:t-1}). \tag{1.9}$$

The probability of ending is therefore treated in exactly the same way as the probability of producing any other token.[1] We will also use a special **beginning-of-sequence** token, written BOS. It is an artificial token placed before the first real token of every sequence: it tells the model that generation is starting. For example, the first prediction can be written $p(w_1 \mid \text{BOS})$. Thus possible next tokens belong to $\mathcal{V} \cup \{\text{EOS}\}$, while a context may also contain BOS.

Example 1.2 (A short sequence)**.** Suppose a model uses one-token contexts and assigns

$$p(\text{the} \mid \text{BOS}) = 0.3, \qquad p(\text{cat} \mid \text{the}) = 0.2, \qquad p(\text{EOS} \mid \text{cat}) = 0.5.$$

The probability that it produces "the cat" and then stops is

$$p(\text{the}, \text{cat}, \text{EOS} \mid \text{BOS}) = 0.3 \times 0.2 \times 0.5 = 0.03.$$

Each conditional uses exactly one preceding token. Their product is the probability of the whole sequence. △

1.3.3 Finite-context approximations

The exact factorization in (1.8) becomes difficult to use as the sequence grows. The conditional distribution at position t may depend on an arbitrarily long prefix, and a separate table row would be needed for every possible prefix. A finite-order model replaces the entire prefix by a fixed-length context.

Definition 1.3 (Order-k approximation)**.** An **order-k autoregressive sequence model**, also called a **kth-order Markov model**, uses only the preceding k tokens to predict the next token. It makes the finite-order Markov approximation

$$p(w_t \mid w_{1:t-1}) \approx p_k(w_t \mid w_{t-k:t-1}). \tag{1.10}$$

Here p denotes an unrestricted distribution over sequences, whereas p_k denotes the order-k model. Within that model, the finite-context relation is exact:

$$p_k(w_t \mid w_{1:t-1}) = p_k(w_t \mid w_{t-k:t-1}).$$

For an order-k model, we prepend k copies of BOS, so that the first few predictions also have a k-token context.

The context $w_{t-k:t-1}$ plays the role of the state. The cases $k = 0, 1, 2$ are especially helpful for understanding the assumption:

Order 0: The next token depends on no previous token: $p(w_t \mid w_{1:t-1}) \approx p_0(w_t)$. Before stopping, its next-token distribution ignores all previous tokens.

Order 1: The next token depends only on the previous token: $p(w_t \mid w_{1:t-1}) \approx p_1(w_t \mid w_{t-1})$. Once the previous token is known, the ones before it say nothing further about what comes next. This is the **Markov property**.

Order 2: The next token depends on the previous two tokens: $p(w_t \mid w_{1:t-1}) \approx p_2(w_t \mid w_{t-2}, w_{t-1})$.

We will use the same context-to-next-token distribution at every position in the sequence. In other words, the conditional probability does not depend on t, which makes the model a **time-homogeneous** Markov

[1] For variable-length generation, we assume that the procedure eventually produces EOS with probability one. Under this condition, the probabilities in (1.9) define a probability distribution over finite sequences. Otherwise, some probability mass could be assigned to sequences that never end.

chain whose state is the last k tokens. Its transition table has one row for each of the V^k contexts.

Finite context makes the model manageable, but its size still grows rapidly. Ignoring boundary symbols, an order-k model has as many as V^k contexts. For every context it must specify a categorical distribution over V possible next tokens. Such a distribution has $V-1$ free parameters: the probabilities of the V possible next tokens must sum to one, so once $V-1$ are chosen, the last is determined. Thus a complete conditional table has as many as

$$V^k(V-1) \tag{1.11}$$

free parameters. This count omits the initial or BOS row. For the binary order-1 rain model, it gives two transition parameters, and the probability of rain on the first day makes three numbers in all. Equivalently, the table contains on the order of V^{k+1} entries. If $V = 10{,}000$, even an order-2 table has up to 10^8 contexts and on the order of 10^{12} entries. Most of these contexts will not appear in a realistically sized corpus.

The approximation also determines which relationships the model can express. For example, in

"The cats that I saw yesterday at the park _____."

the correct choice "were" depends on "cats," which is many words earlier. Any context that does not reach back to "cats" must predict without that information. Increasing k can include more such relationships, but it also increases the number of contexts that must be learned.

1.3.4 Learning conditional tables from counts

Suppose we are given a corpus $\mathcal{D}$ of token sequences. For an order-k model, let $c \in (\mathcal{V} \cup \{\text{BOS}\})^k$ denote a context of length k, and define

$$N(c, w) = \text{the number of times token } w \text{ follows context } c \text{ in } \mathcal{D},$$
$$N(c) = \sum_{u \in \mathcal{V} \cup \{\text{EOS}\}} N(c, u).$$

Here $N(c)$ is the total number of observed continuations of c. For an observed context, define the **empirical conditional distribution**, or **relative-frequency estimate**, by

$$\hat{p}_k(w \mid c) = \frac{N(c, w)}{N(c)}. \tag{1.12}$$

This estimate is defined only for an observed context with $N(c) > 0$, and it assigns probability zero to a continuation with $N(c, w) = 0$.[2] The entries in every observed row form a probability distribution because

$$\sum_{w \in \mathcal{V} \cup \{\text{EOS}\}} \hat{p}_k(w \mid c) = \frac{\sum_{w \in \mathcal{V} \cup \{\text{EOS}\}} N(c, w)}{N(c)} = 1.$$

This is a count-based estimate constructed from the corpus. The next chapter will show that it is also the maximum-likelihood estimate for a categorical next-token distribution.

Example 1.4 (An order-two count table)**.** Consider a corpus consisting of the following four sentences:

the cat sat,	the cat slept,
the dog sat,	a cat sat.

For an order-2 model, pad each sentence with two BOS tokens and add one EOS token. For example, the first sequence becomes

$$(\text{BOS}, \text{BOS}, \text{the}, \text{cat}, \text{sat}, \text{EOS}).$$

[2] Practical count-based sequence models use additional methods to handle unseen contexts and continuations. Smoothing and backoff are not covered here.

Counting each length-two context and its following token gives

Context c	Nonzero counts $N(c, w)$	Estimated distribution $\hat{p}_2(w \mid c)$
(BOS, BOS)	the: 3, a: 1	the: 3/4, a: 1/4
(BOS, the)	cat: 2, dog: 1	cat: 2/3, dog: 1/3
(BOS, a)	cat: 1	cat: 1
(the, cat)	sat: 1, slept: 1	sat: 1/2, slept: 1/2
(the, dog)	sat: 1	sat: 1
(a, cat)	sat: 1	sat: 1
(cat, sat)	EOS: 2	EOS: 1
(cat, slept)	EOS: 1	EOS: 1
(dog, sat)	EOS: 1	EOS: 1

For instance, the context (the, cat) occurs twice. It is followed once by "sat" and once by "slept," so

$$\hat{p}_2(\text{sat} \mid \text{the}, \text{cat}) = \hat{p}_2(\text{slept} \mid \text{the}, \text{cat}) = \frac{1}{2}.$$

This entire model has been learned by counting and normalizing each context row. △

For each context, count its observed continuations and divide by the row total. The result is a categorical next-token distribution for that context.

1.3.5 Generating a sequence one token at a time

Once the conditional table is available, generation is a repeated categorical sampling procedure. For an order-k model:

1. Initialize the context with k copies of BOS.
2. Sample the next token w from $\hat{p}_k(\cdot \mid c)$.
3. If $w = \text{EOS}$, stop. Otherwise append w to the sequence.
4. Form the new context by discarding the oldest context token and appending w, and then return to step 2. For $k = 0$, the context remains empty.

Each draw can be carried out by the categorical inverse-CDF procedure from the probability review: divide $[0, 1)$ into intervals whose lengths are the row probabilities and use a uniform draw to select an interval.

Example 1.5 (Sampling from the table)**.** Use the order-2 model in Example 1.4. Begin with (BOS, BOS) and suppose the successive draws select "the," then "cat," then "slept." The contexts evolve as follows:

$$\begin{aligned} (\text{BOS}, \text{BOS}) &\longrightarrow \text{the}, \\ (\text{BOS}, \text{the}) &\longrightarrow \text{cat}, \\ (\text{the}, \text{cat}) &\longrightarrow \text{slept}, \\ (\text{cat}, \text{slept}) &\longrightarrow \text{EOS}. \end{aligned}$$

Conditioned on the initial BOS context, the probability of this particular generated sequence is the product of the selected probabilities:

$$\frac{3}{4} \times \frac{2}{3} \times \frac{1}{2} \times 1 = \frac{1}{4}.$$

Multiplying these conditional probabilities gives the probability of the whole generated sequence. This is the chain rule applied to the finite-context model. △

Sampling is different from always selecting the most probable next token. Always taking the largest-probability continuation produces a single deterministic path (apart from ties). Sampling preserves the alternatives and their probabilities, so repeated runs can produce different sequences.

Exercise 1.6 (Hand-sampling an order-two model)**.** Use the table in Example 1.4.

1. Starting from (BOS, BOS), suppose the first draw selects "a." Complete the generated sequence and find its probability.
2. Find the probability of generating "the dog sat" and then stopping.
3. Explain why the order-2 model cannot generate "a cat slept," even though every individual word occurs in the corpus.

Solution. After "a," every relevant row is deterministic:

$$(\text{BOS}, \text{a}) \to \text{cat}, \qquad (\text{a}, \text{cat}) \to \text{sat}, \qquad (\text{cat}, \text{sat}) \to \text{EOS}.$$

Thus the generated sentence is "a cat sat," with probability $\frac{1}{4} \times 1 \times 1 \times 1 = \frac{1}{4}$.

For "the dog sat," the probability is

$$\hat{p}_2(\text{the} \mid \text{BOS}, \text{BOS})\hat{p}_2(\text{dog} \mid \text{BOS}, \text{the})\hat{p}_2(\text{sat} \mid \text{the}, \text{dog})\hat{p}_2(\text{EOS} \mid \text{dog}, \text{sat}) = \frac{3}{4} \times \frac{1}{3} \times 1 \times 1 = \frac{1}{4}.$$

Finally, the only observed continuation of (a, cat) is "sat," so $\hat{p}_2(\text{slept} \mid \text{a}, \text{cat}) = 0$. Therefore the model assigns zero probability to "a cat slept."

△

1.3.6 Expressiveness versus data

The order k controls both what the model can represent and how much data it needs. These two effects pull in opposite directions.

If k is small, many distinct histories are mapped to the same context. This allows the model to pool counts and estimate each row from more observations, but it also forces histories with genuinely different next-token behavior to share a distribution. An order-0 word model, for example, can learn that "the" is common but cannot learn that "the cat" is more natural than "the the." A small-context model therefore tends to **underfit**: it ignores useful structure that is present in the sequence.

If k is large, the model can distinguish more histories. It is more expressive, but the V^k possible contexts divide a fixed corpus into many small groups. Most contexts may occur once or not at all. A context seen once has a row that assigns probability one to its single observed continuation, while an unseen context has no estimated row. As the context approaches the length of a training sequence, generation increasingly reproduces pieces of the corpus instead of combining evidence across related contexts. This is memorization caused by sparse data, not proof that the sequence is deterministic.

The small corpus above makes the distinction concrete. An order-1 model pools all occurrences of "cat," regardless of what precedes it. Since "cat" is followed twice by "sat" and once by "slept," it assigns

$$\hat{p}_1(\text{sat} \mid \text{cat}) = \frac{2}{3}, \qquad \hat{p}_1(\text{slept} \mid \text{cat}) = \frac{1}{3}.$$

It can therefore generate the new combination "a cat slept." The order-2 model makes the more specific distinction between (the, cat) and (a, cat); on this small dataset, that extra expressiveness also prevents it from making the new combination. Neither behavior is always correct. We need enough context to express important dependencies and enough observations to estimate the resulting conditional distributions.

Context size	Benefit	Cost with fixed data
Small k	More observations per context	Important history is ignored
Large k	More distinctions between histories	Sparse and unseen contexts

★ **Context must match data**
A larger context lets the model use more information from the preceding tokens. However, longer contexts occur less often in a fixed corpus, so their conditional probabilities may be estimated from very few observations. We must balance a model's ability to represent detailed patterns against the amount of data available to estimate those patterns.

1.3.7 Beyond a separate table for every context

The order-1 and order-2 examples leave us with an uncomfortable choice. The order-1 model combines many histories into the same short context, so it has more observations for each row but forgets useful information. The order-2 model remembers more, but learns the rows for (the, cat) and (a, cat) separately. What if we want to remember a richer context without treating every context as a completely separate case?

The count table cannot give us that. It keeps one row for every context, so its size is not something we choose: it is whatever the vocabulary and the context length make it. With $V = 10{,}000$, the count (1.11) is about 10^{12} free parameters at $k = 2$ and about 10^{24} at $k = 5$. A model whose number of parameters grows in this way, with the number of distinct contexts rather than being fixed in advance, is called **non-parametric**.

Storage is the smaller of the two problems. Each row is estimated from the occurrences of its own context and from nothing else, while a corpus of 10^9 tokens contains at most 10^9 contexts however many rows the table has. At $k = 5$, all but a vanishing fraction of the rows are therefore empty, and an empty row means the model has learned nothing whatsoever about what follows that context. Past a modest k, the table can no longer be learned at all, whatever the storage budget.

Instead of looking up a row in a table, we can use one function for all contexts. We give it a context c, and it returns the probabilities of the possible next tokens:

$$c \longmapsto f_\theta(c) = \boldsymbol{\pi}_\theta(c), \qquad p_\theta(W_t = w \mid c) = \pi_{\theta,w}(c). \tag{1.13}$$

Instead of storing an independent answer for every context, we use one adjustable function for all contexts. Here θ collects all the adjustable numbers in the function, and its length is chosen when the model is designed. A **parametric model** is a model controlled by such a fixed set of parameters. The important difference is how the parameters are used: a lookup table spends a separate row on every context, whereas f_θ answers every context with the same θ. As a result, what the model learns from one context can affect its predictions in another. The trade-off is that the function cannot choose every context row independently. It works well when patterns learned in one context are useful in another.

Example 1.7 (Next-step distributions)**.** The output of the function depends on the quantity we want to generate. For rain or no rain, it could return one number $q_\theta(c)$, interpreted as the probability of rain. For a token, it returns a list of probabilities $\boldsymbol{\pi}_\theta(c)$, one for each possible next token. In both cases the function produces a probability distribution for the next observation, rather than simply guessing one value. △

Two ideas make such a function possible. The first is to stop treating tokens as unrelated symbols. In a count table, the rows for (the, cat) and (a, kitten) have nothing to do with each other, because two symbols are either identical or unrelated and counts collected for one row never inform another. A parametric model instead represents each token by a vector of real numbers, called an **embedding**, so that a context becomes a collection of vectors rather than a collection of symbols. Vectors admit degrees of similarity: "cat" and "kitten" can sit close together, and a function that varies smoothly with its input then returns similar next-token distributions for the two contexts above. The embeddings are themselves part of θ and are learned along with everything else, so the model decides which tokens belong near each other.

The second idea is to build f_θ by composing many simple steps, each a linear map of its input followed by a fixed nonlinear function applied to every coordinate. This is a **neural network**, and we do not need its internal details here. The nonlinearity is what makes the composition worth anything, since composing

linear maps only produces another linear map; with it, a moderate number of parameters describes a rich family of functions of the context vectors. The size of θ is set by the widths of those maps and not by the number of contexts, so the context can be made longer without the parameter count following V^k. That is how a language model conditions on thousands of previous tokens:

Model	Context	Free parameters
Order-2 count table	2 tokens	about 10^{12}
Order-5 count table	5 tokens	about 10^{24}
Neural language model	thousands of tokens	10^9 to 10^{12}

The two tables assume $V = 10{,}000$, and the last row is the range spanned by the language models in use today. The neural model reaches back hundreds of times further than the order-5 table while holding far fewer parameters, because its parameters are shared across contexts rather than divided among them.

Changing how we obtain the next-token probabilities does not change the autoregressive model itself. We still have

$$p_\theta(w_{1:T}) = \prod_{t=1}^{T} p_\theta(w_t \mid w_{1:t-1}), \tag{1.14}$$

and generation still means obtaining a distribution, sampling a token, appending it to the context, and repeating.

One question remains: how should we choose θ? For the table, we counted the observed continuations and normalized each row. For a more general function, there may be no such direct formula. The next chapter develops the general answer: choose parameters that make the observed data probable. The connection to this section comes from

$$\log p_\theta(w_{1:T}) = \sum_{t=1}^{T} \log p_\theta(w_t \mid w_{1:t-1}).$$

The probability of a sequence is a product of next-token probabilities, so its log probability is a sum of next-token terms. We will make use of this fact when we study how to learn the parameters.

★ **What stays the same**
A table and a shared function give us different ways to obtain the next-token probabilities. The chain-rule factorization and the step-by-step generation procedure stay the same.

1.4 Information theory and machine learning

Information theory deals with quantifying information and the rules that govern its transmission, storage, and transformation from one form to another. It has applications in communications, data storage, machine learning, and biology. In machine learning it can be used to help better understand relationships between knowns and unknowns, design loss functions, and establish fundamental limits on how well we can do with a certain amount of data (regardless of the type of algorithm and computational resources).

1.4.1 Quantifying uncertainty

Let X be a Bernoulli random variable that is equal to 1 if it is raining in Seattle and 0 otherwise. Similarly, let Y indicate whether it's raining in Phoenix. How much information do X and Y provide us? Alternatively, before they are revealed, how uncertain are we about X and about Y? Can we measure the information content of a random variable, or equivalently, our uncertainty about it?

Let's look at specific outcomes for each variable:

Raining in Seattle ($X = 1$): This carries a fair amount of information, since rain in Seattle is almost a 50/50 proposition.

Not raining in Phoenix ($Y = 0$): This carries little information, since the outcome is expected and has high probability.

Raining in Phoenix ($Y = 1$): This carries a lot of information, since the outcome is unlikely and surprising.

So as a function of probability, the amount of information of a given statement decreases as the probability increases. If the probability of an outcome is p, what is a good function describing the amount of information we gain from learning that the outcome has occurred? A good choice is the *self-information* function $i(p) = \log \frac{1}{p}$, shown in Figure 1.1 when the base of the log is 2. Then the information content of the statement '$X = x_i$' is

$$i(p(x_i)) = \log \frac{1}{p(x_i)}.$$

And the amount of information *on average* for a random variable X that takes values in the set $\mathcal{X} = \{x_1, \ldots, x_m\}$ is the **entropy**,

$$H(X) = \mathbb{E}\left[\log \frac{1}{p(X)}\right] = \sum_{i=1}^{m} p(x_i) \log \frac{1}{p(x_i)}, \tag{1.15}$$

where for continuous RVs, the sum must be replaced with an integral. If the log is base 2, then the unit is a *bit*.

If there are m different possible outcomes, then

$$0 \leq H(X) \leq \log m,$$

with $H(X) = 0$ when the outcome is certain and $H(X) = \log m$ when all outcomes are equally likely. For example, a fair die has entropy $\log_2 6 \simeq 2.58$ bits, the maximum possible for six outcomes. †

Example 1.8 (Bounds on entropy)**.** Show that $0 \leq H(X) \leq \log m$ for a random variable with at most m possible values, and identify the distributions that attain each bound.

Solution. Write $p_i = p(x_i)$, with the convention that a term with $p_i = 0$ contributes nothing to either sum below.

Lower bound. Since $0 \leq p_i \leq 1$, each term $p_i \log \frac{1}{p_i}$ is non-negative, so $H(X) \geq 0$. The sum vanishes only if every term does, which happens only when each p_i is 0 or 1, that is, when X is deterministic.

Upper bound. Let u be the uniform distribution on $\mathcal{X}$, so that $u(x_i) = 1/m$. Then

$$\log m - H(X) = \sum_{i=1}^{m} p_i \log \frac{p_i}{u(x_i)} = D(p \,\|\, u) \geq 0,$$

where the last step is the non-negativity of relative entropy, (1.17) below, which holds with equality if and only if the two distributions coincide (Exercise 1.18). So $H(X) \leq \log m$, with equality exactly when $p = u$, that is, when X is uniform.

△

Exercise 1.9 (Binomial versus uniform)**.** Which has higher entropy: $X \sim \mathrm{Bin}(2, 1/2)$ or Y uniform on $\{0, 1, 2\}$? Compute both. △

Example 1.10 (Entropy of an English letter)**.** A letter is drawn at random from English text, over the 26 letters A–Z. How much entropy does it carry if every letter is equally likely? How much does it carry under the actual letter frequencies of English, listed in Table 1.1?

Solution. Equally likely letters give the largest possible value, $\log_2 26 \simeq 4.70$ bits. The real frequencies are far from uniform: E is more than a hundred times as common as Z. Weighting by actual probability gives

4.18 bits. Models that also use the dependence of each letter on the ones before it give lower values still: Shannon estimated the entropy of English at around 1 bit per character. Language models exploit exactly this predictability.

△

Example 1.11 (Entropy of a Bernoulli variable)**.** Let $X \sim \text{Ber}(p)$, so that $X = 1$ with probability p and $X = 0$ with probability $1 - p$. What is $H(X)$?

Solution. The sum in (1.15) has only two terms, so

$$H(X) = p \log \frac{1}{p} + (1-p) \log \frac{1}{1-p},$$

which depends on the distribution only through p.

△

This special case is common enough to have its own name and notation: the **binary entropy function**

$$H_b(p) = p \log \frac{1}{p} + (1-p) \log \frac{1}{1-p}$$

is the entropy of any experiment with two outcomes of probabilities p and $1 - p$. For example,

$$\begin{aligned}
H(\text{Fair coin}) &= H_b\left(\frac{1}{2}\right) = 1, \\
H(\text{6 on a die}) &= H_b\left(\frac{1}{6}\right) = 0.65, \\
H(\text{Rain/no rain in Seattle}) &= H_b\left(\frac{150}{365}\right) = 0.977, \\
H(\text{Rain/no rain in Phoenix}) &= H_b\left(\frac{33}{365}\right) = 0.438, \\
H(\text{Rain/no rain in the Sahara}) &= H_b\left(\frac{1}{365}\right) = 0.0273.
\end{aligned}$$

The plot for binary entropy is given in Figure 1.1. The maximum entropy is 1 bit. This makes sense since we can represent the outcome with 1 bit. Random variables with equal chances of 0 and 1 have the highest entropy (and maximum uncertainty). Those with predictable outcomes have lower entropies.

Example 1.12 (Alice and Bob's arrivals)**.** Take the arrival-time problem of Section 1.1.1, but with the arrivals tied more tightly to the traffic than the numbers used there, so that the information quantities are large enough to see:

$$\Pr(T_O) = 0.6, \quad \Pr(A_O|T_O) = 0.95, \quad \Pr(A_O|T_H) = 0.45, \quad \Pr(B_O|T_O) = 0.9, \quad \Pr(B_O|T_H) = 0.15,$$

with Alice and Bob still independent given the traffic. Then $\Pr(A_O) = 0.75$ and $\Pr(B_O) = 0.6$. How much information does each arrival carry?

Solution. Both are two-outcome experiments, so

$$H(A) = H_b(0.75) = 0.811 \text{ bits}, \qquad H(B) = H_b(0.6) = 0.971 \text{ bits}.$$

Bob's arrival is the closer of the two to the maximum of 1 bit: knowing the model, we are nearly as unsure about him as about a fair coin. Alice's arrival is more predictable and carries less information: her marginal

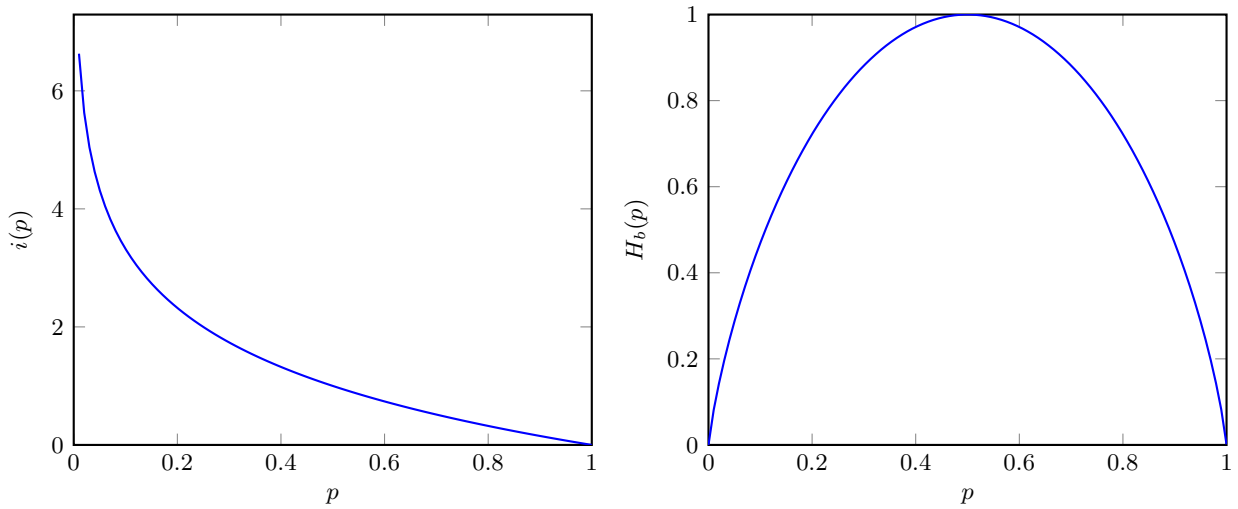


Figure 1.1: Self-information (left) for an event with probability p and binary entropy (right) for a Bernoulli RV with probability of success equal to p.

on-time probability is 0.75, farther from one half than Bob's 0.60. Note that this ranking is a statement about our uncertainty and not about who arrives on time more often.

△

Example 1.13 (Winning the lottery)**.** In a lottery, the jackpot is won by matching five distinct numbers chosen from 70 plus a bonus number chosen from 25. How much information does a ticket holder gain from each of the two outcomes, and what is the entropy of the win/lose variable?

Solution. A ticket wins with probability $1/\left(\binom{70}{5} \cdot 25\right) = 1/302{,}575{,}350$. Learning that a ticket won conveys $\log_2 302{,}575{,}350 \simeq 28.2$ bits; learning that it lost conveys $\simeq 4.8 \times 10^{-9}$ bits (you were almost sure this will happen). The entropy of the win/lose variable is accordingly minute, $H_b\big(1/302{,}575{,}350\big) \simeq 10^{-7}$ bits. This example shows that an outcome can be extremely surprising even when the variable carries almost no information on average.

△

Entropy was introduced by Shannon in his article "A mathematical theory of communication" in 1948, where he also popularized the term *bit* (binary digit).

> "My greatest concern was what to call it. I thought of calling it 'information,' but the word was overly used, so I decided to call it 'uncertainty.' When I discussed it with John von Neumann, he had a better idea. Von Neumann told me, 'You should call it entropy, for two reasons. In the first place your uncertainty function has been used in statistical mechanics under that name, so it already has a name. In the second place, and more important, no one really knows what entropy really is, so in a debate you will always have the advantage." – Claude Shannon, as recounted by M. Tribus and E. C. McIrvine, "Energy and Information," Scientific American 225 (September 1971), p. 180.

1.4.2 Data compression and the meaning of entropy

Entropy measures average uncertainty, but it also has a direct operational meaning: it provides a fundamental bound on how compactly we can represent information. Specifically, entropy bounds the performance of **lossless data compression**: no data compression method, regardless of computational complexity, can beat entropy.

Suppose we want to represent the outcomes of a random variable as binary strings. That is we want to construct a **source code**, consisting of **codewords**, where each codeword represents an outcome. We aim for a code that is as short as possible on average. The strategy is intuitive: assign short codewords to likely outcomes and long codewords to unlikely ones. The question is how well we can do.

Example 1.14 (Coding the alphabet)**.** English text is to be transmitted one letter at a time, over the 26 letters A–Z.

(a) Design a code that spends the same number of binary digits on every letter. How many digits does each letter need, and how many does the word "TEA" take?

(b) Letters are not equally common: E and T occur far more often than Q and Z. Design a code that exploits this, and encode "TEA" with it.

Solution. (a) Four binary digits distinguish only $2^4 = 16$ letters, so each letter needs $\lceil \log_2 26 \rceil = 5$ digits, and any assignment of the 26 letters to distinct 5-digit strings will do. Every letter costs the same 5 bits, so "TEA" takes $3 \times 5 = 15$ bits.

(b) Give the frequent letters the short codewords and the rare ones the long codewords. The best such code is produced by Huffman's algorithm. (But any code that follows the guideline of short codewords for frequent outcomes is fine for this example.) Table 1.1 gives the code it builds from the English letter frequencies.

Table 1.1: A Huffman code for the 26 letters, built from the English letter frequencies (%) shown beside each letter. Frequent letters get the short codewords.

	Codeword	Len	%		Codeword	Len	%		Codeword	Len	%
E	`100`	3	12.7	D	`11111`	5	4.3	P	`110001`	6	1.9
T	`000`	3	9.1	L	`11110`	5	4.0	B	`110000`	6	1.5
A	`1110`	4	8.2	C	`01000`	5	2.8	V	`001110`	6	1.0
O	`1101`	4	7.5	U	`01001`	5	2.8	K	`0011111`	7	0.8
I	`1011`	4	7.0	M	`00101`	5	2.4	X	`00111100`	8	0.2
N	`1010`	4	6.7	W	`00110`	5	2.4	J	`001111010`	9	0.2
S	`0111`	4	6.3	F	`00100`	5	2.2	Q	`0011110110`	10	0.1
H	`0110`	4	6.1	G	`110010`	6	2.0	Z	`0011110111`	10	0.1
R	`0101`	4	6.0	Y	`110011`	6	2.0				

With this code "TEA" is `000 100 1110`, that is 10 bits in place of the 15 of part (a). Averaged over the letter frequencies, the code spends 4.21 bits per letter rather than 5. Samuel Morse arrived at the same principle by hand in the 1830s. His telegraph code spells E as a single dot and saves the long sequences for rare letters such as Q and Z. However, his code is over three symbols: dot, dash, pause.

That 4.21 is already close to the entropy of the letter distribution, i.e., 4.18 bits. The rest of this subsection explains why that is no coincidence, and why no code can average less.

△

It turns out that the ideal code length for an outcome with probability p is $\log_2(1/p)$ bits, exactly the self-information introduced earlier. A likely outcome (p close to 1) needs a short codeword; a surprising outcome (p close to 0) can have a long one. Averaging over outcomes, the minimum achievable average code length is the entropy $H(X)$. This is **Shannon's source coding theorem**.

★ **Shannon's source coding theorem**

If we know the distribution of a source, we can compress close to the entropy limit by assigning roughly $\log_2(1/p)$ bits to an outcome with probability p: short codewords for likely outcomes, long ones for unlikely outcomes. No lossless code averages fewer than $H(X)$ bits per symbol, so nobody

can do better.

Exercise 1.15 (Optimal binary code)**.** A source emits four symbols A, B, C, D with probabilities 1/2, 1/4, 1/8, 1/8.

(a) Compute the entropy $H(X)$.

(b) Consider the binary code A $\to$ `0`, B $\to$ `10`, C $\to$ `110`, D $\to$ `111`. Verify that the average code length equals $H(X)$.

(c) Verify that each codeword length equals $\log_2(1/p)$ for the corresponding probability.

△

1.4.3 Relative entropy, or what if we get the distribution wrong?

The previous subsection showed that knowing the true distribution p lets us compress down to $H(X)$ bits per symbol. But what if we do not know p and instead design our code for a different distribution q?

Let X be a random variable with possible values $\mathcal{X}$ and distribution p, and let q be another distribution over $\mathcal{X}$. A code optimized for q assigns roughly $\log(1/q(x))$ bits to outcome x. When the data actually follow p, the average code length becomes the **cross-entropy**

$$H(p, q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{1}{q(x)}. \tag{1.16}$$

We also write $H(p)$ for the entropy of a distribution p; it is the same number as $H(X)$ whenever p is the distribution of X, and the same as $H(p, p)$. The arguments tell the uses of H apart: with distributions, $H(p, q)$ is the cross-entropy, and with random variables, $H(X, Y)$ is the joint entropy introduced below.

Example 1.16 (A code for DNA)**.** A DNA sequence is written over $\mathcal{X} = \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$, and the four bases occur with probabilities

$$p(\mathrm{A}) = \tfrac{1}{2}, \qquad p(\mathrm{C}) = \tfrac{1}{4}, \qquad p(\mathrm{G}) = p(\mathrm{T}) = \tfrac{1}{8}.$$

Not knowing this, we design our code for the uniform distribution q instead. How many bits per base does that code use, and how many would the best code use?

Solution. The best achievable average is the entropy of p,

$$H(X) = \tfrac{1}{2}\log_2 2 + \tfrac{1}{4}\log_2 4 + 2 \cdot \tfrac{1}{8}\log_2 8 = 1.75 \text{ bits per base.}$$

A code built for the uniform q spends $\log_2 4 = 2$ bits on every base regardless of which one it is, so by (1.16) its average length under the real data is

$$H(p, q) = \sum_{x \in \mathcal{X}} p(x) \log_2 4 = 2 \text{ bits per base.}$$

Ignoring the composition of the sequence therefore costs a quarter of a bit on every base.

△

Since the best possible average length is $H(p)$, the penalty for using the wrong distribution is

$$H(p, q) - H(p) = D(p \| q),$$

the *relative entropy*, or *Kullback–Leibler (KL) divergence*:

$$D(p\|q) = \sum_{x\in\mathcal{X}} p(x)\log\frac{p(x)}{q(x)}. \tag{1.17}$$

★ **Penalty for the wrong distribution**
The relative entropy $D(p\|q)$ is the number of extra bits per symbol we waste by compressing with q instead of the true distribution p, when codeword lengths are the ideal $\log(1/q(x))$. The larger this value, the worse q approximates p.

Example 1.17 (A code for DNA, continued)**.** Take p and q from Example 1.16. Compute $D(p\|q)$ directly from (1.17), and check it against the quarter of a bit that example found wasted.

Solution. Evaluating (1.17) with q uniform,

$$D(p\|q) = \tfrac{1}{2}\log_2\frac{1/2}{1/4} + \tfrac{1}{4}\log_2\frac{1/4}{1/4} + 2\cdot\tfrac{1}{8}\log_2\frac{1/8}{1/4} = \tfrac{1}{2} - \tfrac{1}{4} = 0.25 \text{ bits},$$

which is $H(p,q) - H(p) = 2 - 1.75$, the quarter of a bit expected. The term for A is positive and the terms for G and T are negative, but A is the common base and those two are rare, so the balance is a quarter of a bit lost.

△

The compression picture makes the key properties of relative entropy intuitive.

Non-negativity. No code can beat the entropy bound $H(p)$, so compressing with the wrong distribution q can never be *cheaper* than using the true p. Therefore $D(p\|q) = H(p,q) - H(p) \geq 0$, with equality if and only if $q = p$. †

Exercise 1.18 (Non-negativity of relative entropy)**.** Prove that $D(p\|q) \geq 0$ with equality if and only if $p = q$. *Hint:* use $\ln t \geq 1 - \frac{1}{t}$ for $t > 0$, with equality iff $t = 1$, applied to $t = p(x)/q(x)$. △

When relative entropy is large or infinite. If q assigns a tiny probability to an outcome that p considers common, the code designed for q wastes a huge number of bits on that outcome: $\log(1/q(x))$ is large when $q(x)$ is small. In the extreme case where $q(x) = 0$ for some x with $p(x) > 0$, the code for q has no codeword at all for an outcome that actually occurs; no finite number of bits suffices, and $D(p\|q) = \infty$. In words, if the approximate distribution considers an event impossible but the true distribution does not, no amount of coding can compensate for that mistake.

Example 1.19 (Infinite relative entropy)**.** Keep the DNA distribution p of Example 1.16, and let q assign 1/2 to A, 1/4 to C, 1/4 to G, and 0 to T. What is $D(p\|q)$?

Solution. It is infinite. Every base but T contributes a finite term, while T contributes $\frac{1}{8}\log\frac{1/8}{0} = \infty$, since $p(\mathrm{T}) = 1/8 > 0$ and $q(\mathrm{T}) = 0$. In coding terms, a code designed for q has no codeword at all for T, so a sequence containing one cannot be sent at any rate.

△

Asymmetry: choosing the direction. Relative entropy is not symmetric: $D(p\|q) \neq D(q\|p)$ in general.

Example 1.20 (Fair versus biased coin)**.** Let p be the distribution of a fair coin and q that of a coin with $\Pr(\text{heads}) = 0.9$. Compute both directions and compare them.

Solution. In bits,

$$D(p\|q) = \frac{1}{2}\log_2\frac{1/2}{0.9} + \frac{1}{2}\log_2\frac{1/2}{0.1} \simeq 0.74,$$
$$D(q\|p) = 0.9\log_2\frac{0.9}{1/2} + 0.1\log_2\frac{0.1}{1/2} \simeq 0.53,$$

so the two differ: relative entropy is not symmetric.

△

1.4.3.1 Relative entropy as a cost function

So far $D(p\|q)$ has measured the price of compressing with the wrong distribution. The same number measures how good an approximation q is to p: it is zero exactly when the two agree, it grows as they separate, and it is infinite when q rules out an outcome that p allows. That makes it a natural quantity to minimize when fitting a model q to data drawn from p, which is how it usually reaches machine learning.

The two directions also answer different compression questions. In $D(p\|q)$, we compress data from the true distribution p using a code designed for our estimate q; this is the "forward" direction used above. In $D(q\|p)$, we compress data from our model q using a code designed for the truth p. When does the reverse direction matter?

Suppose we are fitting an approximate distribution q to a true distribution p. Minimizing $D(p\|q)$ forces q to place probability mass wherever p does (otherwise the penalty is infinite), so q tends to cover all of p's support, sometimes spreading mass too broadly. Minimizing $D(q\|p)$ instead penalizes q for placing mass where p does not: if $q(x) > 0$ but $p(x) = 0$, then the term $q(x)\log(q(x)/p(x))$ is $+\infty$, so such a q is never chosen. So q concentrates on a high-probability region of p, potentially ignoring some modes. Both directions arise in machine learning: $D(p\|q)$ as the training objective for generative models via maximum likelihood, and $D(q\|p)$ as the objective in variational inference, where q is a tractable approximation to an intractable posterior p.

For a concrete case, let p be an equal mixture of two Gaussians,

$$p = \tfrac{1}{2}\,\mathcal{N}(-2, 0.7^2) + \tfrac{1}{2}\,\mathcal{N}(2, 0.7^2), \tag{1.18}$$

and let the model q be a single Gaussian $\mathcal{N}(\mu, \sigma^2)$, which cannot reproduce two modes. Minimizing $D(p\|q)$ gives the Gaussian with the same mean and variance as p, namely $\mu = 0$ and $\sigma = 2.12$; it stretches across both modes and puts its own peak at $x = 0$, where p has almost no mass at all. Minimizing $D(q\|p)$ gives $\mu = 1.98$ and $\sigma = 0.73$, a narrow Gaussian sitting on one mode and giving up the other; $\mu = -1.98$ is equivalent by symmetry. Figure 1.2 shows the two fits.

The two answers differ because of which distribution the average is taken under. $D(p\|q)$ averages under p, so it adds to the penalty wherever p has mass and q does not, because of the $\log(p(x)/q(x))$ term, and so q has to cover everything p does. $D(q\|p)$ averages under q, so q is charged only where it places mass itself, and abandoning a mode costs nothing. Both behaviors, and the wider family of divergences that interpolates between them, are analyzed in [2, Fig. 1]. In short, relative entropy penalizes the second distribution being smaller than the first, especially if that smaller value is close to 0. †

Example 1.21 (One Gaussian for two)**.** Confirm the two fits above: find the single Gaussian $q = \mathcal{N}(\mu, \sigma^2)$ that minimizes $D(p\|q)$ for the mixture p of (1.18), the one that minimizes $D(q\|p)$, and the value of each divergence at its minimum.

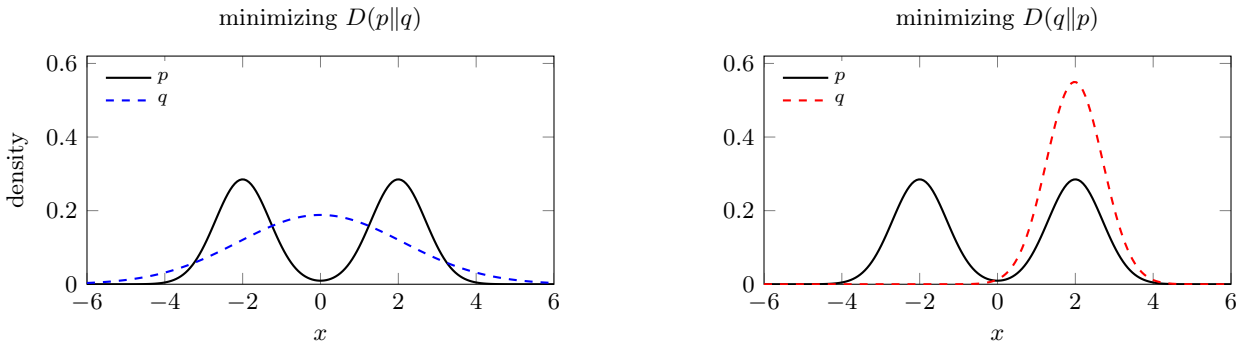


Figure 1.2: Fitting a single Gaussian q to the two-mode density p of (1.18). Minimizing $D(p\|q)$ (left) spreads q over both modes and peaks where p is near zero. Minimizing $D(q\|p)$ (right) puts q on one mode and gives up the other.

Solution. For a Gaussian model, minimizing $D(p\|q)$ matches the mean and the variance of p. Here p has mean 0 and variance $0.7^2 + 2^2 = 4.49$, so

$$\mu = 0, \qquad \sigma = 2.12, \qquad D(p\|q) = 0.61 \text{ bits}.$$

Minimizing $D(q\|p)$ has no closed form, but a numerical search gives

$$\mu = 1.98, \qquad \sigma = 0.73, \qquad D(q\|p) = 0.99 \text{ bits},$$

with $\mu = -1.98$ equivalent by symmetry. Centering at 0 is also a local minimum, with $\sigma = 1.85$ and $D(q\|p) = 1.05$ bits, but a worse one.

△

Since $D(p\|q) = H(p, q) - H(p)$, for a fixed p, minimizing cross-entropy over q is the same as minimizing the relative entropy, the form in which these measures typically appear as loss functions in machine learning.

Exercise 1.22 (Cross-entropy minimization)**.** Let $p = \text{Ber}(3/4)$ and $q = \text{Ber}(q_0)$. Write $H(p, q)$ as a function of q_0 and verify that it is minimized at $q_0 = 3/4$. △

1.4.4 Joint and Conditional Entropy; and Mutual Information

We can also measure the information in multiple random variables using entropy. The information in both X and Y is denoted $H(X, Y)$ and is defined as

$$H(X, Y) = \mathbb{E}\left[\log \frac{1}{p(X, Y)}\right] = \sum_{x\in\mathcal{X}} \sum_{y\in\mathcal{Y}} p(x, y) \log \frac{1}{p(x, y)}.$$

If we know Y, how much information is left in X? This is denoted $H(X|Y)$. If, for example $X = Y + 2$, then $H(X|Y) = 0$ since if we know Y, we also know X. Conditional entropy is defined as

$$H(X|Y) = \sum_{y\in\mathcal{Y}} p(y) H(X|Y = y) = \mathbb{E}\left[\log \frac{1}{p(X|Y)}\right] = H(X, Y) - H(Y).$$

Mutual information, $I(X; Y)$, represents the amount of information that one random variable has about the other, and is defined as

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X).$$

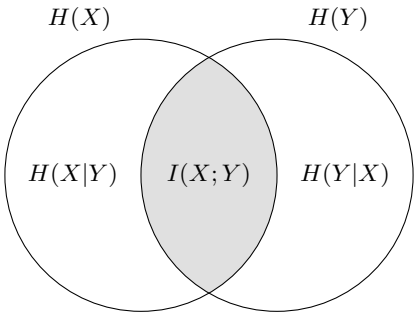


Figure 1.3: The information diagram for two random variables. The left circle is $H(X)$, the right circle is $H(Y)$, the shaded overlap is $I(X;Y)$, and the two circles together cover $H(X,Y)$.

Equivalently, it can be written as a divergence:

$$I(X;Y) = D(p_{X,Y} \,\|\, p_X p_Y).$$

For two random variables these quantities fit together in a diagram, Figure 1.3. Think of the area of the left circle as $H(X)$ and that of the right circle as $H(Y)$. Their overlap is the mutual information $I(X;Y)$, the part that is left of each circle is a conditional entropy, and the total area covered is the joint entropy $H(X,Y)$. Every identity above can be read off the picture: $H(X,Y) = H(Y) + H(X|Y)$ is the right circle plus the left crescent, and $I(X;Y) = H(X) - H(X|Y)$ removes the left crescent from the left circle. The picture also makes it clear that $I(X;Y) \le \min\{H(X), H(Y)\}$, since an overlap cannot be larger than either circle. It should be used as a mnemonic and not as a proof, and it does not extend cleanly to three or more variables.

Three consequences of these definitions are used below. The first is the **chain rule**, immediate from $H(X|Y) = H(X,Y) - H(Y)$:

$$H(X,Y) = H(Y) + H(X|Y), \qquad H(X,Y|Z) = H(Y|Z) + H(X|Y,Z),$$

where the second form is the same statement with every term conditioned on a third variable Z. The second is that $I(X;Y) \ge 0$, since it is a relative entropy and relative entropy is non-negative (Exercise 1.18); equivalently,

$$H(X|Y) \le H(X),$$

with equality exactly when X and Y are independent. Observing Y can only reduce our average uncertainty about X, although a particular outcome $Y = y$ may well increase it. The third is the same fact in the presence of a further variable, $H(X|Y,Z) \le H(X|Z)$: conditioning on more can only help.

Example 1.23 (Alice, Bob, and traffic)**.** Alice and Bob commute through the same traffic and, given the traffic, arrive independently:

$$\Pr(T_O) = 0.6, \quad \Pr(A_O|T_O) = 0.95, \quad \Pr(A_O|T_H) = 0.45, \quad \Pr(B_O|T_O) = 0.9, \quad \Pr(B_O|T_H) = 0.15,$$

the model of Example 1.12, in which $\Pr(A_O) = 0.75$ and $\Pr(B_O) = 0.6$. Find $H(A)$ and $H(B)$, the joint entropy $H(A,B)$, the conditional entropies $H(A|T)$, $H(B|T)$, $H(A|B)$ and $H(B|A)$, and the mutual informations $I(A;T)$, $I(B;T)$ and $I(A;B)$. How much does the traffic tell us about Alice, and how much does Bob tell us about her?

Solution. Each arrival has two outcomes, so $H(A) = H_b(0.75) = 0.811$ bits and $H(B) = H_b(0.6) = 0.971$ bits, as in Example 1.12.

Conditioning on the traffic first. Averaging the two conditional entropies over the traffic,

$$\begin{aligned} H(A|T) &= 0.6\, H_b(0.95) + 0.4\, H_b(0.45) \\ &= 0.6 \times 0.286 + 0.4 \times 0.993 = 0.569 \text{ bits}, \end{aligned}$$

which is below $H(A) = 0.811$, as conditioning requires. The difference is the mutual information,

$$I(A;T) = H(A) - H(A|T) = 0.811 - 0.569 = 0.242 \text{ bits}.$$

Notice how the average hides the two cases. Normal traffic leaves $H_b(0.95) = 0.286$ bits of uncertainty about Alice, far less than 0.811, but heavy traffic leaves $H_b(0.45) = 0.993$ bits, which is *more* than we started with: told that traffic is heavy, we know less about whether Alice will be on time than we did before. Only the average is guaranteed to drop.

The same calculation for Bob gives

$$\begin{aligned} H(B|T) &= 0.6\, H_b(0.9) + 0.4\, H_b(0.15) = 0.525 \text{ bits}, \\ I(B;T) &= 0.971 - 0.525 = 0.446 \text{ bits}. \end{aligned}$$

Traffic says nearly twice as much about Bob as about Alice. Alice's on-time probability changes from 0.95 to 0.45 across traffic conditions, while Bob's changes from 0.90 to 0.15.

For Alice and Bob we need their joint distribution. Summing over the traffic,

$$\Pr(A_O B_O) = 0.6(0.95)(0.9) + 0.4(0.45)(0.15) = 0.54,$$

and the other three entries follow the same way,

$$\Pr(A_O B_L) = 0.21, \qquad \Pr(A_L B_O) = 0.06, \qquad \Pr(A_L B_L) = 0.19,$$

giving $H(A,B) = 1.652$ bits. Separately the two arrivals carry $0.811 + 0.971 = 1.782$ bits; the joint is smaller because it counts what they share only once. The two conditional entropies follow,

$$\begin{aligned} H(A|B) &= H(A,B) - H(B) = 1.652 - 0.971 = 0.681 \text{ bits}, \\ H(B|A) &= H(A,B) - H(A) = 1.652 - 0.811 = 0.840 \text{ bits}, \end{aligned}$$

and either one gives the same mutual information,

$$I(A;B) = H(A) - H(A|B) = 0.811 - 0.681 = 0.13 \text{ bits}.$$

Bob's arrival does carry information about Alice's, as it must, since the two are not independent. But it carries about half of what the traffic does. Alice and Bob are related only through the traffic, so anything Bob says about Alice is something he says by way of T, and Section 1.4.5 makes this precise. Consistently with the assumption of conditional independence, once the traffic is known Bob says nothing more, $I(A;B|T) = 0$.

Figure 1.4 shows the two pairs as information diagrams.

△

1.4.5 Data processing and sufficient statistics

We rarely use data as it arrives. We average it, extract features from it, normalize it, drop the parts that look irrelevant, and pass it through the layers of a network. Every one of these steps replaces a variable by something computed from it. Can any of this increase what the data tells us about the quantity we care about?

The answer is no: Processing does not increase relevant information (although it has other benefits, clearly). This implies that if our input data is irrelevant, no amount of computation can do much. Precisely, for any



Figure 1.4: Information diagrams for Alice and the traffic (left) and Alice and Bob (right), in bits, drawn to scale: each circle's area is that variable's entropy, the shaded overlap is the mutual information, and the two circles together cover the joint entropy. The left crescent of each is the entropy of Alice's arrival given the other variable. The traffic overlaps Alice by almost twice as much as Bob does.

two random variables U and V and any function g,

$$I(g(V);U) \;\leq\; I(V;U), \tag{1.19}$$

which is called the **data processing inequality**. To see why, note that $g(V)$ is determined by V, so knowing V means knowing $g(V)$, and hence $H(U|V) = H(U|V, g(V)) \leq H(U|g(V))$. Subtracting each side from $H(U)$ gives (1.19). The inequality also holds when the processing uses randomness. If its output is Z, the required condition is that U, V, and Z form a **Markov chain**: once V is known, Z provides no further information about U, or equivalently

$$p(z|u,v) = p(z|v).$$

This relationship is denoted $U \to V \to Z$. Whatever Z learns about U, it learns through V, as a message passes along a chain of messengers. A deterministic output $Z = g(V)$ is the simplest example, since g sees nothing but V. In the game of telephone, the first player whispers a message U, the second player hears V and passes it on, and the third player hears Z. The third player hears only the second, so anything lost on the way to V is lost for good. In image editing, U is the scene, V is the photo of it, and Z is the edited version of the photo. Given the photo, the scene and the edited photo are independent.

In examples above V is produced from U and Z from V, and it is tempting to read the arrows as cause and effect. The arrows record only which variable stands in the middle. Let V be the full state of a patient's illness (which disease, and how severe), U their temperature, and Z their blood oxygen level. Assume that, conditional on the illness state, temperature and blood oxygen are independent. Then $U \to V \to Z$ holds, even though the illness drives both readings and neither drives the other. The condition is symmetric, and $Z \to V \to U$ is the same statement. The other orderings need not be Markov chains. $V \to U \to Z$ would mean that once the temperature is known, the illness says nothing more about the blood oxygen, and $U \to Z \to V$ that once the blood oxygen is known, the temperature says nothing more about the illness. Neither condition follows from the assumed independence given the illness state.

★ **Data processing inequality**
If U, V, and Z form a Markov chain $U \to V \to Z$, then

$$I(U;Z) \leq I(U;V).$$

For a learning problem, take $V = X$ to be the features, $U = Y$ the label, and $g = f$ a predictor. Whatever the predictor does with the features, all it ever reports about the label is its output $\hat{Y} = f(X)$, and by (1.19),

$$I(\hat{Y};Y) \leq I(X;Y).$$

So no transformation of the features, no choice of model, and no variation in training on the same data can make that output say more about the label than the features already said.

Example 1.24 (Alice, Bob, and traffic, continued)**.** In the model of Example 1.23, is $A \to T \to B$ a Markov chain? If so, what does the data processing inequality say about $I(A;B)$?

Solution. Once the traffic is known, Bob's arrival provides no further information about Alice's, so $A \to T \to B$ is a Markov chain. The data processing inequality therefore gives $I(A;B) \le I(A;T)$. The two numbers were 0.13 and 0.242 bits. Bob's arrival is a noisy reflection of the traffic, and it is the traffic that Alice actually depends on.

△

The same inequality answers a question in statistics. Take the Bayesian view of Section 1.2.1.2: the parameter is a random variable Θ with a prior distribution, and given $\Theta = \theta$ the sample $X = (X_1, \ldots, X_n)$ has distribution $p(x|\theta)$. Let $S(X)$ be a **statistic**, that is, any function of the sample, such as the sample mean, the sample variance, or the largest observation. Then (1.19) with $V = X$, $U = \Theta$, and $g = S$ gives

$$I(\Theta; S(X)) \le I(\Theta; X) :$$

so summarizing the data cannot increase what it says about the parameter. Usually a summary loses something. Which summaries lose nothing? Those for which the inequality also holds in the other direction, $I(\Theta;X) \le I(\Theta;S(X))$. By the data processing inequality once more, it does if $\Theta \to S(X) \to X$ is a Markov chain, that is, if X and Θ are independent given $S(X)$.

Definition 1.25 (Sufficient statistic)**.** A statistic $S(X)$ is **sufficient** for Θ if X and Θ are independent given $S(X)$, for any prior distribution of Θ.

Equivalently, $p(x|s,\theta) = p(x|s)$: once the summary is known, the parameter has no further effect on the sample. This is the form we usually check. For a sufficient statistic, then, the data processing inequality holds with equality,

$$I(\Theta; S(X)) = I(\Theta; X).$$

So we can keep a sufficient statistic in place of the data and lose no information about the parameter.

Example 1.26 (Estimating the probability of heads)**.** A biased coin comes up heads with probability Θ, unknown to us. Given $\Theta = \theta$, the n tosses $X_1, \ldots, X_n$ are independent and each $\mathrm{Ber}(\theta)$, and we want to estimate θ. Let $S = \sum_{i=1}^{n} X_i$ be the number of heads. Is S sufficient for Θ, or does the order of the tosses say something about the bias as well?

Solution. Given $\Theta = \theta$, any particular sequence with k heads has probability $\theta^k(1-\theta)^{n-k}$, and there are $\binom{n}{k}$ such sequences, so $\Pr[S = k \mid \Theta = \theta] = \binom{n}{k}\theta^k(1-\theta)^{n-k}$ and

$$\Pr[(X_1, \ldots, X_n) = (x_1, \ldots, x_n) \mid S = k, \Theta = \theta] = \frac{\theta^k(1-\theta)^{n-k}}{\binom{n}{k}\theta^k(1-\theta)^{n-k}} = \frac{1}{\binom{n}{k}}$$

whenever $\sum_i x_i = k$, and zero otherwise, for θ with $\Pr(S = k \mid \Theta = \theta) > 0$. The parameter has cancelled out: given the count, every arrangement of the heads is equally likely whatever θ is. So S is sufficient, and

$$I(\Theta; S) = I(\Theta; X_1, \ldots, X_n).$$

To estimate the bias we may throw the sequence away and keep the count; an estimator that also looked at the order of the tosses could not use it to learn anything about θ.

△

The sample proportion S/n is also the maximum-likelihood estimator of θ here, and it is a function of the count. Good estimators are usually functions of a sufficient statistic.

1.4.6 Fano's inequality and the limits of prediction

Mutual information tells us, in a single number, how much the features say about the label. Does that number tell us how accurately the label can be predicted? Section 1.4.5 showed that no algorithm can increase it, but it did not say what a given amount of information buys. Fano's inequality converts $I(X;Y)$, or equivalently the uncertainty $H(Y|X) = H(Y) - I(X;Y)$ left after the features are seen, into a lower bound on the error probability of every predictor. This is the third use of information theory mentioned at the start of the section, a limit on how well we can do regardless of the algorithm and the computational resources.

Let the label Y take one of $m \geq 2$ values in $\mathcal{Y}$. A predictor is any function $\hat{Y} = f(X)$. We assume nothing about how f was obtained, so it may be a lookup table, a decision tree, or a neural network of any size. Its **error probability** is

$$P_e = \Pr[\hat{Y} \neq Y].$$

Theorem 1.27 (Fano's inequality)**.** *For any predictor $\hat{Y} = f(X)$ of a label Y taking $m \geq 2$ values,*

$$H(Y|X) \;\leq\; H_b(P_e) + P_e \log(m-1). \tag{1.20}$$

†

Proof. Let E be the indicator of an error, $E = 1$ if $\hat{Y} \neq Y$ and $E = 0$ otherwise, so that $E \sim \mathrm{Ber}(P_e)$ and $H(E) = H_b(P_e)$. Expand $H(E, Y|\hat{Y})$ with the chain rule in two ways. Grouping Y first,

$$H(E, Y|\hat{Y}) = H(Y|\hat{Y}) + H(E|Y, \hat{Y}) = H(Y|\hat{Y}),$$

because E is determined by Y and $\hat{Y}$. Grouping E first,

$$H(E, Y|\hat{Y}) = H(E|\hat{Y}) + H(Y|E, \hat{Y}) \leq H(E) + H(Y|E, \hat{Y}),$$

since conditioning cannot increase entropy. For the last term,

$$H(Y|E, \hat{Y}) = (1 - P_e)\, H(Y|\hat{Y}, E = 0) + P_e\, H(Y|\hat{Y}, E = 1) \leq 0 + P_e \log(m-1),$$

since without an error $Y = \hat{Y}$ is known, and with an error Y is one of the $m - 1$ values other than $\hat{Y}$, so its entropy is at most $\log(m-1)$. Combining the two expansions gives $H(Y|\hat{Y}) \leq H_b(P_e) + P_e \log(m-1)$. Finally, $H(Y|X) \leq H(Y|\hat{Y})$, as in the proof of (1.19), and this gives (1.20). □

Both terms on the right of (1.20) are near zero when P_e is, so a predictor can be rarely wrong only if the features leave almost no uncertainty about the label. In the other direction, uncertainty that the features do not remove has to show up as errors.

Using $H_b(P_e) \leq 1$ and $\log_2(m-1) \leq \log_2 m$ gives a weaker but more readable form. In bits,

$$P_e \;\geq\; \frac{H(Y|X) - 1}{\log_2 m} \;=\; \frac{H(Y) - I(X;Y) - 1}{\log_2 m}. \tag{1.21}$$

The label carries $H(Y)$ bits of uncertainty and the features provide $I(X;Y)$ of them, so $H(Y|X)$ bits are left over, and all but one of these have to be paid for in errors. If $H(Y|X) < 1$, (1.21) says nothing, and (1.20) itself is needed.

For binary labels the second term of (1.20) vanishes, since $\log(m-1) = \log 1 = 0$, and Fano's inequality becomes

$$H_b(P_e) \;\geq\; H(Y|X). \tag{1.22}$$

Since H_b is increasing on $[0, 1/2]$, this says that P_e is at least the smaller of the two solutions of $H_b(t) = H(Y|X)$. For example, if the features leave half a bit of uncertainty about a binary label, then no predictor can be wrong less than about 11% of the time, because $H_b(0.11) \simeq 0.5$.

Example 1.28 (A ceiling on next-character prediction)**.** Example 1.10 quoted Shannon's estimate that English carries about 1 bit per character once the preceding text is taken into account. Taking that estimate at face value, how often can the next character be predicted correctly, at best?

Solution. Let Y be the next character and X the text before it, so that $H(Y|X) \simeq 1$ bit. Shannon's estimate is for an alphabet of $m = 27$ symbols, the 26 letters and a space. By (1.20), the error probability of any predictor satisfies

$$H_b(P_e) + P_e \log_2 26 \;\geq\; 1,$$

and the smallest P_e that meets this is 0.108. So no predictor, however large the model and however much text it was trained on, can guess the next character correctly more than about 89% of the time.

The ceiling depends on the entropy estimate alone. A better language model leaves it unchanged, and a better estimate of the entropy of English moves it. Shannon's own figure is a range. In "Prediction and Entropy of Printed English" (1951) he bounded the entropy of English between 0.6 and 1.3 bits per letter for a context of 100 letters, which puts the ceiling on accuracy between 94% and 85%. In one of Shannon's experiments, a subject performed below this ceiling, correctly predicting 89 of 129 symbols on the first attempt, or 69%. After each attempt, the correct symbol was known before the subject predicted the next one.

△

1.4.7 Fisher information: when the unknown is not random

Mutual information $I(X;Y)$ measures what X tells us about Y, and exists only because Y is a random variable. In the frequentist view of Section 1.2.1.1 the unknown is not random: the parameter θ^* is a fixed number that we happen not to know, and the data $\boldsymbol{X} \sim p(\boldsymbol{x}; \theta^*)$ are the only random objects in the problem. How much does the data tell us about θ? In the Bayesian view Θ has a distribution and $I(\Theta; \boldsymbol{X})$ is available, which is how Section 1.4.5 defined sufficiency. The frequentist view needs a different measure of information.

What makes θ^* easy or hard to determine is how much the distribution of the data changes as θ moves. If $p(\cdot\,;\theta)$ and $p(\cdot\,;\theta')$ are the same for two different values θ and θ', no procedure can tell the two apart, however much data it sees. If they are very similar, many observations are needed to tell θ from θ', and the closer the two distributions, the more it takes. If a small change in θ visibly reshapes the distribution, a modest sample is enough.

Relative entropy measures how distinguishable two distributions are. Fix θ and consider $D(p_\theta \| p_{\theta+\delta})$ as a function of δ, using natural logarithms, as is conventional here. It is zero at $\delta = 0$ and non-negative everywhere, so $\delta = 0$ is a minimum, its first derivative vanishes there, and its leading term is quadratic.[3] The **Fisher information** at θ is twice the coefficient of that quadratic term,

$$\begin{aligned} \mathcal{I}(\theta) &= \lim_{\delta \to 0} \frac{2}{\delta^2}\, D(p_\theta \,\|\, p_{\theta+\delta}), \\ \text{equivalently} \quad D(p_\theta \,\|\, p_{\theta+\delta}) &= \tfrac{1}{2}\, \mathcal{I}(\theta)\, \delta^2 + o(\delta^2) \end{aligned} \tag{1.23}$$

as $\delta \to 0$, written $\mathcal{I}$ to keep it apart from the mutual information $I(X;Y)$. So $\mathcal{I}(\theta)$ is the curvature of the divergence at its minimum: the narrower the valley around θ, the more a small change in the parameter changes the distribution of the data, and the more each observation tells us about θ. Figure 1.5 shows the valley for a coin.

[3] Provided $p(x;\theta)$ is smooth enough in θ for the expansion to hold. The uniform distribution on $[0, \theta]$ is the standard case where it is not.

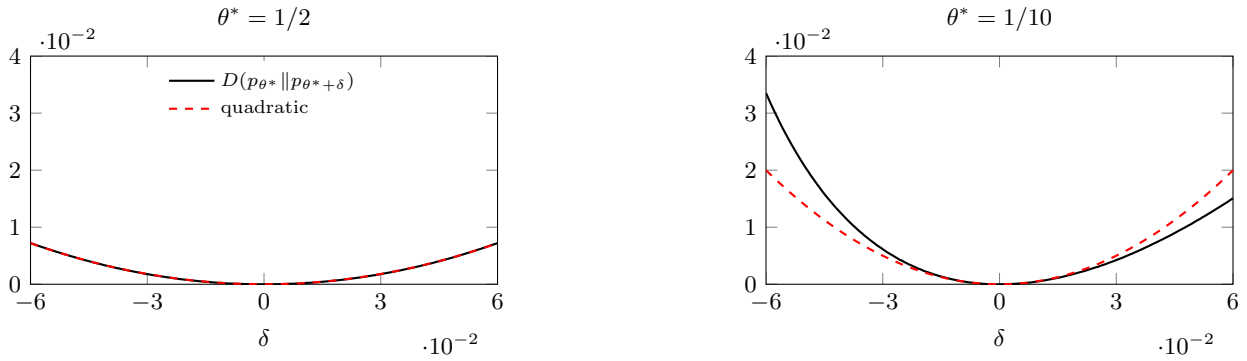


Figure 1.5: Relative entropy between two coins whose biases differ by δ, in nats, with the quadratic approximation (1.23) dashed. Both panels use the same scales. The valley is steeper around $\theta^* = 1/10$ than around $\theta^* = 1/2$, so the same change in bias is easier to detect in a lopsided coin, and its Fisher information is larger. The left panel is symmetric in δ and the right one is not, a reminder that (1.23) describes only the immediate neighborhood of θ^*.

In the chapter on parameter estimation, we will see how Fisher information can be used to bound the accuracy of estimators quantitatively.

While this overview is sufficient for our purposes in this course, if you are interested, you can check out the slides for this Short Lecture on Information Theory, or the course Mathematics of Information. The standard reference for the subject is Cover and Thomas [1].

References

[1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. 2nd. Wiley-Interscience, 2006.
[2] Thomas Minka. *Divergence measures and message passing*. Tech. rep. MSR-TR-2005-173. Microsoft Research, 2005.

Chapter 2

Frequentist Parameter Estimation

2.1 Overview

Parameter estimation can be used to infer unknowns about the real world (e.g, the frequency of a given disease among individuals with a certain genetic mutation) and to estimate the distribution of the data in machine learning problems.

There are two main frameworks for parameter estimation:

- Frequentist methods: In the frequentists' perspective, the true parameter value θ^* is unknown and fixed. The estimate $\hat{\theta}$ is a function of the data, which provides a single "best" estimate of θ^*. Frequentists have different methods for estimation including *maximum likelihood*, which we will discuss in detail, and the *moment method*, which finds the parameters by solving equations obtained by equating empirical moments and theoretical moments.
- Bayesian methods: Parameters are considered to be random and are treated as such. The Bayesian method provides a unified approach consisting of the following steps:
  1. Start with the prior distribution for the parameter
  2. Collect data
  3. Obtain posterior distribution by updating the prior distribution using data and Bayes' theorem

Let us consider a problem about *inferring* unknown values and making decisions and use probability to solve it, using both frequentist and Bayesian views. Suppose that the probability that someone with a given allele of a gene will develop a certain disease is θ. We are interested in determining θ. Different interpretations lead to different approaches to problems. But to determine θ, both frequentists and Bayesians need data.

Data ($\mathcal{D}$). Among a sample of 100 people with this allele, 2 had the disease.

- A Frequentist thinks of θ as unknown non-random parameter. She starts by asking "What is the probability of the observation as a function of θ?" We can view each of the 100 people chosen to be an independent Bernoulli trial with probability θ. So the distribution is Binomial and the probability of the observation as a function of θ is

$$L(\theta) = \binom{100}{2}\theta^2(1-\theta)^{98}.$$

  Probability of the observation as a function of the parameter is called the *likelihood function*. So what value for θ makes the most sense? Since the observation has actually happened, we would expect it to have a high probability so we find θ that maximizes the likelihood. This method is called *maximum*

likelihood estimation, and we'll discuss it in much more detail later. In this case, we estimate θ to be

$$\hat{\theta} = \arg\max_{\theta} L(\theta) = \frac{2}{100},$$

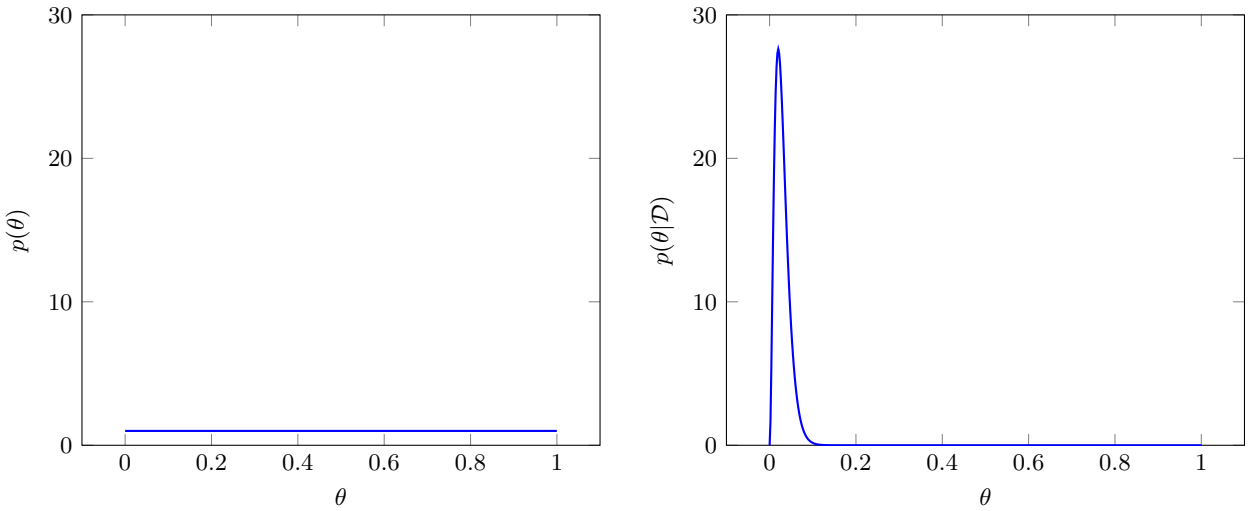
which is a reasonable estimate. But how close is the estimate to the true value? For frequentists, this is a tricky question to answer probabilistically since the true value and the estimate are both deterministic at this point. With some clever reasoning (some would say mental gymnastics), frequentists come up with *confidence intervals* and *confidence levels* to quantify the accuracy of estimators.

- A Bayesian thinks of θ as random and assigns to it a distribution, called the *prior*, before seeing the data. Thinking of θ as random is imaginative (some would say questionable) since there is no repeatable experiment and there is a single value that is true. One way to justify randomness of θ is to think of our universe being drawn from a set of possible universes. Regardless, the Bayesian view is used widely in practice.

  Our Bayesian statistician then looks at the data and updates her distribution for θ, thus obtaining the *posterior* distribution. Assume that before seeing the data, we believe that the distribution for θ is uniform, i.e., $p(\theta) \sim \text{Uni}[0,1] = \text{Beta}(1,1)$. This means that while we do not know what θ is, we believe it is equally likely to be any value between 0 and 1. When we see the data, we can update this belief,

  $$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})} \qquad \text{(Bayes' rule)}$$

  It turns out $p(\theta|\mathcal{D}) \sim \text{Beta}(3, 99)$.



In contrast to the frequentist view, the Bayesian view is consistent and flexible: any question about θ is answered by a computation with the posterior.

What is the source of uncertainty in this problem? It is the finite sample size. If we know the status of a very large number of people with the allele, we would know the distribution/ the value of θ.

2.2 Maximum likelihood estimation

Suppose data x is collected. We model this data as a realization of a random variable X with distribution p_X, which has an unknown parameter θ^*. The probability of observing x, assuming θ, is $p_X(x;\theta)$. To estimate θ^*, **Maximum likelihood estimation** (MLE) chooses the parameter that assigns the highest probability to the data:

$$\hat{\theta}_{\text{mle}} = \arg\max_{\theta} p_X(x;\theta).$$

The expression $p(x;\theta)$, viewed as a function of θ, is called the **likelihood**; hence the name maximum likelihood estimation. As shorthand, we use $L(\theta) = p_X(x;\theta)$ and $\ell(\theta) = \ln L(\theta)$, where $\ell(\theta)$ is the **log-likelihood**. Clearly, the value of θ that maximizes $L(\theta)$ is the same as the one that maximizes $\ell(\theta)$:

$$\hat{\theta}_{\text{mle}} = \arg\max_{\theta} \ell(\theta) = \arg\max_{\theta} \ln p_X(x;\theta)$$

Example 2.1 (Heavy or light traffic)**.** In this example, we attempt to show the intuition behind maximum likelihood. Suppose that a given road has heavy traffic or light traffic. We denote the probability of light traffic by θ^*. To estimate data, we count the number of times X that the road has light traffic in a period of 100 days. After collecting this data, we observe that $X = 65$. We have

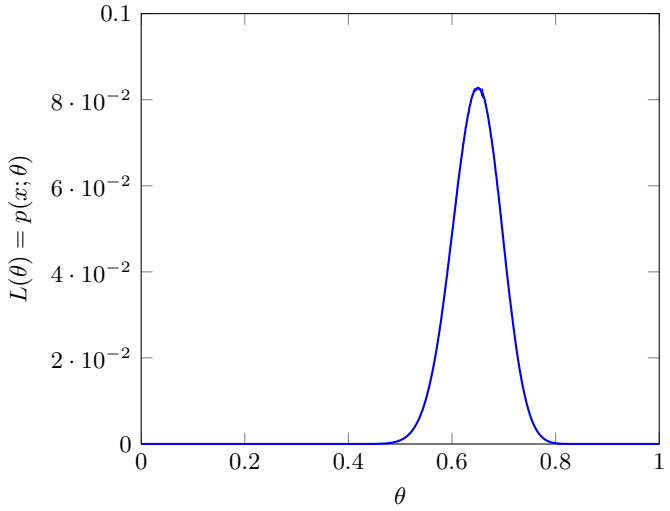
$$p_X(65;\theta) = \binom{100}{65}\theta^{65}(1-\theta)^{35}$$

Let's try a few different choices for θ, e.g., $\theta \in \{0.2, 0.4, 0.6, 0.8\}$, and see which one makes more sense:

$$\begin{aligned} p(65;\theta = 0.2) &= 1.6 \times 10^{-22}, \\ p(65;\theta = 0.4) &= 0.00000026, \\ p(65;\theta = 0.6) &= 0.0491, \\ p(65;\theta = 0.8) &= 0.00019, \end{aligned}$$

If $\theta = 0.2$, the probability of 65 days with light traffic is extremely small. So observing $x = 65$ would be very unlikely, which in turn would make $\theta = 0.2$ an unreasonable guess. Among the presented choices, $\theta = 0.6$ appears the most reasonable. This reasoning suggests the following: *The value of the parameter that assigns a higher probability to the observation is a better choice.*

Since we are not limited to a specific set of choices, we can find the parameter that *maximizes* the probability of the observation. In the figure below, $L(\theta) = p(x;\theta)$ is plotted as a function of θ. This is the likelihood function.



We can see that $\theta = 0.65$ maximizes the likelihood and hence is the maximum-likelihood estimate. We can also show this analytically. First, the likelihood is given as

$$L(\theta) = p(x;\theta) = \binom{100}{65}\theta^{65}(1-\theta)^{35}.$$

We usually use the log-likelihood as the function to optimize:

$$\ell(\theta) = \log L(\theta) = \log\left(\binom{100}{65}\theta^{65}(1-\theta)^{35}\right) \doteq 65\log\theta + 35\log(1-\theta), \tag{2.1}$$

where $\doteq$ denotes equality but with ignoring additive terms that are constant in θ (and thus do not alter the value of θ that maximize the log-likelihood). We differentiate $\ell(\theta)$ to find the value of θ that maximizes $l(\theta)$.

$$\frac{d\ell(\theta)}{d\theta} = \frac{65}{\theta} - \frac{35}{1-\theta} = 0 \implies 65 - 65\theta = 35\theta \implies \hat{\theta}_{\text{mle}} = \frac{65}{100}. \tag{2.2}$$

Note that this result is intuitive as it agrees with our observation that 65% of the days had light traffic. △

A note on notation. We indicate the parameter usually as an extra argument of the probability function, e.g., $p(x;\theta)$, separated by a semicolon to indicate that it is a parameter not a realization of a random variable. Furthermore, in general, our data is a vector, which we denote by bold symbols such as $\boldsymbol{x}$. The corresponding random variable is $\boldsymbol{X}$.

2.2.1 MLE with a finite parameter space

In many problems the parameter can only take values in a finite set $\Theta = \{\theta_1, \ldots, \theta_m\}$. In this setting, maximum likelihood is conceptually simple: evaluate the likelihood for each candidate and pick the one with the largest value

$$\hat{\theta}_{\text{mle}} = \arg\max_{\theta\in\Theta} L(\theta) = \arg\max_{\theta\in\Theta} \ell(\theta).$$

Example 2.2 (N-sided die with finite candidates)**.** An N-sided die is rolled once and we observe $X = x$. The parameter $N \in \Theta = \{4, 6, 8, 12, 20\}$. For each possible value of x, let us find the maximum-likelihood estimate of N.

Solution. Under the model of a fair die,

$$p(x;N) = \begin{cases} \frac{1}{N}, & 1 \le x \le N, \\ 0, & \text{otherwise.} \end{cases}$$

Hence the likelihood is zero for any $N < x$, and equals $1/N$ for $N \ge x$, so MLE selects the smallest candidate N that is still feasible:

$$\hat{N}_{\text{mle}} = \min\{N \in \Theta : N \ge x\}.$$

For example, if $x = 7$ then $\hat{N}_{\text{mle}} = 8$ and if $x = 1$ then $\hat{N}_{\text{mle}} = 4$.

△

Example 2.3 (Fair vs. bent coin)**.** The coin is either fair or bent with $p(H) = 0.6$, so $\Theta = \{0.5, 0.6\}$. Consider two experiments.

Solution.

- One flip ($n = 1$): If $X = H$, compare 0.6 vs. 0.5 and choose $\hat{\theta}_{\text{mle}} = 0.6$; if $X = T$, compare 0.4 vs. 0.5 and choose $\hat{\theta}_{\text{mle}} = 0.5$.
- Ten flips ($n = 10$): Let k be the number of heads. Compare the log-likelihoods using an annotated comparison:

  $$\ell(\theta{=}0.6) \underset{\theta=0.5}{\overset{\theta=0.6}{\lessgtr}} \ell(\theta{=}0.5) \iff k\log\tfrac{0.6}{0.5} + (10-k)\log\tfrac{0.4}{0.5} \underset{\theta=0.5}{\overset{\theta=0.6}{\lessgtr}} 0$$

  Equivalently,

  $$k \underset{\theta=0.5}{\overset{\theta=0.6}{\gtrless}} \frac{10\log(5/4)}{\log(3/2)} \approx 5.5,$$

so we choose the bent coin if $k \geq 6$; otherwise the fair coin.

△

We note that MLE for finite parameter spaces is closely related to hypothesis testing, which we will not study in depth. Additional concepts in hypothesis testing include likelihood ratio, type I and type II errors, trade-off between these errors, and the Neyman-Pearson lemma.

2.2.2 MLE with a continuous parameter space

Example 2.4 (Parameters of the normal distribution)**.** A device for measuring an unknown quantity μ^* (e.g., the mass of an electron) is used n times producing values $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$. Each measurement is independent and for each i we have $Y_i = \mu^* + Z_i$, where Z_i is the measurement noise satisfying $Z_i \sim \mathcal{N}(0, (\sigma^*)^2)$. Note that this implies $Y_i \sim \mathcal{N}(\mu^*, (\sigma^*)^2)$.

Suppose we have collected data $\boldsymbol{y} = (y_1, \ldots, y_n)$. We consider the problem in two cases: μ^* is unknown but σ^* is known; and both μ^* and σ^* are unknown.

Solution.

- Known σ^*, unknown μ^*: We have

$$p_{Y_i}(y_i; \mu) = \frac{1}{\sigma^* \sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{y_i - \mu}{\sigma^*} \right)^2 \right)$$

$$L(\mu) = p_{\boldsymbol{Y}}(\boldsymbol{y}; \mu) = \prod_{i=1}^{n} p_{Y_i}(y_i; \mu)$$

$$\ell(\mu) = \sum_{i=1}^{n} \ln p_{Y_i}(y_i; \mu) = \sum_{i=1}^{n} \left(-\ln(\sigma^* \sqrt{2\pi}) - \frac{1}{2} \left(\frac{y_i - \mu}{\sigma^*} \right)^2 \right) \doteq -\frac{1}{2} \sum_{i=1}^{n} \left(\frac{y_i - \mu}{\sigma^*} \right)^2$$

  and so

$$\frac{d\ell}{d\mu} = \sum_{i=1}^{n} \frac{y_i - \mu}{\sigma^*} = 0 \implies \hat{\mu}_{\text{mle}} = \frac{1}{n} \sum_{i=1}^{n} y_i = \bar{y}.$$

- Unknown σ^*, μ^*: We have

$$\ell(\mu, \sigma) = \sum_{i=1}^{n} \left(-\ln(\sigma \sqrt{2\pi}) - \frac{1}{2} \left(\frac{y_i - \mu}{\sigma} \right)^2 \right) \doteq -n \ln \sigma - \frac{1}{2} \sum_{i=1}^{n} \left(\frac{y_i - \mu}{\sigma} \right)^2$$

  and so

$$\frac{\partial \ell}{\partial \mu} = \sum_{i=1}^{n} \frac{y_i - \mu}{\sigma} = 0,$$

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \sum_{i=1}^{n} \frac{(y_i - \mu)^2}{\sigma^3} = 0.$$

Solving this system of equations for μ and σ yields

$$\hat{\mu}_{\text{mle}} = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{y},$$
$$\hat{\sigma}^2_{\text{mle}} = \frac{1}{n}\sum_{i=1}^{n} (y_i - \bar{y})^2.$$

△

2.2.3 Maximum likelihood and the closest distribution

We have described maximum likelihood as aiming to find a distribution that gives a high probability to the observed data. An alternative view relates it to the empirical distribution of the data, denoted $p_{\boldsymbol{x}}$. Given $\boldsymbol{x} = \{x_1, \ldots, x_n\}$, let $\#_x$ denote the number of times x appears in $\boldsymbol{x}$. The empirical distribution is given as

$$p_{\boldsymbol{x}}(x) = \frac{\#_x}{n} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(\boldsymbol{x} = \boldsymbol{x}_i)$$

where $\mathbb{1}(\cdot)$ equals 1 if the enclosed condition is true and 0 otherwise.

Now consider a parameterized family of distributions p_θ. It makes sense to choose θ such that p_θ is close to $p_{\boldsymbol{x}}$. In other words, we would like θ to be chosen such that p_θ describes the observed data well. A standard way of measuring the "closeness" of p_θ to the empirical distribution $p_{\boldsymbol{x}}$ is relative entropy (reviewed in Chapter 1), $D(p_{\boldsymbol{x}}||p_\theta)$.

It turns out the closest distribution is in fact given by maximum likelihood, i.e.,

$$\hat{\theta}_{\text{mle}} = \arg\min_\theta D(p_{\boldsymbol{x}}||p_\theta). \tag{2.3}$$

This fact provide further evidence for the soundness of MLE strategy. Note in particular that if there exists θ such that $p_\theta = p_{\boldsymbol{x}}$, it will be chosen by MLE. This is because relative entropy is always non-negative and equals to 0 if and only if the two distributions are the same. So choosing $p_\theta = p_{\boldsymbol{x}}$, if possible, provides the smallest value for the relative entropy, i.e., 0.

Exercise 2.5 (Likelihood and relative entropy)**.** Prove (2.3). △

†

Exercise 2.6 (The other direction)**.** Note that relative entropy is not symmetric. Instead of $D(p_{\boldsymbol{x}}||p_\theta)$, we could minimize $D(p_\theta||p_{\boldsymbol{x}})$. What are the differences between the two formulations and which one is more suitable for parameter estimation? △

Plug-in prediction. The estimated distribution is directly useful for prediction. In a supervised problem with features X and target Y, model the joint distribution as $p_\theta(x, y)$, estimate $\hat{\theta}$ by maximum likelihood, and then predict by treating $p_{\hat{\theta}}$ as the truth: for squared loss, $\hat{f}(x) = \mathbb{E}_{\hat{\theta}}[Y \mid X = x]$, and for 0-1 loss, $\hat{f}(x) = \arg\max_y p_{\hat{\theta}}(y \mid x)$, the optimal rules of Chapter 1, computed under the estimated distribution. This *plug-in* approach is one of the two main routes from data to predictions, the other being empirical risk minimization (see the linear regression chapter).

2.3 Properties of estimators

Maximum likelihood is just one way of estimating parameters. We can choose any function of the data as the estimate. For instance, in Example 2.4, we could choose the middle (median) value among $y_1, \ldots, y_n$ as the estimate for μ^*. Given the fact that there are many estimators, how do we evaluate them and select one?

Clearly, we would like the estimate to be close to the true value. But stating this condition in a rigorous probabilistic way is a bit challenging in the frequentist framework. We are specifically interested in the error:

$$\hat{\theta}(\boldsymbol{x}) - \theta^*,$$

where $\hat{\theta}(\boldsymbol{x})$ is the estimate based on data $\boldsymbol{x}$ and θ^* is the true value[1]. Evaluating $\hat{\theta}(\boldsymbol{x})$ is difficult because, obviously, the true value is unknown.

So instead of finding the specific error, we may try to find the probability that the true value θ^* is within say 10% of the estimate $\hat{\theta}$. But after the estimate is produced based on a given data set, the estimate is a deterministic value. For instance, in Example 2.1, the MLE is given as $\hat{\theta}_{\text{mle}} = 0.65$. So questions such as "What is the probability that the difference between θ^* and $\hat{\theta}(\boldsymbol{x})$ is larger than 0.05?" are not meaningful because, while θ^* is unknown, both θ^* and $\hat{\theta}(\boldsymbol{x})$ are deterministic after data is collected and the estimation task is performed.

The solution to these difficulties is to study the properties of the estimator not based on a specific realization $\boldsymbol{x}$ of the data but in general, over all possible data sets that could be produced and all the resulting estimated values. We can think of the thought experiment in which many, many, data sets are collected and the estimation task is performed based on each. The estimate itself is a random variable because each time we perform the estimation task, new data samples are obtained and these are random, following a certain distribution. In other words, instead of considering a single estimate $\hat{\theta}(\boldsymbol{x})$ for a specific realization $\boldsymbol{x}$, we study the estimator $\hat{\theta}(\boldsymbol{X})$, i.e., a random variable. Then it makes sense to ask "What is the probability that the difference between θ^* and $\hat{\theta}(\boldsymbol{X})$ is larger than 0.05?" since $\hat{\theta}(\boldsymbol{X})$ is a random variable with some distribution. It may be difficult to find the distribution of $\hat{\theta}(\boldsymbol{x})$ and it may depend on the unknown parameter θ^* but at least the question is meaningful. In this section, we will see some of the evaluation criteria based on this view.

A note on notation. Typically, we use θ as the generic parameter, with θ^* denoting its true value, according to which $\boldsymbol{X}$ is distributed. For a given data $\boldsymbol{x}$, the estimate is shown by $\hat{\theta}(\boldsymbol{x})$ or $\hat{\theta}$. So, $\hat{\theta}$ denotes both the estimator, i.e., a function that produces the estimate given the data, and the estimate; the intent should be clear from the context. Finally, we may use $\hat{\Theta} = \hat{\theta}(\boldsymbol{X})$ to denote the estimate as a random variable.

2.3.1 Bias

Bias is the expected estimation error,

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}(\boldsymbol{X}) - \theta^*] = \mathbb{E}[\hat{\theta}(\boldsymbol{X})] - \theta^* \tag{2.4}$$

As discussed, the expected value is taken over the randomness in $\boldsymbol{X}$. Bias of the estimator tells us whether in general the estimator over- or under-estimates the true value. If bias is equal to 0, then the estimator is called **unbiased**.

Example 2.7 (Heavy or light traffic, continued)**.** Previously, we obtained the maximum likelihood estimate for the probability θ of having light traffic. Let us find its bias. Again we collect data over 100 days and let X denote the number of days when there is light traffic. We know that $\hat{\theta}_{\text{mle}}$, as a function of data, is given by

$$\hat{\theta}_{\text{mle}}(X) = \frac{X}{100},$$

Note that instead of using a specific value for the number of days with light traffic, such as 65, we use a random variable X representing this quantity. Dropping the dependence on X for simplicity, the expected value of $\hat{\theta}_{\text{mle}}$ is given by

$$\mathbb{E}\Big[\hat{\theta}_{\text{mle}}(\boldsymbol{X})\Big] = \frac{\mathbb{E}[X]}{100}.$$

[1]Note the slight abuse of notation: sometimes θ is used as the generic parameter, e.g., as the argument of the likelihood function, and sometimes as the true value of the parameter. The distinction should be clear from the context

Assuming θ^* to be the true value, the number X of days when there is light traffic follows $\text{Bin}(100, \theta^*)$, and so $\mathbb{E}[X] = 100\theta^*$. It follows that

$$\mathbb{E}\Big[\hat{\theta}_{\text{mle}}(\boldsymbol{X})\Big] = \frac{100\theta^*}{100} = \theta^*.$$

Hence, the maximum likelihood estimate is an unbiased estimator. △

Example 2.8 (Three estimators of a mean)**.** Given iid data $\boldsymbol{y} = (y_1, \ldots, y_n), n \geq 3$, with mean θ^*, let us find the bias of each of the following estimators,

$$\begin{aligned}
\hat{\theta}_1(\boldsymbol{y}) &= \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \\
\hat{\theta}_2(\boldsymbol{y}) &= y_1, \\
\hat{\theta}_3(\boldsymbol{y}) &= \frac{2y_2 + y_3}{3}.
\end{aligned}$$

Let Y_i be the random variable corresponding to observation y_i and $\bar{Y} = \sum_{i=1}^{n} Y$. We have

$$\begin{aligned}
\mathbb{E}\,\hat{\theta}_1(\boldsymbol{Y}) &= \mathbb{E}\,\bar{Y} = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\,Y_i = \frac{1}{n}\sum_{i=1}^{n}\theta^* = \theta^*, \\
\mathbb{E}\,\hat{\theta}_2(\boldsymbol{Y}) &= \mathbb{E}\,Y_1 = \theta^*, \\
\mathbb{E}\,\hat{\theta}_3(\boldsymbol{Y}) &= \mathbb{E}\left[\frac{2Y_2 + Y_3}{3}\right] = \frac{2\,\mathbb{E}\,Y_2 + \mathbb{E}\,Y_3}{3} = \theta^*.
\end{aligned}$$

So all of these estimators are unbiased. △

Example 2.9 (Sample mean and variance)**.** Given n samples $\boldsymbol{y} = (y_1, \ldots, y_n)$ from a distribution with mean μ^* and variance $(\sigma^*)^2$, are the estimators

$$\hat{\mu} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

for the mean and variance, respectively, unbiased?

For $\hat{\mu}$, we have

$$\mathbb{E}[\hat{\mu}(\boldsymbol{Y})] = \mathbb{E}[\bar{Y}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} Y_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[Y_i] = \frac{1}{n}n\,\mathbb{E}[Y_1] = \mu^*$$

and so the estimator for the mean is unbiased. We can show (how?) that

$$\mathbb{E}\big[\hat{\sigma}^2(\boldsymbol{Y})\big] = \frac{n-1}{n}(\sigma^*)^2$$

and the bias of estimating $(\sigma^*)^2$ is

$$\mathbb{E}\big[\hat{\sigma}^2(\boldsymbol{Y})\big] - (\sigma^*)^2 = -\frac{1}{n}(\sigma^*)^2.$$

Based on this, we can create an unbiased estimator for the variance as

$$\hat{\sigma}_u^2(\boldsymbol{y}) = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2.$$

△

Example 2.10 (Balls in the urn)**.** [1, Example 2.8.2] An urn has m^* balls, numbered $1, 2, ..., m^*$. Suppose however that m^* is unknown to us. We pick one random ball from the urn and the number on the ball is y. We estimate m^* using maximum likelihood. First, let Y be the random variable corresponding to observation y, with distribution $p_Y(y; m^*)$. We have

$$p_Y(y; m) = \begin{cases} \frac{1}{m} & y \leq m, \\ 0 & y > m. \end{cases}$$

and thus

$$L(m) = \begin{cases} \frac{1}{m} & m \geq y, \\ 0 & m < y. \end{cases}$$

Hence, $L(m)$ is maximized by choosing $m(y) = y$ and so $\hat{m}_{\text{mle}} = y$. To find the bias of $\hat{m}_{\text{mle}}$,

$$\mathbb{E}[\hat{m}_{\text{mle}}(Y)] = \mathbb{E}[Y] = \sum_{i=1}^{m^*} i \cdot \frac{1}{m^*} = \frac{m^* + 1}{2},$$
$$\text{Bias}(\hat{m}_{\text{mle}}) = \frac{m^* + 1}{2} - m^* = -\frac{m^* - 1}{2},$$

which means that the ML estimator tends to underestimates m^* by almost a factor of 2. △

Example 2.11 (Linear unbiased estimator)**.** Can we design an unbiased estimator for Example 2.10? There are many options, but for simplicity we may choose an estimator that is linear in the data, in particular, one of the form

$$\hat{m}_L(y) = ay + b.$$

We find a and b such that $\hat{m}_L$ is unbiased. We have

$$\mathbb{E}[\hat{m}_L(Y)] = a\,\mathbb{E}\,Y + b = a\frac{m^* + 1}{2} + b.$$

Setting this equal to m^* (equality should hold for any m^*) yields $a = 2$ and $b = -1$, i.e.,

$$\hat{m}_L(y) = 2y - 1.$$

△

Example 2.12 (Survival of humanity (!))**.** The human species will eventually die out. We use two methods to estimate the total number of humans m who will ever live. Let humans be enumerated by birth order as $h_1, h_2, ..., h_y, ..., h_m$, where h_1 represents Adam, h_2 represents Eve, h_y represents you, and h_n represents the last human to live. Assuming that your birth order y is random, the problem is similar to estimating the number of balls in an urn in Example 2.10.

Assuming that 100 billion humans have been born so far, we have $\hat{m}_{\text{mle}} = 100$ billion and $\hat{m}_L = 200$ billion. The ML estimate predicts that the end is here. Further, assuming that there will be 140 million births each year, the unbiased estimator predicts the end of humanity to occur in around 700 years. △

2.3.2 Mean squared error and variance

Example 2.13 (Unbiased but useless)**.** Consider an unbiased estimator $\hat{\theta}$ and define $\hat{\theta}' = \hat{\theta} + W$, where W is a zero-mean random variable with a large variance. Now, $\hat{\theta}'$ is unbiased, similar to $\hat{\theta}$, but it is not a good estimator (regardless of how good $\hat{\theta}$ is). So clearly, being unbiased alone is not sufficient to ensure that an estimator is "good." △

For an estimator $\hat{\theta}$, where the random variable describing data is denoted by $\boldsymbol{X}$, the mean squared error (MSE) is defined as

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\left[\left(\hat{\theta}(\boldsymbol{X}) - \theta^*\right)^2\right].$$

The smaller the MSE, the more accurate the estimator.

Let $\hat{\Theta} = \hat{\theta}(\boldsymbol{X})$. Note that

$$\begin{aligned}
\mathrm{MSE}(\hat{\theta}) &= \mathbb{E}\left[\left(\hat{\Theta} - \theta^*\right)^2\right] \\
&= \mathbb{E}\left[\left(\left(\hat{\Theta} - \mathbb{E}\,\hat{\Theta}\right) + \left(\mathbb{E}\,\hat{\Theta} - \theta^*\right)\right)^2\right] \\
&= \mathbb{E}\left[\left(\hat{\Theta} - \mathbb{E}\,\hat{\Theta}\right)^2\right] + \left(\mathbb{E}\,\hat{\Theta} - \theta^*\right)^2 + 2\,\mathbb{E}\left[\left(\hat{\Theta} - \mathbb{E}\,\hat{\Theta}\right)\right]\left(\mathbb{E}\,\hat{\Theta} - \theta^*\right) \\
&= \mathbb{E}\left[\left(\hat{\Theta} - \mathbb{E}\,\hat{\Theta}\right)^2\right] + \left(\mathbb{E}\,\hat{\Theta} - \theta^*\right)^2,
\end{aligned}$$

where, the third equality uses the fact that $\mathbb{E}\,\hat{\Theta} - \theta^*$ is a deterministic constant and the fourth equality the fact that $\mathbb{E}\left[\left(\hat{\Theta} - \mathbb{E}\,\hat{\Theta}\right)\right] = 0$. Hence,

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + (\mathrm{Bias}(\hat{\theta}))^2.$$

For unbiased estimators, the variance is an important quantity since it is equal to the MSE.

Example 2.14 (Heavy or light traffic, revisited)**.** We saw in Example 2.7 that the maximum likelihood estimate for the probability of traffic θ^* is unbiased. Now, let us find its variance. Again, we write $\hat{\theta}_{\mathrm{mle}}(\boldsymbol{X}) = \frac{X}{100}$ and

$$\mathrm{Var}(\hat{\theta}_{\mathrm{mle}}) = \frac{\mathrm{Var}(X)}{100^2} = \frac{\theta^*(1-\theta^*)}{100},$$

where X is the number of days without traffic, which follows $\mathrm{Bin}(100, \theta^*)$ with variance $100\theta^*(1-\theta^*)$. As we can see, the variance (hence, MSE) increases as the true value of θ^* approaches $1/2$, i.e., every data point contains more uncertainty. Furthermore, we can extend this result to the more general case where we collect data for n days. By the same argument, we get

$$\mathrm{MSE}(\hat{\theta}_{\mathrm{mle}}) = \mathrm{Var}(\hat{\theta}_{\mathrm{mle}}) = \frac{\theta^*(1-\theta^*)}{n}.$$

△

Example 2.15 (The sample mean is unbiased)**.** Consider data $\boldsymbol{y} = (y_1, ..., y_n)$, where the corresponding random variables Y_i are iid with distribution $\mathcal{N}(\mu, \sigma^2)$. The ML estimator for the mean μ is $\hat{\theta}_{\mathrm{mle}}(\boldsymbol{y}) = \bar{y} = \frac{1}{n}\sum_{i=1}^n y_i$ is unbiased. We have

$$\mathrm{MSE}(\hat{\theta}_{\mathrm{mle}}) = \mathrm{Var}(\bar{Y}) = \frac{\sigma^2}{n}.$$

△

Note that as n increases, the MSE decreases and the estimate becomes more accurate, as would be expected. This property is studied next.

Exercise 2.16 (MSE of the mean estimators)**.** For the estimators in Example 2.8, find the MSE, assuming the variance is $(\sigma^*)^2$. △

Exercise 2.17 (Bias-variance trade-off)**.** Given iid data $\boldsymbol{y} = (y_1, \ldots, y_n), n \geq 3$, with mean θ^* and variance

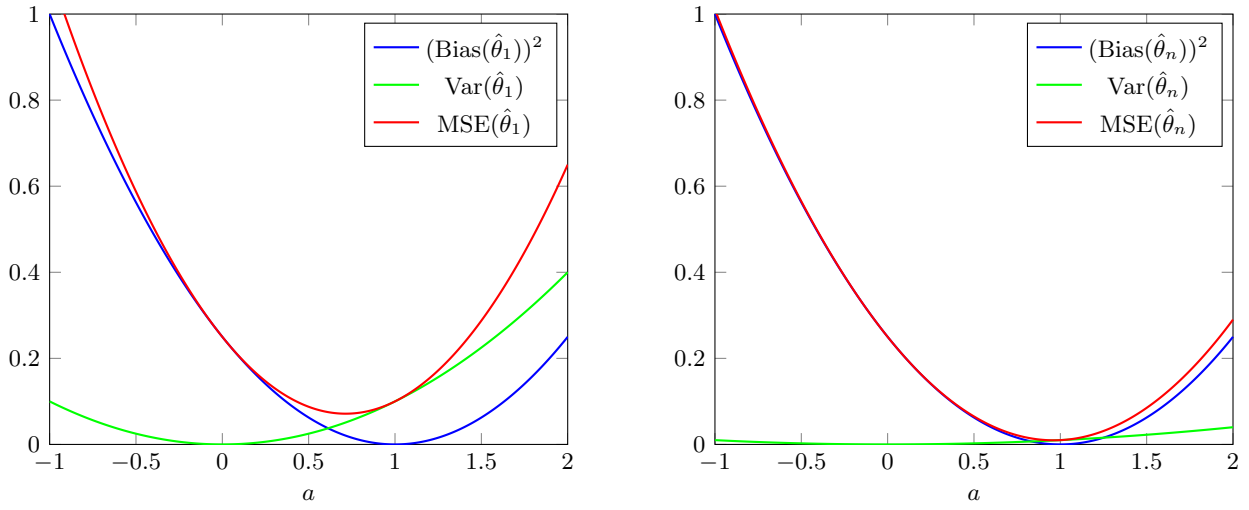
σ^2, show that the MSE of

$$\hat{\theta}_1 = ay_1,$$
$$\hat{\theta}_n = a\bar{y} = \frac{a}{n}\sum_{i=1}^{n} y_i,$$

for some constant $a \in \mathbb{R}$ is given as

$$\text{MSE}(\hat{\theta}_1) = (a-1)^2(\theta^*)^2 + a^2\sigma^2,$$
$$\text{MSE}(\hat{\theta}_n) = (a-1)^2(\theta^*)^2 + a^2\sigma^2/n.$$

What is a good value for a? Does anything other than $a = 1$ make sense? The components of the MSE are given in the plots below for $\hat{\theta}_1$ and $\hat{\theta}_n$ with $n = 10$, for $\theta^* = 0.5, \sigma^2 = 0.1$. A trade-off between the bias and variance is evident. Why is it not feasible to design an estimator by optimizing for a? What is the difference between estimation based on little data ($\hat{\theta}_1$) and a lot of data ($\hat{\theta}_n, n = 10$)?



△

2.3.3 Consistency

Consider an estimator $\hat{\theta}_n(\boldsymbol{x})$ based on n samples $\boldsymbol{x} = (x_1, \ldots, x_n)$. Let $\boldsymbol{X} = (X_1, \ldots, X_n)$ be the random variables that describe the n data samples and let $\hat{\Theta}_n = \hat{\theta}_n(\boldsymbol{X})$ be the random variable that corresponds to the estimate. The estimator $\hat{\theta}_n$ is said to be **consistent** if $\hat{\Theta}_n \to \theta^*$ as $n \to \infty$. More precisely, for all $\epsilon > 0$, we need

$$\lim_{n\to\infty} \Pr(|\hat{\Theta}_n - \theta^*| \geq \epsilon) = 0.$$

That is, $\hat{\Theta}_n$ must converge in probability to θ^* (the mode of convergence named in the probability review chapter): the estimator is accurate if the size of the data is large.

Example 2.18 (Both estimators with more data)**.** The ML and linear estimators described in Examples 2.10 and 2.11 are very different for a single data point. But how do they behave if we have a lot of data. First we need to define these for n data samples. Suppose that we take n samples from the urn with replacement, resulting in $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)$. Define

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

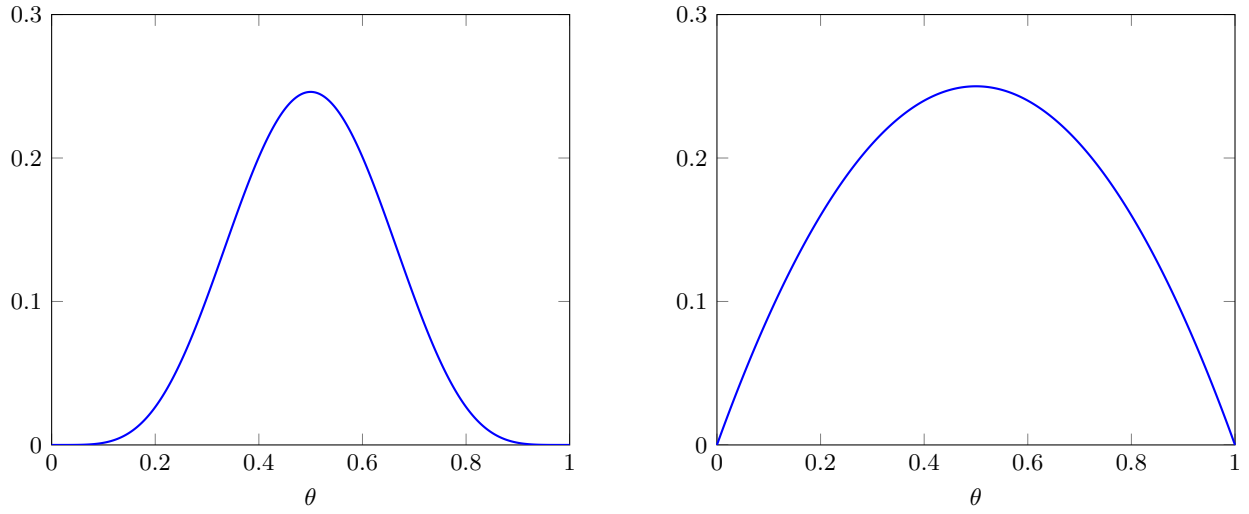


Figure 2.1: The likelihood function on the left demonstrates strong dependence on θ compared to the one on the right.

To extend the linear estimator to n data points, we can choose

$$\hat{m}_{L,n} = 2\bar{y} - 1.$$

For the ML estimator, we have (why?)

$$\hat{m}_{\text{mle},n} = \max_i y_i.$$

Both of these, although they look very different, are consistent and converge to m^* as $n \to \infty$.

- As $n \to \infty$, by LLN, $\bar{Y}$ converges to the mean of the distribution, i.e., $\mathbb{E}[Y_1] = \frac{m^*+1}{2}$. Hence, $\hat{m}_{L,n} \to 2 \cdot \frac{m^*+1}{2} - 1 = m^*$.
- For the ML estimator, as $n \to \infty$, at some point, we will pick the ball numbered m^* and so we will eventually have $\hat{m}_{\text{mle}} = m^*$.

Given the two estimators, the bad news is that the estimators disagree significantly for small data. However, as the size of the sample data increases, the two estimators agree. △

2.4 The Cramer-Rao lower bound (graduate only)

For an unbiased estimator, the MSE is equal to the variance, and thus the variance represents the accuracy of the estimator. This leads to the following question: *For a given distribution of data, what is the smallest possible variance of an unbiased estimator?*

The accuracy of estimating a parameter θ depends on how strongly the distribution of the data $\boldsymbol{X}$ depends on θ. If the dependence is strong, i.e., for values of θ other than the true value θ^*, the probability of the observed data falls sharply, then we may expect to find θ^* with accuracy. On the other hand, if the dependence is week, then it will be difficult to find θ^* with precision. These two cases are shown in Figure 2.1.

Let the data be encoded as a vector $\boldsymbol{X}$, whose distribution is given by p with parameter θ^*. Assuming $\boldsymbol{X} = \boldsymbol{x}$, the log-likelihood is $p(\boldsymbol{x}; \theta)$. The sharpness of the log-likelihood $\ell(\theta)$ at the true value θ^* can be quantified as

$$-\frac{\partial^2 \ell(\theta)}{\partial \theta^2}\bigg|_{\theta=\theta^*} = -\frac{\partial^2 \ln p(\boldsymbol{x}; \theta)}{\partial \theta^2}\bigg|_{\theta=\theta^*}. \tag{2.5}$$

Given the randomness of the data $\boldsymbol{X}$, the above quantity is random,

$$-\frac{\partial^2 \ln p(\boldsymbol{X};\theta)}{\partial\theta^2}\bigg|_{\theta=\theta^*}$$

So to average over the data, we define

$$\mathcal{I}(\theta^*) = -\,\mathbb{E}\left[\frac{\partial^2 \ln p(\boldsymbol{X};\theta)}{\partial\theta^2}\bigg|_{\theta=\theta^*}\right] = -\int \frac{\partial^2 \ln p(\boldsymbol{x};\theta)}{\partial\theta^2}\bigg|_{\theta=\theta^*} p(\boldsymbol{x};\theta^*)d\boldsymbol{x},$$

which is called the **Fisher information**.

The following theorem provides a lower bound on the variance, which is referred to as the Cramer-Rao lower bound (CRLB).

Theorem 2.19 (Cramer-Rao lower bound)**.** *Given that the log-likelihood $\ell(\theta)$ satisfies certain regularity conditions, the variance of any unbiased estimator $\hat{\theta}$ of θ^* satisfies*

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta^*)}.$$

If an estimator achieves the CRLB, i.e., $\mathrm{Var}(\hat{\theta}) = 1/\mathcal{I}(\theta^*)$, then it is called **efficient**.

As a special case, consider when we have n iid data points, and denote the estimator based on this data as $\hat{\theta}_n$. Denote the Fisher information based on n data points as $I_n(\theta^*)$ and based on one data point as $I_1(\theta^*) = I(\theta^*)$. Since the Fisher information is additive (Why? Hint: definition), we have $I_n(\theta^*) = nI(\theta^*)$. Thus, the variance of an unbiased estimator $\hat{\theta}_n$ based on n independent observations satisfies

$$\mathrm{Var}(\hat{\theta}_n) \geq \frac{1}{n\mathcal{I}(\theta^*)}. \tag{2.6}$$

Example 2.20 (Fisher information for a Gaussian)**.** In Example 2.4, where we estimated the mean μ^* of a Gaussian distribution with known σ^2 based on n iid samples $y_1, \ldots, y_n$, the log-likelihood, ignoring constant terms, was given as

$$\ell(\mu) \doteq -\sum_{i=1}^{n} \frac{(y_i-\mu)^2}{2\sigma^2}.$$

And,

$$\frac{\partial \ell(\mu)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i-\mu). \tag{2.7}$$

Observe that

$$\frac{\partial^2 \ell(\mu)}{\partial \mu^2} = -\frac{n}{\sigma^2} \implies I(\mu^*) = -\,\mathbb{E}\left[\frac{\partial^2 \ell(\mu^*)}{\partial \mu^2}\right] = \frac{n}{\sigma^2}.$$

Based on the CRLB, the variance of the estimator satisfies

$$\mathrm{Var}(\hat{\mu}) \geq \frac{\sigma^2}{n}.$$

The variance of the estimator is $\mathrm{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$. Hence, the ML estimator is efficient in this case. △

2.5 Asymptotic normality of the MLE

As shown before, the maximum-likelihood estimator is not necessarily unbiased. However, if we have a large amount of data, under some regularity conditions, the ML estimator $\hat{\Theta}_n$ based on n iid data points satisfies

$$\sqrt{n}(\hat{\Theta}_n - \theta^*) \to \mathcal{N}(0, I^{-1}(\theta^*)).$$

So for large data, $\hat{\Theta}_n$ is nearly normally distributed with mean θ^* (hence unbiased) and variance $I^{-1}(\theta^*)/n$ (efficient).

While we stated the CRLB and the asymptotic normality of the MLE for scalar parameters, almost identical results also hold for a vector of parameters.

References

[1] Bruce Hajek. *Random Processes for Engineers*. Illinois, 2014. URL: http://hajek.ece.illinois.edu/Papers/randomprocJuly14.pdf (visited on 01/30/2017).

Chapter 3

Bayesian Parameter Estimation

Learning Objectives

After studying this chapter, you should be able to:

- Apply Bayes' theorem to update prior beliefs with observed data
- Identify and use conjugate priors for common likelihood functions and interpret hyperparameters as pseudo-observations
- Compute Bayesian point estimates (mean, median, mode) and identify the loss functions they minimize
- Derive posterior predictive distributions for future observations
- Compare Bayesian and frequentist estimators from both perspectives
- Recognize exponential family distributions and their sufficient statistics

3.1 From Prior to Posterior

In the Bayesian philosophy, unknown parameters are viewed as random. So, our knowledge about the parameter can be encoded as a distribution. The distribution representing our belief before observing the data is called the **prior distribution**. After we observe the data, our belief changes, resulting in the **posterior distribution**.

Specifically, the components of a Bayesian estimation problem are:

- Data $\boldsymbol{x}$: The data is a realization of a random variable $\boldsymbol{X}$. The distribution of $\boldsymbol{X}$ depends on a parameter Θ.
- Parameter Θ: The parameter of the distribution of $\boldsymbol{X}$, which is unknown, and hence a random variable in the Bayesian framework.
- Joint and marginal distributions p: A joint distribution $p_{\boldsymbol{X},\Theta}$ and its marginals $p_{\boldsymbol{X}}$ and p_{Θ}.

The steps of Bayesian estimation of a parameter θ are:

1. Identifying the prior distribution, $p_{\Theta}(\theta)$. This is called the prior because it encodes our beliefs about Θ before seeing any data.
2. Collecting *data* $\boldsymbol{x}$ and forming the likelihood: $p_{\boldsymbol{X}|\Theta}(\boldsymbol{x}|\theta)$
3. Finding the posterior distribution $p_{\Theta|\boldsymbol{X}}(\theta|\boldsymbol{x})$ as

$$p_{\Theta|\boldsymbol{X}}(\theta|\boldsymbol{x}) = \frac{p_{\Theta}(\theta)p_{\boldsymbol{X}|\Theta}(\boldsymbol{x}|\theta)}{p_{\boldsymbol{X}}(\boldsymbol{x})}, \tag{3.1}$$

The distribution $p_{\Theta|\boldsymbol{X}}$ is called the posterior distribution since it encodes our knowledge about the parameter after observing the data. Usually, since the distribution is clear from the argument, we drop the subscripts of p, writing the above equation as

$$p(\theta|\boldsymbol{x}) = \frac{p(\theta)p(\boldsymbol{x}|\theta)}{p(\boldsymbol{x})}, \tag{3.2}$$

Normalizing distributions. Finding the posterior distribution requires computing the integral $p(\boldsymbol{x}) = \int_\theta p(\theta)p(\boldsymbol{x}|\theta)d\theta$. Since we have to compute an integral anyway, we might as well drop all multiplicative terms that are constant in θ and then normalize the final distribution. In particular, $p(\boldsymbol{x})$ is one such term. So we often first find a function *proportional* to $p(\theta|\boldsymbol{x})$ as

$$p(\theta|\boldsymbol{x}) \propto p(\theta)p(\boldsymbol{x}|\theta), \tag{3.3}$$

where we can also drop constant terms in θ from $p(\theta)$ and $p(\boldsymbol{x}|\theta)$. We can then normalize the result by integration. This is often difficult to do. Sometimes, given this function, we can identify the distribution. More generally, we can use computational methods, such as Markov Chain Monte Carlo, or approximation methods, such as variational inference, as we will see later. Finally, in certain cases, we can find what we need without any integration. For example, if our goal is to find the value of θ maximizing $p(\theta|\boldsymbol{x})$.

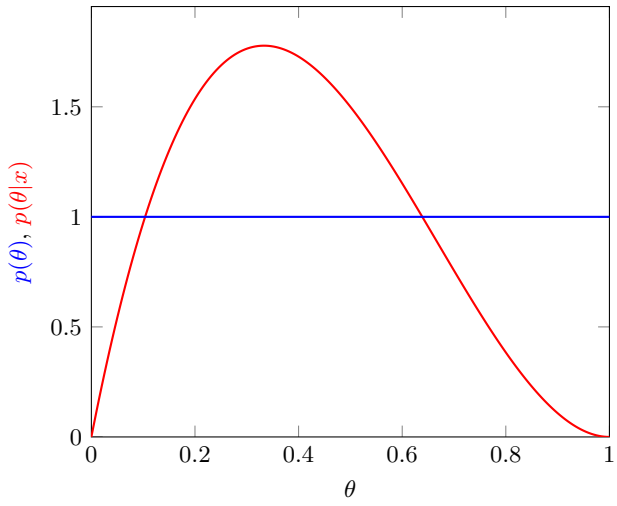
Example 3.1 (Estimating a geometric parameter)**.** Let Θ denote the unknown parameter of a geometric random variable X, where

$$p_{X|\Theta}(x|\theta) = \theta(1-\theta)^{x-1}.$$

Suppose we observe $X = x$. We would like to estimate Θ based on this observation. If all possible values of Θ are equally likely, we may choose $\Theta \sim \text{Uni}(0, 1)$. We then have

$$p(\theta) = 1 \tag{3.4}$$
$$p(x|\theta) = \theta(1-\theta)^{x-1} \tag{3.5}$$
$$p(\theta|x) \propto p(\theta)p(x|\theta) \propto \theta(1-\theta)^{x-1} \tag{3.6}$$

The expression $\theta(1-\theta)^{x-1}$ as a function of x is the geometric distribution. But as a function of θ, it is proportional to $\text{Beta}(2, x)$. As an example, if $x = 3$, then $\Theta|x \sim \text{Beta}(2, 3)$:



△

Exercise 3.2 (One Bernoulli trial)**.** The probability of 1 (success) in a Bernoulli experiment (e.g., flipping a coin, a system working or not working, etc) is Θ, which we would like to estimate. Suppose that the experiment is performed once and the outcome x is observed to be $x = 1$. Assuming a uniform prior, find

the posterior distribution of Θ, i.e., $p_{\Theta|X}(\theta|1)$. $\triangle$

Example 3.3 (Bernoulli trials, beta posterior)**.** The probability of success in a Bernoulli experiment is Θ, which we would like to estimate. We show success in the ith trial with $y_i = 1$ and failure by $y_i = 0$.

- Prior distribution: Assuming that a priori we do not know anything about Θ, it is appropriate to choose $p_\Theta \sim \text{Uni}[0, 1]$, i.e., $p(\theta) = 1$ in the interval $[0, 1]$.
- Likelihood: We then perform the experiment n times. Suppose that we observe s successes and f failures. Let us denote this observation as $\boldsymbol{x} = (s, f)$. The likelihood is

$$p(\boldsymbol{x}|\theta) = \binom{n}{s}\theta^s(1-\theta)^f \tag{3.7}$$

- The posterior distribution:

$$p(\theta|\boldsymbol{x}) \propto 1 \cdot \theta^s(1-\theta)^f = \theta^s(1-\theta)^f \tag{3.8}$$

  We observe that this distribution is of the form of a beta distribution, $\text{Beta}(y; \alpha, \beta) \sim y^{\alpha-1}(1-y)^{\beta-1}$. Hence,

$$p(\theta|\boldsymbol{x}) \sim \text{Beta}(s+1, f+1). \tag{3.9}$$

$\triangle$

Note that since we are interested in Θ, we can drop multiplicative terms that are constant with respect to θ, such as $\binom{n}{s}$, in the above example.

Now that we have the posterior distribution, we can answer questions about the parameter, for example, What is the probability that $0.4 < \Theta < 0.6$?

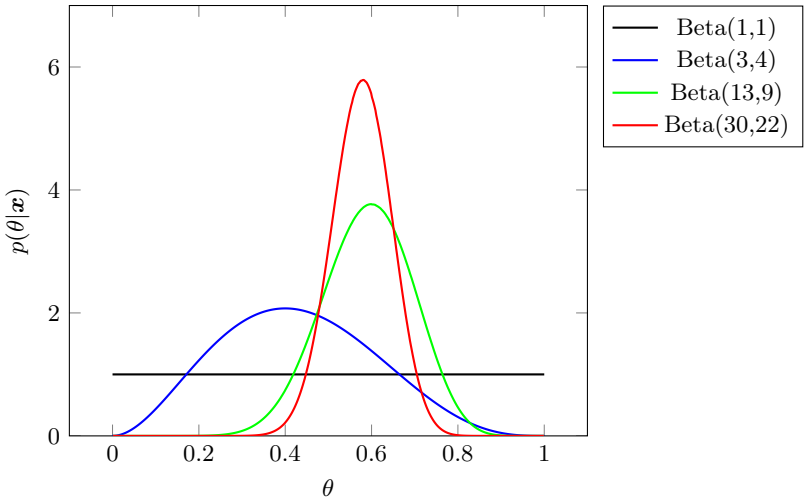
$$\int_{0.4}^{0.6} p(\theta|\boldsymbol{x})d\theta \tag{3.10}$$

Example 3.4 (Consecutive Bayesian updating)**.** Continuing the previous example, suppose that we collect more data $\boldsymbol{x}' = (s', f')$, consisting of s' successes and f' failures. Our prior distribution now is the posterior of the previous example, $p(\theta) \propto \theta^s(1-\theta)^f$. We have

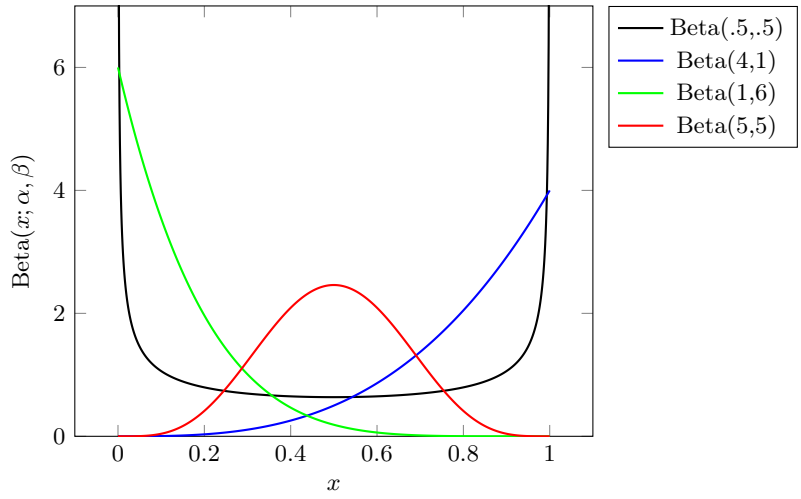
$$\begin{aligned} p(\boldsymbol{x}'|\theta) &= \binom{s'+f'}{s'}\theta^{s'}(1-\theta)^{f'} \\ p(\theta|\boldsymbol{x}') &\propto \theta^s(1-\theta)^f\theta^{s'}(1-\theta)^{f'} \\ &= \theta^{s+s'}(1-\theta)^{f+f'} \\ \Theta|\boldsymbol{x}' &\sim \text{Beta}(s+s'+1, f+f'+1). \end{aligned} \tag{3.11}$$

Equivalently, we can update our uniform prior $p(\theta) \propto 1$ with data $(s+s', f+f')$ to obtain $p(\theta|(s+s', f+f')) \sim \text{Beta}(s+s'+1, f+f'+1)$. As we can see, the Bayesian approach provides a way to update our belief in a consistent manner.

The figure below provides an example of the posterior with 0, 5, 20, and 50 samples. It can be observed that the posterior becomes sharper as more data is collected. $\triangle$



Example 3.5 (Interpreting beta priors)**.** Beta is a common prior for the probability of Bernoulli experiments. Based on the discussion above, one way to interpret a Beta prior with parameters $\alpha \geq 1, \beta \geq 1$ is to imagine that, starting with the uniform prior, we have already collected $\alpha + \beta - 2$ samples, with $\alpha - 1$ successes. The following plot shows the Beta distribution with different parameters to give a sense of the range of possible priors. △



Example 3.6 (Poisson likelihood, gamma prior)**.** Suppose that Y has a Poisson distribution with paramter Λ. That is, $p_{Y|\Lambda}(y|\lambda) = \text{Poi}(y;\lambda)$. Hence,

$$p(y|\lambda) = \frac{\lambda^y e^{-\lambda}}{y!}, \quad y \in \{0, 1, \dots\}$$

We intend to estimate Λ based on n iid samples $y_1^n = (y_1, \dots, y_n)$ of Y.

We assume that the prior for Λ is given as $p(\lambda) = \text{Gamma}(\lambda;\alpha,\beta) \propto \lambda^{\alpha-1}e^{-\beta\lambda}$. We have

$$p(\lambda) \propto \lambda^{\alpha-1}e^{-\beta\lambda} \tag{3.12}$$

$$p(y_1^n|\lambda) = \prod_{i=1}^n \frac{\lambda^{y_i}e^{-\lambda}}{y_i!} \propto \prod_{i=1}^n \lambda^{y_i}e^{-\lambda} = e^{-n\lambda}\lambda^{n\bar{y}}, \tag{3.13}$$

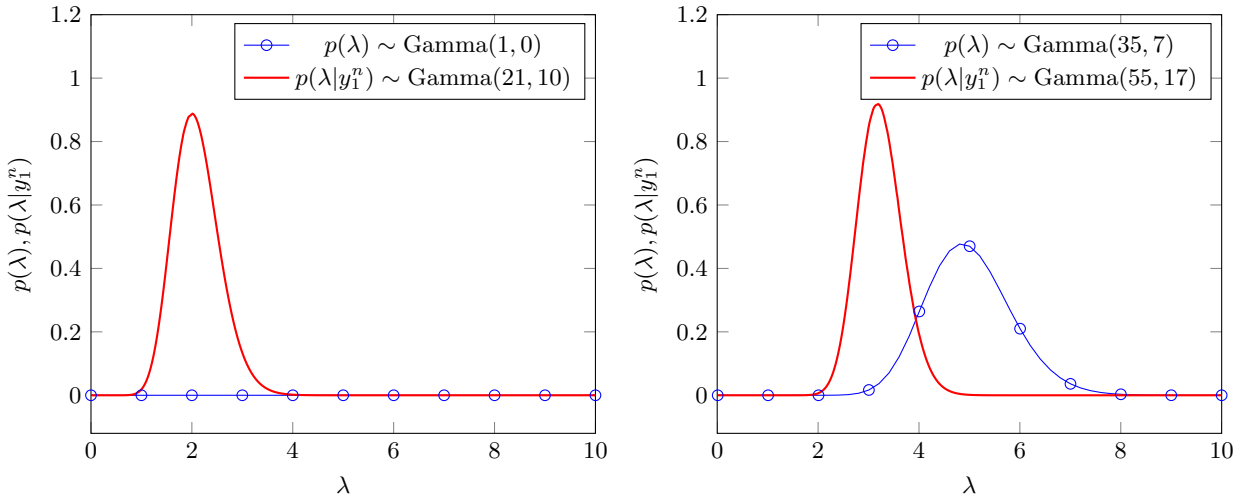
where $\bar{y} = \frac{1}{n}\sum_{i=1}^n y_i$. Note that while $p(y_1^n|\lambda)$ is a distribution in y_1^n, we still dropped the $y_i!$ from its

expression since our final goal is to find a distribution in λ and for this purpose terms that are independent of λ can be viewed as constant. The posterior is

$$p(\lambda|y_1^n) \propto \lambda^{\alpha-1}e^{-\beta\lambda}e^{-n\lambda}\lambda^{n\bar{y}} = \lambda^{\alpha+n\bar{y}-1}e^{-\lambda(n+\beta)} \propto \text{Gamma}(\lambda; \alpha + n\bar{y}, n + \beta). \tag{3.14}$$

If we choose $\alpha = 1, \beta = 0$, then the Gamma prior is flat, giving all possible values the same prior probability. But this is not a proper distribution. However, as long as the final posterior is a proper distribution, an **improper prior** is deemed acceptable.

Suppose that $n = 10$ and $\bar{y} = 2$. The figure below shows the posterior distribution with different priors. The prior on the left is called a **non-informative prior** because it is flat and the one on the right is an **informative prior** given that it represents a prior belief that certain values have a higher probability.



△

3.2 Bayesian Point Estimates

Having the complete distribution for $p_{\Theta|\boldsymbol{X}}(\theta|\boldsymbol{x})$ is useful since it provides the probability for different values for θ. But sometimes we want to estimate Θ with a single value $\hat{\theta} = \hat{\theta}(\boldsymbol{x})$ as a function of the data, similar to maximum likelihood. It is very important to note that in the Bayesian framework, the data is given and the estimate is known (and not random). The best choice for $\hat{\theta}$ then depends on how we characterize the estimation error:

Average Error	Optimal Estimator
$\mathbb{E}[(\Theta - \hat{\theta})^2\|\boldsymbol{x}]$	$\hat{\theta} = \mathbb{E}[\Theta\|\boldsymbol{x}]$ (**mean**)
$\mathbb{E}[\|\Theta - \hat{\theta}\|\|\boldsymbol{x}]$	$\hat{\theta} =$ **median** of $p(\theta\|\boldsymbol{x})$
$\Pr(\Theta \neq \hat{\theta}\|\boldsymbol{x}) = \mathbb{E}[I(\Theta \neq \hat{\theta})\|\boldsymbol{x}]$	$\hat{\theta} = \arg\max_\theta p(\theta\|\boldsymbol{x})$ (**mode**)

In the table, $I(condition)$ is 1 if the condition is satisfied and is 0 otherwise.

We prove the first case in the table. Let $\bar{\theta} = \mathbb{E}[\Theta|\boldsymbol{x}]$. We have

$$\begin{aligned}
\mathbb{E}[(\hat{\theta} - \Theta)^2|\boldsymbol{x}] &= \mathbb{E}[((\hat{\theta} - \bar{\theta}) - (\Theta - \bar{\theta}))^2|\boldsymbol{x}] && (3.15)\\
&= \mathbb{E}[(\hat{\theta} - \bar{\theta})^2 - 2(\hat{\theta} - \bar{\theta})(\Theta - \bar{\theta}) + (\Theta - \bar{\theta})^2|\boldsymbol{x}] && (3.16)\\
&= (\hat{\theta} - \bar{\theta})^2 - 2(\hat{\theta} - \bar{\theta})\,\mathbb{E}[(\Theta - \bar{\theta})|\boldsymbol{x}] + \mathbb{E}[(\Theta - \bar{\theta})^2|\boldsymbol{x}] && (3.17)\\
&= (\hat{\theta} - \bar{\theta})^2 + \mathbb{E}[(\Theta - \bar{\theta})^2|\boldsymbol{x}] && (3.18)\\
&= (\hat{\theta} - \bar{\theta})^2 + \mathrm{Var}(\Theta|\boldsymbol{x}) && (3.19)\\
&\geq \mathrm{Var}(\Theta|\boldsymbol{x}), && (3.20)
\end{aligned}$$

and the lower bound on the error is achieved when $\hat{\theta} = \bar{\theta}$.

Example 3.7 (Beta posterior point estimates)**.** Generalizing Example 3.3 by assuming $p(\theta) = \mathrm{Beta}(\theta; \alpha, \beta)$, we obtain $p(\theta|\boldsymbol{x}) = \mathrm{Beta}(\theta; \alpha + s, \beta + f)$ (for Uniform, $\alpha = \beta = 1$). We have

$$\begin{aligned}
\text{Mean} &= \frac{s + \alpha}{s + f + \alpha + \beta}, && (3.21)\\
\text{Median} &\simeq \frac{s + \alpha - 1/3}{s + f + \alpha + \beta - 2/3}, && (3.22)\\
\text{Mode} &= \frac{s + \alpha - 1}{s + f + \alpha + \beta - 2}. && (3.23)
\end{aligned}$$

Generally speaking, Bayesian point estimates are between what is suggested only using the prior and what would be obtained using only the likelihood. For example, the mean of the prior is $\frac{\alpha}{\alpha+\beta}$ and the maximum likelihood solution is $\frac{s}{s+f}$. The mean of the posterior, $\frac{s+\alpha}{s+f+\alpha+\beta}$, is between these two. △

3.3 Posterior Predictive Distribution

Given n iid samples, $y_1^n = (y_1, \ldots, y_n)$, we are often interested in the distribution of the next (unobserved) value, $p_{Y_{n+1}|Y_1^n}(y_{n+1}|y_1^n)$. This distribution is referred to as *predictive posterior*. We have

$$\begin{aligned}
p(y_{n+1}|y_1^n) &= \int p(y_{n+1}, \theta|y_1^n)d\theta && (3.24)\\
&= \int p(\theta|y_1^n)p(y_{n+1}|\theta, y_1^n)d\theta && (3.25)\\
&= \int p(\theta|y_1^n)p(y_{n+1}|\theta)d\theta, && (3.26)
\end{aligned}$$

where we have used the fact that $Y_{n+1} \perp\!\!\!\perp Y_1^n|\Theta$. We have thus written the predictive posterior in terms of two known distributions.

Example 3.8 (Bernoulli predictive probability)**.** Continuing Example 3.3, let success in the $n+1$st experiment be denoted by $Y_{n+1} = 1$ and failure by $Y_{n+1} = 0$. We have

$$p_{Y_{n+1}|Y_1^n}(1|y_1^n) = \int \theta p(\Theta|y_1^n) = \mathbb{E}[\Theta|y_1^n] = \frac{s+1}{s+f+2}, \tag{3.27}$$

where we have used the facts that $p_{Y_{n+1}|\Theta}(1|\theta) = \theta$ and that the mean of $\mathrm{Beta}(s+1, f+1)$ is $\frac{s+1}{s+f+2}$. △

We may also ask about the expected value of Y_{n+1} given y_1^n, i.e., $\mathbb{E}[Y_{n+1}|y_1^n]$. We can find this by first finding $p(y_{n+1}|y_1^n)$ explicitly. But it is often easier to use the law of iterated expectations, since y_1^n influences Y_{n+1} through Θ. Recall that

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y], \qquad \mathbb{E}[\mathbb{E}[Y|X, Z = z]|Z = z] = \mathbb{E}[Y|Z = z]. \tag{3.28}$$

Hence,

$$\mathbb{E}[Y_{n+1}|y_1^n] = \mathbb{E}[\mathbb{E}[Y_{n+1}|\Theta, y_1^n]|y_1^n] = \mathbb{E}[\mathbb{E}[Y_{n+1}|\Theta]|y_1^n], \tag{3.29}$$

where the last step follows from the fact that $Y_{n+1} \perp\!\!\!\perp Y_1^n|\Theta$, implying that $\mathbb{E}[Y_{n+1}|\Theta, y_1^n] = \mathbb{E}[Y_{n+1}|\Theta]$.

Example 3.9 (Poisson predictive mean)**.** Let's find $\mathbb{E}[Y_{n+1}|y_1^n]$ in Example 3.6. First, observe that $\mathbb{E}[Y_{n+1}|\Lambda] = \Lambda$. We hence need to find $\mathbb{E}[\Lambda|y_1^n]$. We know from before that $\Lambda|y_1^n$ is distributed according to Gamma$(\alpha + n\bar{y}, \beta + n)$. Therefore, $\mathbb{E}[Y_{n+1}|y_1^n] = \mathbb{E}[\Lambda|y_1^n] = \frac{\alpha+n\bar{y}}{\beta+n}$. △

3.4 Gaussian Prior and Likelihood

Suppose that we want to estimate the mean of a Gaussian distribution with known variance,

$$p(y_i|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i-\theta)^2}{2\sigma^2}} \tag{3.30}$$

given iid data $\{y_1, \ldots, y_n\}$.

Improper priors. Assuming that we have no information about this mean, it makes sense to choose the prior

$$p(\theta) \propto 1. \tag{3.31}$$

But since the integral $\int_{-\infty}^{\infty} 1 d\theta = \infty$, this does not lead to a valid distribution. Nevertheless, such a choice is acceptable, if the posterior is a valid distribution. Such priors are called *improper priors*. An improper prior does not necessarily have to be uniform.

Example 3.10 (Gaussian mean, improper prior)**.** Consider the above likelihood and prior and let $\bar{y} = \frac{1}{n}\sum_{i=1}^n y_i$. We have

$$\begin{aligned}
p(\theta|y_1^n) &\propto p(y_1^n|\theta) \cdot 1 && (3.32)\\
&\propto \exp\left(-\frac{\sum_{i=1}^n (y_i-\theta)^2}{2\sigma^2}\right) && (3.33)\\
&\propto \exp\left(-\frac{\sum_{i=1}^n (\theta^2 - 2y_i\theta + y_i^2)}{2\sigma^2}\right) && (3.34)\\
&\propto \exp\left(-\frac{\theta^2 - 2\bar{y}\theta}{2\sigma^2/n}\right) && (3.35)\\
&\propto \exp\left(-\frac{(\theta-\bar{y})^2}{2\sigma^2/n}\right) && (3.36)\\
\Theta|y_1^n &\sim \mathcal{N}(\bar{y}, \sigma^2/n). && (3.37)
\end{aligned}$$

For the expected value of the next sample, we have

$$\mathbb{E}[Y_{n+1}|y_1^n] = \mathbb{E}[\mathbb{E}[Y_{n+1}|\Theta]|y_1^n] = \mathbb{E}[\Theta|y_1^n] = \bar{y}. \tag{3.38}$$

We can see more explicitly as well,

$$\begin{aligned}
\mathbb{E}[Y_{n+1}|y_1^n] &= \int y_{n+1} p(y_{n+1}|y_1^n) dy_{n+1} && (3.39)\\
&= \int y_{n+1} \int p(y_{n+1}, \theta|y_1^n) d\theta dy_{n+1} && (3.40)\\
&= \int y_{n+1} \int p(y_{n+1}|\theta) p(\theta|y_1^n) d\theta dy_{n+1} && (3.41)\\
&= \int p(\theta|y_1^n) \int y_{n+1} p(y_{n+1}|\theta) dy_{n+1} d\theta && (3.42)\\
&= \int \theta p(\theta|y_1^n) d\theta && (3.43)\\
&= \mathbb{E}[\Theta|y_1^n] && (3.44)\\
&= \bar{y}. && (3.45)
\end{aligned}$$

△

For the posterior predictive variance, we have

$$\mathrm{Var}(Y_{n+1}|y_1^n) = \sigma^2 + \sigma^2/n. \tag{3.46}$$

It can be shown that $Y_{n+1}|y_1^n$ has a Gaussian distribution, and as we know its conditional mean and variance, we have

$$Y_{n+1}|y_1^n \sim \mathcal{N}(\bar{y}, \sigma^2 + \sigma^2/n) \tag{3.47}$$

From the variance, we can see that there are two sources of uncertainty. One is the inherent randomness in Y, quantified by σ^2 and the other is the result of the uncertainty of our estimate of the mean, quantified by σ^2/n. †

Example 3.11 (Predictive variance)**.** Let us prove that $\mathrm{Var}(Y_{n+1}|y_1^n) = \sigma^2 + \sigma^2/n$:

$$\begin{aligned}
\mathrm{Var}(Y_{n+1}|y_1^n) &= \mathbb{E}\Big[(Y_{n+1} - \bar{y})^2 | y_1^n\Big]\\
&= \mathbb{E}\Big[\mathbb{E}\Big[(Y_{n+1} - \bar{y})^2 | \Theta, y_1^n\Big] | y_1^n\Big]\\
&= \mathbb{E}\Big[\sigma^2 + (\Theta - \bar{y})^2 | y_1^n\Big]\\
&= \sigma^2 + \mathbb{E}\Big[(\Theta - \bar{y})^2 | y_1^n\Big]\\
&= \sigma^2 + \sigma^2/n
\end{aligned}$$

△

We now consider the same problem with a proper Gaussian prior. Note that below as $\tau_0 \to \infty$, the proper prior below tends to the improper prior $p(\theta) \propto 1$.

Example 3.12 (Gaussian mean, proper prior)**.** We would like to estimate the mean Θ of normally distributed independent values $y_1^n = (y_1, \ldots, y_n)$. Let $\bar{y} = \sum y_i/n$. We assume

$$\Theta \sim \mathcal{N}\big(\theta_0, \tau_0^2\big) \tag{3.48}$$

$$Y_i|\theta \sim \mathcal{N}\big(\theta, \sigma^2\big) \tag{3.49}$$

where θ_0 and τ_0^2 are the prior mean and variance, respectively, and σ^2 is known. We have

$$p(\theta|y_1^n) \propto p(\theta)p(y_1^n|\theta) \tag{3.50}$$

$$\propto \frac{1}{\sigma\tau_0}\exp\left(-\frac{\sum_{i=1}^n (y_i-\theta)^2}{2\sigma^2} - \frac{(\theta-\theta_0)^2}{2\tau_0^2}\right) \tag{3.51}$$

With some algebra, it can be shown that conditioned on y_1^n, Θ is normally distributed,

$$\Theta|y_1^n \sim \mathcal{N}\left(\frac{\frac{n\bar{y}}{\sigma^2}+\frac{\theta_0}{\tau_0^2}}{\frac{n}{\sigma^2}+\frac{1}{\tau_0^2}}, \frac{1}{\frac{n}{\sigma^2}+\frac{1}{\tau_0^2}}\right). \tag{3.52}$$

△

†

Example 3.13 (Completing the square)**.** Let us prove (3.52) using (3.51). We start with the following **claim**: If $p_X(x) \propto e^{-f(x)}$, where $f(x) = ax^2 - bx + c$ with $a > 0$, then $X \sim \mathcal{N}\left(\frac{b}{2a}, \frac{1}{2a}\right)$. Observe that

$$ax^2 - bx + c = \frac{x^2 - bx/a + c/a}{1/a} = \frac{\left(x-\frac{b}{2a}\right)^2 - (\frac{b}{2a})^2 + \frac{c}{a}}{2(1/(2a))} = \frac{(x-b/(2a))^2}{2(1/(2a))} + C, \tag{3.53}$$

where C is a constant independent from x. Hence,

$$p_X(x) \propto \exp\left(\frac{(x-b/(2a))^2}{2(1/(2a))}\right), \tag{3.54}$$

proving the claim. Then, (3.52) can be proven by setting

$$a = \frac{n}{2\sigma^2} + \frac{1}{2\tau_0^2}, \quad b = \frac{n\bar{y}}{\sigma^2} + \frac{\theta_0}{\tau_0^2}. \tag{3.55}$$

△

Example 3.14 (Bayesian bias-variance trade-off)**.** Suppose that the prior for Θ is $\Theta \sim \mathcal{N}(0, \tau_0^2)$. Then, from (3.52), the mean (also the mode and median) Bayesian point estimator for Θ is

$$\hat{\theta}_B = \bar{y}\left(\frac{\tau_0^2}{\tau_0^2 + \sigma^2/n}\right), \tag{3.56}$$

while the maximum-likelihood estimator is $\hat{\theta}_{mle} = \bar{y}$. We can evaluate both estimators in the frequentist framework, finding their MSE.

Note that the frequentist framework requires us to assume a true value θ^* and view $\hat{\theta}_B$ and $\hat{\theta}_{mlue}$ as functions of random data Y_1^n. So they are random variables (however, we won't switch to capital letters to represent them here). First, as the MLE is unbiased,

$$\text{MSE}(\hat{\theta}_{mle}) = \text{Var}(\hat{\theta}_{mle}) = \text{Var}(\bar{Y}) = \sigma^2/n, \tag{3.57}$$

and by CRLB, this is the best unbiased estimator.

For the Bayesian estimator, we have

$$\text{Bias}(\hat{\theta}_B) = \mathbb{E}[\hat{\theta}_B] - \theta^* = \theta^*\left(\frac{\tau_0^2}{\tau_0^2 + \sigma^2/n}\right) - \theta^* = -\theta^*\left(\frac{\sigma^2/n}{\tau_0^2 + \sigma^2/n}\right) \tag{3.58}$$

$$\text{Var}(\hat{\theta}_B) = \frac{\sigma^2}{n}\left(\frac{\tau_0^2}{\tau_0^2 + \sigma^2/n}\right)^2 \tag{3.59}$$

$$\text{MSE}(\hat{\theta}_B) = (\theta^*)^2\left(\frac{\sigma^2/n}{\tau_0^2 + \sigma^2/n}\right)^2 + \frac{\sigma^2}{n}\left(\frac{\tau_0^2}{\tau_0^2 + \sigma^2/n}\right)^2 \tag{3.60}$$

We can see that the bias term is decreasing in τ_0^2 while the variance term is increasing. So, there is a trade-off between the two types of error. Smaller values of τ_0 mean we have a strong prior, thus leading to bias. A strong prior is also less sensitive to data, thus leading to a smaller variance.

In particular, for $\tau_0^2 = (\theta^*)^2$, we have

$$\text{MSE}(\hat{\theta}_B) = \frac{(\theta^*)^2\sigma^2/n}{(\theta^*)^2 + \sigma^2/n} < \sigma^2/n = \text{MSE}(\hat{\theta}_{mle}). \tag{3.61}$$

So, with the right prior, $\hat{\theta}_B$ has lower MSE than the maximum-likelihood estimator. Of course, this requires knowledge of (θ^*), which is not available. However, a good prior found based on experience or intuition can provide good results.

△

3.5 Conjugate Priors

Given a likelihood function, a **conjugate prior** is a family of distributions such that the posterior is in the same family. The posterior is proportional to prior times likelihood,

$$p(\theta|y_1^n) \propto p(\theta)\, p(y_1^n|\theta), \tag{3.62}$$

so when this product, as a function of θ, has the form of a member of the family, the posterior is that member. The family then also gives the normalizing constant, and the integral $\int p(\theta)p(y_1^n|\theta)\, d\theta$ never has to be computed. Conjugate priors also often let us read the parameters of an informative prior as pseudo-observations added to a flat, possibly improper, prior. We have already met three conjugate pairs: the Beta prior for Bernoulli trials (Example 3.3), the Gamma prior for Poisson counts (Example 3.6), and the Gaussian prior for a Gaussian mean with known variance (3.52).

In each pair below, the posterior's parameters combine the prior's parameters with a summary of the data.

- Bernoulli/Beta: θ is the probability of success, and $y_i = 1$ marks a success and $y_i = 0$ a failure, so $y = \sum_{i=1}^n y_i$ is the number of successes and $n - y$ the number of failures.

$$p(y_i|\theta) = \theta^{y_i}(1-\theta)^{1-y_i} \qquad \text{Ber}(\theta) \tag{3.63}$$

$$p(y_1^n|\theta) = \theta^y(1-\theta)^{n-y} \tag{3.64}$$

$$p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1} \qquad \text{Beta}(\alpha,\beta) \tag{3.65}$$

$$p(\theta|y_1^n) \propto \theta^{y+\alpha-1}(1-\theta)^{n-y+\beta-1} \qquad \text{Beta}(y+\alpha, n-y+\beta) \tag{3.66}$$

  A $\text{Beta}(\alpha,\beta)$ prior is therefore equivalent to starting from the uniform prior $\text{Beta}(1,1)$ and having observed $\alpha - 1$ successes and $\beta - 1$ failures.

- Poisson/Gamma: y_i is the number of incidents at a fire station on day i and θ the average number of

incidents per day, so $y = \sum_{i=1}^{n} y_i$ is the total number of incidents in n days.

$$p(y_i|\theta) = \frac{\theta^{y_i} e^{-\theta}}{y_i!} \qquad \text{Poi}(\theta) \tag{3.67}$$

$$p(y_1^n|\theta) \propto \theta^y \exp(-n\theta) \tag{3.68}$$

$$p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} \exp(-\beta\theta) \qquad \text{Gamma}(\alpha, \beta) \tag{3.69}$$

$$p(\theta|y_1^n) \propto \theta^{y+\alpha-1} \exp(-(n+\beta)\theta) \qquad \text{Gamma}(y+\alpha, n+\beta) \tag{3.70}$$

A Gamma(α, β) prior is therefore equivalent to starting from the flat improper prior Gamma$(1, 0)$ and having observed $\alpha - 1$ incidents in β days. In the Gamma density Gamma(α, β), α is the shape and β the rate, and the mean is α/β.

- Exponential/Gamma: y_i is the time, in days, from the $(i-1)$th to the ith incident at the same fire station and θ the average number of incidents per day, so $y = \sum_{i=1}^{n} y_i$ is the time taken by n incidents.

$$p(y_i|\theta) = \theta \exp(-\theta y_i) \qquad \text{Exp}(\theta) = \text{Gamma}(1, \theta) \tag{3.71}$$

$$p(y_1^n|\theta) = \theta^n \exp(-\theta y) \tag{3.72}$$

$$p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} \exp(-\beta\theta) \qquad \text{Gamma}(\alpha, \beta) \tag{3.73}$$

$$p(\theta|y_1^n) \propto \theta^{n+\alpha-1} \exp(-(y+\beta)\theta) \qquad \text{Gamma}(n+\alpha, y+\beta) \tag{3.74}$$

A Gamma(α, β) prior is therefore equivalent to starting from the flat improper prior Gamma$(1, 0)$ and having observed $\alpha - 1$ incidents in β days.

- Gaussian/Gaussian: y_i is the ith of n measurements of an unknown quantity θ, each with noise of known variance σ^2, and $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ is their average.

$$p(y_i|\theta) \propto \exp\left(-\frac{(y_i-\theta)^2}{2\sigma^2}\right) \qquad \mathcal{N}(\theta, \sigma^2) \tag{3.75}$$

$$p(y_1^n|\theta) \propto \exp\left(-\frac{\sum_{i=1}^{n}(y_i-\theta)^2}{2\sigma^2}\right) \tag{3.76}$$

$$p(\theta) \propto \exp\left(-\frac{(\theta-\theta_0)^2}{2\tau_0^2}\right) \qquad \mathcal{N}(\theta_0, \tau_0^2) \tag{3.77}$$

$$p(\theta|y_1^n) \propto \exp\left(-\frac{(\theta-\theta_1)^2}{2\tau_1^2}\right) \qquad \mathcal{N}(\theta_1, \tau_1^2), \tag{3.78}$$

where

$$\theta_1 = \frac{\frac{1}{\tau_0^2}\theta_0 + \frac{1}{\sigma^2/n}\bar{y}}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2/n}}, \tag{3.79}$$

$$\frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2/n}. \tag{3.80}$$

The precisions add, and the posterior mean is the precision-weighted average of the prior mean and $\bar{y}$. A $\mathcal{N}(\theta_0, \tau_0^2)$ prior is therefore equivalent to starting from the flat improper prior $p(\theta) \propto 1$ and having observed σ^2/τ_0^2 measurements with average θ_0.

If a prior is conjugate for the likelihood of a single observation, it is also conjugate for the likelihood of many iid observations. Updating with n iid observations is equivalent to updating n times with one observation at a time: the posterior after y_1 is in the family, so it serves as the prior for y_2, and so on.

Example 3.15 (Waiting times between incidents)**.** The waiting times (in days) between incidents at a fire station are modeled as $Y_i|\Theta = \theta \sim \text{Exp}(\theta)$, where Θ is the rate of incidents per day. The prior is

$\Theta \sim \text{Gamma}(2,2)$, with mean $2/2 = 1$ per day. We observe $n = 5$ waits with total $y = \sum_{i=1}^{5} y_i = 10$ days. Find the posterior and the posterior mean.

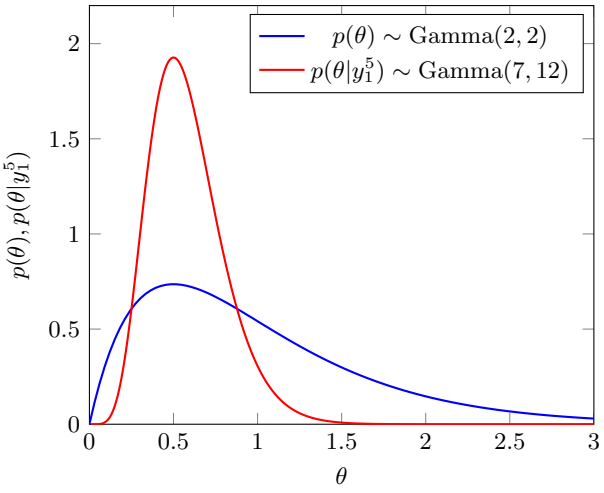
Solution. Multiplying prior and likelihood adds the exponents of θ and of e:

$$\text{Prior:} \quad p(\theta) \propto \theta^{2-1} e^{-2\theta}, \tag{3.81}$$

$$\text{Likelihood:} \quad p(y_1^5|\theta) = \theta^5 e^{-10\theta}, \tag{3.82}$$

$$\text{Posterior:} \quad p(\theta|y_1^5) \propto \theta^{2+5-1} e^{-(2+10)\theta}, \tag{3.83}$$

so $\Theta|y_1^5 \sim \text{Gamma}(7,12)$. The posterior mean is $(2+5)/(2+10) = 7/12 \approx 0.58$ per day, between the prior mean 1 and the maximum-likelihood estimate $n/y = 5/10 = 0.5$. The posterior variance $7/12^2 \approx 0.05$ is about a tenth of the prior variance $2/2^2 = 0.5$.



△

3.6 The Exponential Family (EF)

For a random variable $\boldsymbol{Y}$ with parameter $\boldsymbol{\Theta}$, $p(\boldsymbol{y}|\boldsymbol{\theta})$ is said to be from the **exponential family** if it has the form

$$p(\boldsymbol{y}|\boldsymbol{\theta}) = \exp(\langle \boldsymbol{a}(\boldsymbol{y}), \boldsymbol{b}(\boldsymbol{\theta}) \rangle + f(\boldsymbol{y}) + g(\boldsymbol{\theta})), \tag{3.84}$$

where $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{y}, \boldsymbol{\theta}$ can be vectors (in bold) and f, g are scalar functions. The function $\boldsymbol{b}(\boldsymbol{\theta})$ is the **natural parameter**. The data and the parameter meet only through the inner product $\langle \boldsymbol{a}(\boldsymbol{y}), \boldsymbol{b}(\boldsymbol{\theta}) \rangle$: f depends on $\boldsymbol{y}$ alone and g on $\boldsymbol{\theta}$ alone. Below, quantities computed from the data are in green, the prior's parameters in blue, and the posterior in red.

Example 3.16 (The exponential distribution)**.** For $y \geq 0$,

$$p(y|\theta) = \theta e^{-\theta y} = \exp\big(y \cdot (-\theta) + \ln \theta\big), \tag{3.85}$$

so $a(y) = y$, $b(\theta) = -\theta$, $f(y) = 0$, and $g(\theta) = \ln \theta$. △

The exponential family includes many common distributions, among them the Bernoulli, binomial (with a known number of trials), Poisson, exponential, Gaussian, Beta, and Gamma. For likelihoods in this family, the data can be summarized in a statistic of fixed size, and the conjugate prior can be written down directly.

Sufficient statistics. Suppose that we have n iid observations. The likelihood is a product, and a product of exponentials is the exponential of the sum:

$$p(\boldsymbol{y}_1^n|\boldsymbol{\theta}) = \prod_{i=1}^n p(\boldsymbol{y}_i|\boldsymbol{\theta}) = \exp\left(\left\langle \sum_{i=1}^n \boldsymbol{a}(\boldsymbol{y}_i), \boldsymbol{b}(\boldsymbol{\theta})\right\rangle + \sum_{i=1}^n f(\boldsymbol{y}_i) + ng(\boldsymbol{\theta})\right) \tag{3.86}$$

$$\propto \exp(\langle \boldsymbol{t}(\boldsymbol{y}_1^n), \boldsymbol{b}(\boldsymbol{\theta})\rangle + ng(\boldsymbol{\theta})), \tag{3.87}$$

where $\boldsymbol{t}(\boldsymbol{y}_1^n) = \sum_{i=1}^n \boldsymbol{a}(\boldsymbol{y}_i)$ is the **sufficient statistic**. The factor $\exp(\sum_i f(\boldsymbol{y}_i))$ does not involve $\boldsymbol{\theta}$, so it is absorbed into the proportionality constant. The maximum-likelihood estimate maximizes the likelihood over $\boldsymbol{\theta}$, and the posterior multiplies the likelihood by the prior and normalizes over $\boldsymbol{\theta}$. Both use the likelihood only as a function of $\boldsymbol{\theta}$, so a factor free of $\boldsymbol{\theta}$ changes neither, and both depend on the data only through $\boldsymbol{t}(\boldsymbol{y}_1^n)$. Within the model, the rest of the information in $\boldsymbol{y}_1^n$ is irrelevant for estimating $\boldsymbol{\theta}$. The size of $\boldsymbol{t}(\boldsymbol{y}_1^n)$ does not grow with n.

For the exponential distribution, $a(y) = y$ (Example 3.16), so $t(y_1^n) = \sum_{i=1}^n y_i$, the total waiting time. This is why the posterior in Example 3.15 depends on the data only through n and y.

Example 3.17 (Gaussian mean with known variance)**.** In $p(y|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\theta)^2/(2\sigma^2)}$, the exponent expands as

$$-\frac{(y-\theta)^2}{2\sigma^2} = y\,\frac{\theta}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{\theta^2}{2\sigma^2}. \tag{3.88}$$

So $a(y) = y$, $b(\theta) = \theta/\sigma^2$, $f(y) = -\frac{y^2}{2\sigma^2} - \frac{1}{2}\ln(2\pi\sigma^2)$ (the prefactor joins f), and $g(\theta) = -\frac{\theta^2}{2\sigma^2}$. The sufficient statistic is $t(y_1^n) = \sum_{i=1}^n y_i = n\bar{y}$, which is why the posterior (3.52) depends on the data only through $\bar{y}$. △

Bayesian estimation with conjugate priors. The conjugate prior of an exponential family has the form of the likelihood in θ, with two parameters $\boldsymbol{\nu}$ and m in the places of $\boldsymbol{t}(\boldsymbol{y}_1^n)$ and n:

$$p(\boldsymbol{y}_i|\boldsymbol{\theta}) \propto \exp(\langle \boldsymbol{a}(\boldsymbol{y}_i), \boldsymbol{b}(\boldsymbol{\theta})\rangle + g(\boldsymbol{\theta})) \tag{3.89}$$

$$p(\boldsymbol{y}_1^n|\boldsymbol{\theta}) \propto \exp(\langle \boldsymbol{t}(\boldsymbol{y}_1^n), \boldsymbol{b}(\boldsymbol{\theta})\rangle + ng(\boldsymbol{\theta})) \tag{3.90}$$

$$p(\boldsymbol{\theta}) \propto \exp(\langle \boldsymbol{\nu}, \boldsymbol{b}(\boldsymbol{\theta})\rangle + mg(\boldsymbol{\theta})) \qquad Dist(\boldsymbol{\nu}, m) \tag{3.91}$$

$$p(\boldsymbol{\theta}|\boldsymbol{y}_1^n) \propto \exp(\langle \boldsymbol{\nu} + \boldsymbol{t}(\boldsymbol{y}_1^n), \boldsymbol{b}(\boldsymbol{\theta})\rangle + (m+n)g(\boldsymbol{\theta})) \qquad Dist(\boldsymbol{\nu} + \boldsymbol{t}(\boldsymbol{y}_1^n), m+n), \tag{3.92}$$

where $Dist$ refers to a specific type of distribution, determined by $\boldsymbol{b}$ and g. Multiplying two exponentials of this form adds their coefficients of $\boldsymbol{b}(\boldsymbol{\theta})$ and of $g(\boldsymbol{\theta})$, so the posterior has the same form as the prior, and the prior is conjugate. The posterior adds the sufficient statistic $\boldsymbol{t}(\boldsymbol{y}_1^n)$ to $\boldsymbol{\nu}$ and the sample size n to m.

Pseudo-observations. The parameters in conjugate priors can be interpreted as representing pseudo-observations by comparing the forms of $p(\boldsymbol{y}_1^n|\boldsymbol{\theta})$ and $p(\boldsymbol{\theta})$. In particular, $\boldsymbol{\nu}$ plays the same role as $\boldsymbol{t}(\boldsymbol{y}_1^n)$ and m the same role as n: the prior acts like m earlier observations whose sufficient statistic is $\boldsymbol{\nu}$.

Example 3.18 (Bernoulli exponential family)**.** The likelihood for a Bernoulli observation is

$$p(y_i|\theta) = \theta^{y_i}(1-\theta)^{1-y_i} \tag{3.93}$$

$$= \exp(y_i \ln\theta + (1-y_i)\ln(1-\theta)) \tag{3.94}$$

$$= \exp\left(y_i \ln\frac{\theta}{1-\theta} + \ln(1-\theta)\right). \tag{3.95}$$

We thus let $a(y_i) = y_i$, $b(\theta) = \ln\frac{\theta}{1-\theta}$, and $g(\theta) = \ln(1-\theta)$. Furthermore, let $y = t(y_1^n) = \sum_{i=1}^n a(y_i) = \sum_{i=1}^n y_i$,

the number of successes. Then,

$$p(y_1^n|\theta) = \exp\left(y \ln \frac{\theta}{1-\theta} + n \ln(1-\theta)\right), \tag{3.96}$$

$$p(\theta) \propto \exp\left(\nu \ln \frac{\theta}{1-\theta} + m \ln(1-\theta)\right) = \left(\frac{\theta}{1-\theta}\right)^{\nu} (1-\theta)^m \tag{3.97}$$

$$= \theta^{\nu}(1-\theta)^{m-\nu}, \qquad \text{Beta}(\nu+1, m-\nu+1), \tag{3.98}$$

$$p(\theta|y_1^n) \propto \exp\left((\nu+y) \ln \frac{\theta}{1-\theta} + (m+n)\ln(1-\theta)\right) = \theta^{\nu+y}(1-\theta)^{m+n-\nu-y}, \tag{3.99}$$

$$\text{Beta}(\nu+y+1, m+n-\nu-y+1). \tag{3.100}$$

The Beta distributions are read off by matching the exponents to $\theta^{\alpha-1}(1-\theta)^{\beta-1}$. With $\alpha = \nu+1$ and $\beta = m-\nu+1$, the posterior is $\text{Beta}(\alpha+y, \beta+n-y)$, the Bernoulli/Beta pair of Section 3.5. For $\alpha, \beta \geq 1$, the prior acts like $\nu = \alpha - 1$ earlier successes in $m = \alpha + \beta - 2$ earlier trials. △

Example 3.19 (Exponential distribution, gamma prior)**.** From Example 3.16, $b(\theta) = -\theta$ and $g(\theta) = \ln\theta$, and $y = t(y_1^n) = \sum_{i=1}^n y_i$ is the total waiting time. Then,

$$p(\theta) \propto \exp\left(\nu \cdot (-\theta) + m \ln\theta\right) = \theta^m e^{-\nu\theta}, \qquad \text{Gamma}(m+1, \nu), \tag{3.101}$$

$$p(\theta|y_1^n) \propto \exp\left((\nu+y)\cdot(-\theta) + (m+n)\ln\theta\right) = \theta^{m+n}e^{-(\nu+y)\theta}, \qquad \text{Gamma}(m+n+1, \nu+y). \tag{3.102}$$

Matching the exponents to $\theta^{\alpha-1}e^{-\beta\theta}$, $\alpha = m+1$ and $\beta = \nu$, so the posterior is $\text{Gamma}(\alpha+n, \beta+y)$, the Exponential/Gamma pair of Section 3.5. For $\alpha \geq 1$, the prior acts like $m = \alpha - 1$ earlier waits totalling $\nu = \beta$ days. The Gamma(2, 2) prior of Example 3.15 is one earlier wait of 2 days. △

Exercise 3.20 (Poisson through the recipe)**.** A Poisson observation counts the events in one interval, $p(y|\lambda) = \lambda^y e^{-\lambda}/y!$ for $y = 0, 1, 2, \ldots$

(a) Write $p(y|\lambda)$ in the form (3.84). What are a, b, f, and g?

(b) Find the conjugate prior and the posterior after n iid counts, and compare the result with Example 3.6.

Solution. (a) $p(y|\lambda) = \exp(y\ln\lambda - \lambda - \ln y!)$, so $a(y) = y$, $b(\lambda) = \ln\lambda$, $f(y) = -\ln y!$, and $g(\lambda) = -\lambda$.

(b) The sufficient statistic is $t(y_1^n) = \sum_{i=1}^n y_i = n\bar{y}$, the total count. The conjugate prior is

$$p(\lambda) \propto \exp(\nu\ln\lambda - m\lambda) = \lambda^{\nu}e^{-m\lambda}, \qquad \text{Gamma}(\nu+1, m), \tag{3.103}$$

and the posterior is

$$p(\lambda|y_1^n) \propto \lambda^{\nu+n\bar{y}}e^{-(m+n)\lambda}, \qquad \text{Gamma}(\nu+n\bar{y}+1, m+n). \tag{3.104}$$

With $\alpha = \nu+1$ and $\beta = m$, the posterior is $\text{Gamma}(\alpha+n\bar{y}, \beta+n)$, as in Example 3.6. The prior acts like m earlier intervals in which ν events were counted.

△

Chapter 4

Multivariate Random Variables

In this chapter, we will review some topics related to random vectors, which will be of use in the following chapters.

4.1 Random vectors

A **random vector** is a vector of random variables.[1] Consider the random vectors $\boldsymbol{X}$ and $\boldsymbol{Y}$

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_m \end{pmatrix}, \quad \boldsymbol{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}. \tag{4.1}$$

The **expected value** of $\boldsymbol{X}$ is

$$\mathbb{E}\,\boldsymbol{X} = \begin{pmatrix} \mathbb{E}\,X_1 \\ \vdots \\ \mathbb{E}\,X_m \end{pmatrix}. \tag{4.2}$$

The **correlation matrix** of $\boldsymbol{X}$ and $\boldsymbol{Y}$ is the $m \times n$ matrix $\mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T]$, whose i,jth element is $\mathbb{E}[X_i Y_j]$. The **cross-covariance matrix** $\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y})$ of $\boldsymbol{X}$ and $\boldsymbol{Y}$ is the matrix $\mathbb{E}[(\boldsymbol{X} - \mathbb{E}\,\boldsymbol{X})(\boldsymbol{Y} - \mathbb{E}\,\boldsymbol{Y})^T]$, whose i,jth element is $\mathrm{Cov}(X_i, Y_j)$. The covariance of a vector $\boldsymbol{X}$ is $\mathrm{Cov}(\boldsymbol{X}) = \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{X})$. The **conditional expectation** $\mathbb{E}[\boldsymbol{X}|\boldsymbol{Y}]$ of $\boldsymbol{X}$ given $\boldsymbol{Y}$ is a vector whose ith element is $\mathbb{E}[X_i|\boldsymbol{Y}]$.

If the elements of $\boldsymbol{X}$ are uncorrelated, then $\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$ and the covariance matrix becomes diagonal. If, in addition, $\mathrm{Cov}(X_i, X_i) = \mathrm{Var}(X_i) = \sigma^2$, i.e., all elements of $\boldsymbol{X}$ have the same variance σ^2, then $\mathrm{Cov}(\boldsymbol{X}) = \sigma^2 I$.

Example 4.1 (Multinomial Distribution)**.** Consider rolling a fair six-sided die n times, with each roll being independent. Let

$$\boldsymbol{X} = (X_1, X_2, X_3, X_4, X_5, X_6)^T$$

be a random vector where X_i represents the number of times face i appears. The vector $\boldsymbol{X}$ follows a multinomial distribution with parameters n and $\boldsymbol{p} = (1/6, 1/6, 1/6, 1/6, 1/6, 1/6)^T$, where p_i is the probability of face i appearing on a single roll.

Since the faces must sum to n rolls, we have the constraint $\sum_{i=1}^{6} X_i = n$, which means the components

[1] We use lowercase bold letters to denote deterministic vectors, uppercase bold letters to denote random vectors, and uppercase sans serif letters, such as A, to denote matrices.

cannot be independent. For a fair die:

$$\mathbb{E}[X_i] = \frac{n}{6} \quad \text{for all } i, \tag{4.3}$$

$$\mathrm{Var}(X_i) = n \cdot \frac{1}{6} \cdot \frac{5}{6} = \frac{5n}{36}, \tag{4.4}$$

$$\mathrm{Cov}(X_i, X_j) = -n \cdot \frac{1}{6} \cdot \frac{1}{6} = -\frac{n}{36} \quad \text{for } i \neq j. \tag{4.5}$$

Hence, we have

$$\mathbb{E}[\boldsymbol{X}] = \frac{n}{6}\begin{pmatrix}1\\1\\1\\1\\1\\1\end{pmatrix}, \qquad \mathrm{Cov}(\boldsymbol{X}) = \frac{n}{36}\begin{pmatrix}5 & -1 & -1 & -1 & -1 & -1\\ -1 & 5 & -1 & -1 & -1 & -1\\ -1 & -1 & 5 & -1 & -1 & -1\\ -1 & -1 & -1 & 5 & -1 & -1\\ -1 & -1 & -1 & -1 & 5 & -1\\ -1 & -1 & -1 & -1 & -1 & 5\end{pmatrix}. \tag{4.6}$$

Note that the off-diagonal covariances are *negative*: if we observe more occurrences of one face, there must be fewer rolls available for the other faces. This is a natural example of negative correlation arising from a constraint. △

4.1.1 Properties of expectation and covariance

For deterministic matrices A, B, deterministic vectors $\boldsymbol{a}, \boldsymbol{b}$, and random vectors $\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{W}, \boldsymbol{Z}$, we have [1]

1. $\mathbb{E}[\mathsf{A}\boldsymbol{X} + \boldsymbol{a}] = \mathsf{A}\,\mathbb{E}\,\boldsymbol{X} + \boldsymbol{a}$
2. $\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y}) = \mathbb{E}[\boldsymbol{X}(\boldsymbol{Y} - \mathbb{E}\,\boldsymbol{Y})^T] = \mathbb{E}[(\boldsymbol{X} - \mathbb{E}\,\boldsymbol{X})\boldsymbol{Y}^T] = \mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T] - \mathbb{E}\,\boldsymbol{X}\,\mathbb{E}\,\boldsymbol{Y}^T$
3. $\mathbb{E}[(\mathsf{A}\boldsymbol{X})(\mathsf{B}\boldsymbol{Y})^T] = \mathsf{A}\,\mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^T]\mathsf{B}^T$
4. $\mathrm{Cov}(\mathsf{A}\boldsymbol{X} + \boldsymbol{a}, \mathsf{B}\boldsymbol{Y} + \boldsymbol{b}) = \mathsf{A}\,\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y})\mathsf{B}^T$
5. $\mathrm{Cov}(\mathsf{A}\boldsymbol{X} + \boldsymbol{a}) = \mathsf{A}\,\mathrm{Cov}(\boldsymbol{X})\mathsf{A}^T$
6. $\mathrm{Cov}(\boldsymbol{W} + \boldsymbol{X}, \boldsymbol{Y} + \boldsymbol{Z}) = \mathrm{Cov}(\boldsymbol{W}, \boldsymbol{Y}) + \mathrm{Cov}(\boldsymbol{W}, \boldsymbol{Z}) + \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y}) + \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Z})$

Example 4.2 (Covariance under scaling)**.** For a random vector $\boldsymbol{X}$ and constants $a, \boldsymbol{b}$, from property 5, we have $\mathrm{Cov}(a\boldsymbol{X} + \boldsymbol{b}) = a^2\,\mathrm{Cov}(\boldsymbol{X})$. We also prove this using the other properties. The relevant properties are given in each step.

$$\begin{aligned}\mathrm{Cov}(a\boldsymbol{X} + \boldsymbol{b}) &= \mathrm{Cov}(a\boldsymbol{X} + \boldsymbol{b}, a\boldsymbol{X} + \boldsymbol{b}) && (4.7)\\ &\overset{6}{=} \mathrm{Cov}(a\boldsymbol{X}, a\boldsymbol{X}) + \mathrm{Cov}(a\boldsymbol{X}, \boldsymbol{b}) + \mathrm{Cov}(\boldsymbol{b}, a\boldsymbol{X}) + \mathrm{Cov}(\boldsymbol{b}, \boldsymbol{b}) && (4.8)\\ &\overset{2}{=} a^2\,\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{X}) + 0 + 0 + 0 && (4.9)\end{aligned}$$

△

Example 4.3 (Prime and odd counts)**.** Continuing Example 4.1, let us consider a linear transformation that counts specific types of outcomes. Define:

- Y_1 = number of prime outcomes (faces showing 2, 3, or 5)
- Y_2 = number of odd outcomes (faces showing 1, 3, or 5)

We can express $\boldsymbol{Y} = (Y_1, Y_2)^T$ as a linear transformation of $\boldsymbol{X}$:

$$\boldsymbol{Y} = \mathsf{A}\boldsymbol{X}, \quad \text{where } \mathsf{A} = \begin{pmatrix}0 & 1 & 1 & 0 & 1 & 0\\ 1 & 0 & 1 & 0 & 1 & 0\end{pmatrix}. \tag{4.10}$$

The first row of A selects faces 2, 3, and 5 (primes), while the second row selects faces 1, 3, and 5 (odd numbers). Using property 1 and property 5, we can compute:

$$\mathbb{E}[\boldsymbol{Y}] = \mathsf{A}\,\mathbb{E}[\boldsymbol{X}] = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} n/6 \\ \vdots \\ n/6 \end{pmatrix} = \begin{pmatrix} n/2 \\ n/2 \end{pmatrix}, \tag{4.11}$$

$$\mathrm{Cov}(\boldsymbol{Y}) = \mathsf{A}\,\mathrm{Cov}(\boldsymbol{X})\mathsf{A}^T. \tag{4.12}$$

Computing $\mathsf{A}\,\mathrm{Cov}(\boldsymbol{X})\mathsf{A}^T$:

$$\mathrm{Cov}(\boldsymbol{Y}) = \frac{n}{36} \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 5 & -1 & -1 & -1 & -1 & -1 \\ -1 & 5 & -1 & -1 & -1 & -1 \\ -1 & -1 & 5 & -1 & -1 & -1 \\ -1 & -1 & -1 & 5 & -1 & -1 \\ -1 & -1 & -1 & -1 & 5 & -1 \\ -1 & -1 & -1 & -1 & -1 & 5 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \\ 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{pmatrix} \tag{4.13}$$

$$= \frac{n}{36} \begin{pmatrix} 9 & 3 \\ 3 & 9 \end{pmatrix} = \begin{pmatrix} n/4 & n/12 \\ n/12 & n/4 \end{pmatrix}. \tag{4.14}$$

Thus, $\mathrm{Var}(Y_1) = \mathrm{Var}(Y_2) = n/4$ and $\mathrm{Cov}(Y_1, Y_2) = n/12 > 0$. The positive covariance makes sense: outcomes 3 and 5 contribute to both counts, so observing more primes tends to mean observing more odd numbers. △

4.2 Gaussian Random Vectors (Multivariate Normal Distribution)

Recall that a random variable X is Gaussian (normal) with mean μ and variance $\sigma^2 > 0$ if the pdf of X is given by

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{4.15}$$

Definition 4.4 (Jointly Gaussian random variables)**.** A collection of random variables is **jointly Gaussian** if any linear combination of these variables is Gaussian. A **Gaussian random vector**, also known as a *multivariate normal vector*, is a vector whose elements are jointly Gaussian. A collection of random vectors is jointly Gaussian if the vector obtained by concatenating them is jointly Gaussian.

Example 4.5 (Linear combination of Gaussians)**.** If $\begin{pmatrix} X \\ Y \end{pmatrix}$ is a Gaussian vector, then $Z = 2X + 3Y$ is Gaussian. Furthermore,

$$\mathbb{E}[Z] = 2\,\mathbb{E}[X] + 3\,\mathbb{E}[Y], \tag{4.16}$$

$$\mathrm{Var}(Z) = \mathrm{Cov}(2X+3Y, 2X+3Y) = 4\,\mathrm{Cov}(X,X) + 12\,\mathrm{Cov}(X,Y) + 9\,\mathrm{Cov}(Y,Y) \tag{4.17}$$

$$= 4\,\mathrm{Var}(X) + 12\,\mathrm{Cov}(X,Y) + 9\,\mathrm{Var}(Y), \tag{4.18}$$

which completely characterizes the distribution of Z as $Z \sim \mathcal{N}(\mathbb{E}[Z], \mathrm{Var}(Z))$. △

For a Gaussian random vector $\boldsymbol{X}$ of dimension d, with mean $\mathbb{E}[\boldsymbol{X}] = \boldsymbol{\mu}$ and covariance matrix $\mathsf{K} = \mathrm{Cov}(\boldsymbol{X}) = \mathbb{E}[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^T]$, we have

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{d/2}|\mathsf{K}|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \mathsf{K}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right), \tag{4.19}$$

provided that the covariance matrix is invertible.

For a Gaussian random vector, the elements of $\boldsymbol{X}$ are *independent* if and only if the covariance matrix is diagonal.

4.2.1 Mahalanobis Distance

The **Mahalanobis distance** is a distance metric that measures the distance between a point and a distribution. For a distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix K, the Mahalanobis distance between a point $\boldsymbol{x}$ and the distribution is defined as:

$$\Delta(\boldsymbol{x}, \boldsymbol{\mu}) = \sqrt{(\boldsymbol{x} - \boldsymbol{\mu})^T \mathsf{K}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})} \tag{4.20}$$

With this definition, the Gaussian pdf in (4.19) can be rewritten as

$$\log p_{\boldsymbol{X}}(\boldsymbol{x}) \doteq -\frac{1}{2}\Delta^2(\boldsymbol{x}, \boldsymbol{\mu}), \tag{4.21}$$

where $\doteq$ indicates equality up to an additive constant that does not depend on $\boldsymbol{x}$.

This distance measure differs from the Euclidean distance by accounting for the correlation structure between variables through the covariance matrix K. It is invariant under full-rank linear transformations of the data.

The Mahalanobis distance provides geometric intuition for multivariate Gaussian distributions. Points with the same probability density lie on ellipsoids of constant Mahalanobis distance from the mean. The principal axes of these ellipsoids align with the eigenvectors of the covariance matrix, and their lengths are proportional to the square roots of the eigenvalues.

Example 4.6 (Correlation and ellipse shape)**.** Recall that the **correlation coefficient** ρ between two random variables X_1 and X_2 is defined as

$$\rho = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)\,\mathrm{Var}(X_2)}} \tag{4.22}$$

The covariance matrix of a bivariate Gaussian random vector $\boldsymbol{X} = (X_1, X_2)^T$ can be expressed in terms of the variances $\sigma_1^2 = \mathrm{Var}(X_1)$, $\sigma_2^2 = \mathrm{Var}(X_2)$, and the correlation coefficient ρ as

$$\mathsf{K} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

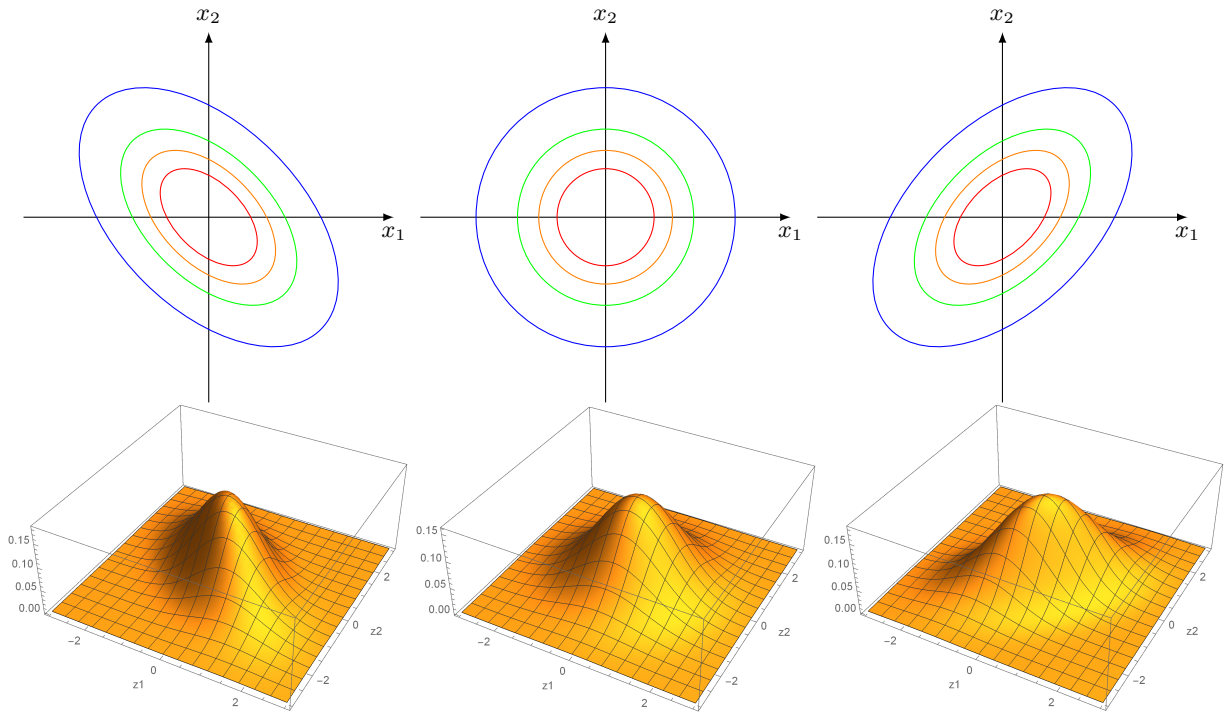
Consider three cases with $\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and variances $\sigma_1^2 = \sigma_2^2 = 1$, but different correlation coefficients:

- Case 1: $\rho = -0.5$, giving $\mathsf{K}_1 = \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix}$
- Case 2: $\rho = 0$, giving $\mathsf{K}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
- Case 3: $\rho = 0.5$, giving $\mathsf{K}_3 = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$

Plots for these distributions are given in Figure 4.1 illustrate how the correlation affects the shape of the distribution for a bivariate Gaussian, when the variances are the same:

- Negative correlation ($\rho < 0$): The ellipse is oriented along the line $y = -x$, indicating that as one variable increases, the other tends to decrease.
- Zero correlation ($\rho = 0$): The ellipse becomes a circle, indicating that the variables are independent.
- Positive correlation ($\rho > 0$): The ellipse is oriented along the line $y = x$, indicating that both variables tend to increase or decrease together.

△



(a) $\rho = -0.5$ (b) $\rho = 0$ (c) $\rho = 0.5$

Figure 4.1: Contours of equal probability density and 3D pdfs for bivariate Gaussian distributions with the same variance but different correlation values. When $\rho < 0$, the ellipse is oriented along $y = -x$; when $\rho = 0$, we get a circle; when $\rho > 0$, the ellipse is oriented along $y = x$.

If $\mathsf{K} = \sigma^2 \mathsf{I}$ (i.e., the variables are uncorrelated with equal variances), the Mahalanobis distance is proportional to the Euclidean distance. The Mahalanobis distance can be extended to measure the **distance between two normal distributions** $\mathcal{N}(\boldsymbol{\mu}_1, \mathsf{K})$ and $\mathcal{N}(\boldsymbol{\mu}_2, \mathsf{K})$ with the same covariance:

$$\Delta(\boldsymbol{\mu}_1, \boldsymbol{\mu}_2) = \sqrt{(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \mathsf{K}^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)} \tag{4.23}$$

These properties make the Mahalanobis distance particularly useful in multivariate analysis, pattern recognition, and classification problems where the correlation structure of the data is important.

4.2.2 Conditional distributions of a Gaussian vector

Suppose $\boldsymbol{X}$ is a d-dimensional Gaussian random vector with mean $\boldsymbol{\mu}$ and covariance matrix K, i.e., $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathsf{K})$. Partition $\boldsymbol{X}$ as

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_a \\ \boldsymbol{X}_b \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}, \quad \mathsf{K} = \begin{pmatrix} \mathsf{K}_{aa} & \mathsf{K}_{ab} \\ \mathsf{K}_{ba} & \mathsf{K}_{bb} \end{pmatrix},$$

where $\boldsymbol{X}_a$ is k-dimensional and $\boldsymbol{X}_b$ is $(d-k)$-dimensional.

The conditional distribution of $\boldsymbol{X}_a$ given $\boldsymbol{X}_b = \boldsymbol{x}_b$ is also Gaussian:

$$\boldsymbol{X}_a \mid \boldsymbol{X}_b = \boldsymbol{x}_b \sim \mathcal{N}(\boldsymbol{\mu}_{a|b}, \mathsf{K}_{a|b}),$$

where

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \mathsf{K}_{ab}\mathsf{K}_{bb}^{-1}(\boldsymbol{x}_b - \boldsymbol{\mu}_b), \tag{4.24}$$

$$\mathsf{K}_{a|b} = \mathsf{K}_{aa} - \mathsf{K}_{ab}\mathsf{K}_{bb}^{-1}\mathsf{K}_{ba}. \tag{4.25}$$

This result is fundamental in regression, inference, and Gaussian process modeling, as it describes how observing some components of a Gaussian vector affects the distribution of the remaining components.

Example 4.7 (Bivariate Gaussian conditioning)**.** Let $\boldsymbol{X} = (X_1, X_2)^T \sim \mathcal{N}(\boldsymbol{\mu}, \mathsf{K})$, with

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \mathsf{K} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

Then the conditional distribution of X_1 given $X_2 = x_2$ is

$$X_1 \mid X_2 = x_2 \sim \mathcal{N}\Big(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2),\ (1-\rho^2)\sigma_1^2\Big).$$

We thus have $\mathbb{E}[X_1|X_2 = x_2] = \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$ and $\mathrm{Var}(X_1|X_2 = x_2) = (1-\rho^2)\sigma_1^2$. △

Example 4.8 (Standardized bivariate Gaussian)**.** Let us standardize $\boldsymbol{X}$ of the previous example. The standardized variables are given by

$$Z_1 = \frac{X_1 - \mu_1}{\sigma_1}, \quad Z_2 = \frac{X_2 - \mu_2}{\sigma_2}.$$

It follows that (Z_1, Z_2) are jointly Gaussian with

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \sim \mathcal{N}\Big(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\Big).$$

The conditional distribution of Z_1 given $Z_2 = z_2$ is

$$Z_1 \mid Z_2 = z_2 \sim \mathcal{N}(\rho z_2, 1 - \rho^2).$$

We thus have $\mathbb{E}[Z_1|Z_2 = z_2] = \rho z_2$ and $\mathrm{Var}(Z_1|Z_2 = z_2) = 1 - \rho^2$. This gives a clear interpretation of the

correlation coefficient ρ: On average, for each unit increase in Z_2, Z_1 increases by ρ units. The variance of Z_1 given Z_2 is reduced by a factor of $1-\rho^2$, reflecting the information gained about Z_1 from knowing Z_2. △

Exercise 4.9 (Regression coefficient symmetry)**.** We said above that for every unit of increase in Z_2, Z_1 increases by ρ units on average. Does that mean that for every unit of increase in Z_1, Z_2 increases by $1/\rho$ units on average? Explain. △

4.2.3 Maximum likelihood estimation

Consider a d-dimensional random vector $\boldsymbol{X} = (X_1, \ldots, X_d)$ with distribution $\mathcal{N}(\boldsymbol{\theta}^*, \mathsf{K}^*)$ given in (4.19), where $\boldsymbol{\theta}^*, \mathsf{K}^*$ are unknown. Suppose we are interested in the relationship between X_d and $X_1, \ldots, X_{d-1}$. For example, for $\boldsymbol{X}^T = (X_1, X_2, X_3)$, X_1 and X_2 could indicate the heights of the parents and X_3 could be the height of the child. We may, for example, be interested in finding $\mathbb{E}[X_d | X_1, \ldots, X_{d-1}]$, thus estimating X_d based on $X_1, \ldots, X_{d-1}$. If we find the distribution, in other words, $\boldsymbol{\theta}^*, \mathsf{K}^*$, we can do so. Furthermore, the matrix K^* can indicate which dimensions are more strongly correlated.

Consider a set of n iid samples $\mathcal{D} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$, where each $\boldsymbol{x}_i$ is a sample of $\boldsymbol{X}$. We denote the elements of $\boldsymbol{x}_i$ as $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{id})$.

To estimate $\boldsymbol{\theta}^*$ and K^*, we write

$$\ell(\boldsymbol{\theta}, \mathsf{K}) = \ln p(\mathcal{D}; \boldsymbol{\theta}, \mathsf{K}) = \sum_{i=1}^{n} \ln p(\boldsymbol{x}_i; \boldsymbol{\theta}, \mathsf{K}) \tag{4.26}$$

$$\doteq \frac{n}{2} \ln |\mathsf{K}^{-1}| - \frac{1}{2} \sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{\theta})^T \mathsf{K}^{-1} (\boldsymbol{x}_i - \boldsymbol{\theta}), \tag{4.27}$$

where we have used the fact that $|\mathsf{K}^{-1}| = \frac{1}{|\mathsf{K}|}$.

As seen in the appendix (last chapter), for a symmetric matrix A, we have $\frac{d}{d\boldsymbol{v}}(\boldsymbol{y}^T \mathsf{A} \boldsymbol{y}) = 2\boldsymbol{y}^T \mathsf{A} \frac{d\boldsymbol{y}}{d\boldsymbol{v}}$. Hence,

$$\frac{\partial \ell}{\partial \boldsymbol{\theta}} = -\frac{1}{2} \sum_{i=1}^{n} 2(\boldsymbol{x}_i - \boldsymbol{\theta})^T \mathsf{K}^{-1} (-\mathsf{I}) = \sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{\theta})^T \mathsf{K}^{-1}. \tag{4.28}$$

Setting this equal to zero yields

$$\hat{\boldsymbol{\theta}}_{ML} = \bar{\boldsymbol{x}} = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{x}_i. \tag{4.29}$$

Exercise 4.10 (Maximum likelihood covariance estimate)**.** Using the facts

$$\frac{\partial}{\partial \mathsf{A}} \boldsymbol{x}^T \mathsf{A} \boldsymbol{x} = \boldsymbol{x}\boldsymbol{x}^T, \quad \frac{\partial}{\partial \mathsf{A}} \ln |\mathsf{A}| = \mathsf{A}^{-T} \tag{4.30}$$

prove that

$$\hat{\mathsf{K}}_{ML} = \frac{1}{n} \sum_{i=1}^{n} (\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^T \tag{4.31}$$

△

4.2.4 Bayesian estimation

We now solve the same problem using Bayesian estimation, with the following likelihood

$$\boldsymbol{X}|\boldsymbol{\Theta} \sim \mathcal{N}(\boldsymbol{\Theta}, \mathsf{K}), \tag{4.32}$$

$$p(\boldsymbol{x}_1^n|\boldsymbol{\theta}) \propto \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\theta})^T\mathsf{K}^{-1}(\boldsymbol{x}_i-\boldsymbol{\theta})\right), \tag{4.33}$$

where, for simplicity, we assume K is known and we only need to estimate $\boldsymbol{\Theta}$. As the prior, we choose

$$\boldsymbol{\Theta} \sim \mathcal{N}(\boldsymbol{\mu}_0, \mathsf{S}_0) \tag{4.34}$$

$$p(\boldsymbol{\theta}) \propto \exp\left(-\frac{1}{2}(\boldsymbol{\theta}-\boldsymbol{\mu}_0)^T\mathsf{S}_0^{-1}(\boldsymbol{\theta}-\boldsymbol{\mu}_0)\right). \tag{4.35}$$

Hence,

$$p(\boldsymbol{\theta}|\boldsymbol{x}_1^n) \propto \exp\left(-\frac{1}{2}(\boldsymbol{\theta}-\boldsymbol{\mu}_0)^T\mathsf{S}_0^{-1}(\boldsymbol{\theta}-\boldsymbol{\mu}_0) - \frac{1}{2}\sum_{i=1}^{n}(\boldsymbol{x}_i-\boldsymbol{\theta})^T\mathsf{K}^{-1}(\boldsymbol{x}_i-\boldsymbol{\theta})\right). \tag{4.36}$$

The exponent in the posterior is quadratic in $\boldsymbol{\theta}$, indicating that $\boldsymbol{\Theta}$ has a Gaussian distribution. So $\boldsymbol{\Theta}|\boldsymbol{x}_1^n \sim \mathcal{N}(\hat{\boldsymbol{\theta}}_n, \mathsf{S}_n)$, for appropriate choices of $\hat{\boldsymbol{\theta}}_n$ and S_n,

$$p(\boldsymbol{\theta}|\boldsymbol{x}_1^n) \propto \exp\left(-\frac{1}{2}(\boldsymbol{\theta}-\hat{\boldsymbol{\theta}}_n)^T\mathsf{S}_n^{-1}(\boldsymbol{\theta}-\hat{\boldsymbol{\theta}}_n)\right). \tag{4.37}$$

To find $\hat{\boldsymbol{\theta}}_n$ and S_n, we equate (4.36) and (4.37), ignoring constant multiplicative factors, which leads to

$$(\boldsymbol{\theta}-\boldsymbol{\mu}_0)^T\mathsf{S}_0^{-1}(\boldsymbol{\theta}-\boldsymbol{\mu}_0) + \sum_{i=1}^{n}(\boldsymbol{\theta}-\boldsymbol{x}_i)^T\mathsf{K}^{-1}(\boldsymbol{\theta}-\boldsymbol{x}_i) \doteq (\boldsymbol{\theta}-\hat{\boldsymbol{\theta}}_n)^T\mathsf{S}_n^{-1}(\boldsymbol{\theta}-\hat{\boldsymbol{\theta}}_n), \tag{4.38}$$

$$\boldsymbol{\theta}^T\mathsf{S}_0^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T\mathsf{S}_0^{-1}\boldsymbol{\mu}_0 + n\boldsymbol{\theta}^T\mathsf{K}^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T\mathsf{K}^{-1}\sum_{i=1}^{n}\boldsymbol{x}_i \doteq \boldsymbol{\theta}^T\mathsf{S}_n^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T\mathsf{S}_n^{-1}\hat{\boldsymbol{\theta}}_n. \tag{4.39}$$

Here, we have used the fact that

$$(\boldsymbol{a}-\boldsymbol{b})^T\mathsf{A}(\boldsymbol{a}-\boldsymbol{b}) = \boldsymbol{a}^T\mathsf{A}\boldsymbol{a} - \boldsymbol{a}^T\mathsf{A}\boldsymbol{b} - \boldsymbol{b}^T\mathsf{A}\boldsymbol{a} + \boldsymbol{b}^T\mathsf{A}\boldsymbol{b} = \boldsymbol{a}^T\mathsf{A}\boldsymbol{a} - 2\boldsymbol{a}^T\mathsf{A}\boldsymbol{b} + \boldsymbol{b}^T\mathsf{A}\boldsymbol{b},$$

for vectors $\boldsymbol{a}, \boldsymbol{b}$ and a symmetric matrix A. Note that $\boldsymbol{a}^T\mathsf{A}\boldsymbol{b} = \boldsymbol{b}^T\mathsf{A}\boldsymbol{a}$, as both sides are scalars and $\boldsymbol{a}^T\mathsf{A}\boldsymbol{b} = (\boldsymbol{a}^T\mathsf{A}\boldsymbol{b})^T = \boldsymbol{b}^T\mathsf{A}\boldsymbol{a}$.

We now collect the terms of the form $\boldsymbol{\theta}^T\mathsf{A}\boldsymbol{\theta}$,

$$\boldsymbol{\theta}^T(\mathsf{S}_0^{-1} + n\mathsf{K}^{-1})\boldsymbol{\theta} - 2\boldsymbol{\theta}^T(\mathsf{S}_0^{-1}\boldsymbol{\mu}_0 + \mathsf{K}^{-1}\sum_{i=1}^{n}\boldsymbol{x}_i) \doteq \boldsymbol{\theta}^T\mathsf{S}_n^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T\mathsf{S}_n^{-1}\hat{\boldsymbol{\theta}}_n, \tag{4.40}$$

leading to the following values for the parameters of the posterior distribution $\boldsymbol{\Theta}|\boldsymbol{x}_1^n \sim \mathcal{N}(\hat{\boldsymbol{\theta}}_n, \mathsf{S}_n)$,

$$\mathsf{S}_n^{-1} = \mathsf{S}_0^{-1} + n\mathsf{K}^{-1}, \tag{4.41}$$

$$\hat{\boldsymbol{\theta}}_n = \mathsf{S}_n(\mathsf{S}_0^{-1}\boldsymbol{\mu}_0 + n\mathsf{K}^{-1}\bar{\boldsymbol{x}}) \tag{4.42}$$

$$= (\mathsf{S}_0^{-1} + n\mathsf{K}^{-1})^{-1}(\mathsf{S}_0^{-1}\boldsymbol{\mu}_0 + n\mathsf{K}^{-1}\bar{\boldsymbol{x}}), \tag{4.43}$$

where $\bar{\boldsymbol{x}}$ is $\sum_{i=1}^{n}\boldsymbol{x}_i/n$. The posterior mean, $\hat{\boldsymbol{\theta}}_n$, which we can also view as a point estimate, is the weighted average of the prior mean $\boldsymbol{\mu}_0$ and what is suggested by the data $\bar{\boldsymbol{x}}$.

Exercise 4.11 (Isotropic covariance case)**.** Find $\hat{\boldsymbol{\theta}}_n$ and S_n^{-1} when $\mathsf{S}_0 = s^2\mathsf{I}$ and $\mathsf{K} = \sigma^2\mathsf{I}$ and interpret the results. △

References

[1] Bruce Hajek. *Random Processes for Engineers*. Illinois, 2014. URL: http://hajek.ece.illinois.edu/Papers/randomprocJuly14.pdf (visited on 01/30/2017).

Chapter 5

Linear Regression

5.1 Introduction

The goal of *regression* is to predict a real value y as a function of the input variable $\boldsymbol{x}$. (The vector $\boldsymbol{x}$ is referred to as the feature vector, while y is called the target variable.) For example, in a marketing campaign, we may be interested in predicting total sales, given ad budgets in various platforms based on prior experience. Our data is a set of pairs $\mathcal{D} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$, collected from n prior ad campaigns for n previous products,

Product		TV ads	Print ads	Web ads	Sale
1	$\boldsymbol{x}_1^T =$	\$20k	\$10k	\$10k	$y_1 =$ \$500k
$\vdots$			$\vdots$		$\vdots$
i	$\boldsymbol{x}_i^T =$	x_{i1}	x_{i2}	x_{i3}	y_i
$\vdots$			$\vdots$		$\vdots$
n	$\boldsymbol{x}_n^T =$	x_{n1}	x_{n2}	x_{n3}	y_n

If we can predict y for any given value of $\boldsymbol{x}$, we can predict the outcome of a marketing campaign or optimize the marketing budget.We can also study what types of ads are more helpful, etc.

In *linear regression* our prediction for y is $\boldsymbol{x}^T\boldsymbol{\theta}$, where $\boldsymbol{x}$ and $\boldsymbol{\theta}$ are elements of $\mathbb{R}^d$. In our marketing example, our goal becomes to find $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)$ such that $\boldsymbol{x}^T\boldsymbol{\theta} = \boldsymbol{\theta}^T\boldsymbol{x} = \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$ is a good predictor for y.

Empirical Risk Minimization (ERM). Since we usually do not know the joint distribution but have access to data $\mathcal{D} = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, we cannot directly minimize the expected loss. Instead we can minimize the **empirical risk**, i.e., the loss on observed data points,

$$f^* = \arg\min_f \mathbb{E}[L(Y, f(X))] \quad \rightarrow \quad \hat{f}_n = \arg\min_f \frac{1}{n}\sum_{i=1}^{n} L(y_i, f(x_i)). \tag{5.1}$$

So instead of the best possible solution based on the distribution, f^*, we should try to find $\hat{f}_n$ based on n data points. But finding $\hat{f}_n$ is still problematic, as it only provides a way for us to determine the value of $f(x)$ for $x \in \{x_1, \ldots, x_n\}$. In other words, it is not able to extrapolate or generalize. To improve generalization and avoid overfitting, we restrict f to a hypothesis set $\mathcal{H}$ and often add regularization.

A common solution, which is also helpful from a practical point of view, is to restrict the choices for f to a

set $\mathcal{H}$, called the **hypothesis set**. This leads to the ERM formulation of the learning problem

$$f^*_{\mathcal{H}} = \arg\min_{f\in\mathcal{H}} \mathbb{E}[L(Y, f(X))] \quad \rightarrow \quad \hat{f}_{n,\mathcal{H}} = \arg\min_{f\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} L(y_i, f(x_i)). \tag{5.2}$$

For example, we may choose $\mathcal{H}$ to be the set of linear or sigmoid functions. By restricting predictors to the hypothesis set $\mathcal{H}$, we have introduced our prior knowledge, or *bias* towards the learning task.

From a probabilistic standpoint, we may consider each $(\boldsymbol{x}_i, y_i)$ to be an independent realization of the random pair $(\boldsymbol{X}, Y)$ with some joint distribution $p_{\boldsymbol{X},Y}$. We then formulate the linear regression problem as follows: Find

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathbb{E}[L(Y, \boldsymbol{X}^T\boldsymbol{\theta})], \tag{5.3}$$

for a given loss function L. As we typically do not have the joint distribution for $\boldsymbol{X}, Y$, we aim to find

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \frac{1}{n}\sum_{i=1}^{n} L(y_i, \boldsymbol{x}_i^T\boldsymbol{\theta}). \tag{5.4}$$

The linear form, $\hat{y} = \boldsymbol{x}^T\hat{\boldsymbol{\theta}} = \sum_{j=1}^{d}\hat{\theta}_j x_j$, may appear restrictive since it apparently excludes dependence on, for example, x_j^2. Imagine, in our marketing example, that buyers are likely to purchase a product if they see both TV ads and Web ads. In other words, y is large when x_1x_3 is large. It seems that this case is not covered well by linear regression. However, this is not the case since we can transform the input variable using a set of functions $g_1, \ldots, g_e$ and reformulate our prediction as $\sum_{j=1}^{e}\theta_j g_j(\boldsymbol{x})$, where g_j are any functions of $\boldsymbol{x}$, such as x_1^2 and x_1x_3. (But *feature extraction*, i.e., finding approrpiate fatures is a challenging problem.) Note that the expression $\sum_{j=1}^{e}\theta_j g_j(\boldsymbol{x})$ is still linear in $\boldsymbol{\theta}$, which is what matters, since we need to optimize $\boldsymbol{\theta}$.

Notation. Define $\mathsf{X} \in \mathbb{R}^{n\times d}$ and $\boldsymbol{y}$ as

$$\mathsf{X} = \begin{pmatrix} \boldsymbol{x}_1^T \\ \vdots \\ \boldsymbol{x}_n^T \end{pmatrix}, \qquad \boldsymbol{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \tag{5.5}$$

For a given value of $\boldsymbol{\theta}$, we let $\hat{\boldsymbol{y}} = \mathsf{X}\boldsymbol{\theta}$ to be the predicted value. Furthermore, let $\boldsymbol{\epsilon}$ be the error vector such that

$$\boldsymbol{y} = \hat{\boldsymbol{y}} + \boldsymbol{\epsilon} = \mathsf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}. \tag{5.6}$$

Example 5.1 (Three data points)**.** Suppose

$$\boldsymbol{x}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tag{5.7}$$

$$y_1 = -1, \qquad y_2 = 1, \qquad y_3 = 0. \tag{5.8}$$

Then

$$\mathsf{X} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \hat{\boldsymbol{y}} = \mathsf{X}\boldsymbol{\theta} = \theta_1\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + \theta_2\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad \boldsymbol{\epsilon} = \boldsymbol{y} - \hat{\boldsymbol{y}} = \begin{pmatrix} -1-\theta_2 \\ 1-2\theta_1 \\ -\theta_1-\theta_2 \end{pmatrix}. \tag{5.9}$$

△

5.2 Least-squares

A common choice for the loss function is

$$L(y_i, \boldsymbol{x}_i^T \boldsymbol{\theta}) = (y_i - \boldsymbol{x}_i^T \boldsymbol{\theta})^2. \tag{5.10}$$

The empirical risk can be written as

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{n} (y_i - \boldsymbol{x}_i^T \boldsymbol{\theta})^2 = \|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2, \tag{5.11}$$

where we have dropped the $1/n$ factor present in (5.4) as it does not affect our choice of $\boldsymbol{\theta}$. Denote

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}), \tag{5.12}$$

and define $\hat{\boldsymbol{y}} = \mathsf{X}\hat{\boldsymbol{\theta}}$ as the predicted value or estimate based on the model.

Least-squares is relatively easy to deal with from a computational perspective. It also has the same solution as the MLE for a common probabilistic model as we will see, thus providing an additional rationale for the resulting approach.

Projection onto the column space of X. Our first observation is that $\hat{\boldsymbol{y}}$ is in the column space of X, i.e., it is a linear combination of the columns of X. We can thus restate our goal as finding $\hat{\boldsymbol{y}}$ in the column space of X such that $\|\boldsymbol{y} - \hat{\boldsymbol{y}}\|$ is minimized. Hence, $\hat{\boldsymbol{y}}$ is the projection of $\boldsymbol{y}$ onto the column space of X as shown in Figure 5.1. Then, from the Projection Lemma in the Appendix, $\boldsymbol{y} - \hat{\boldsymbol{y}}$ is orthogonal to each column of X.

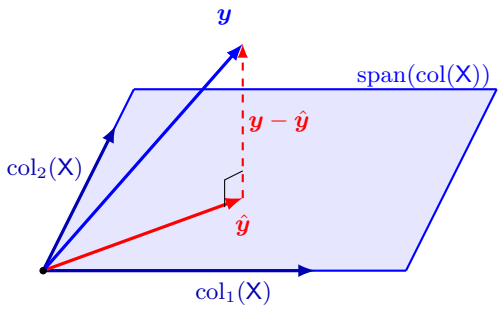


Figure 5.1: Error is minimized by projecting $\boldsymbol{y}$ onto the column space of X, Span(col(X)).

This orthogonality of the error to columns of X can be written as $\mathsf{X}^T(\boldsymbol{y} - \hat{\boldsymbol{y}}) = 0$. We have

$$\mathsf{X}^T(\boldsymbol{y} - \hat{\boldsymbol{y}}) = 0 \iff \mathsf{X}^T(\boldsymbol{y} - \mathsf{X}\hat{\boldsymbol{\theta}}) = 0 \tag{5.13}$$

$$\iff \mathsf{X}^T\boldsymbol{y} = \mathsf{X}^T\mathsf{X}\hat{\boldsymbol{\theta}} \tag{5.14}$$

$$\iff \hat{\boldsymbol{\theta}} = (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\boldsymbol{y} \tag{5.15}$$

Here we have assumed that $\mathsf{X}^T\mathsf{X}$ is invertible. This holds if the columns of X are linearly independent. To see this, we will show $\mathsf{X}^T\mathsf{X}\boldsymbol{\alpha} = \mathbf{0}$ implies that $\boldsymbol{\alpha} = \mathbf{0}$ if the columns of X are linearly independent:

$$\mathsf{X}^T\mathsf{X}\boldsymbol{\alpha} = \mathbf{0} \Rightarrow \boldsymbol{\alpha}^T\mathsf{X}^T\mathsf{X}\boldsymbol{\alpha} = 0 \Rightarrow (\mathsf{X}\boldsymbol{\alpha})^T(\mathsf{X}\boldsymbol{\alpha}) = 0 \Rightarrow \mathsf{X}\boldsymbol{\alpha} = \mathbf{0} \Rightarrow \boldsymbol{\alpha} = \mathbf{0}, \tag{5.16}$$

where the last step follows from the fact that the columns of X are linearly independent.

Example 5.2 (Least-squares solution)**.** From Example 5.1, we have

$$\mathsf{X} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \boldsymbol{y} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \tag{5.17}$$

and so

$$\mathsf{X}^T\mathsf{X} = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix}, \qquad (\mathsf{X}^T\mathsf{X})^{-1} = \frac{1}{9}\begin{pmatrix} 2 & -1 \\ -1 & 5 \end{pmatrix} \tag{5.18}$$

$$(\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T = \frac{1}{9}\begin{pmatrix} -1 & 4 & 1 \\ 5 & -2 & 4 \end{pmatrix} \qquad \hat{\boldsymbol{\theta}} = \begin{pmatrix} 5/9 \\ -7/9 \end{pmatrix} \tag{5.19}$$

$$\hat{\boldsymbol{y}} = \begin{pmatrix} -7/9 \\ 10/9 \\ -2/9 \end{pmatrix}, \qquad \boldsymbol{y} - \hat{\boldsymbol{y}} = \begin{pmatrix} -2/9 \\ -1/9 \\ 2/9 \end{pmatrix}. \tag{5.20}$$

△

Gradient descent. The closed-form solution provided in (5.15) for finding $\hat{\boldsymbol{\theta}}$ requires taking matrix inverses of possibly very large matrices, which could be computationally expensive. A less expensive solution is gradient descent, where we take the derivative of the loss to minimize it. Let $\nabla\mathcal{L}(\boldsymbol{\theta}) = \left(\frac{d\mathcal{L}}{d\boldsymbol{\theta}}\right)^T$ be the gradient of $\mathcal{L}$. Recall that the direction of the gradient indicates the direction of maximum increase and its magnitude represents the slope of the increase. We have

$$\mathcal{L} = (\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta})^T(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}), \tag{5.21}$$

$$\nabla\mathcal{L} = 2\big[(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta})^T(-\mathsf{X})\big]^T = -2\mathsf{X}^T(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}). \tag{5.22}$$

(Setting the gradient equal to 0 again gives $\hat{\boldsymbol{\theta}} = (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\boldsymbol{y}$. Note that the Hessian is $\mathsf{X}^T\mathsf{X}$, which is positive-semi-definite.) In gradient descent, we start from an arbitrary value $\boldsymbol{\theta}^{(0)}$ and move towards the solution in steps:

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} + \rho\mathsf{X}^T(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}^{(t)}) \tag{5.23}$$

$$= \boldsymbol{\theta}^{(t)} + \rho\sum_{i=1}^{n}\boldsymbol{x}_i(y_i - \boldsymbol{x}_i^T\boldsymbol{\theta}^{(t)}), \tag{5.24}$$

where ρ is the learning rate. This approach gets to the lowest point by moving in the direction of the *steepest descent* as shown in figure below for Example 5.1.

Standardization

We sometimes assume that X is standardized, meaning that each column $\boldsymbol{v}$ is shifted and scaled such that $\boldsymbol{v}^T\mathbf{1} = 0$ and $\boldsymbol{v}^T\boldsymbol{v} = 1$ and that $\boldsymbol{y}$ is centered so that $\boldsymbol{y}^T\mathbf{1} = 0$. Standardization of the inputs puts different features under the same scale and can help to reduce the correlation between features when having polynomial/interaction terms. Standardizing inputs can also be shown to be equivalent to minimizing the squared loss with an intercept term. †

Example 5.3 (Standardized inputs)**.** We show that standardizing inputs and then finding the solution with no intercept term is equivalent to minimizing the squared loss with an intercept term. By including an intercept term, the loss becomes

$$\mathcal{L}(\boldsymbol{\theta}, \theta_0) = \sum_{i=1}^{n}\big(y_i - \boldsymbol{x}_i^T\boldsymbol{\theta} - \theta_0\big)^2. \tag{5.25}$$

Such formulation has the advantage of allowing y to be nonzero when $\boldsymbol{x}$ is zero. Let the solution to this new

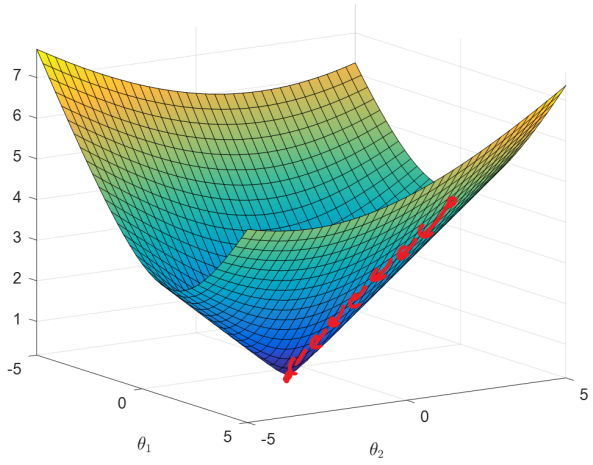


Figure 5.2: Gradient descent for linear regression

loss formulation be $\hat{\boldsymbol{\theta}}, \hat{\theta}_0$:

$$\hat{\theta}_0, \hat{\boldsymbol{\theta}} = \arg\min_{\theta_0, \theta} \mathcal{L}(\boldsymbol{\theta}, \theta_0) \tag{5.26}$$

and let $\hat{\boldsymbol{y}} = \mathsf{X}\hat{\boldsymbol{\theta}} + \hat{\theta}_0 \mathbf{1}$ be the predictions.

Let the columns of X be $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$. We provide an analysis that the optimal prediction $\hat{\boldsymbol{y}}$ found by considering the intercept term is the same as minimizing the normal squared loss (5.11) over the standardized inputs $\tilde{\mathsf{X}}$ and $\tilde{\boldsymbol{y}}$:

$$\check{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} \left(\tilde{y}_i - \tilde{\boldsymbol{x}}_i^T \boldsymbol{\theta}\right)^2, \tag{5.27}$$

$$\check{\boldsymbol{y}} = \tilde{\mathsf{X}}\check{\boldsymbol{\theta}} + \bar{y}\mathbf{1}, \tag{5.28}$$

where

$$\tilde{\mathsf{X}} = \begin{pmatrix} \tilde{\boldsymbol{x}}_1^T \\ \vdots \\ \tilde{\boldsymbol{x}}_n^T \end{pmatrix} = (\tilde{\boldsymbol{v}}_1, \ldots, \tilde{\boldsymbol{v}}_m), \quad \tilde{\boldsymbol{v}}_j = (\boldsymbol{v}_j - \beta_j \mathbf{1})/\alpha_j, \quad \beta_j = \frac{1}{n}\sum_{i=1}^{n} v_{ji}, \quad \alpha_j = \|\boldsymbol{v}_j - \beta_j \mathbf{1}\|_2, \tag{5.29}$$

$$\tilde{\boldsymbol{y}} = \boldsymbol{y} - \bar{y}\mathbf{1}, \qquad \bar{y} = \frac{1}{n}\sum_{j=1}^{n} y_j. \tag{5.30}$$

Let's first minimize the new loss (5.25). We can fix $\boldsymbol{\theta}$ for now. Then, the loss function is quadratic in θ_0 and the optimal choice for θ_0 can be found as

$$\hat{\theta}_0(\boldsymbol{\theta}) = \bar{y} - \sum_{j=1}^{m} \beta_j \theta_j. \tag{5.31}$$

Substituting θ_0 by $\hat{\theta}_0(\boldsymbol{\theta})$ in the loss (5.25) gives

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} (y_i - \boldsymbol{x}_i^T \boldsymbol{\theta} - \bar{y} + \sum_{j=1}^{m} \beta_j \theta_j)^2. \tag{5.32}$$

After rewriting (5.27) as

$$\check{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} (\tilde{y}_i - \tilde{\boldsymbol{x}}_i^T \boldsymbol{\theta})^2 = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} (y_i - \bar{y} - \sum_{j=1}^{m} \frac{(x_{ij} - \beta_j)}{\alpha_j} \theta_j)^2 \tag{5.33}$$

$$= \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} (y_i - \sum_{j=1}^{m} x_{ij} \frac{\theta_j}{\alpha_j} - \bar{y} + \sum_{j=1}^{m} \beta_j \frac{\theta_j}{\alpha_j})^2, \tag{5.34}$$

we can observe an analogy between (5.32) and (5.34). It follows that $\check{\theta}_j = \alpha_j \hat{\theta}_j$ for all $1 \leq j \leq m$.

Hence, for all $1 \leq i \leq n$,

$$\check{y}_i = \tilde{\boldsymbol{x}}_i^T \check{\boldsymbol{\theta}} + \bar{y} = \sum_{j=1}^{m} \tilde{x}_{ij} \check{\theta}_j + \bar{y} = \sum_{j=1}^{m} (x_{ij} - \beta_j) \hat{\theta}_j + \bar{y} = \boldsymbol{x}_i^T \hat{\boldsymbol{\theta}} + \hat{\theta}_0 = \hat{y}_i, \tag{5.35}$$

i.e., the predictions we obtained from (5.26) and (5.27) are equal. Note that the second last equality follows from $\hat{\theta}_0(\hat{\boldsymbol{\theta}}) = \hat{\theta}_0$. △

5.3 Probabilistic Models for Regression

So far we haven't made any assumptions regarding the statistics of the data. In this section, we consider two models: i) a model that only characterizes the mean and covariance of the error vector and ii) a Gaussian model.

5.3.1 General model

Let us now assume that

$$Y = \boldsymbol{x}^T \boldsymbol{\theta}^* + \epsilon, \qquad \mathbb{E}[\epsilon] = 0, \ \mathrm{Var}(\epsilon) = \sigma^2.$$

We have n samples $(\boldsymbol{x}_i, y_i)$. For simplicity, we will assume that $\boldsymbol{x}_i$ are deterministic. We will further assume for any $i, j, i \neq j$, ϵ_i and ϵ_j are uncorrelated. In vector form, we have

$$\boldsymbol{Y} = \mathsf{X}\boldsymbol{\theta}^* + \boldsymbol{\epsilon}, \qquad \mathbb{E}[\boldsymbol{\epsilon}] = \mathbf{0}, \ \mathrm{Cov}(\boldsymbol{\epsilon}) = \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] = \sigma^2 I.$$

At this point, we are not making any other assumptions related to the distribution of $\boldsymbol{\epsilon}$.[1]

Frequentist evaluation: Consider the estimator $\hat{\boldsymbol{\theta}} = (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\boldsymbol{y}$. Let us investigate the behavior of $\hat{\boldsymbol{\theta}}$ by viewing it as a random variable $\hat{\boldsymbol{\Theta}}$ under this model. We have

$$\mathbb{E}[\hat{\boldsymbol{\Theta}}] = \mathbb{E}[(\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\boldsymbol{Y}] \tag{5.36}$$

$$= (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\,\mathbb{E}[\boldsymbol{Y}] \tag{5.37}$$

$$= (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\,\mathbb{E}[\mathsf{X}\boldsymbol{\theta}^* + \boldsymbol{\epsilon}] \tag{5.38}$$

$$= \boldsymbol{\theta}^*, \tag{5.39}$$

indicating that $\hat{\boldsymbol{\theta}}$ is an unbiased estimate of $\boldsymbol{\theta}^*$. In particular, each dimension is estimated without bias, i.e., $\mathbb{E}[\hat{\Theta}_i] = \theta_i^*$.

[1] For ϵ, we will not follow the convention that random variables are shown as capital letters since capital ϵ can be confused with Latin E.

We also find

$$\begin{aligned}\mathrm{Cov}(\hat{\boldsymbol{\Theta}}) &= \mathrm{Cov}((\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\boldsymbol{Y}) && (5.40)\\ &= (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\,\mathrm{Cov}(\boldsymbol{Y})\mathsf{X}(\mathsf{X}^T\mathsf{X})^{-1} && (5.41)\\ &= (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\,\mathrm{Cov}(\boldsymbol{\epsilon})\mathsf{X}(\mathsf{X}^T\mathsf{X})^{-1} && (5.42)\\ &= (\mathsf{X}^T\mathsf{X})^{-1}\sigma^2. && (5.43)\end{aligned}$$

In particular, element i of the diagonal of $\mathrm{Cov}(\hat{\boldsymbol{\Theta}})$ is the variance of $\hat{\Theta}_i$ and also its MSE, as the estimator is unbiased.

The Gauss-Markov theorem. The Gauss-Markov theorem states that under the assumptions that $\mathbb{E}[\boldsymbol{\epsilon}] = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\epsilon}) = \sigma^2 I$, $\hat{\boldsymbol{\theta}}$ is the best *linear* unbiased estimator. Here, linear means that $\hat{\boldsymbol{\theta}}$ is linear in $\boldsymbol{y}$, i.e., $\hat{\boldsymbol{\theta}} = a_1 y_1 + a_2 y_2 + \cdots + a_m y_m$ for some scalars a_1^m. The Gauss-Markov theorem implies that for any[2] vector $\boldsymbol{u}$, $\boldsymbol{u}^T\hat{\boldsymbol{\theta}}$ is an unbiased estimator of $\boldsymbol{u}^T\boldsymbol{\theta}^*$ with the smallest possible variance.

5.3.2 Gaussian model

Let us further assume that ϵ_i are iid, with distribution $\mathcal{N}(0, \sigma^2)$, i.e., $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$. In other words, we have:

$$\begin{aligned}\boldsymbol{Y} &= \mathsf{X}\boldsymbol{\theta}^* + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I) && (5.44)\\ p_{\boldsymbol{Y}}(\boldsymbol{y}; \boldsymbol{\theta}^*, \sigma^2) &\sim \mathcal{N}(\mathsf{X}\boldsymbol{\theta}^*, \sigma^2 I). && (5.45)\end{aligned}$$

Exercise 5.4 (Independent Gaussian components)**.** Prove that if $p(\boldsymbol{y}; \boldsymbol{\theta}, \sigma^2) \sim \mathcal{N}(\mathsf{X}\boldsymbol{\theta}, \sigma^2 I)$, then for all i, $p(y_i; \boldsymbol{\theta}, \sigma^2) \sim \mathcal{N}(\boldsymbol{x}_i^T\boldsymbol{\theta}, \sigma^2)$ and the y_i are independent. △

Now we have a probabilistic model with unknown parameters $\boldsymbol{\theta}$ and σ^2.

Maximum Likelihood

Given that the covariance matrix is $\sigma^2 I$ and assuming that $\boldsymbol{y}$ is n-dimensional, the density and the likelihood are

$$\begin{aligned}p(\boldsymbol{y}; \boldsymbol{\theta}, \sigma^2) &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta})^T(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta})\right) && (5.46)\\ &\propto \frac{1}{\sigma^n} \exp\left(-\frac{\|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2}{2\sigma^2}\right) && (5.47)\\ \ell(\boldsymbol{\theta}, \sigma^2) &\doteq -n\ln(\sigma) - \frac{1}{2\sigma^2}\|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2. && (5.48)\end{aligned}$$

So maximizing for $\boldsymbol{\theta}$ leads to minimizing $\|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2 = \sum_{i=1}^n (y_i - \boldsymbol{x}_i^T\boldsymbol{\theta})^2$ which we already know the solution to:

$$\hat{\boldsymbol{\theta}}_{ML} = \hat{\boldsymbol{\theta}} = (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\boldsymbol{y}. \tag{5.49}$$

We can similarly show that

$$\hat{\sigma}^2_{ML} = \frac{1}{n}\sum_{i=1}^n (y_i - \boldsymbol{x}_i^T\hat{\boldsymbol{\theta}})^2. \tag{5.50}$$

The mean and covariance of $\hat{\boldsymbol{\theta}}$ are the same as in Section 5.3.1. But now we also know that $\hat{\boldsymbol{\theta}}$ *is Gaussian.*

[2] This isn't entirely precise!

This is because the linear combination of Gaussian variables is Gaussian. Hence,

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2(\mathsf{X}^T\mathsf{X})^{-1}). \tag{5.51}$$

Cramer-Rao Lower Bound. With the additional Gaussian assumption in this section, using Cramer-Rao lower bound, a stronger result compared to the Gauss-Markov theorem can be obtained. Namely, $\hat{\boldsymbol{\theta}}$ is the best unbiased estimator (not just the best linear unbiased estimator). †

Example 5.5 (Cramer-Rao lower bound)**.** For an unbiased vector estimator $\hat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$, the CRLB has the form

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \succcurlyeq I^{-1}(\boldsymbol{\theta}), \tag{5.52}$$

where $A \succcurlyeq B$ denotes that $A - B$ positive semidefinite. Let us find the CRLB for $\hat{\boldsymbol{\theta}}$ of (5.49). We have

$$\ell(\boldsymbol{\theta}, \sigma^2) \doteq -n\ln(\sigma) - \frac{1}{2\sigma^2}(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta})^T(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}). \tag{5.53}$$

$$\nabla_{\boldsymbol{\theta}}\ell = \left(-\frac{1}{\sigma^2}(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta})^T(-\mathsf{X})\right)^T \tag{5.54}$$

$$= \frac{1}{\sigma^2}\mathsf{X}^T(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}) \tag{5.55}$$

$$\mathsf{H}_{\boldsymbol{\theta}}\ell = \frac{d\nabla_{\boldsymbol{\theta}}\ell}{d\boldsymbol{\theta}} = -\frac{1}{\sigma^2}\mathsf{X}^T\mathsf{X}. \tag{5.56}$$

and so $I(\boldsymbol{\theta}) = \frac{1}{\sigma^2}\mathsf{X}^T\mathsf{X}$. Hence, $I(\boldsymbol{\theta})^{-1} = \sigma^2(\mathsf{X}^T\mathsf{X})^{-1}$, which matches the covariance of $\hat{\boldsymbol{\theta}}$ found in (5.43). △

Bayesian Linear Regression

In Bayesian linear regression, the Gaussian likelihood

$$\boldsymbol{y}|\boldsymbol{\theta}, \sigma^2 \sim \mathcal{N}(\mathsf{X}\boldsymbol{\theta}, \sigma^2 I) \tag{5.57}$$

is a common choice. But we also need to choose priors for $\boldsymbol{\theta}$ and σ^2. A possible non-informative choice is

$$p(\boldsymbol{\theta}, \sigma^2) \propto 1/\sigma^2, \tag{5.58}$$

or equivalently, $p(\sigma^2) \propto \frac{1}{\sigma^2}$, $p(\boldsymbol{\theta}) \propto 1$ and σ^2, and $\boldsymbol{\theta}$ are independent.

We are interested in finding

$$p(\boldsymbol{\theta}, \sigma^2|\boldsymbol{y}) = p(\boldsymbol{\theta}|\sigma^2, \boldsymbol{y})p(\sigma^2|\boldsymbol{y}) \tag{5.59}$$

We start with

$$p(\boldsymbol{\theta}|\boldsymbol{y}, \sigma^2) = \frac{p(\boldsymbol{\theta}, \boldsymbol{y}|\sigma^2)}{p(\boldsymbol{y}|\sigma^2)} \propto p(\boldsymbol{\theta}, \boldsymbol{y}|\sigma^2) = p(\boldsymbol{y}|\boldsymbol{\theta}, \sigma^2)p(\boldsymbol{\theta}|\sigma^2) \propto \exp\left(-\frac{(\mathsf{X}\boldsymbol{\theta} - \boldsymbol{y})^T(\mathsf{X}\boldsymbol{\theta} - \boldsymbol{y})}{2\sigma^2}\right). \tag{5.60}$$

Note that in the above expression, we are viewing $\boldsymbol{y}$ and σ^2 as given. So the expression $p(\boldsymbol{y}|\sigma^2)$ is treated as a constant and discarded. Furthermore, $p(\boldsymbol{\theta}|\sigma^2) = p(\boldsymbol{\theta}) \propto 1$.

The right-hand expression in (5.60) is quadratic in $\boldsymbol{\theta}$. So we'll try to see if we can write it in terms of a Gaussian distribution. With foresight, let the mean and the covariance of this distribution be denoted $\hat{\boldsymbol{\theta}}$ and $K\sigma^2$. We need

$$(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T K^{-1}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \doteq (\mathsf{X}\boldsymbol{\theta} - \boldsymbol{y})^T(\mathsf{X}\boldsymbol{\theta} - \boldsymbol{y}). \tag{5.61}$$

Ignoring terms that are constant in $\boldsymbol{\theta}$, we require

$$\boldsymbol{\theta}^T K^{-1}\boldsymbol{\theta} - 2\boldsymbol{\theta}^T K^{-1}\hat{\boldsymbol{\theta}} \doteq \boldsymbol{\theta}^T X^T X\boldsymbol{\theta} - 2\boldsymbol{\theta}^T X^T\boldsymbol{y}, \tag{5.62}$$

which is satisfied by $K^{-1} = \mathsf{X}^T\mathsf{X}$ and

$$-2\boldsymbol{\theta}^T K^{-1}\hat{\boldsymbol{\theta}} = -2\boldsymbol{\theta}^T X^T \boldsymbol{y}, \tag{5.63}$$

$$-2\boldsymbol{\theta}^T X^T X\hat{\boldsymbol{\theta}} = -2\boldsymbol{\theta}^T X^T \boldsymbol{y}, \tag{5.64}$$

$$\mathsf{X}^T\mathsf{X}\hat{\boldsymbol{\theta}} = \mathsf{X}^T\boldsymbol{y}, \tag{5.65}$$

$$\hat{\boldsymbol{\theta}} = (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\boldsymbol{y}. \tag{5.66}$$

So it suffices to set $\hat{\boldsymbol{\theta}} = (\mathsf{X}^T\mathsf{X})^{-1}\mathsf{X}^T\boldsymbol{y}$ and $K = (\mathsf{X}^T\mathsf{X})^{-1}$,

$$p(\boldsymbol{\theta}|\boldsymbol{y}, \sigma^2) \sim \mathcal{N}(\boldsymbol{\theta}; \hat{\boldsymbol{\theta}}, K\sigma^2). \tag{5.67}$$

Now we need to find $p(\sigma^2|\boldsymbol{y})$. Using the fact that $p(\sigma^2|\boldsymbol{y}) = p(\boldsymbol{\theta}, \sigma^2|\boldsymbol{y})/p(\boldsymbol{\theta}|\sigma^2, \boldsymbol{y})$, it can be shown that $p(\sigma^2|\boldsymbol{y})$ has a scaled inverse-χ^2 distribution,

$$p(\sigma^2|\boldsymbol{y}) \sim \text{Inv-}\chi^2(n-m, s^2), \tag{5.68}$$

where m is the dimension of $\boldsymbol{x}_i$ and

$$s^2 = \frac{1}{n-m}(\boldsymbol{y} - \mathsf{X}\hat{\boldsymbol{\theta}})^T(\boldsymbol{y} - \mathsf{X}\hat{\boldsymbol{\theta}}). \tag{5.69}$$

While we can continue analytically and find $p(\boldsymbol{\theta}|\boldsymbol{y})$, in practice, we proceed computationally by generating samples from $p(\sigma^2|\boldsymbol{y})$ and then $p(\boldsymbol{\theta}|\boldsymbol{y}, \sigma^2)$. With this sampling approach we can also perform prediction for a given input vector $\boldsymbol{x}_{n+1}$ of by producing samples from $p(y_{n+1}|\boldsymbol{\theta}, \sigma^2) \sim \mathcal{N}(\boldsymbol{x}_{n+1}^T\boldsymbol{\theta}, \sigma^2)$.

5.4 Regularized Linear Regression

Sometimes we are interested in reducing the flexibility of the model to avoid over-fitting, especially when the size of the data set is small. Alternatively, we may be interested in putting restrictions (e.g., forcing small coefficients to become 0) so that only the most important aspects of the data appear in the learned model, thus increasing its interpretability. These can be done by altering the loss function by adding a regularization term.

Ridge Regression

Ridge regression adds a penalty for the magnitude of the coefficients. Specifically, the loss function is

$$\mathcal{L}(\boldsymbol{\theta}) = \|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2 + \lambda\|\boldsymbol{\theta}\|_2^2, \tag{5.70}$$

where λ is a parameter determining the relative importance of the square error versus the regularization loss term $\|\boldsymbol{\theta}\|_2^2$. The problem of minimizing this loss,

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2 + \lambda\|\boldsymbol{\theta}\|_2^2, \tag{5.71}$$

can be shown to be equivalent to

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2, \tag{5.72}$$

$$\text{subject to : } \|\boldsymbol{\theta}\|_2^2 \leq t, \tag{5.73}$$

for some t. There is a one-to-one correspondence between λ and t. The second form is perhaps easier to understand because of the explicit constraints on $\|\boldsymbol{\theta}\|_2^2$.

From (5.71),

$$\nabla \mathcal{L}(\boldsymbol{\theta}) = -2\mathsf{X}^T(\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}) + 2\lambda\boldsymbol{\theta}, \tag{5.74}$$

$$\nabla \mathcal{L}(\hat{\boldsymbol{\theta}}) = 0 \iff \mathsf{X}^T(\boldsymbol{y} - \mathsf{X}\hat{\boldsymbol{\theta}}) = \lambda\hat{\boldsymbol{\theta}} \tag{5.75}$$

$$\iff \hat{\boldsymbol{\theta}} = (\mathsf{X}^T\mathsf{X} + \lambda I)^{-1}\mathsf{X}^T\boldsymbol{y}. \tag{5.76}$$

Exercise 5.6 (Ridge regression invertibility)**.** Prove that for $\lambda > 0$, $\mathsf{X}^T\mathsf{X} + \lambda I$ is invertible, even if the columns of X are not linearly independent. △

Bayesian Interpretation

We will now view the regularization penalty from a Bayesian point of view. As before, assume the Gaussian likelihood

$$\boldsymbol{y}|\boldsymbol{\theta}, \sigma^2 \sim \mathcal{N}(\mathsf{X}\boldsymbol{\theta}, \sigma^2 I). \tag{5.77}$$

For simplicity, we focus on estimating only $\boldsymbol{\theta}$ and not σ^2. For the prior on $\boldsymbol{\theta}$, let

$$p(\boldsymbol{\theta}|\sigma^2) \sim \mathcal{N}(0, (\sigma^2/\lambda)I) \propto e^{-\frac{\lambda\boldsymbol{\theta}^T\boldsymbol{\theta}}{2\sigma^2}}. \tag{5.78}$$

Then

$$p(\boldsymbol{\theta}|\boldsymbol{y}, \sigma^2) \propto p(\boldsymbol{y}|\boldsymbol{\theta}, \sigma^2)p(\boldsymbol{\theta}|\sigma^2) \propto \exp\left(-\frac{(\mathsf{X}\boldsymbol{\theta} - \boldsymbol{y})^T(\mathsf{X}\boldsymbol{\theta} - \boldsymbol{y}) + \lambda\boldsymbol{\theta}^T\boldsymbol{\theta}}{2\sigma^2}\right). \tag{5.79}$$

Based on the previous discussion, it is immediately clear that the mode of the posterior distribution for $\boldsymbol{\theta}$ is $(\mathsf{X}^T\mathsf{X} + \lambda I)^{-1}\mathsf{X}^T\boldsymbol{y}$. Furthermore, since the distribution is quadratic, and hence Gaussian, this is also the mean of the posterior. Hence the formulation for ridge regression is equivalent to assuming a zero-mean Gaussian distribution for $\boldsymbol{\theta}$, which assigns high prior probabilities to smaller length of $\boldsymbol{\theta}$.

Lasso

In lasso, the regularization penalty has the form of the ℓ_1 norm,

$$\|\boldsymbol{\theta}\|_1 = \sum_{i=1}^{m} |\theta_i|, \tag{5.80}$$

where m is the length of $\boldsymbol{\theta}$. The problem is to find

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2 + \lambda\|\boldsymbol{\theta}\|_1, \tag{5.81}$$

or equivalently

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \|\boldsymbol{y} - \mathsf{X}\boldsymbol{\theta}\|_2^2, \tag{5.82}$$

$$\text{subject to : } \|\boldsymbol{\theta}\|_1 \leq t. \tag{5.83}$$

Lasso does not have a closed form solution but efficient computational methods exist.

From a Bayesian point of view, lasso is equivalent to finding the *mode* of the posterior for $\boldsymbol{\theta}$ assuming the same model as above but with the double exponential (Laplace) prior

$$p(\boldsymbol{\theta}|\sigma^2) \propto e^{-\frac{\lambda\|\boldsymbol{\theta}\|_1}{2\sigma^2}}. \tag{5.84}$$

Discussion and generalization

In general we could choose the regularization penalty to be of the form[3]

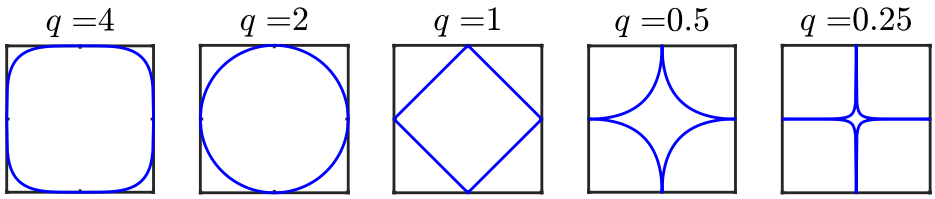
$$\|\boldsymbol{\theta}\|_q^q = \sum_{i=1}^{m} |\theta_i|^q, \tag{5.85}$$

where m is the length of $\boldsymbol{\theta}$. For $q = 1$ and $q = 2$, we get lasso and ridge regression, respectively.

The effect of the regularization can be viewed from a Bayesian framework, by setting the prior

$$\exp\left(-\frac{\lambda}{2\sigma^2}\|\boldsymbol{\theta}\|_q^q\right). \tag{5.86}$$

The contours for the priors for different values of q are given below.



In all cases, as we get further from the origin, the prior probability drops. But when q is small, the probability falls slower along the axes, encouraging solutions in which some of the coordinates are small or zero.

5.5 Error analysis and model selection

If our goal is to minimize the square of the prediction error, why would we use a different loss function for empirical risk minimization, as we did for ridge regression and lasso? How does this choice affect the error? Given that we have different choices for the form of the model and its parameters, how do we choose?

5.5.1 Bias-variance trade-off for quadratic error

Let us consider a general regression problem where we want to predict a value Y given an input vector $\boldsymbol{x}$. Let the prediction/estimate for Y given $\boldsymbol{x}$ be denoted by $f(\boldsymbol{x})$, where f is the function predicting Y given $\boldsymbol{x}$. For linear regression this is of the form $f(\boldsymbol{x}) = \boldsymbol{x}^T\boldsymbol{\theta}$, so finding the predictor is the same as finding an appropriate value for $\boldsymbol{\theta}$.

For a *specific* estimator f (e.g., one found based on a given data set $\{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$), the expected value of the quadratic loss for the next data point is

$$\mathcal{L}(f) = \mathbb{E}\big[(Y - f(\boldsymbol{X}))^2 | \boldsymbol{X} = \boldsymbol{x}\big] \tag{5.87}$$

$$= \mathbb{E}\big[(Y - f(\boldsymbol{x}))^2 | \boldsymbol{X} = \boldsymbol{x}\big], \tag{5.88}$$

which is called the **test error** for $\boldsymbol{x}$. We can view this as the loss for the $n+1$st data point, where we are given $\boldsymbol{x} = \boldsymbol{x}_{n+1}$ and are interested in the loss of predicting Y_{n+1}. So, we are interested in evaluating f for a given input. For instance, in our marketing example from the beginning of the chapter, $\boldsymbol{x}_{n+1}$ would indicate a specific budget, e.g., $\boldsymbol{x}_{n+1} = (\$10k, \$20k, \$5k)$.

[3] $\|\boldsymbol{\theta}\|_q = \left(\sum_{i=1}^m |\theta_i|^q\right)^{1/q}$ is called the ℓ_q-*norm* of $\boldsymbol{\theta}$.

Let $\bar{y}(\boldsymbol{x}) = \mathbb{E}[Y|\boldsymbol{X} = \boldsymbol{x}]$. Then, using the fact that $\mathbb{E}[(Z-c)^2] = \mathrm{Var}(Z) + (\mathbb{E}[Z] - c)^2$, we have

$$\mathcal{L}(f) = \mathbb{E}\big[(Y - f(\boldsymbol{x}))^2 | \boldsymbol{X} = \boldsymbol{x}\big] \tag{5.89}$$

$$= \mathrm{Var}(Y|\boldsymbol{X} = \boldsymbol{x}) + \big(\bar{y}(\boldsymbol{x}) - f(\boldsymbol{x})\big)^2. \tag{5.90}$$

Note that the error has two parts: an irreducible part, referred to as intrinsic error, which is not under our control, and a part that depends on the choice of the predictor. The intrinsic error results from the noise in "nature," i.e., the fact that $\boldsymbol{X}$ does not have enough information to fully determine Y. In other words, this term can be viewed as the accumulated effect of all factors that are not included in $\boldsymbol{X}$. Having a larger dataset or choosing a better f does not affect this term. The reducible part compares the performance of our predictor with the best possible. This error is minimized by setting $f(\boldsymbol{x}) = \bar{y}(\boldsymbol{x}) = \mathbb{E}[Y|\boldsymbol{X} = \boldsymbol{x}]$. However, doing so exactly is only possible if we have the distribution or an infinite amount of data.

To summarize, we can write *the test error for given f and $\boldsymbol{x}$* as

$$\mathcal{L}(f) = \text{irred.} + \big(f(\boldsymbol{x}) - \bar{y}(\boldsymbol{x})\big)^2. \tag{5.91}$$

We should choose f to minimize the above quantity. Let us consider how f is chosen through empirical risk minimization.

1. Determine a set $\mathcal{F}$ from which f can be chosen, e.g., all linear functions.
2. Define an empirical loss. Typically, this reflects the loss function in the expected loss (5.88), but may include a regularization term, i.e., $\sum_{i=1}^{n}(y_i - f(\boldsymbol{x}_i))^2 + \text{Reg}$.
3. Collect data, $\mathcal{D} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$, and find $f \in \mathcal{F}$ that minimizes the empirical loss.

Consider a thought experiment in which this process is repeated many times. In each trial, the set $\mathcal{F}$ and the definition of the empirical loss stay the same, while $\mathcal{D}$ and consequently, f are random. Since f is a function of $\mathcal{D}$, let us denote it as $f_{\mathcal{D}}$. Let $\mathcal{M}$ denote the fixed components of this process, i.e., the set $\mathcal{F}$ and the definition of the empirical loss. We are interested to find the loss as a function of $\mathcal{M}$, which is under our control, averaged over all possible datasets (which is outside our control). This is called the **expected test error**

$$\mathcal{L}(\mathcal{M}) = \mathbb{E}[\mathcal{L}(f_{\mathcal{D}})] = \text{irred.} + \mathbb{E}\big[(f_{\mathcal{D}}(\boldsymbol{x}) - \bar{y}(\boldsymbol{x}))^2\big], \tag{5.92}$$

where the expectation is taken over all possible datasets. Note that the irreducible part, $\mathrm{Var}(y|\boldsymbol{x})$ is a constant.With a similar trick as above, we have

$$\mathcal{L}(\mathcal{M}) - \text{irred.} = \mathbb{E}\big[(f_{\mathcal{D}}(\boldsymbol{x}) - \bar{y}(\boldsymbol{x}))^2\big] \tag{5.93}$$

$$= \mathbb{E}\big[(f_{\mathcal{D}}(\boldsymbol{x}) - \mathbb{E}\, f_{\mathcal{D}}(\boldsymbol{x}))^2\big] + (\mathbb{E}\, f_{\mathcal{D}}(\boldsymbol{x}) - \bar{y}(\boldsymbol{x}))^2, \tag{5.94}$$

where the last equality follows from the fact that $\mathbb{E}\, f_{\mathcal{D}}(\boldsymbol{x})$ and $\bar{y}(\boldsymbol{x})$ are constants for a given $\boldsymbol{x}$. The first term on the last line is the square of the bias and the second term is the variance of $f_{\mathcal{D}}(\boldsymbol{x})$. So

$$\mathcal{L}(\mathcal{M}) = \text{irred.} + (\text{bias})^2 + \text{variance} \tag{5.95}$$

Now, the loss is written as the sum of squared bias term, which compares the average prediction across all possible datasets with the best possible predictor, and a variance term, which quantifies how different the estimate for each dataset is from the average, across all datasets.

Typically, as model complexity/flexibility[4] increases, bias decreases, while variance increases, since it has more freedom to vary based on the dataset. Simple/rigid models on the other hand typically have high bias and low variance. The bottom line is that neither unbiased models nor low variance predictors are necessarily the best in terms of minimizing prediction error.

[4]By flexibility, I mean its responsiveness to changes in the data, i.e., the extent to which the results change when data changes.

Another factor that affects the error is the size of the data. In many situations, the size of the data does affect the bias term but it may not significant since this term is averaged over data sets. The variance terms is affected because with more data, we get more information, approaching the situation in which $p(y|x)$ is known. So for very large datasets, the prediction reflects the distribution, which is constant.

5.5.2 Model Selection

As discussed above, the complexity of the model can affect the error. How can we choose the best model? In the past, we have used minimizing the empirical risk (training error) to optimize the parameters of a model. Can we again use the same strategy?

The **training error** for a given predictor f, i.e.,

$$\frac{1}{n}\sum L(y_i, f(\boldsymbol{x}_i)). \tag{5.96}$$

The training error itself is difficult to study. Averaged over all datasets, the **expected training error** is

$$\frac{1}{n}\sum \mathbb{E}[L(Y_i, f_{\mathcal{D}}(\boldsymbol{x}_i))], \tag{5.97}$$

where for simplicity, we have assumed that that the $\boldsymbol{x}_i$ are fixed.

A typically behavior is in the table below, (this is not universally true). Training error is usually smaller for more complex models but this is not necessarily true for the test error. This makes choosing the best model based on training error difficult.

	Expected Train Err	Expected Test Err			
		Irred.	Bias^2	Var.	Total
More complex model	↓	–	↓	↑	?
More data	↑	–	↓	↓	↓

5.5.2.1 Overfitting and Underfitting

Suppose the true relationship between two scalar variables x and Y is

$$y = ax + w, \quad w \sim \mathcal{N}(0, \sigma^2). \tag{5.98}$$

We assume that $\sigma < ax$ for typical values of x since otherwise, we cannot predict Y accurately even if a is known (the irreducible error is large relative to the best predicted value).

The data available to us consists of two points

$$\mathcal{D} = \{(x_1 = 1, y_1), (x_2 = 2, y_2)\}. \tag{5.99}$$

We consider predictors of the forms

- $\hat{y}(x) = 0$,
- $\hat{y}(x) = \theta x$,
- $\hat{y}(x) = \theta_1 x + \theta_2 x^2$.

For each predictor, we will find the parameter values $(\theta, \theta_1, \theta_2)$ that minimize the square loss for our data,

$$\frac{1}{2}\big[(y_1 - \hat{y}(x_1))^2 + (y_2 - \hat{y}(x_2))^2\big]. \tag{5.100}$$

For the third predictor, we will also consider the regularized version with loss

$$\frac{1}{2}\big[(y_1 - \hat{y}(x_1))^2 + (y_2 - \hat{y}(x_2))^2\big] + b\theta_2^2, \tag{5.101}$$

where b is a constant. Here, I chose the form $b\theta_2^2$ instead of the ℓ_2 norm, $b(\theta_1^2+\theta_2^2)$ to simplify the derivation. We will still be able to see the effect of regularization.

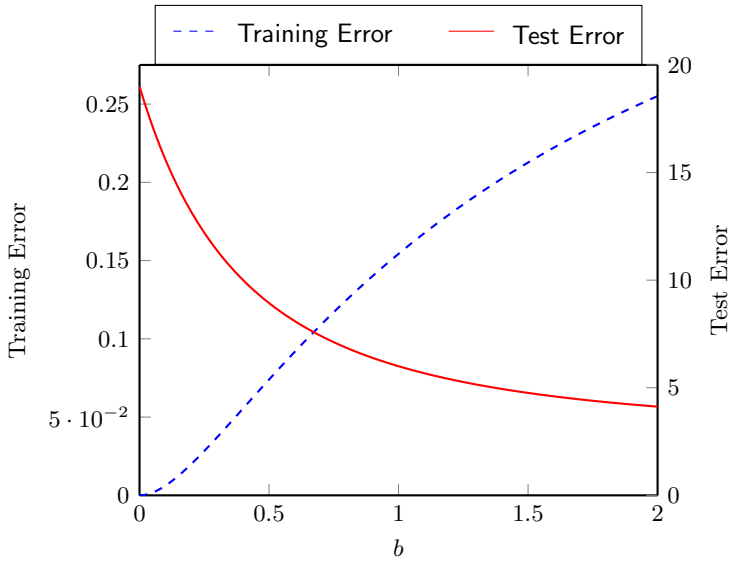
We then find the expected error for the training data and for a test data point (x_3, y_3), where we assume $x_3 = 3$. The expectation is taken over the randomness in y_1, y_2, y_3. The results are given in the table below.

Prediction	Expected Train Err	Expected Test Error for $x_3 = 3$			
		Irr.	Bias^2	Var.	Total
$\hat{y}(x) = 0$	$\frac{5a^2}{2} + \sigma^2$	σ^2	$9a^2$	0	$9a^2 + \sigma^2$
$\hat{y}(x) = \frac{y_1+2y_2}{5}x$	$\frac{\sigma^2}{2}$	σ^2	0	$\frac{9}{5}\sigma^2$	$\frac{14}{5}\sigma^2$
$\hat{y}(x) = \frac{4y_1-y_2}{2}x - \frac{2y_1-y_2}{2}x^2$	0	σ^2	0	$18\sigma^2$	$19\sigma^2$
$\hat{y}(x) = \frac{b(y_1+2y_2)+8y_1-2y_2}{5b+4}x - \frac{4y_1-2y_2}{5b+4}x^2$	$\frac{\sigma^2}{2}\left(1 - \frac{1}{5b/4+1}\right)^2$	σ^2	0	$\frac{9\sigma^2}{5}\left(1 + \frac{9}{(5b/4+1)^2}\right)$	$\frac{\sigma^2}{5}\left(14 + \frac{81}{(5b/4+1)^2}\right)$

As we go down the table, the model complexity increases. This allows the model to fit the training data better, leading to smaller expected training (square) error. The irreducible component of the test error stays the same, regardless of the model. The prediction bias for the test data point decreases, while its variance increases.

Given the assumption that σ is small relative to a, the smallest total error is obtained by the middle predictor. The zero predictor is not complex enough to be able to fit even the training data well. This situation is referred to as **underfitting**. The quadratic predictor is so complex that it can fit the training data, including the noise in the data, perfectly. But it does not generalize well due to its susceptibility to noise and high variance. This is called **overfitting**. In other words, the model memorizes this specific dataset rather than looking for patterns in it.

For the predictor with regularization, the graph below shows how the training and test errors change as a function of b. In this case, the predictor without regularization is overfitting the data. As b increases, overfitting decreases and we obtain a better test error. Note however that here the specific form of regularization prevents underfitting for large b, something that may occur in practice.



Example 5.7 (Regularization bias-variance trade-off)**.** Regularization allows us to control the flexibility of the model. In ridge regression as λ increases, the model becomes more constrained. For $\lambda > 0$ it can be

shown to be biased. With $\mathcal{D} = (\mathsf{X}, \boldsymbol{y})$,

$$\mathbb{E}[\hat{y}_{n+1}] = \mathbb{E}[\boldsymbol{x}_{n+1}^T \hat{\boldsymbol{\theta}}] \tag{5.102}$$
$$= \boldsymbol{x}_{n+1}^T (\mathsf{X}^T\mathsf{X} + \lambda I)^{-1} \mathsf{X}^T \, \mathbb{E}[\boldsymbol{y}] \tag{5.103}$$
$$= \boldsymbol{x}_{n+1}^T (\mathsf{X}^T\mathsf{X} + \lambda I)^{-1} \mathsf{X}^T \mathsf{X} \boldsymbol{\theta}. \tag{5.104}$$

Noting that $\mathbb{E}[y_{n+1}] = \boldsymbol{x}_{n+1}^T \boldsymbol{\theta}$, we see that the estimate of $\hat{y}_{n+1}$ is biased. In particular, if $\mathsf{X}^T\mathsf{X} = I$, then

$$\mathbb{E}[\hat{y}_{n+1}] = \frac{\boldsymbol{x}_{n+1}^T \boldsymbol{\theta}}{1+\lambda} < \boldsymbol{x}_{n+1}^T \boldsymbol{\theta} = \mathbb{E}[y_{n+1}]. \tag{5.105}$$

But it can be shown to have lower variance. If the choice of λ is appropriate, it will have a smaller total loss. △

It is important to note models could perform poorly for reasons other than over- and under-fitting. For example, if the true distribution of the data is $y = a \sin x + w$, no polynomial predictor will perform well for a wide range of inputs due to the poor match between the true distribution and the learning model.

5.5.2.2 Training, validation, and test sets

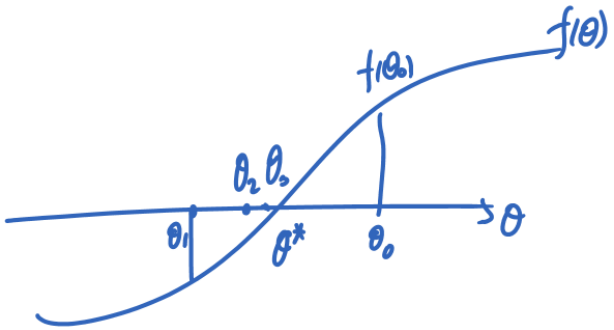
In the previous subsection, we could identify the best model because we knew the true model for $(\boldsymbol{x}, y)$, which nature uses to produce them. In practice, however, the true model is not known and we cannot compute the expected test error. We have also seen that the training error is not necessarily a good estimate for the test error.

If we have sufficient data, a good solution is to divide it into three parts, a **training set**, a **validation set**, and a **test set**. For each model, the training set is used to optimize its parameters. Then all optimized models are evaluated on the validation set. Since the validation set is not used in training, this reduces the risk of over-fitting and, so, the errors for the validation set are better estimates for the test error. We choose the best model based on the validation set. We perform a final assessment using the test set, which should provide a good estimate of the error of the selected model for future practical use. Note that the test set cannot be used for any other purpose. If it is used in training or validation, it will not provide a reliable estimate of the error in the wild.

5.6 Stochastic Gradient Descent

Even though that gradient descent is sometimes less computationally expensive than directly finding the solution, its cost may still be high. In such cases, using *stochastic gradient descent* (SGD) may be helpful. SGD tries to improve the estimate by considering one data point (or a small batch of data points) at a time.

First, let's consider finding the root of a function $f(\theta)$ with a simple method. We assume that $f(\theta)$ is bounded and there is a unique root θ^* such that f is increasing at θ^*.



Suppose that we start from a point $\theta^{(0)}$ that is appropriately close to θ^*. We proceed iteratively as

$$\theta^{(t+1)} = \theta^{(t)} - a_t f(\theta^{(t)}), \tag{5.106}$$

where a_t satisfies

$$\sum_{t=1}^{\infty} a_t = \infty, \qquad \sum_{t=1}^{\infty} a_t^2 < \infty. \tag{5.107}$$

For example, $a_t = 1/t$ is a good choice while $a_t = 1/t^2$ isn't. It can then be shown that $\theta^{(t)}$ converges to θ^*.

But what if we cannot compute $f(\theta)$ but instead we have access to a noisy version $F(\theta)$ that satisfies $f(\theta) = \mathbb{E}[F(\theta)]$, where $F(\theta)$ is bounded. It turns out that if we let

$$\theta^{(t+1)} = \theta^{(t)} - a_t F(\theta^{(t)}), \tag{5.108}$$

where in each iteration we sample $F(\theta)$, then $\theta^{(t)}$ again converges to θ^*.

Now let us consider the loss function for linear regression (note that we are using the expected loss as opposed to empirical loss)

$$\mathcal{L}(\boldsymbol{\theta}) = \mathbb{E}[(y - \boldsymbol{x}^T\boldsymbol{\theta})^2], \tag{5.109}$$

where we are also assuming that $\boldsymbol{x}$ is random with some distribution. To minimize this loss, we compute the gradient:

$$\nabla\mathcal{L}(\boldsymbol{\theta}) = \mathbb{E}[-2(y - \boldsymbol{x}^T\boldsymbol{\theta})\boldsymbol{x}] \tag{5.110}$$

We would like to find $\boldsymbol{\theta}$ such that the gradient above is zero.

Let

$$f(\boldsymbol{\theta}) = \mathbb{E}[-2(y - \boldsymbol{x}^T\boldsymbol{\theta})\boldsymbol{x}] \tag{5.111}$$

$$F(\boldsymbol{\theta}) = -2(y - \boldsymbol{x}^T\boldsymbol{\theta})\boldsymbol{x}, \tag{5.112}$$

so that $f(\boldsymbol{\theta}) = \mathbb{E}[F(\boldsymbol{\theta})]$. Now the elements of the data set $\{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$ can be used to produce samples for $F(\boldsymbol{\theta})$. So we let

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} + a_t(y_i - \boldsymbol{x}_i^T\boldsymbol{\theta}^{(t)})\boldsymbol{x}_i, \tag{5.113}$$

which is the stochastic gradient descent algorithm for linear regression.

Chapter 6

Linear Classification

In a classification problem, we have an input vector $\boldsymbol{x}$ together with a corresponding label y. Based on a data set $\{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_n, y_n)\}$, our goal is to predict y given a new value for $\boldsymbol{x}$. If y is a continuous variable the problem is that of regression, whereas in classification problems, y will represent a set of discrete class labels. For example, we may wish to classify images of handwritten digits. In this case, $\boldsymbol{x}$ is a vector providing the values of pixels of the image and $y \in \{0, 1, \ldots, 9\}$ is the label indicating what digit the image represents.

6.1 Overview of probabilistic models

The probabilistic approach to classification requires us to learn the distribution $p(y|\boldsymbol{x})$, which for any given $\boldsymbol{x}$ provides the probability of belonging to different classes. We can identify the class for a given $\boldsymbol{x}$ as the class that has the maximum probability,

$$\hat{y}(\boldsymbol{x}) = \arg\max_j p(y = j|\boldsymbol{x}).$$

This choice *minimizes the probability of predicting the wrong class*

$$\mathcal{L} = \Pr(\hat{y}(\boldsymbol{x}) \neq Y) = \mathbb{E}[I(Y \neq \hat{y}(\boldsymbol{x}))].$$

To find the distribution $p(y|\boldsymbol{x})$, our first step is developing a model that relates $\boldsymbol{x}$ and y. There are two possible approaches.

We may develop a **generative model**, i.e., a model that is capable of *generating* data and also helping us predict y for a given $\boldsymbol{x}$. A generative model has two components, both of which must be learned from data:

- Prior class probabilities: $p(y)$
- Class-conditional probabilities: $p(\boldsymbol{x}|y)$

From these, using Bayes' theorem we can find $p(y|\boldsymbol{x})$ as

$$p(y|\boldsymbol{x}) = \frac{p(y)p(\boldsymbol{x}|y)}{p(\boldsymbol{x})} \propto p(y)p(\boldsymbol{x}|y).$$

We can often estimate $p(y = j)$ simply by computing the fraction of class j in our training data. For $p(\boldsymbol{x}|y)$, a common approach is to represent it parametrically and then learn the parameters from the data. For example, we may assume that given class j, $\boldsymbol{x}$ is distributed normally with mean $\boldsymbol{\mu}_j$ and covariance matrix K_j and then learn these parameters from data.

Alternatively, we can develop a **discriminative model**. In this case, we directly model $p(y|\boldsymbol{x})$ since this is the distribution that we need to decide which class $\boldsymbol{x}$ belongs to.

6.2 Generative Probabilistic Models

6.2.1 Gaussian Class-Conditionals

Let us denote

$$p(Y = j) = \pi_j.$$

We further assume $p(\boldsymbol{x}|Y = j)$ is Gaussian with mean $\boldsymbol{\mu}_j$ and covariance matrix Σ_j,

$$p(\boldsymbol{x}|Y = j) = \frac{1}{\sqrt{(2\pi)^d|\Sigma_j|}} \exp\left(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_j)^T\Sigma_j^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_j)\right)$$

For our purpose, it suffices to consider $\ln p(\boldsymbol{x}|y = j)$, which, after dropping the constant terms, becomes

$$\ln p(\boldsymbol{x}|Y = j) \doteq -\frac{1}{2}\ln|\Sigma_j| - \frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_j)^T\Sigma_j^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_j).$$

From these, we can find $\ln p(y = j|\boldsymbol{x})$ and then decide $\hat{y}(\boldsymbol{x})$ as

$$\hat{y}(\boldsymbol{x}) = \arg\max_j \ln p(y = j|\boldsymbol{x}).$$

More specifically,

$$\begin{aligned}\ln p(y = j|\boldsymbol{x}) &\doteq \ln \pi_j - \frac{1}{2}\ln|\Sigma_j| - \frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_j)^T\Sigma_j^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_j) \\ &= \boldsymbol{x}^T\left(-\frac{1}{2}\Sigma_j^{-1}\right)\boldsymbol{x} + (\boldsymbol{\mu}_j^T\Sigma_j^{-1})\boldsymbol{x} + \left(-\frac{1}{2}\boldsymbol{\mu}_j^T\Sigma_j^{-1}\boldsymbol{\mu}_j - \frac{1}{2}\ln|\Sigma_j| + \ln\pi_j\right) \\ &= \boldsymbol{x}^T A_j\boldsymbol{x} + \boldsymbol{\beta}_j^T\boldsymbol{x} + \gamma_j,\end{aligned} \tag{6.1}$$

For an appropriately defined symmetric matrix A_j, vector $\boldsymbol{\beta}_j$, and scalar γ_j. In the next two sections, we will consider two cases based on whether the covariance matrix depends on the class.

6.2.2 Linear Discriminant Analysis

First, let us suppose all classes have the same covariance matrix $\Sigma_j = \Sigma$. Then, the terms $\boldsymbol{x}^T\left(-\frac{1}{2}\Sigma^{-1}\right)\boldsymbol{x}$ and $-\frac{1}{2}\ln|\Sigma_j|$ in (6.1) become independent of the class and we thus have

$$\ln p(y = j|\boldsymbol{x}) \doteq \boldsymbol{\beta}_j^T\boldsymbol{x} + \gamma_j, \tag{6.2}$$

where

$$\boldsymbol{\beta}_j^T = \boldsymbol{\mu}_j^T\Sigma^{-1}, \qquad \gamma_j = -\frac{1}{2}\boldsymbol{\mu}_j^T\Sigma^{-1}\boldsymbol{\mu}_j + \ln\pi_j,$$

Suppose we have only two classes, $y = 0$ and $y = 1$, with $p(y = 1) = \pi = 1 - p(y = 0)$. Equivalent to finding $\arg\max_j \ln p(y = j|\boldsymbol{x})$ for each $\boldsymbol{x}$, we can divide the space into two regions,

$$\ln p(y = 1|\boldsymbol{x}) \underset{\hat{y}=0}{\overset{\hat{y}=1}{\gtrless}} \ln p(y = 0|\boldsymbol{x}).$$

What is the decision boundary between them? We can find it by solving $\ln p(y = 1|\boldsymbol{x}) = \ln p(y = 0|\boldsymbol{x})$,

$$\boldsymbol{\beta}_1^T\boldsymbol{x} + \gamma_1 = \boldsymbol{\beta}_0^T\boldsymbol{x} + \gamma_0 \iff (\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0)^T\boldsymbol{x} + \gamma_1 - \gamma_0 = 0 \iff \boldsymbol{\beta}^T\boldsymbol{x} + \gamma = 0,$$

where

$$\boldsymbol{\beta}^T = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T\Sigma^{-1}, \qquad \gamma = \ln\frac{\pi}{1-\pi} - \frac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T\Sigma^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0). \tag{6.3}$$

Hence, the decision boundary is the hyperplane $\boldsymbol{\beta}^T\boldsymbol{x} + \gamma = 0$. On one side of this plane, we predict class 1

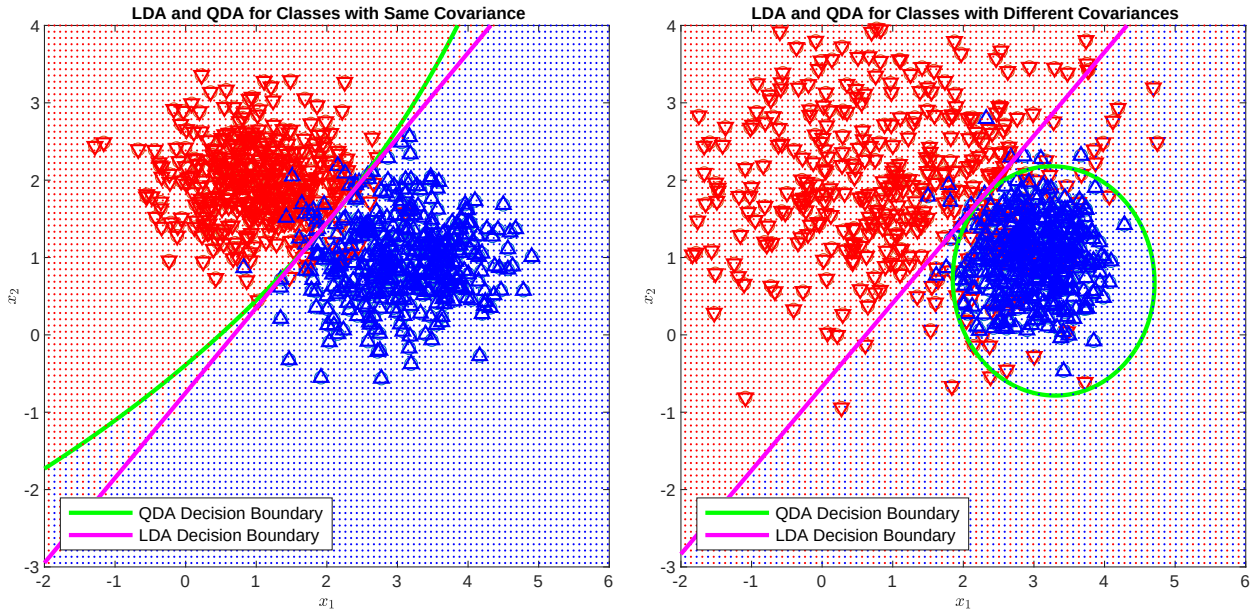


Figure 6.1: LDA vs QDA for when $\Sigma_1 = \Sigma_2$ (left) and when $\Sigma_1 \neq \Sigma_2$ (right).

(we let $\hat{y}(\boldsymbol{x}) = 1$) and on the other side, we declare class 0:

$$\hat{y}(\boldsymbol{x}) = \begin{cases} 1, & \boldsymbol{\beta}^T \boldsymbol{x} + \gamma > 0 \\ 0, & \boldsymbol{\beta}^T \boldsymbol{x} + \gamma < 0 \end{cases}$$

Since the boundary is linear (i.e., a hyperplane such as a line, 2-D plane, etc), this method is called **Linear Discriminant Analysis** (LDA).

As a special case, consider, $\pi = \frac{1}{2}, \Sigma = I$. Then the boundary becomes

$$(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \left(\boldsymbol{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0}{2} \right) = 0,$$

which implies that the boundary is the plane that passes through the midpoint of the line connecting $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_0$ and is perpendicular to it.

What about the probability $p(y|\boldsymbol{x})$ of each class for a given $\boldsymbol{x}$, which can tell us about the certainty of belonging to each class? From (6.2), we have $p(y = j|\boldsymbol{x}) \propto e^{\boldsymbol{\beta}_j^T \boldsymbol{x} + \gamma_j}$ and so for two classes

$$p(y = 1|\boldsymbol{x}) = \frac{e^{\boldsymbol{\beta}_1^T \boldsymbol{x} + \gamma_1}}{e^{\boldsymbol{\beta}_1^T \boldsymbol{x} + \gamma_1} + e^{\boldsymbol{\beta}_0^T \boldsymbol{x} + \gamma_0}} = \frac{1}{1 + e^{-(\boldsymbol{\beta}^T \boldsymbol{x} + \gamma)}} = \sigma(\boldsymbol{\beta}^T \boldsymbol{x} + \gamma),$$

where $\boldsymbol{\beta}$ and γ are given in (6.3), and $\sigma(u) = \frac{1}{1+e^{-u}}$ is the sigmoid (logistic) function.

If there are $c > 2$ classes, decision hyperplanes between pairs of classes will divide the space into c regions. And the conditional probability of class j is given by

$$p(y = j|\boldsymbol{x}) = \frac{e^{\boldsymbol{\beta}_j^T \boldsymbol{x} + \gamma_j}}{\sum_{k=1}^{c} e^{\boldsymbol{\beta}_k^T \boldsymbol{x} + \gamma_k}} = \sigma_j(\boldsymbol{\beta}_1^T \boldsymbol{x} + \gamma_1, \ldots, \boldsymbol{\beta}_c^T \boldsymbol{x} + \gamma_c),$$

where $\sigma_j(\boldsymbol{v}) = \frac{e^{v_j}}{\sum_k e^{v_k}}$ is the softmax function.

6.2.3 Quadratic Discriminant Analysis

Let us now assume that each class has a different covariance matrix Σ_j. To decide between two classes, say $y = 0$ and $y = 1$, the decision boundary is given by $\ln p(y = 1|\boldsymbol{x}) = \ln p(y = 0|\boldsymbol{x})$. This will lead to a quadratic equation of the form $\boldsymbol{x}^T A\boldsymbol{x} + \boldsymbol{\beta}^T\boldsymbol{x} + \gamma = 0$, which leads to a nonlinear decision boundary. As a result, this method is called **Quadratic Discriminant Analysis** (QDA).

Figure 6.1 demonstrates LDA and QDA when $\Sigma_1 = \Sigma_2$ (left) and when $\Sigma_1 \neq \Sigma_2$ (right). Here the boundaries are learned from data (see Section 6.2.2). On the left the data is generated by distributions that match the assumption made by LDA and so LDA and QDA perform similarly. However, on the right the covariances are different and so, as expected, QDA performs better. Note however that we could augment our feature vectors as $(x_1, x_2, x_1x_2, x_1^2, x_2^2)$ instead of just (x_1, x_2) and then apply LDA, allowing a decision boundary that is not linear in x_1, x_2. In that case, the performance of LDA would generally be similar to that of QDA (Hastie et al,. Elements of Statistical Learning).

6.2.4 Maximum Likelihood Solution to LDA

Once we specified a parametric form for the class-conditional densities $p(\boldsymbol{x}|y = j)$, we can determine the values of the parameters, together with the prior class probabilities $p(y = j)$, using maximum likelihood.

Data: Our data set comprises of observations of $\boldsymbol{x}$ along with their corresponding class labels. Let the n independent samples be denoted by $\mathcal{D} = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_n, y_n)\}$, where $\boldsymbol{x}_i \in \mathbb{R}^m$ and $y_i \in \{0, 1\}$ for all i.

Model:

$$p(y = j) = \begin{cases} \pi, & j = 1 \\ 1 - \pi, & j = 0 \end{cases}$$

$$p(\boldsymbol{x}|y = 0) \sim \mathcal{N}(\boldsymbol{\mu}_0, \Sigma),$$
$$p(\boldsymbol{x}|y = 1) \sim \mathcal{N}(\boldsymbol{\mu}_1, \Sigma),$$

for some $\pi, \boldsymbol{\mu}_0, \boldsymbol{\mu}_1$ and diagonal matrix $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_m^2)$. The diagonal covariance matrix implies conditional independence among the elements of $\boldsymbol{x}$. That is, for a given data point $(\boldsymbol{x}_i, y_i)$, depending on the value of y_i, we have one of the following cases

$$p(\boldsymbol{x}_i|y_i = 0) = \prod_{j=1}^{m} p(x_{ij}|y_i = 0) = \prod_{j=1}^{m} \frac{1}{\sqrt{2\pi\sigma_j^2}} \exp\left(\frac{-(x_{ij} - \mu_{0j})^2}{2\sigma_j^2}\right) \tag{6.4}$$

$$p(\boldsymbol{x}_i|y_i = 1) = \prod_{j=1}^{m} p(x_{ij}|y_i = 1) = \prod_{j=1}^{m} \frac{1}{\sqrt{2\pi\sigma_j^2}} \exp\left(\frac{-(x_{ij} - \mu_{1j})^2}{2\sigma_j^2}\right) \tag{6.5}$$

Note that since we assume both classes have the same covariance matrix, the decision boundary will be linear (i.e., LDA). Also, we have assumed given the class, features are independent (since Σ is diagonal); this is called the **Naive Bayes** model.

Likelihood:

$$\begin{aligned}
p(\mathcal{D}|\pi, \boldsymbol{\mu}_0, \boldsymbol{\mu}_1, \Sigma) &= \prod_{i=1}^{n} p(y_i)p(\boldsymbol{x}_i|y_i) \\
&= \left(\prod_{i:y_i=0} p(y_i=0)p(\boldsymbol{x}_i|y_i=0)\right)\left(\prod_{i:y_i=1} p(y_i=1)p(\boldsymbol{x}_i|y_i=1)\right) \\
&= \left(\prod_{i:y_i=0} (1-\pi)\prod_{j=1}^{m}\frac{1}{\sqrt{2\pi\sigma_j^2}}\exp\left(\frac{-(x_{ij}-\mu_{0j})^2}{2\sigma_j^2}\right)\right)\left(\prod_{i:y_i=1}\pi\prod_{j=1}^{m}\frac{1}{\sqrt{2\pi\sigma_j^2}}\exp\left(\frac{-(x_{ij}-\mu_{1j})^2}{2\sigma_j^2}\right)\right),
\end{aligned}$$

where $x_{i,j}$ is the jth component of $\boldsymbol{x}_i$ and $\mu_{0,j}, \mu_{1,j}$ are the jth components of $\boldsymbol{\mu}_0$ and $\boldsymbol{\mu}_1$, respectively. The maximum likelihood solution is (exercise)

$$\begin{aligned}
\hat{\pi}_{ML} &= \frac{\sum y_i}{n}, \\
(\hat{\mu}_{0,j})_{ML} &= \frac{\sum_{i:y_i=0} x_{i,j}}{\sum(1-y_i)}, \quad (\hat{\mu}_{1,j})_{ML} = \frac{\sum_{i:y_i=1} x_{i,j}}{\sum y_i}, \\
(\hat{\sigma_j^2})_{ML} &= \frac{1}{n}\left(\sum_{i:y_i=0}(x_{i,j}-\hat{\mu}_{0,j})^2 + \sum_{i:y_i=1}(x_{i,j}-\hat{\mu}_{1,j})^2\right)
\end{aligned}$$

6.2.5 Generative Model for Discrete Features **

If a features is categorical, for example, type of a vehicle or genre of a movie, we can encode them as binary vectors. For example, if there are three categories, with the vector $(1,0,0)$ we can indicate belonging to the first category. This is called *one-hot* or *dummy encoding*. In this case, our data is still denoted by $(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)$, where each $\boldsymbol{x}_i$ is composed of vectors, that is[1]

$$\boldsymbol{x}_i = (\boldsymbol{x}_{i1}, \ldots, \boldsymbol{x}_{im}),$$

and each $\boldsymbol{x}_{ij} = (x_{ij1}, \ldots, x_{ijl}, \ldots)$ is a binary vector of finite length which represents a one-hot encoding of a feature.

Example 6.1 (One-hot encoding)**.** Suppose $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_m$ provide information about a set of movies, where $\boldsymbol{x}_1 = (\boldsymbol{x}_{11}, \ldots, \boldsymbol{x}_{1m}), \boldsymbol{x}_2 = (\boldsymbol{x}_{21}, \ldots, \boldsymbol{x}_{2m}), \ldots$, with $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_m$ denoting in order the genre of the movie, the director of the movie, etc. Explicitly, for the genre, if we order them as (comedy, horror, drama, scifi, action), and for five directors A, B, C, D, E, order them as (A, B, C, D, E), then $\boldsymbol{x}_{11} = (0,0,1,0,0), \boldsymbol{x}_{12} = (0,0,0,1,0)$ means movie 1 is a drama directed by director D, and $\boldsymbol{x}_{21} = (1,0,0,0,0), \boldsymbol{x}_{22} = (1,0,0,0,0)$ means that movie 2 is a comedy directed by director A. △

Model: We model this classification problem in the following way:

$$p(x_{ijl} = 1|y_i = k) = \eta_{kjl}, \qquad \sum_l \eta_{kjl} = 1,$$

and all x_{ijl} are independent from one another. For two vectors $\boldsymbol{a}, \boldsymbol{b}$ with the same length, we define $\boldsymbol{a}^{\boldsymbol{b}} = \prod_{i=1}^{|\boldsymbol{a}|} a_i^{b_i}$. Let $\boldsymbol{\eta}_{kj} = (\eta_{kj1}, \ldots)$. We have

$$p(\boldsymbol{x}_{ij}|y_i = k) = \boldsymbol{\eta}_{kj}^{\boldsymbol{x}_{ij}},$$

[1] All vectors in this section are column vectors and all concatenations are also along the vertical dimension. However, for simplicity of notation, we write $\boldsymbol{x}_i = (\boldsymbol{x}_{i1}, \ldots, \boldsymbol{x}_{im})$ instead of $\boldsymbol{x}_i^T = (\boldsymbol{x}_{i1}^T, \ldots, \boldsymbol{x}_{im}^T)^T$

and then

$$p(\boldsymbol{x}_i|y_i = k) = \prod_{j=1}^{m} p(\boldsymbol{x}_{ij}|y_i = k) = \prod_{j=1}^{m} \boldsymbol{\eta}_{kj}^{\boldsymbol{x}_{ij}} = \boldsymbol{\eta}_k^{\boldsymbol{x}_i},$$

where $\boldsymbol{\eta}_k = (\boldsymbol{\eta}_{k1}, \ldots, \boldsymbol{\eta}_{km})$. It follows that

$$p(y_i = k|\boldsymbol{x}_i) \propto p(\boldsymbol{x}_i|y_i = k)p(y_i = k) = \pi_k \boldsymbol{\eta}_k^{\boldsymbol{x}_i} \propto \exp(\ln \pi_k + \boldsymbol{x}_i^T \ln \boldsymbol{\eta}_k).$$

For a new data point $(\boldsymbol{x}, y)$, we similarly have

$$\ln p(y = k|\boldsymbol{x}) \doteq \boldsymbol{\beta}_k^T \boldsymbol{x} + \gamma_k, \tag{6.6}$$

where $\boldsymbol{\beta}_k = \ln \boldsymbol{\eta}_k$ and $\gamma_k = \ln \pi_k$. The log-probabilities are again linear in $\boldsymbol{x}$, an fact that as we will see contributes to the motivation for logistic regression.

6.2.6 Class-conditionals from the exponential family

The exponential family of distributions includes common distributions such as Gaussian, exponential, gamma, beta, Dirichlet, Bernoulli, Poisson, and geometric. Distributions from this family have the following form

$$p(\boldsymbol{x}|\boldsymbol{\theta}) = \exp[\boldsymbol{b}(\boldsymbol{\theta})^T \boldsymbol{a}(\boldsymbol{x}) + f(\boldsymbol{x}) + g(\boldsymbol{\theta})].$$

Let us consider the case in which $\boldsymbol{a}(\boldsymbol{x}) = \boldsymbol{x}$, and parameters are functions of class y. So instead of $\boldsymbol{\theta}$ we write $\boldsymbol{\theta}_j$, when considering the jth class. Then the class-conditional distribution will become

$$p(\boldsymbol{x}|y = j) = \exp[\boldsymbol{b}(\boldsymbol{\theta}_j)^T \boldsymbol{x} + f(\boldsymbol{x}) + g(\boldsymbol{\theta}_j)].$$

Furthermore, let $p(y = j) = \pi_j$. Given $\boldsymbol{x}$, the log-probability of each class is given as

$$\ln p(y = j|\boldsymbol{x}) \doteq \ln \pi_j + \ln p(\boldsymbol{x}|y = j) \doteq \ln \pi_j + \boldsymbol{b}(\boldsymbol{\theta}_j)^T \boldsymbol{x} + g(\boldsymbol{\theta}_j) \doteq \boldsymbol{\beta}_j^T \boldsymbol{x} + \gamma_j, \tag{6.7}$$

where $\boldsymbol{\beta}_j = \boldsymbol{b}(\boldsymbol{\theta}_j)$ and $\gamma_j = \ln \pi_j + g(\boldsymbol{\theta}_j)$. So for a large class of class-conditional probabilities, the log-probabilities of classes given the feature vector $\boldsymbol{x}$ is linear in $\boldsymbol{x}$.

6.3 Discriminative Models and Logistic Regression

In the discriminative approach, we model $p(y = j|\boldsymbol{x})$ directly. But what is a good model for this conditional distribution? As we have seen in (6.2), (6.6) and (6.7), in many generative cases, the log-probabilities of classes given data is linear in $\boldsymbol{x}$,

$$\ln p(y = j|\boldsymbol{x}) \doteq \boldsymbol{\beta}_j^T \boldsymbol{x} + \gamma_j.$$

And based on Section 6.2.2, this form leads to linear class boundaries and posterior class probabilities of the logistic form for two classes,

$$p(y = 1|\boldsymbol{x}) = \frac{1}{1 + e^{-(\boldsymbol{\beta}^T \boldsymbol{x} + \gamma)}}, \qquad p(y = 0|\boldsymbol{x}) = \frac{e^{-(\boldsymbol{\beta}^T \boldsymbol{x} + \gamma)}}{1 + e^{-(\boldsymbol{\beta}^T \boldsymbol{x} + \gamma)}},$$

where $\boldsymbol{\beta} = \boldsymbol{\beta}_1 - \boldsymbol{\beta}_2$ and $\gamma = \gamma_1 - \gamma_2$.

Limiting ourselves to two classes, this observation raises the following question: 'Why not assume from the beginning that $p(y|\boldsymbol{x})$ is of the logistic form and learn this distribution instead of learning first $p(\boldsymbol{x}|y)$ and $p(y)$?' Doing so leads to a *discriminative model* resulting in *logistic regression*.

Let $h(\boldsymbol{x}) = p(y = 1|\boldsymbol{x})$ and assume that the data consists of n iid samples, $\mathcal{D} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$. We have

$$p(\mathcal{D}; \boldsymbol{\beta}, \gamma) = \prod_{i=1}^{n} h(\boldsymbol{x}_i)^{y_i} (1 - h(\boldsymbol{x}_i))^{1-y_i}$$

and the negative log-likelihood loss is given by

$$-\mathcal{L}(\boldsymbol{\beta},\gamma)=\sum_{i=1}^{n}\left(y_i\ln\frac{1}{h(\boldsymbol{x}_i)}+(1-y_i)\ln\frac{1}{1-h(\boldsymbol{x}_i)}\right). \tag{6.8}$$

We can use gradient descent to minimize this loss (maximize the likelihood). For simplicity, let $\boldsymbol{\theta}=\begin{pmatrix}\boldsymbol{\beta}\\ \gamma\end{pmatrix}$, $\tilde{\boldsymbol{x}}=\begin{pmatrix}\boldsymbol{x}\\ 1\end{pmatrix}$, and $h_{\boldsymbol{\theta}}=p(y=1|\boldsymbol{x})=\frac{1}{1+e^{-\boldsymbol{\theta}^T\tilde{\boldsymbol{x}}}}$. Then

$$\boldsymbol{\theta}^{(t+1)}=\boldsymbol{\theta}^{(t)}+\rho_t\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}),$$

where

$$\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta})=\sum_{i=1}^{n}(y_i-h_{\boldsymbol{\theta}}(\tilde{\boldsymbol{x}}_i))\tilde{\boldsymbol{x}}_i.$$

When we find $\boldsymbol{\theta}$, and thus $\boldsymbol{\beta},\gamma$, we have the decision boundary as $\boldsymbol{\beta}^T\boldsymbol{x}+\gamma=0$. Points $\boldsymbol{x}$ for which $\boldsymbol{\beta}^T\boldsymbol{x}+\gamma>0$ are classified as class $y=1$.

6.4 Risk minimization and loss functions for classification

An alternative approach to generative models and logistic regression we discussed before is directly minimizing an empirical loss,

$$\frac{1}{n}\sum_{i=1}^{n}L(y_i,\boldsymbol{x}_i,\hat{y}(\boldsymbol{x}_i)),$$

where $\hat{y}(\boldsymbol{x})$ is the predictor of the class for input vector $\boldsymbol{x}$. For ease of exposition, instead of assuming $y\in\{0,1\}$, we assume $y\in\{-1,1\}$.

Our attention will be limited to linear classifiers, determined by a vector $\boldsymbol{\beta}$ and a constant γ, which define the hyperplane $\boldsymbol{\beta}^T\boldsymbol{x}+\gamma=0$. On one side of the hyperplane, we decide class 1 and the other side class -1,

$$\hat{y}(\boldsymbol{x})=\text{sign}(\boldsymbol{\beta}^T\boldsymbol{x}+\gamma)=\begin{cases}1, & \text{if } \boldsymbol{\beta}^T\boldsymbol{x}+\gamma>0,\\ -1, & \text{if } \boldsymbol{\beta}^T\boldsymbol{x}+\gamma<0,\end{cases}$$

where the dependence of $\hat{y}$ on $\boldsymbol{\beta},\gamma$ is implicit.

One such linear classifier is shown in Figure 6.2. Below, we will use the fact that for any point $\boldsymbol{x}_i$ with label y_i and prediction $\hat{y}(\boldsymbol{x}_i)$, the loss contributed by it can often be viewed as a function of its signed distance d_i to the decision hyperplane. Without loss of generality, assume that $y_i=1$ and $\boldsymbol{x}_i=\boldsymbol{x}_0+d_i\boldsymbol{\beta}/\|\boldsymbol{\beta}\|$ for some $\boldsymbol{x}_0$ on the decision boundary. If d_i is positive, then this point is classified correctly, since $\boldsymbol{\beta}^T\boldsymbol{x}_i+\gamma>0$. The distance between $\boldsymbol{x}_i$ and the decision boundary equals $|d_i|$.

6.4.1 Zero-one loss

The most natural loss function for classification is the **0-1 loss**,

$$L_{01}(y,\hat{y}(\boldsymbol{x}))=\left\{\begin{cases}1, & \text{if } y\neq\hat{y}(\boldsymbol{x})\\ 0, & \text{if } y=\hat{y}(\boldsymbol{x})\end{cases}\right\}=\begin{cases}1, & \text{if } y(\boldsymbol{\beta}^T\boldsymbol{x}+\gamma)<0,\\ 0, & \text{if } y(\boldsymbol{\beta}^T\boldsymbol{x}+\gamma)>0.\end{cases}$$

Figure 6.3 shows the 0-1 loss for a point in the positive class. Note that how far the point is from the boundary does not affect how much it contributes to the loss.

Unfortunately, minimizing this loss function is computationally difficult (NP-hard) [1]. So in practice, we use differentiable loss-functions for which efficient algorithms exist. Here we will consider two such loss functions.

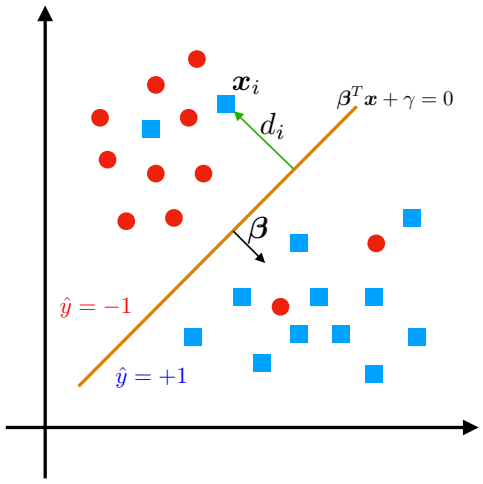


Figure 6.2: A linear classifier defined by the vector $\boldsymbol{\beta}$ and scalar γ. Squares represents points with $y = +1$ and circles $y = -1$. For a point $\boldsymbol{x}_i$, many loss functions can be viewed as a function of the signed distance d_i of $\boldsymbol{x}_i$ to the decision hyperplane.

First, we view logistic regression in terms of empirical risk minimization and then we will consider the hinge loss in the context of support vector machine (SVM) classifiers.

6.4.2 Logistic regression

Let us re-examine the logistic regression loss function (6.8). The loss incurred by a data point $\boldsymbol{x}_i$ at signed distance d_i from the decision hyperplane (i.e., $\boldsymbol{x}_i = \boldsymbol{x}_0 + d_i\boldsymbol{\beta}/\|\boldsymbol{\beta}\|$) is

$$\ln(1 + e^{-(\boldsymbol{\beta}^T\boldsymbol{x}_i+\gamma)}) = \ln(1 + e^{-d_i\|\boldsymbol{\beta}\|}).$$

The figure below shows this loss: For $d_i < 0$, where the input is misclassified, the loss is larger, and it increases as the point gets farther from the boundary. But even for points that are classified correctly, there is a loss, which decreases as we get farther from the boundary.

6.4.3 Hinge loss (SVM)

Hinge loss results from penalizing misclassified points as well as those that are classified correctly, but are within a certain margin close to the decision boundary. The expression for hinge loss is

$$\max(0, 1 - y_i(\boldsymbol{\beta}^T\boldsymbol{x}_i + \gamma)).$$

Letting $y_i = 1$ and $\boldsymbol{x}_i = \boldsymbol{x}_0 + d_i\boldsymbol{\beta}/\|\boldsymbol{\beta}\|$ as before, results in

$$\max(0, 1 - d_i\|\boldsymbol{\beta}\|).$$

which is shown in Figure 6.3. So the penalty for misclassified points is larger the farther away they are from the boundary. In addition, even points classified correctly are penalized if they are within a **margin** of width $1/\|\boldsymbol{\beta}\|$ of the decision boundary.

In addition to penalizing points within the margin, we would like to ensure that the margin is not very small. This can be done by ensuring $1/\|\boldsymbol{\beta}\|$ is large or equivalently $\|\boldsymbol{\beta}\|^2$ is small. Both of these goals can be achieved with the loss

$$\frac{1}{n}\sum_{i=1}^{n}\max(0, 1 - y_i(\boldsymbol{\beta}^T\boldsymbol{x}_i + \gamma)) + \lambda\|\boldsymbol{\beta}\|^2, \tag{6.9}$$

where λ is a constant that balances the two components of the loss. This results in the so called **support**

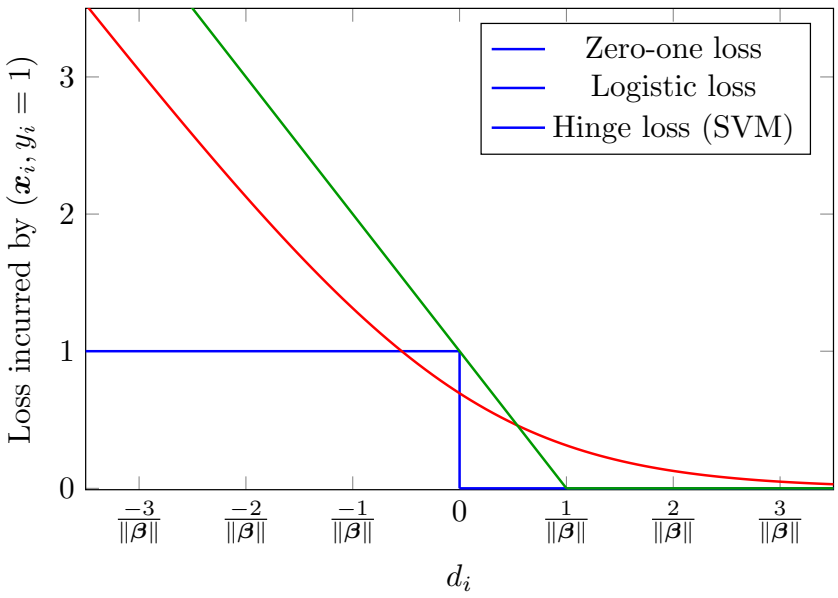


Figure 6.3: Loss functions for a data point $(\boldsymbol{x}_i, y_i = 1)$ as a function of the distance of $\boldsymbol{x}_i$ from the boundary (in terms of the length of $\boldsymbol{\beta}$).

vector machine classifier (SVM).

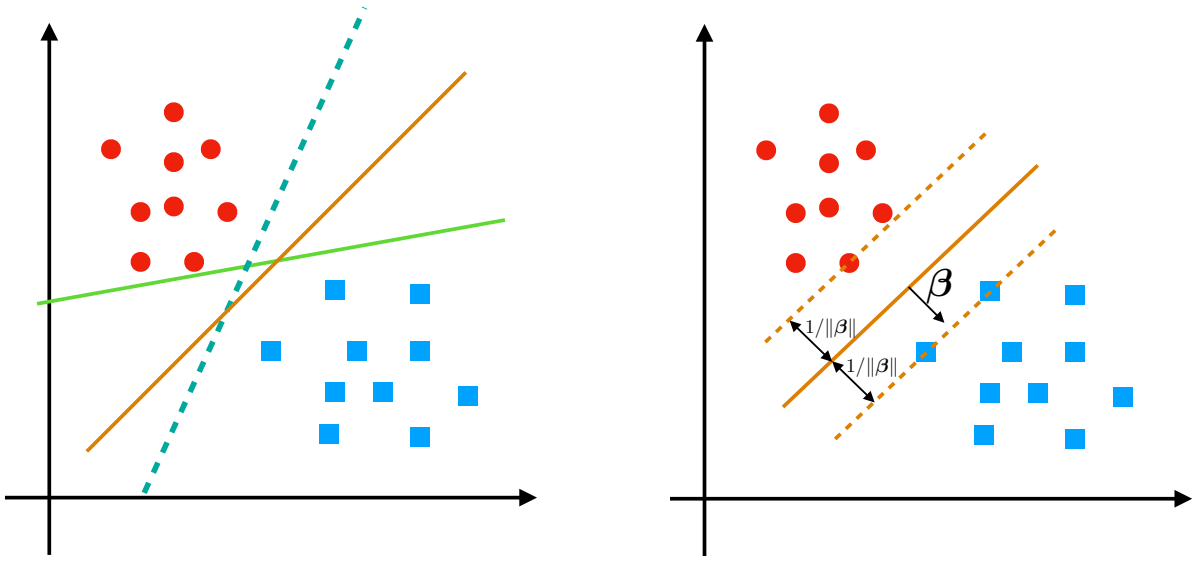
SVM as maximum-margin classifier. Let us consider the case in which the data is linearly separable, i.e., there exists a hyperplane that correctly classifies all data points. In such a case, as shown in Figure 6.4a, there are typically an infinite number of separating hyperplanes. This leads to the question of which one should be chosen. The SVM loss given in (6.9) provides a solution. Assume λ is positive but very small. So we are primarily concerned about the first term in the loss, i.e., the hinge loss. Between choices that incur the same hinge loss, we must pick the one that maximizes the margin, i.e., minimizes $\|\boldsymbol{\beta}\|^2$. Thus:

- We can make the hinge loss term zero by choosing any separating hyperplane that makes no mistakes and choosing any margin (length of $\|\boldsymbol{\beta}\|$) that is small enough such that there no points within the margin.
- Now the second term ensures that among the hyperplanes that perfectly separate the data, we should pick the one that has the maximum margin, as shown in Figure 6.4b.

In nonlinear cases, SVM can use "kernel functions" to transform the input space into a higher-dimensional space where it is easier to find a linear separating hyperplane. This, along with a related computational technique, is known as the kernel trick, allowing SVM to effectively perform in complex, nonlinear classification tasks.

References

[1] Tan T. Nguyen and Scott Sanner. "Algorithms for direct 0-1 loss optimization in binary classification." In: *Proceedings of the 30th International Conference on International Conference on Machine Learning - Volume 28*. ICML'13. Atlanta, GA, USA: JMLR.org, June 2013, pp. III–1085–III–1093. (Visited on 03/23/2020).



(a) For two linearly separable classes, there are an infinite number of classifiers that perfectly separate the training data. Which one should we pick?

(b) The maximum-margin classifier is the classifier that maximizes the distance between the decision boundary and the closest points to it.

Figure 6.4: SVM for separable data

Chapter 7

Expectation-Maximization *

7.1 Overview

Expectation-maximization (EM) is a method for dealing with missing data/hidden variables. In other words, part of the variables in the assumed model do not have associated data points. For example, for classification, the complete data consists of the features $\boldsymbol{x}$ and labels y, as shown in the left panel of Figure 7.1. With a probabilistic model for this data, we can find the parameters for each class through maximum likelihood, where the log-likelihood function is

$$\log p(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\theta}),$$

where $\boldsymbol{x} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$ and $\boldsymbol{y} = (y_1, \ldots, y_n)$ and $\boldsymbol{\theta}$ represents the parameters of class-conditional distributions for each of the classes.

But what if the class labels are not given as in the right panel of Figure 7.1? The problem becomes more difficult, but doesn't seem hopeless as we can still distinguish two clusters and assign points to these with various degrees of confidence.

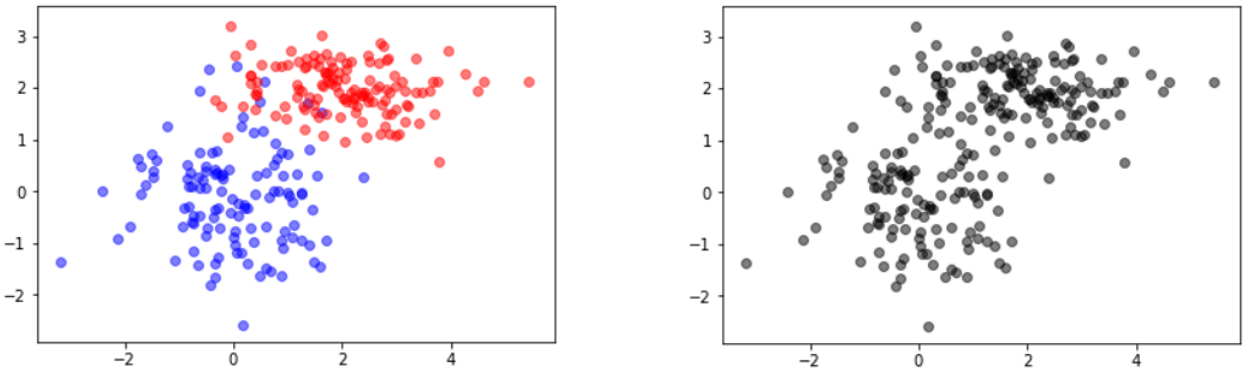


Figure 7.1: Data from two classes, with labels given as colors (left) and not given (right).

We thus formulate this problem as finding $\boldsymbol{\theta}$ that maximizes

$$\log p(\boldsymbol{x}; \boldsymbol{\theta}) = \log \sum_{\boldsymbol{y}} p(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\theta})$$

In this case, $(\boldsymbol{x}, \boldsymbol{y})$ is the complete data, for which computing the likelihood is easy, but a component of this data, namely $\boldsymbol{y}$, is missing. Now computing the likelihood is difficult because of the summation, which is typically over a large number of possibilities. Expectation-maximization is a method for handling missing

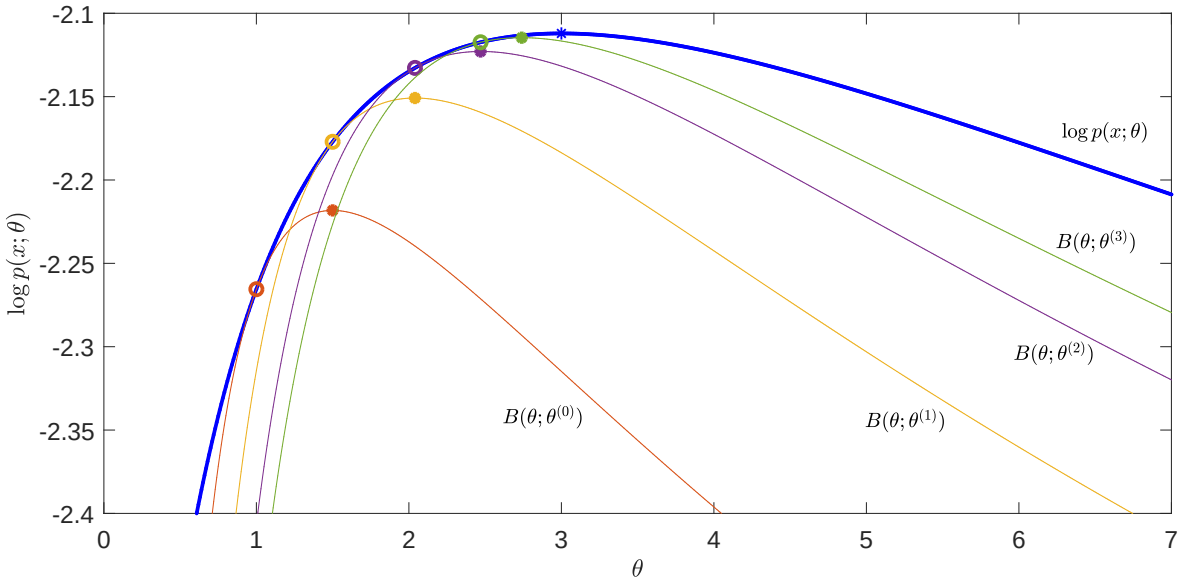


Figure 7.2: The log-likelihood of the observation and consecutive EM lower bounds and estimates. In each iteration, the current value of θ is denoted by $\circ$ and the new value by $*$. Here, $\theta^{(0)} = 1, \theta^{(1)} = 1.5, \theta^{(2)} = 2.04, \theta^{(3)} = 2.472$. Continuing in the same manner, we would obtain estimates $2.740, 2.880, 2.946, 2.976, \ldots$, where 3 is the true maximum.

data.

EM is an iterative method that given the current estimate for the parameter, finds a new estimate. The idea behind EM is finding lower bounds on the log-likelihood of the observed data and maximizing these lower bounds. This is illustrated in Figure 7.2 (see Example 7.1). Suppose our current estimate of $\boldsymbol{\theta}$ is $\boldsymbol{\theta}'$. In each iteration, we find a lower bound $B(\boldsymbol{\theta}, \boldsymbol{\theta}')$ on $\log p(\boldsymbol{x}; \boldsymbol{\theta})$ that coincides with it at $\boldsymbol{\theta} = \boldsymbol{\theta}'$, i.e.,

$$\begin{aligned} \log p(\boldsymbol{x}; \boldsymbol{\theta}) &\geq B(\boldsymbol{\theta}, \boldsymbol{\theta}'), \qquad \text{for all } \boldsymbol{\theta}, \\ \log p(\boldsymbol{x}; \boldsymbol{\theta}) &= B(\boldsymbol{\theta}, \boldsymbol{\theta}'), \qquad \text{for } \boldsymbol{\theta} = \boldsymbol{\theta}'. \end{aligned} \tag{7.1}$$

Now let our new estimate be

$$\boldsymbol{\theta}'' = \arg\max_{\boldsymbol{\theta}} B(\boldsymbol{\theta}, \boldsymbol{\theta}').$$

Note that we have not used $\log p(\boldsymbol{x}; \boldsymbol{\theta})$ to find $\boldsymbol{\theta}''$. Since

$$\log p(\boldsymbol{x}, \boldsymbol{\theta}'') \geq B(\boldsymbol{\theta}'', \boldsymbol{\theta}') \geq B(\boldsymbol{\theta}', \boldsymbol{\theta}') \geq \log p(\boldsymbol{x}, \boldsymbol{\theta}'),$$

we have

$$\log p(\boldsymbol{x}; \boldsymbol{\theta}'') \geq \log p(\boldsymbol{x}; \boldsymbol{\theta}').$$

So our new estimate is at least as good as the old one, and under certain conditions, it is going to be strictly better. We then use $\boldsymbol{\theta}''$ in place of $\boldsymbol{\theta}'$ and repeat. Note that if $\log p(\boldsymbol{x}; \boldsymbol{\theta})$ is bounded, since the sequence $\log p(\boldsymbol{x}; \boldsymbol{\theta}')$ is non-decreasing, it will converge. Under appropriate conditions, this means that $\boldsymbol{\theta}'$ also converges to a stationary point of $p(\boldsymbol{x}; \boldsymbol{\theta})$. See [1] for details.

It remains to find a lower bound that satisfies (7.1). For any $\boldsymbol{y}$ such that $p(\boldsymbol{y}|\boldsymbol{x}; \boldsymbol{\theta}) > 0$,

$$\ell(\boldsymbol{\theta}) = \ln p(\boldsymbol{x}; \boldsymbol{\theta}) = \ln \frac{p(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\theta})}{p(\boldsymbol{y}|\boldsymbol{x}; \boldsymbol{\theta})}.$$

Then, for any distribution q for the missing data $\boldsymbol{y}$,

$$\begin{aligned}
\ell(\boldsymbol{\theta}) &= \sum_y q(\boldsymbol{y}) \ln p(\boldsymbol{x};\boldsymbol{\theta}) \\
&= \sum_y q(\boldsymbol{y}) \ln \frac{p(\boldsymbol{x},\boldsymbol{y};\boldsymbol{\theta})}{p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta})} \\
&\geq \sum_y q(\boldsymbol{y}) \ln \frac{p(\boldsymbol{x},\boldsymbol{y};\boldsymbol{\theta})}{p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta})} - D(q(\boldsymbol{y})||p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta})) \\
&= \sum_y q(\boldsymbol{y}) \ln \frac{p(\boldsymbol{x},\boldsymbol{y};\boldsymbol{\theta})}{p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta})} - \sum_y q(\boldsymbol{y}) \ln \frac{q(\boldsymbol{y})}{p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta})} \\
&= \sum_y q(\boldsymbol{y}) \ln p(\boldsymbol{x},\boldsymbol{y};\boldsymbol{\theta}) - \sum_y q(\boldsymbol{y}) \ln q(\boldsymbol{y}),
\end{aligned}$$

where for two distributions p_1 and p_2, $D(p_1(z)\|p_2(z)) = \sum_z p_1(z) \log \frac{p_1(z)}{p_2(z)}$ is the *relative entropy* (Kullback–Leibler or KL divergence), reviewed in Chapter 1. Recall that relative entropy is non-negative and equals 0 if and only if $p_1 = p_2$.

Thus for any distribution q, we have a lower bound on $\ell(\boldsymbol{\theta})$. Suppose our current guess for $\boldsymbol{\theta}$ is $\boldsymbol{\theta}^{(t)}$. We would like this lower bound to be equal to $\ell(\boldsymbol{\theta})$ at $\boldsymbol{\theta} = \boldsymbol{\theta}^{(t)}$. For this to occur, we need

$$D\Big(q(\boldsymbol{y})||p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta}^{(t)})\Big) = 0 \iff q(\boldsymbol{y}) = p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta}^{(t)}),$$

resulting in

$$\ell(\boldsymbol{\theta}) \geq \sum_y p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta}^{(t)}) \ln p(\boldsymbol{x},\boldsymbol{y};\boldsymbol{\theta}) - \sum_{\boldsymbol{y}} p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta}^{(t)}) \ln p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta}^{(t)}) = B(\boldsymbol{\theta},\boldsymbol{\theta}^{(t)}).$$

Now instead of maximizing ℓ, we can maximize B. We note however that the second term in B is not a function of $\boldsymbol{\theta}$. So we instead define the following expectation

$$Q(\boldsymbol{\theta},\boldsymbol{\theta}^{(t)}) = \sum_y p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta}^{(t)}) \ln p(\boldsymbol{x},\boldsymbol{y};\boldsymbol{\theta}),$$

and find

$$\boldsymbol{\theta}^{(t+1)} = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta},\boldsymbol{\theta}^{(t)}).$$

For simplicity of notation, I often use $\boldsymbol{\theta}'$ to denote $\boldsymbol{\theta}^{(t)}$ and $\boldsymbol{\theta}''$ to denote $\boldsymbol{\theta}^{(t+1)}$. Also, let $\mathbb{E}'$ be expected value assuming the value of $\boldsymbol{\theta}'$. We can then describe the EM algorithm as

- The E-step:

$$Q(\boldsymbol{\theta};\boldsymbol{\theta}') = \sum_{\boldsymbol{y}} p(\boldsymbol{y}|\boldsymbol{x};\boldsymbol{\theta}') \ln p(\boldsymbol{x},\boldsymbol{y};\boldsymbol{\theta}) = \mathbb{E}[\ln p(\boldsymbol{x},\boldsymbol{Y};\boldsymbol{\theta})|\boldsymbol{x};\boldsymbol{\theta}'] = \mathbb{E}'[\ln p(\boldsymbol{x},\boldsymbol{Y};\boldsymbol{\theta})|\boldsymbol{x}]$$

- The M-step:

$$\boldsymbol{\theta}'' = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta};\boldsymbol{\theta}').$$

Update $\boldsymbol{\theta}' \leftarrow \boldsymbol{\theta}''$ and repeat.

Roughly speaking, EM can be viewed as alternatively finding an estimate for the missing data through expectation by assuming a value for the parameters (the E-step) and finding a new estimate for the parameter based on the estimate of the data.

7.2 Clustering with EM

For classification the complete data is $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$. When the labels y_i are missing, the problem becomes *clustering*.

We assume Gaussian class-conditionals:

$$\begin{aligned} p(Y_i = 1) &= \pi, & (\boldsymbol{x}_i | Y_i = 1) &\sim \mathcal{N}(\boldsymbol{\mu}_1, \mathsf{K}_1) \\ p(Y_i = 0) &= 1 - \pi, & (\boldsymbol{x}_i | Y_i = 0) &\sim \mathcal{N}(\boldsymbol{\mu}_0, \mathsf{K}_0) \end{aligned}$$

Let $\boldsymbol{\theta} = (\pi, \boldsymbol{\mu}_0, \boldsymbol{\mu}_1, \mathsf{K}_0, \mathsf{K}_1)$. Ideally, we would want to maximize the likelihood for the observed data $\{(\boldsymbol{x}_i)\}_{i=1}^n$,

$$\ell(\boldsymbol{\theta}) = \ln p(\boldsymbol{x}_1^n | \boldsymbol{\theta}) = \ln \sum_{y_1^n} p(\boldsymbol{x}_1^n, y_1^n | \boldsymbol{\theta}).$$

But this is difficult to do because of a lack of an analytical solution due to the summation. Instead, we can use a computational method such as EM.

We will proceed as follows:

- **Set-up:** It is helpful to start with the log-likelihood of the complete data and simplify it before proceeding to the EM algorithm. We have

  $$\ln p(\boldsymbol{x}_1^n, y_1^n; \boldsymbol{\theta}) = \sum_{i=1}^n \ln p(\boldsymbol{x}_i, y_i; \boldsymbol{\theta}),$$

  and for each term in this sum,

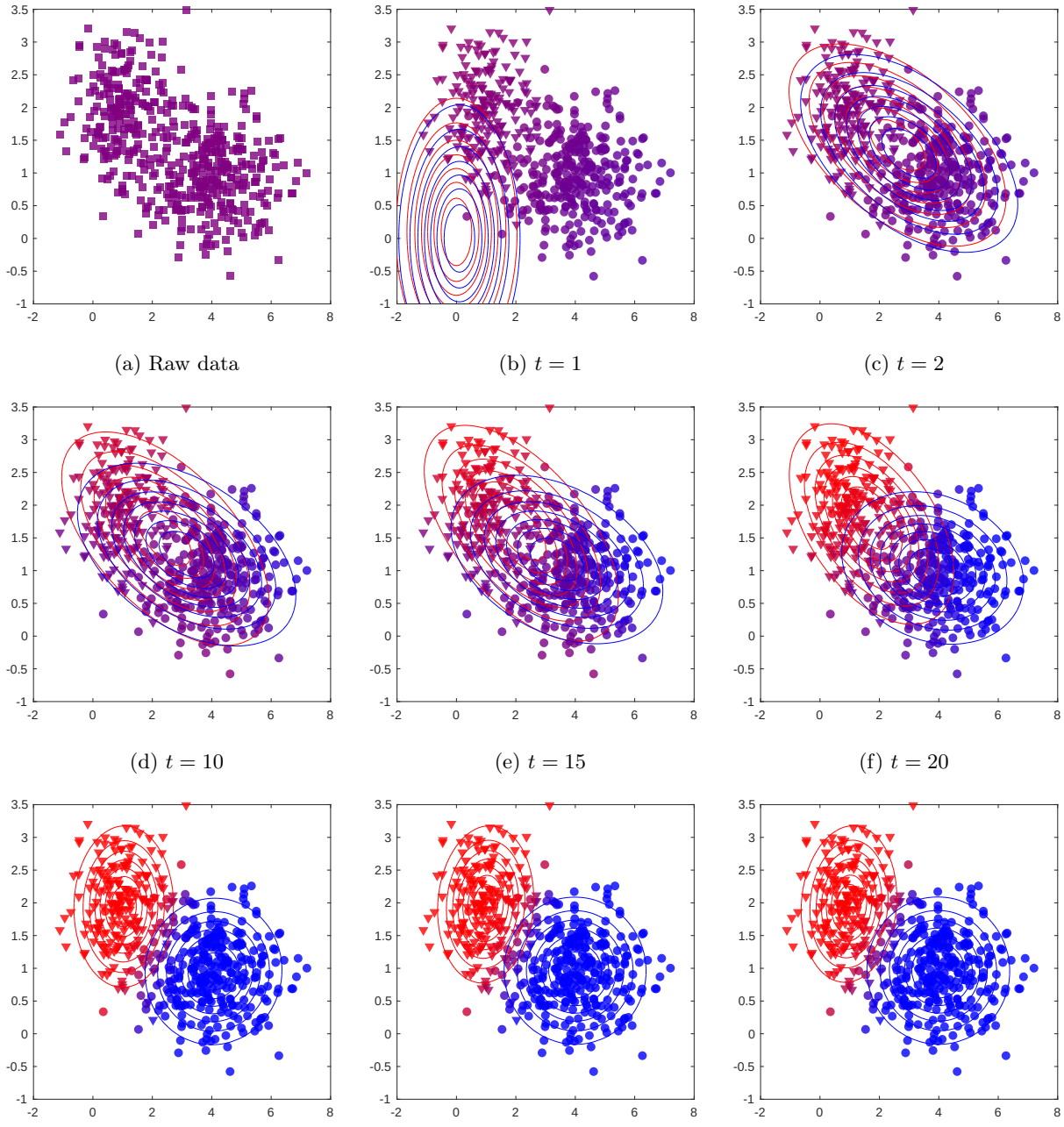
  $$\begin{aligned} \ln p(\boldsymbol{x}_i, y_i; \boldsymbol{\theta}) &= \ln\Big((\pi p(\boldsymbol{x}_i | Y_i = 1; \boldsymbol{\theta}))^{y_i} ((1-\pi) p(\boldsymbol{x}_i | Y_i = 0; \boldsymbol{\theta}))^{1-y_i} \Big) \\ &= y_i \ln(\pi p(\boldsymbol{x}_i | Y_i = 1; \boldsymbol{\theta})) + (1 - y_i) \ln((1-\pi) p(\boldsymbol{x}_i | Y_i = 0; \boldsymbol{\theta})). \end{aligned}$$

- **The E-step:** Let $\boldsymbol{\theta}'$ be the current estimate for $\boldsymbol{\theta}$ and let $\mathbb{E}'$ denote expected value operator with respect to the distribution $p(\boldsymbol{y} | \boldsymbol{x}; \boldsymbol{\theta}')$. We have

  $$\begin{aligned} Q(\boldsymbol{\theta}; \boldsymbol{\theta}') &= \mathbb{E}'[\ln p(\boldsymbol{x}_1^n, Y_1^n; \boldsymbol{\theta}) | \boldsymbol{x}_1^n] \\ &= \mathbb{E}'\left[\sum_{i=1}^n \ln p(\boldsymbol{x}_i, Y_i; \boldsymbol{\theta}) | \boldsymbol{x}_1^n \right] \\ &= \sum_{i=1}^n \mathbb{E}'[\ln p(\boldsymbol{x}_i, Y_i; \boldsymbol{\theta}) | \boldsymbol{x}_i] \end{aligned}$$

  And for each term in the sum,

  $$\begin{aligned} \mathbb{E}'[\ln p(\boldsymbol{x}_i, Y_i; \boldsymbol{\theta}) | \boldsymbol{x}_i] &= \mathbb{E}'[Y_i \ln(\pi p(\boldsymbol{x}_i | Y_i = 1; \boldsymbol{\theta})) + (1 - Y_i) \ln((1-\pi) p(\boldsymbol{x}_i | Y_i = 0; \boldsymbol{\theta})) | \boldsymbol{x}_i] \\ &= \mathbb{E}'[Y_i | \boldsymbol{x}_i] \ln(\pi p(\boldsymbol{x}_i | Y_i = 1; \boldsymbol{\theta})) + \mathbb{E}'[1 - Y_i | \boldsymbol{x}_i] \ln((1-\pi) p(\boldsymbol{x}_i | Y_i = 0; \boldsymbol{\theta})) \\ &= \gamma_i' (\ln \pi + \ln p(\boldsymbol{x}_i | Y_i = 1; \boldsymbol{\theta})) + (1 - \gamma_i')(\ln(1-\pi) + \ln p(\boldsymbol{x}_i | Y_i = 0; \boldsymbol{\theta})), \end{aligned}$$



(a) Raw data (b) $t = 1$ (c) $t = 2$

(d) $t = 10$ (e) $t = 15$ (f) $t = 20$

(g) $t = 30$ (h) $t = 40$ (i) $t = 50$

Figure 7.3: EM clustering of a mixture of two Gaussian datasets. In (a) the raw data is shown and in (b-i), steps of the EM algorithm are shown. To compare with the underlying distributions and clusters, the points from each of the Gaussian distributions are shown with triangles and circles. However, the EM algorithm does not have access to this data. The contour plots represent the current estimate for the parameters of each of the Gaussian distributions and the color of each data point represents the estimate of the EM algorithm for the probability that the point belongs to the clusters ($\gamma_i' = p(Y_i = 1|\boldsymbol{x}_i; \boldsymbol{\theta}')$). A video of the clustering can be found here.

where

$$\begin{aligned}\gamma_i' &= \mathbb{E}'[Y_i|\boldsymbol{x}_i] \\ &= p(Y_i = 1|\boldsymbol{x}_i;\boldsymbol{\theta}') \\ &= \frac{p(\boldsymbol{x}_i, Y_i = 1;\boldsymbol{\theta}')}{p(\boldsymbol{x}_i, Y_i = 1;\boldsymbol{\theta}') + p(\boldsymbol{x}_i, Y_i = 0;\boldsymbol{\theta}')} \\ &= \frac{\pi'\mathcal{N}(\boldsymbol{x}_i;\mu_1', \mathsf{K}_1')}{\pi'\mathcal{N}(\boldsymbol{x}_i;\mu_1', \mathsf{K}_1') + (1-\pi')\mathcal{N}(\boldsymbol{x}_i;\mu_0', \mathsf{K}_0')}.\end{aligned}$$

Here, γ_i' has a significant meaning. It represents the probability that a given point $\boldsymbol{x}_i$ belongs to class 1 given the current estimate of the parameters. Instead of computing the likelihood based on a known value for y_i, in the E-step, we compute the likelihood by partially assigning $\boldsymbol{x}_i$ to class 1 and to class 0.

- **The M-step:** To find π'':

$$\frac{\partial Q}{\partial \pi} = \sum_{i=1}^{n}\left(\frac{\gamma_i'}{\pi} - \frac{1-\gamma_i'}{1-\pi}\right) = 0 \Rightarrow \pi'' = \frac{\sum_{i=1}^{n}\gamma_i'}{n}.$$

To find $\boldsymbol{\mu}_1''$:

$$\begin{aligned}\frac{\partial Q}{\partial \boldsymbol{\mu}_1} &= \frac{\partial}{\partial \boldsymbol{\mu}_1}\sum_{i=1}^{n}\gamma_i' \ln p(\boldsymbol{x}_i|Y_i = 1;\boldsymbol{\theta}) \\ &= \frac{\partial}{\partial \boldsymbol{\mu}_1}\sum_{i=1}^{n}\gamma_i'\left(-\frac{1}{2}(\boldsymbol{x}_i - \boldsymbol{\mu}_1)^T\mathsf{K}_1^{-1}(\boldsymbol{x}_i - \boldsymbol{\mu}_1)\right) \\ &= \sum_{i=1}^{n}\gamma_i'\mathsf{K}_1^{-1}(\boldsymbol{x}_i - \boldsymbol{\mu}_1) = 0 \Rightarrow \boldsymbol{\mu}_1'' = \frac{\sum_{i=1}^{n}\gamma_i'\boldsymbol{x}_i}{\sum_{i=1}^{n}\gamma_i'}.\end{aligned}$$

To find K_1'':

$$\begin{aligned}\frac{\partial Q}{\partial \mathsf{K}_1^{-1}} &= \frac{\partial}{\partial \mathsf{K}_1^{-1}}\sum_{i=1}^{n}\gamma_i'\left(\frac{1}{2}\ln|\mathsf{K}_1^{-1}| - \frac{1}{2}(\boldsymbol{x}_i - \boldsymbol{\mu}_1)^T\mathsf{K}_1^{-1}(\boldsymbol{x}_i - \boldsymbol{\mu}_1)\right) \\ &= \frac{1}{2}\mathsf{K}_1\sum_{i=1}^{n}\gamma_i' - \frac{1}{2}\sum_{i=1}^{n}\gamma_i'(\boldsymbol{x}_i - \boldsymbol{\mu}_1)(\boldsymbol{x}_i - \boldsymbol{\mu}_1)^T = 0 \\ \Rightarrow \mathsf{K}_1'' &= \frac{\sum_{i=1}^{n}\gamma_i'(\boldsymbol{x}_i - \boldsymbol{\mu}_1'')(\boldsymbol{x}_i - \boldsymbol{\mu}_1'')^T}{\sum_{i=1}^{n}\gamma_i'}.\end{aligned}$$

Several steps of an EM clustering of a dataset are shown in Figure 7.3. In essence, the EM algorithm uses the current estimates of posterior class probabilities of a point as labels and updates the distributions. Having updated the distributions, it updates the posterior class probabilities and repeats.

7.3 EM with general missing data **

So far, we have considered problems in which data can be divided into an observed component x and a hidden component y, with the expectation given by

$$Q(\boldsymbol{\theta};\boldsymbol{\theta}') = \sum_{y} p(y|x;\boldsymbol{\theta}')\ln p(\boldsymbol{x}, y;\boldsymbol{\theta})$$

But we can use EM to solve a more general class of problems, where this division may not be possible. Specifically, we assume that the complete data is given by z and the observed data is given by x, where x is a function of z. In this case, the expectation is given by

$$Q(\boldsymbol{\theta};\boldsymbol{\theta}') = \sum_z p(z|x;\boldsymbol{\theta}') \ln p(z;\boldsymbol{\theta})$$

Example 7.1 ([1])**.** Let

$$\begin{aligned} &x = s + \epsilon, \\ &s \sim \mathcal{N}(0,\theta), \qquad \theta \geq 0 \\ &\epsilon \sim \mathcal{N}(0,\sigma^2) \qquad \sigma^2 > 0, \end{aligned}$$

where s and ϵ are independent, σ is known, and θ is unknown. Our goal is to estimate θ. In this case, the complete data is $\boldsymbol{z} = (s,\epsilon)$ and observed data is $x = s + \epsilon$.

We can solve this problem directly by noting that

$$x \sim \mathcal{N}(0, \theta + \sigma^2),$$

where we have used

$$\mathrm{Var}(x) = \mathrm{Cov}(s+\epsilon, s+\epsilon) = \sigma^2 + \theta.$$

The maximum likelihood estimate for the variance of x is then

$$\hat{\theta}_{ML} = \begin{cases} x^2 - \sigma^2 & \text{if } x^2 \geq \sigma^2, \\ 0 & \text{if } x^2 < \sigma^2. \end{cases}$$

With EM:

- The E-step:

$$\begin{aligned} Q(\theta;\theta') &= \mathbb{E}'[\ln p(\boldsymbol{z};\theta)|x] \\ &= \mathbb{E}'[\ln p(s;\theta) + \ln p(\epsilon;\theta)|x] \\ &\doteq \mathbb{E}'[\ln p(s;\theta)|x] \\ &\doteq \mathbb{E}'\left[-\frac{\ln\theta}{2} - \frac{s^2}{2\theta}|x\right] \\ &= -\frac{\ln\theta}{2} - \frac{\mathbb{E}'[s^2|x]}{2\theta} \end{aligned}$$

- The M-step:

$$\frac{\partial Q}{\partial \theta} = -\frac{1}{2\theta} + \frac{\mathbb{E}'[s^2|x]}{2\theta^2} = 0 \Rightarrow \theta'' = \mathbb{E}'[s^2|x].$$

  This is a very intuitive result.

  With some manipulation (HW), this results in

$$\theta'' = \left(\frac{\theta'}{\theta' + \sigma^2}\right)^2 x^2 + \frac{\theta'\sigma^2}{\theta' + \sigma^2}.$$

The plot for the log-likelihood and the EM estimates, starting from $\theta^{(0)} = 1$, is given in Figure 7.2, where $\sigma^2 = 1$ and $x = 2$ and thus $\hat{\theta}_{ML} = 3$. △

7.4 The MM Algorithm **

The idea behind the EM algorithm, i.e., finding a lower bound with certain properties, can be generalized, leading to the Minorization-Maximization (MM) algorithm. Specifically, EM provides a certain way of finding a lower bound, but if we find a lower bound by another method that still satisfies appropriate equality and inequality conditions, we can still maximize the function we are interested in. We illustrate this by applying MM to rank aggregation.

7.4.1 Rank Aggregation from Pairwise Comparisons via MM

Rank aggregation refers to combining a set of full or partial rankings of a set of alternatives in order to obtain a consensus ranking. For example, we may be interested in ranking sport teams based on match results. In this case, the input data is a set of pairwise comparisons (i.e., a partial ranking involving two items) and the desired output is a ranking of all the teams.

The data: There are n teams. We are given a dataset $\mathcal{D} = \{w_{12}, w_{13}, \ldots, w_{n-1,n}\}$, where w_{ij} is the number of times team i beats team j. It will be helpful to assume $w_{ii} = 0$ rather than leaving it undefined.

The model: For two teams i and j, we assume

$$\Pr(i \text{ beats } j) = \frac{e^{s_i}}{e^{s_i} + e^{s_j}},$$

where s_i is a score reflecting the strength of team i. Denote $\boldsymbol{s} = (s_1, \ldots, s_n)$.

This leads to the log-likelihood

$$\mathcal{L}(\boldsymbol{s}) = \sum_{i,j} w_{ij}(s_i - \ln(e^{s_i} + e^{s_j}))$$

As an aside, note that for a differentiable convex function $f(x)$, we have

$$\begin{aligned} f(x) &\geq f(x') + f'(x')(x - x'), && \text{for all } x', \\ f(x) &= f(x') + f'(x')(x - x'), && \text{for } x' = x. \end{aligned}$$

Since $-\ln x$ is a convex function,

$$-\ln x \geq -\ln x' - \frac{x - x'}{x'} = -\ln x' - \frac{x}{x'} + 1.$$

Hence, if we define

$$Q(\boldsymbol{s}, \boldsymbol{s}') = \sum_{i,j} w_{ij}\left(s_i - \ln\left(e^{s_i'} + e^{s_j'}\right) - \frac{e^{s_i} + e^{s_j}}{e^{s_i'} + e^{s_j'}} + 1\right),$$

then,

$$\begin{aligned} \mathcal{L}(\boldsymbol{s}) &\geq Q(\boldsymbol{s}, \boldsymbol{s}'), && \text{for all } \boldsymbol{s}', \\ \mathcal{L}(\boldsymbol{s}) &= Q(\boldsymbol{s}, \boldsymbol{s}'), && \text{for } \boldsymbol{s} = \boldsymbol{s}'. \end{aligned}$$

We can simplify Q by ignoring terms that do not involve $\boldsymbol{s}$, and then separating the parameters (the latter

was not possible for $\mathcal{L}$)

$$\begin{aligned} Q(\boldsymbol{s}, \boldsymbol{s}') &= \sum_{i,j} w_{ij} \left(s_i - \frac{e^{s_i} + e^{s_j}}{e^{s'_i} + e^{s'_j}} \right) \\ &= \sum_i s_i \sum_j w_{ij} - \sum_i e^{s_i} \sum_j \frac{w_{ij}}{e^{s'_i} + e^{s'_j}} - \sum_i \sum_j w_{ij} \frac{e^{s_j}}{e^{s'_i} + e^{s'_j}} \\ &= \sum_i s_i \sum_j w_{ij} - \sum_i e^{s_i} \sum_j \frac{w_{ij}}{e^{s'_i} + e^{s'_j}} - \sum_i \sum_j w_{ji} \frac{e^{s_i}}{e^{s'_i} + e^{s'_j}} \\ &= \sum_i s_i \sum_j w_{ij} - \sum_i e^{s_i} \sum_j \frac{w_{ij} + w_{ji}}{e^{s'_i} + e^{s'_j}}. \end{aligned}$$

Given the current estimate $\boldsymbol{s}'$, we can now find the next estimate $\boldsymbol{s}''$ by differentiating Q, and setting it equal to 0,

$$\begin{aligned} \frac{\partial Q}{\partial s_i} &= \sum_j w_{ij} - e^{s_i} \sum_j \frac{w_{ij} + w_{ji}}{e^{s'_i} + e^{s'_j}} = 0 \\ s''_i &= \ln \frac{\sum_j w_{ij}}{\sum_j \frac{w_{ij} + w_{ji}}{e^{s'_i} + e^{s'_j}}}. \end{aligned}$$

This allows us to estimate the scores s_i. When convergence is achieved or after a set number of iterations, we sort the scores and thus find a ranking of the n teams. [1]

References

[1] Bruce Hajek. *Random Processes for Engineers*. Illinois, 2014. URL: http://hajek.ece.illinois.edu/Papers/randomprocJuly14.pdf (visited on 01/30/2017).

Chapter 8

Basics of Graphical Models

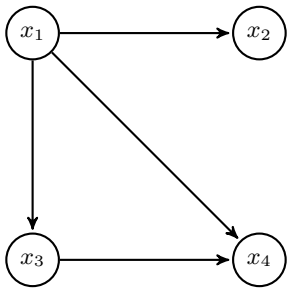
8.1 Introduction

Graphical models (GMs) are used to represent distributions on graphs. They enable us to *represent conditional independencies* and *factorization of distributions* facilitate probabilistic inference through message passing algorithms. There are different types of GMs:

- Bayesian Networks (BN, aka Directed Graphical Models): Natural for representing causal relationships
- Markov Random Fields (MRF, aka Undirected Graphical Models): Suitable for representing co-influence or non-causal relationships among a subsets of variables, e.g., friendship in social networks and pixels in an image (adjacent pixels are likely to have similar colors).
- Factor Graphs: A flexible type of GM that can represent distributions represented by BNs and MRFs.

8.2 Bayesian Networks

A Bayesian network is a **directed acyclic graph** (DAG) with some additional attributes. A DAG is a graph whose edges have direction and in which there is no cycle if one follows the edges based on their direction. In a DAG, a **parent** of a node y is a node x such that there is an edge from x to y. A **child** of y is a node z such that y is the parent of z. An **ancestor** is a parent, parent of a parent, etc., and a **descendant** is a child, child of a child, etc. A **complete DAG** is a DAG such that with an edge between each pair of vertices. An example of a DAG with four nodes is shown below.



In a Bayesian network represented by a DAG G:

- Nodes $x_1, \ldots, x_m$ represent variables or quantities (can be scalar or vector)
- Edges represent causal relationships

- The probability distribution over $x_1^m = x_1, \ldots, x_m$ can be expressed as:

$$p(x_1^m) = \prod_{i=1}^{m} p(x_i | \text{ pa}(x_i))$$

  where $\text{pa}(x_i)$ are the parents of x_i in G, i.e., nodes with an edge *to* x_i.

We then say that the distribution p **factorizes** with respect to G. For example, for a distribution p that factorizes with respect to the graph shown above, we have

$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_1)p(x_4|x_1, x_3). \tag{8.1}$$

What does (8.1) tell us about the distribution? Recall that based on the chain rule of probability, we always have

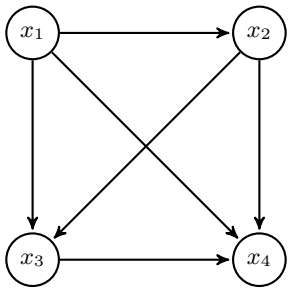
$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3). \tag{8.2}$$

It is straightforward to show that (8.1) is equivalent to

$$\begin{aligned} p(x_3|x_1, x_2) &= p(x_3|x_1), \\ p(x_4|x_1, x_2, x_3) &= p(x_4|x_1, x_3). \end{aligned} \tag{8.3}$$

These two expressions are conditional independence statements, which we can restate as $x_3 \perp\!\!\!\perp x_2 \,|\, x_1$ and $x_4 \perp\!\!\!\perp x_2 \,|\, x_1, x_3$. Thus saying that p factorizes with respect to the graph above is equivalent to assuming (8.3). This is in general true. The set of missing incoming edges for each node in the graph represents a conditional independence assumption.

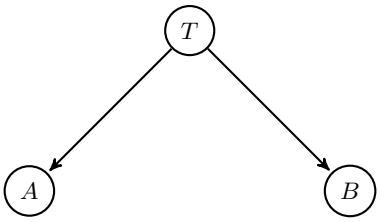
The complete graph, shown for four nodes below, represents the factorization given in (8.2), which holds for any distribution and thus the graph can represent any distribution. But such a graph is not particularly useful since the power of graphical models results from the independence assumptions that they encode.



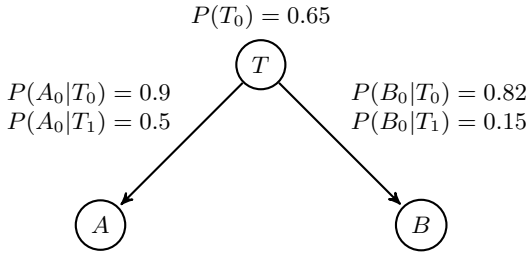
Note that the complete graph is acyclic as it imposes an ordering over the nodes (in this case, x_1, x_2, x_3, x_4). *We can view any Bayesian network as being obtained from a complete DAG by removing edges.* So every Bayesian network is also acyclic.

Example 8.1 (Alice, Bob, and traffic)**.** Alice and Bob are employees of a business in Charlottesville, both of whom take 29S to get to work. We are interested in whether they arrive on time or late. We assume their arrival time is affected by traffic, which leads to dependence, but there aren't any other factors that can affect both of them. Let $A = 0$ and $A = 1$ denote Alice being on time and being late A_1, respectively and similarly for Bob ($B = 0$ and B_1). Traffic is either normal ($T = 0$) or heavy ($T = 1$). We use X_0 and X_1 as shorthand for $X = 0$ and $X = 1$ for our random variables.

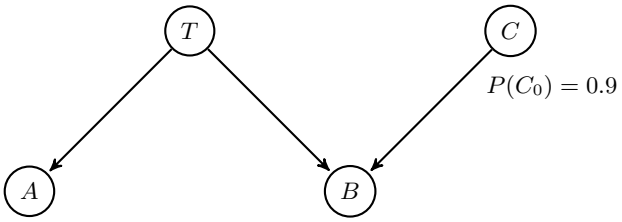
The Bayesian Network that models the probability distribution is shown below.



This graph implies that $A \perp\!\!\!\perp B|T$ and that $p(ABT) = p(T)p(A|T)p(B|T)$. We now have the *structure* of our model. But we still need the *conditional probability distributions* to complete the model. Suppose these distributions are as below:



Taking the example a step further, suppose that Bob has a son, Charlie (C_0 and C_1) who has to be dropped off at school. Charlie being late has an effect on Bob being late. We will adjust the Bayesian Network below and use the joint probability distribution in the following table.



CT	$P(B_0\|CT)$	$P(B_1\|CT)$
C_0T_0	0.9	0.1
C_0T_1	1/6	5/6
C_1T_0	0.1	0.9
C_1T_1	0	1

Note that this new conditional distribution does not change any previously calculated probabilities involving Traffic, Alice, and Bob, but the numbers were chosen specifically to achieve this; this is not always the case.

Based on this graph, the joint probability distribution is:

$$p(ABTC) = p(T)p(C)p(A|T)p(B|CT).$$

It is easy to show that $T \perp\!\!\!\perp C$ but as we will see below $T \not\perp\!\!\!\perp C|B$.

Bayesian networks facilitate certain kinds of reasoning. In **causal reasoning**, we draw conclusions about unobserved effects base on observed causes. For example, if we know there was heavy traffic, then it is more likely that Bob was late, $p(B_1|T_1) = 0.85 > p(B_1) = 0.41$. **Evidential reasoning** allows us to say something about the cause by observing the effects. For example,

$$p(T_1|B_1) = \frac{p(B_1|T_1)p(T_1)}{p(B_1)} = 0.7177 > p(T_1) = 0.35,$$

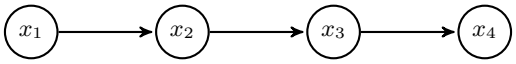
tells us that heavy traffic is more likely when Bob is late, even though we have no direct information about the traffic.

We also have $p(T_1|B_1C_1) < P(T_1|B_1)$, which makes intuitive sense. Bob being late provides evidence for traffic being heavy. But if we know Charlie is late, then we have an alternative explanation for Bob being late, lessening the need for traffic being heavy as a reason for Bob's tardiness. This type of reasoning, where given an effect, occurrence of one cause lessens the probability of another cause, is called **explaining away**. △

8.2.1 Markov Model

A **Markov Model** or a **Markov chain** is a Bayesian network whose graph consists of a single path. Such a model can, for example, represent the total winning of a gambler as a function of time, where each game is independent. The main assumption is that *given the present, the future is independent of the past*: how much money you'll have after the next game is independent of past games, if your current worth is known. Another, idealized example is weather forecast: Given that we know today's weather, past weather is irrelevant for the purpose of forecasting tomorrow's weather.

A Markov chain with four nodes is given below

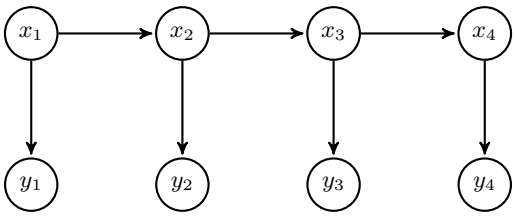


with an associated joint probability distribution that factorizes as

$$p(x_1^4) = p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3).$$

Consider a set of n random variables each of which can take on k different values. The most general probability distribution over these variables will have $k^n - 1$ parameters (the -1 comes from the fact that we know the probabilities must sum to one). In practice, this is such a huge number even for $k = 2$ and relatively small n, e.g., $n = 100$, that we can't even store the distribution, let alone learn it from data. The Markov model, however, has $(k-1) + (n-1)k(k-1)$ parameters, which is much more manageable. This is an example of graphical models making modeling more feasible.

A closely related model is the **hidden Markov model (HMM)**:



An HMM is used when the true state of the system cannot be directly observed but we can observe some function of the state. For example, x_i can represent if cancer is in remission or not and y_i can represent observations from medical tests.

Like Markov random fields, Markov and hidden Markov models are named after Russian mathematician Andrey Markov, but Markov models are Bayesian Networks and not Markov Random Fields.

8.2.2 Why graphical models?

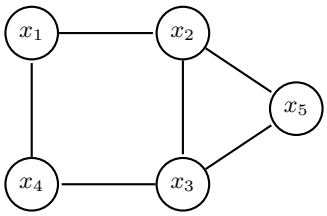
Graphical models, such as Bayesian networks are useful for several reasons.

- They provide a simple but flexible way to encode conditional independencies, enabling us to answer questions about independence based on graphs.
- GMs help constructing tractable models. As an example, see the number of parameters for a Markov chain versus an unrestricted model described above.
- Restriction to GMs has computational benefits, allowing us to draw conclusions about hidden quantities based on observations efficiently using algorithms such as belief propagation.

8.3 Markov Random Fields

Definition 8.2 (Clique and maximal clique)**.** The following definitions from graph theory will be used in this section. In an undirected graph, a *clique* is a subset of nodes such that there is an edge between any two of them. A *maximal clique* is a clique such that there are no nodes not in the clique that connected to all the nodes already in the clique.

Suppose that we are interested in developing a political party affiliation model for a group of 5 people (or millions of people if we have social network data). Let's assume their friendships are given by the following graph



in which each node x_i represents the party of person i and an edge between x_i and x_j means that i and j are friends. How can we develop a probability distribution that can help us in this task?

We would like to encode the following observations in our distribution. We know that if two people are friends (e.g., 1 and 2), then it is more likely for them to have a common political alignment. Furthermore, for three people who are all friends (2,3,5), it is perhaps even more likely that they share the same political views. Let party affiliation be denoted by 0 or 1. We define

$$\psi_{ij}(x_i, x_j) = \begin{cases} 1, & x_i = x_j \\ 1/2, & x_i \neq x_j \end{cases} \tag{8.4}$$

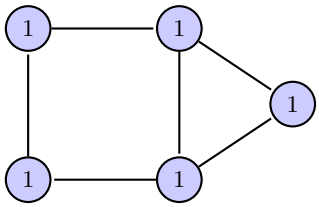
and

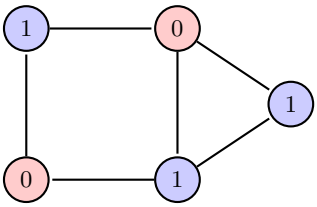
$$\psi_{ijk}(x_i, x_j, x_k) = \begin{cases} 1, & x_i = x_j = x_k \\ 1/2, & \text{if two of the three are equal} \end{cases} \tag{8.5}$$

So agreements are assigned a higher value. Now we can define a probability distribution as

$$p(x_1, \ldots, x_5) \propto \psi_{12}(x_1, x_2)\psi_{14}(x_1, x_4)\psi_{34}(x_3, x_4)\psi_{235}(x_2, x_3, x_5), \tag{8.6}$$

which assigns higher probability to configurations in which cliques of friends are in the same parties, as we wanted. For example, the probability of the left configuration is 16 times as likely to occur as the one on the right.





Note that there is no guarantee that the right side of (8.6) sums to 1 when going over all possible configurations so we need a normalization factor, which in this context is called the partition function,

$$Z = \sum_{x_1^5} \psi_{12}(x_1, x_2)\psi_{14}(x_1, x_4)\psi_{34}(x_3, x_4)\psi_{235}(x_2, x_3, x_5).$$

We can then write

$$p(x_1, \ldots, x_5) = \frac{1}{Z}\psi_{12}(x_1, x_2)\psi_{14}(x_1, x_4)\psi_{34}(x_3, x_4)\psi_{235}(x_2, x_3, x_5).$$

In our example, it turns out that $Z = 8.5$, and thus $p(1, 1, 1, 1, 1) = 0.11765$ while $p(1, 0, 1, 0, 1) = 0.0073529$.

Finally, we note while we chose the potential function for each pair and triple to be the same regardless of the identity of the nodes, this is not a necessity; for example, we could have chosen different functions for ψ_{12} and ψ_{34}.

We can now consider the general case. A **Markov random field (MRF)** or an undirected graphical model consists of an undirected graph G with nodes $x_1^m = x_1, \ldots, x_m$, and a probability distribution p that *factorizes with respect to* G, i.e.,

$$p(x_1^m) = \frac{1}{Z} \prod_{C \text{ is a clique in } G} \psi_C(x_C), \tag{8.7}$$

where for each clique C in G, x_C is the set of nodes in that clique, ψ_C is a *potential function*, which assigns non-negative values to all configurations of x_C, and Z is the *partition function*, which ensures that the right side is a proper distribution. Without loss of generality, we may assume the cliques are maximal by absorbing the potential functions for smaller cliques into the maximal clique. For our political party example above, for the clique with nodes x_2, x_3, x_5, we can either have 4 potential functions over all the sub-cliques,

$$\psi'(x_2, x_3)\psi'(x_3, x_5)\psi'(x_2, x_5)\psi'(x_2, x_3, x_5)$$

or a single potential function

$$\psi(x_2, x_3, x_5).$$

Both are valid and equally powerful in terms of representation.

When designing an MRF we incorporate local information into the potential functions, but the final result is that we learn about the global view of the entire system. Also, in an MRF, the relationships between nodes are symmetric rather than causal or directed.

8.3.1 Energy-based models

When for all configurations $\boldsymbol{x} = x_1^m$, the probability $p(\boldsymbol{x})$ is positive, it is helpful to represent the distribution as

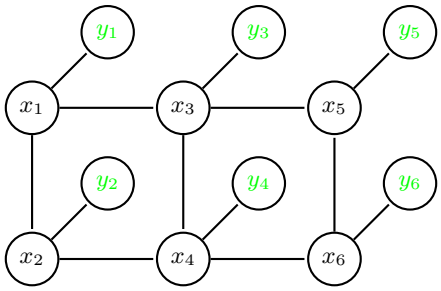
$$p(\boldsymbol{x}) \propto e^{-E(\boldsymbol{x})},$$

where $E(\cdot)$ is called the **energy function**. Such a distribution is also called a **Boltzmann distribution**. The terminology comes from statistical physics. In that context, lower energy corresponds to higher stability and thus higher probability for a system. For a graphical model, the energy function can be written as the sum of terms each of which correspond to a clique in the graph,

$$E(\boldsymbol{x}) = \sum_{C \text{ is a clique in } G} -\phi_C(\boldsymbol{x}_C) \quad \Rightarrow \quad p(\boldsymbol{x}) \propto \prod e^{\phi_C(\boldsymbol{x}_C)}$$

A **Boltzmann machine** is such a graphical model, typically with both nodes that can be observed and nodes that are hidden (latent).

Example 8.3 (**An MRF for denoising Images**)**.** The figure below shows an MRF for a noisy black and white image. Here $x_1, x_2, \cdots, x_6$ represent the true B/W status of the pixels and $y_1, y_2, \cdots, y_6$ the noisy values (e.g., due to noise of a camera). We denote 'Black'=-1 and 'White' = 1.



The energy function can be written as

$$E(\boldsymbol{x}, \boldsymbol{y}) = -\sum_{i}^{m} \alpha_i x_i - \sum_{(i,j)\in\mathcal{E}(G)} \beta_{i,j} x_i x_j - \sum_{i}^{m} \zeta_i x_i y_i,$$

where $\mathcal{E}(G)$ is the set of edges between neighboring pixels and $\beta_{i,j} > 0$ and $\zeta_i > 0$. The α_i control how likely a pixel is to be white without considering other pixels. The interaction between neighboring pixels is controlled by β_{ij}; since each is positive, it is more likely for adjacent pixels to have the same status. We assume that it is more likely for the noisy pixel to match the true pixel and so $\zeta_i > 0$ as well.

In a denoising task, we are given $\boldsymbol{y}$ and our goal is to recover $\boldsymbol{x}$. A reasonable solution is

$$\arg\max_{\boldsymbol{x}} p(\boldsymbol{x}, \boldsymbol{y}).$$

If we can output fractional values (if the denoised image can be grayscale), another possible solution is

$$\mathbb{E}[\boldsymbol{X}|\boldsymbol{y}].$$

△

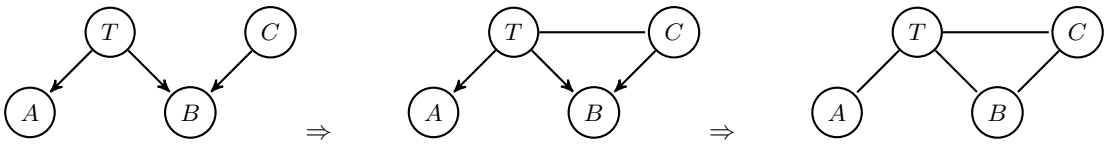
8.4 Moralization: Converting BNs to MRFs

In a BN, there is a term for each node x_i of the form

$$p(x_i|\,\mathrm{pa}(x_i)).$$

To be able to have the same term in an MRF, we need to have a clique containing x_i and its parents. So to design an MRF that can represent the same distribution as the BN, we first connect all the parents of each nodes with each other and then remove all directions from the edges.

Example 8.4 (Moralizing the traffic graph)**.** As an example, consider:



△

We have

$$p(A, B, T, C) = p(T)p(C)p(A|T)p(B|T, C) \qquad \Rightarrow \qquad p(A, B, T, C) = \psi(T)\psi(C)\psi(A, T)\psi(B, T, C),$$

where, for example, $\psi(B, T, C) = p(B|T, C)$.

Chapter 9

Independence in Graphical Models

Graphical models encode independence assumptions. In this chapter, we will study algorithms that enable us to answer questions of the form "Is $S_1 \perp\!\!\!\perp S_2 \mid S_3$?" where S_1, S_2, S_3 are subsets of the nodes in the graph.

Recall that we construct Bayesian network by assuming certain independence assumptions that allow us to remove edges from a complete DAG. The topic of this section is study of all independence properties, which is more general than assumptions used to construct Bayesian networks.

9.1 Independence for sets of random variables

We know that for three random variables x, y, z, x is independent of z given y, denoted $x \perp\!\!\!\perp z \mid y$, if and only if

$$p(x, z \mid y) = p(x|y)p(z|y).$$

This extends to sets of random variables and random vectors. For example, $\{x, y\} \perp\!\!\!\perp \{z, w\} \mid \{t, u\}$, or simply $x, y \perp\!\!\!\perp z, w \mid t, u$, if and only if

$$p(x, y, z, w \mid t, u) = p(x, y \mid t, u)p(z, w \mid t, u)$$

Using this we can show that if $x, y \perp\!\!\!\perp z, w \mid t, u$, then $x \perp\!\!\!\perp z \mid t, u$ and $x, y \perp\!\!\!\perp z \mid t, u$. For example,

$$\begin{aligned} p(x, z \mid t, u) &= \sum_{y', w'} p(x, y', z, w' \mid t, u) \\ &= \sum_{y', w'} p(x, y' \mid t, u)p(z, w' \mid t, u) \\ &= \sum_{y'} p(x, y' \mid t, u) \sum_{w'} p(z, w' \mid t, u) \\ &= p(x \mid t, u)p(z \mid t, u). \end{aligned}$$

Note however that if $x \perp\!\!\!\perp z$ and $y \perp\!\!\!\perp z$ it does not follow that $x, y \perp\!\!\!\perp z$. For a counter-example, set $x \sim \text{Ber}(1/2), y \sim \text{Ber}(1/2)$ and $z = x + y$.

Exercise 9.1 (Independence via factorization)**.** Show that for three disjoint sets of random variables A,B,C, if for some functions f and g,

$$p(A, B \mid C) \propto f(A, C)g(B, C),$$

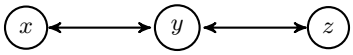
where the constant of proportionality may depend on C, then $A \perp\!\!\!\perp B \mid C$. △

9.2 Independence in Bayesian Networks

In the last chapter we saw that we can obtain a Bayesian Network by starting from a complete graph, representing the chain rule of probability, and then relying on independence assumptions, remove certain edges. Conversely, these independence assumptions are implied by the graphical model. But, in addition, to these, many other independence statements are implied by the network. In this section, we will introduce the concept of *d-separation*, using which we can find all independence statements satisfied by every distribution that factorizes with respect to the Bayesian network. We start by considering several simple networks that will help us describe d-separation.

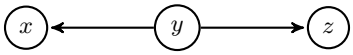
9.2.1 Simple Bayesian networks

Independence analysis in BNs relies on determining when information flows along paths in the graph. As a preliminary step, we study whether information about x affects our belief about z in the graphs of the form given below



with various directions on the edges and with y or one of its descendants being known or unknown.

Example 9.2 (Common cause, y unobserved)**.** Given three random variables x, y, and z with relationships shown below, is $x \perp\!\!\!\perp z$?

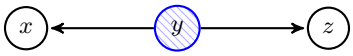


The answer: not in general. The only thing we know from the GM is $p(x, y, z) = p(y)p(x|y)p(z|y)$. We thus have

$$p(x, z) = \sum_y p(x, y, z) = \sum_y p(y)p(x|y)p(z|y)$$

and this is not necessarily equal to $p(x)p(z)$. *Exercise:* Find a counter example, i.e., find p such that it factorizes with respect to the graph but $x \not\perp\!\!\!\perp z$. △

Example 9.3 (Common cause, y observed)**.** Is $x \perp\!\!\!\perp z \mid y$ in the graph below?

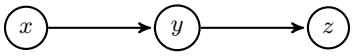


The answer: yes. We need to show $p(x, z \mid y) = p(x|y)p(z|y)$,

$$p(x, z \mid y) = \frac{p(x, y, z)}{p(y)} = \frac{p(y)p(x|y)p(z|y)}{p(y)} = p(x|y)p(z|y).$$

△

Example 9.4 (Mediator, y unobserved)**.** Is $x \perp\!\!\!\perp z$ in the graph below?

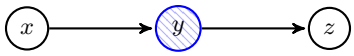


The answer: not in general since

$$p(x, z) = \sum_y p(x)p(y|x)p(z|y) = p(x) \sum_y p(y|x)p(z|y)$$

is not necessarily equal to $p(x)p(z)$. *Exercise:* Provide a counter example for $x \perp\!\!\!\perp z$. △

Example 9.5 (Mediator, y observed)**.** Is $x \perp\!\!\!\perp z \mid y$ in the graph below?

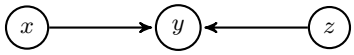


The answer: yes. We have

$$p(x, z \mid y) = \frac{p(x, y, z)}{p(y)} = \frac{p(x)p(y|x)p(z|y)}{p(y)} = \frac{p(y)p(x|y)p(z|y)}{p(y)} = p(x|y)p(z|y).$$

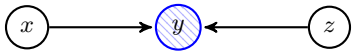
△

Example 9.6 (Collider, y unobserved)**.** Is $x \perp\!\!\!\perp z$ in the graph below?



Yes: $p(x, z) = \sum_y p(x, y, z) = \sum_y p(x)p(z)p(y \mid x, z) = p(x)p(z) \sum_y p(y \mid x, z) = p(x)p(z)$. △

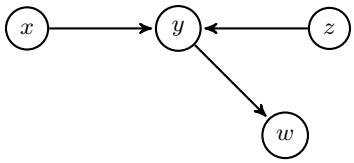
Example 9.7 (Explaining away)**.** Is $x \perp\!\!\!\perp z \mid y$ in the graph below?



Not in general. *Exercise:* Verify that for $x \sim \text{Ber}(\frac{1}{2}), z \sim \text{Ber}(\frac{1}{2})$ and $y = x + z$, $p(x, y, z)$ factorizes with respect to the graph above and $x \not\perp\!\!\!\perp z \mid y$. △

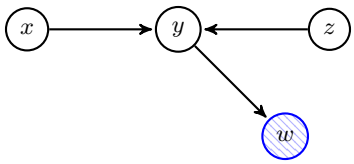
In graphs of Examples 9.6 and 9.7, if y has a descendant, that will also affect the independence relationship between x and z. These cases are considered next.

Example 9.8 (Collider, descendant unobserved)**.** Is $x \perp\!\!\!\perp z$ in the graph below?



Yes: $p(x, z) = \sum_{y,w} p(x, y, z, w) = \sum_{y,w} p(x)p(z)p(y \mid x, z)p(w|y) = p(x)p(z) \sum_{y,w} p(y \mid x, z)p(w|y) = p(x)p(z)$. △

Example 9.9 (Explaining away via descendant)**.** Is $x \perp\!\!\!\perp z \mid y$ in the graph below?

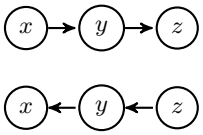
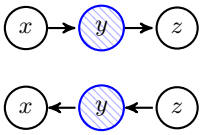
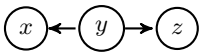
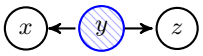
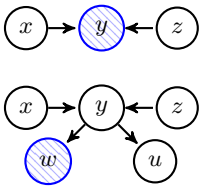
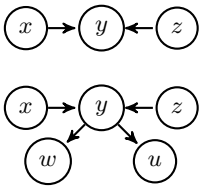


Not in general. Exercise: Verify that for $x \sim \text{Ber}(\frac{1}{2}), z \sim \text{Ber}(\frac{1}{2})$, $y = x+z$, and $w = y$, $p(x, y, z, w)$ factorizes with respect to the graph above and $x \not\perp\!\!\!\perp z \mid y$. △

9.2.2 d-separation

Based on our analysis in the previous section, we can summarize whether information flows from x to z in a graph of the form $x - y - z$ in Table 9.1. The table is organized by the direction of edges at y, with H (Head) representing an incoming edge and T (Tail) representing an outgoing edge. We can see that for the HT, TH, and TT configurations, y blocks the path from x to z if it is known (given) and for HH, it blocks the path if it is not known and neither are any of its descendants.

Table 9.1: Flow of information between x and z. Nodes with style y are knwon.

	Passing through	Blocked
HT/TH		
TT		
HH		

We can generalize this observation to decide, for three disjoint sets A, B, and C, of nodes, whether $A \perp\!\!\!\perp B \mid C$.

Definition 9.10 (Blocked path)**.** For a set C of known/observed nodes, a path is said to be blocked if it has a node v such that the nodes incident to v are:

- HT,TH, or TT and $v \in C$;
- HH, and neither v nor its descendants are in C.

Definition 9.11 (d-separated sets)**.** For disjoint sets A, B, and C, we say that A and B are **d-separated** given C if every path between a node in A and a node in B is blocked if we assume the nodes in C are known.

Theorem 9.12 (Soundness of d-separation)**.** *For three disjoint sets of nodes, A, B, and C, in a graph G, such that A and B are d-separated given C, then $A \perp\!\!\!\perp B \mid C$ according to any probability p that factorize with respect to G.*

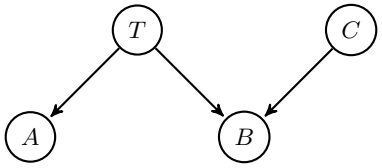
Remark ** The converse of the theorem also holds in the sense that any distribution p that satisfies all independencies implied by d-separation factorizes with respect to the graph.

Remark ** Could a distribution p that factorizes with respect to G satisfy independencies that are not implied by d-separation? Indeed, yes. The distribution $p = \prod_{i=1}^{n} p(x_i)$ factorizes with respect to any graph G and for any non-trivial G, p satisfies independencies that are not implied by d-separation in G. However, for any independency $A \perp\!\!\!\perp B \mid C$ not implied by d-separation, there is a probability distribution factorizing with respect to G for which $A \not\perp\!\!\!\perp B \mid C$.

Example 9.13 (Traffic graph independencies)**.** In the traffic graphic from last chapter, shown below, we want to find all independences of the form $x \perp\!\!\!\perp y$ and $x \perp\!\!\!\perp y \mid z$ for vertices x, y, z. For those that do not follow form d-separation, we write $x \not\perp\!\!\!\perp y$ and $x \not\perp\!\!\!\perp y \mid z$. We have

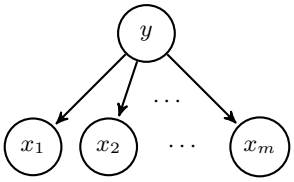
- No conditioning: $T \perp\!\!\!\perp C$, $T \not\perp\!\!\!\perp A$, $T \not\perp\!\!\!\perp B$, $C \perp\!\!\!\perp A$, $C \not\perp\!\!\!\perp B$, $A \not\perp\!\!\!\perp B$.
- Given T: $A \perp\!\!\!\perp B \mid T$, $A \perp\!\!\!\perp C \mid T$, $B \not\perp\!\!\!\perp C \mid T$.
- Given C: $A \not\perp\!\!\!\perp B \mid C$, $A \not\perp\!\!\!\perp T \mid C$, $B \not\perp\!\!\!\perp T \mid C$.
- Given A: $T \not\perp\!\!\!\perp B \mid A$, $T \perp\!\!\!\perp C \mid A$, $B \not\perp\!\!\!\perp C \mid A$.

- Given B: $T \not\perp\!\!\!\perp A \mid B$, $T \not\perp\!\!\!\perp C \mid B$, $A \not\perp\!\!\!\perp C \mid B$.



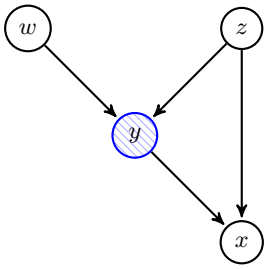
In addition, we have $A \perp\!\!\!\perp \{B, C\} \mid T$ but $\{T, A\} \not\perp\!\!\!\perp C \mid B$. △

Example 9.14 (**The Naive Bayes model**)**.** The graph for the naive Bayes classification model is



where y denotes the class and $x_1, \ldots, x_m$ denote the dimensions of the input vector. Given y the dimensions are independent, i.e., $x_i \perp\!\!\!\perp x_j \mid y$ for $i \neq j$. But if the class y is not known, generally speaking, $x_i \not\perp\!\!\!\perp x_j$. △

Example 9.15 (Four-node counterexample)**.** For four nodes w, x, y, and z, shown below, assume y is given. We can determine that none of the independencies $w \perp\!\!\!\perp z \mid y, x \perp\!\!\!\perp z \mid y, x \perp\!\!\!\perp w \mid y$ follow from d-separation. In fact, we can find a counter example, i.e., a distribution that factorizes with respect to the graph below and does not satisfy these independencies. Specifically, let $w \sim \text{Ber}(1/2)$, $z \sim \text{Ber}(1/2)$, $y = w + z$ and $x = y + z$. Note however that $y \perp\!\!\!\perp n \mid y$ for $n \in \{x, w, z\}$ by the definition of independence.



△

9.2.3 Markov Blanket in Bayesian Networks

In a graphical model, the **Markov blanket** of a node y is the set of nodes S such that $y \perp\!\!\!\perp U \mid S$ for any set U. In other words, the set S isolates y from the rest of the graph. In a Bayesian network, the Markov blanket of y consists of its parents, its children, and the immediate parents of its children. The proof of this statement is left as an exercise. An example is shown in Figure 9.1.

9.3 Independence in MRFs

The set of independencies implied by an MRF are more straightforward as separation is the naive graph-theoretic separation. As an example, consider the friendship graph of the previous chapter and assume we know the political affiliation of x_2,x_3.

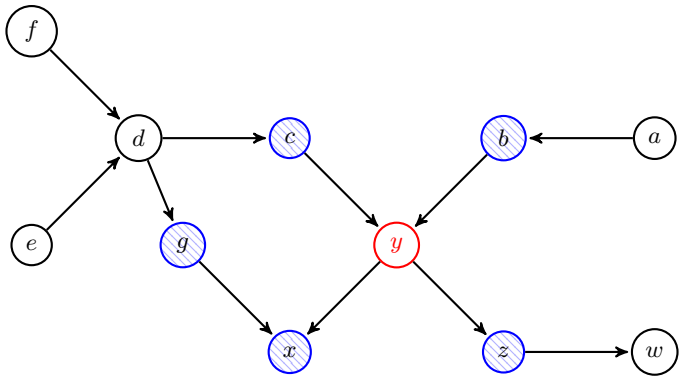


Figure 9.1: The Markov blanket of node y are the set of nodes colored red.

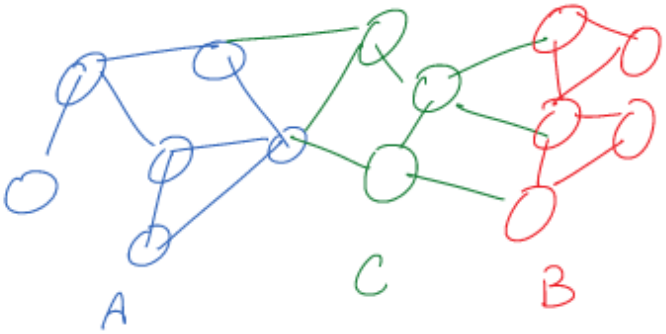
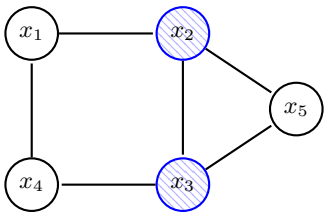


Figure 9.2: MRF Theorem



Then intuitively, we can expect that knowing x_5 does not provide any relevant information about x_1,x_4 and so we must have $x_1, x_4 \perp\!\!\!\perp x_5 \mid x_2, x_3$.

In an MRF G, suppose x_A, x_B, and x_C are disjoint subsets of vertices such that $x_A \cup x_B \cup x_C = G$, as shown in Figure 9.2. *If every path from x_A to x_B travels through x_C, then $x_A \perp\!\!\!\perp x_B \mid x_C$.* To see that this is the case, note that

$$\begin{aligned} p(x_A, x_B \mid x_C) &= P(x_A \mid x_C) P(x_B \mid x_C) \\ &= \frac{p(x_A, x_B, x_C)}{p(x_C)} \propto p(x_A, x_B, x_C) \\ &\propto \prod_{Q \text{ is a clique in } G} \psi_Q(x_Q) = \prod_{Q \in x_A \cup x_C} \psi(x_Q) \prod_{Q \in x_B \cup x_C} \psi(x_Q). \end{aligned}$$

The last equality follows from the fact that there is no clique in G that has a node in both x_A, and x_B since x_C separates x_A and x_B. The result follows from Exercise 9.1.

Examples are given in Figure 9.3.

The Markov Blanket of a node in an MRF is the set of neighbors as shown in Figure 9.4.

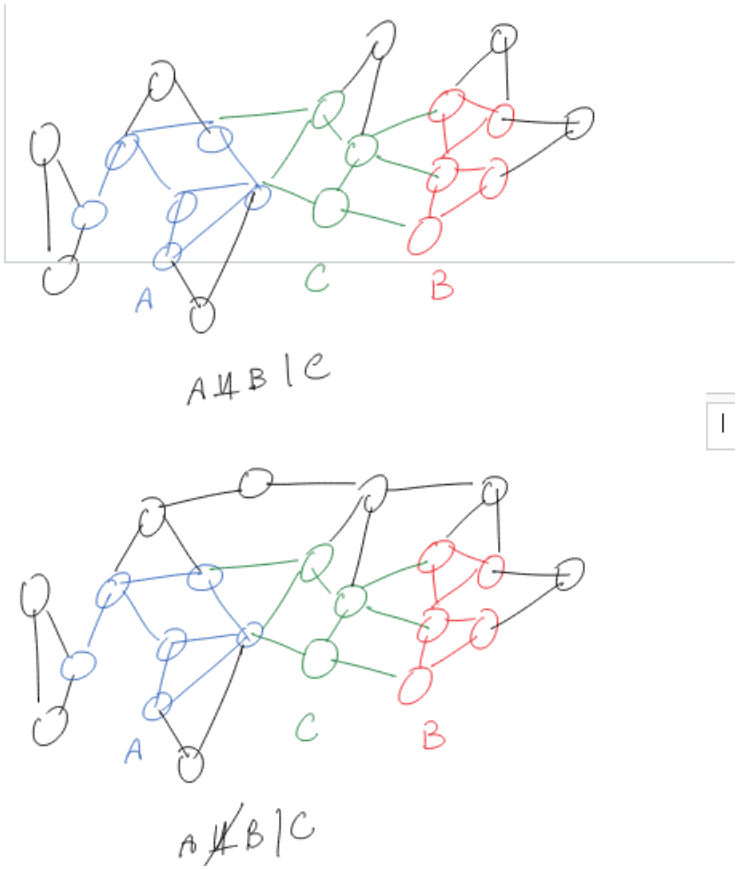


Figure 9.3: Two examples of MRFs

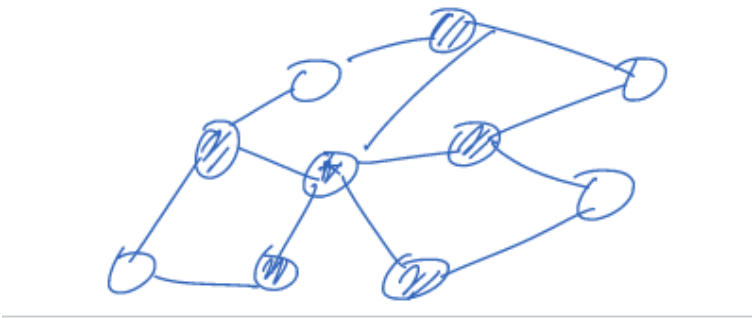


Figure 9.4: As example of a Markov Blanket

Chapter 10

Parameter Estimation in Graphical Models

A graphical model has two components: the graph structure (the nodes and their connections), and the conditional probability distributions/potential functions, which are usually expressed in parametric form. In this chapter:

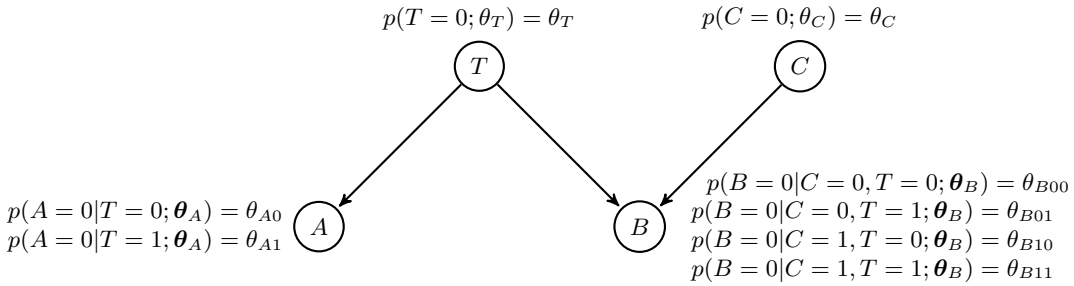
- We will consider the problem of estimating the parameters in graphical models. The problem is simpler in the case of Bayesian networks and for simplicity, that is were our attention will be focused.
- We will not consider the more challenging problem of learning the structure of a network. The best case scenario is that you have good reason to design a graph in a certain way, e.g., based on causality.

Consider a BN with a known graph with m nodes $x_1, \ldots, x_m$ in which the parameters of the conditional distribution are unknown. There are m conditional probability distributions (**CPD**s)[1], one for each node, and each of these has an unknown parameter vector. We denote the concatenated vector of all parameters as $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_m)$. We collect a dataset $\mathcal{D} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ of n iid samples, where $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{im})$. Our goal is to estimate $\boldsymbol{\theta}$ and possibly also to predict the next outcome $\boldsymbol{x}_{n+1} = (\boldsymbol{x}_{n+1,1}, \ldots, \boldsymbol{x}_{n+1,m})$.

10.1 MLE for Parameters of Bayesian Networks

We will start with maximum likelihood estimation via an example.

Example 10.1 (The TABC network)**.** Consider the network from previous chapters with the vector of parameters $\boldsymbol{\theta} = (\theta_T, \theta_C, \boldsymbol{\theta}_A, \boldsymbol{\theta}_B)$.



[1]Some of the nodes do not have any parents so their distribution is not conditioned on any other nodes. We view these as conditioned on the empty set and thus refer to all probability distributions in a Bayesian Network as *conditional* probability distributions.

To collect data, on n days, we record whether there is heavy traffic and whether Alice, Bob, and/or Charlie are late, resulting in $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$, where $\boldsymbol{x}_i = (T_i, A_i, B_i, C_i)$. Then we maximize the likelihood

$$\arg\max_{\boldsymbol{\theta}} p(\mathcal{D}; \boldsymbol{\theta}) = \arg \max_{\theta_T, \boldsymbol{\theta}_A, \boldsymbol{\theta}_B, \theta_C} p(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n; (\theta_T, \boldsymbol{\theta}_A, \boldsymbol{\theta}_B, \theta_C)) \tag{10.1}$$

△

Note that in the above example, the maximization evidently involve 6 dimensions. In real-world problems the networks have many more parameters. This would create computational difficulties since it would require maximizing a function of many variables. Fortunately, in the case of Bayesian networks, the problem decomposes to estimating the parameters for each nodes separately, as we will see.

Decomposability of likelihood. For a network with m nodes, parameters $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_m)$ and data $\mathcal{D} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$, the likelihood function is

$$p(\mathcal{D}; \boldsymbol{\theta}) = \prod_{i=1}^{n} p(\boldsymbol{x}_i; \boldsymbol{\theta}),$$

where for the ith data sample, we have

$$p(\boldsymbol{x}_i; \boldsymbol{\theta}) = \prod_{j=1}^{m} p(x_{ij} | \operatorname{pa}(x_{ij}); \boldsymbol{\theta}_j)$$

and thus the log-likelihood of the whole dataset is

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \ln p(\boldsymbol{x}_i; \boldsymbol{\theta}) = \sum_{i=1}^{n} \sum_{j=1}^{m} \ln p(x_{ij} | \operatorname{pa}(x_{ij}); \boldsymbol{\theta}_j) = \sum_{j=1}^{m} \sum_{i=1}^{n} \ln p(x_{ij} | \operatorname{pa}(x_{ij}); \boldsymbol{\theta}_j).$$

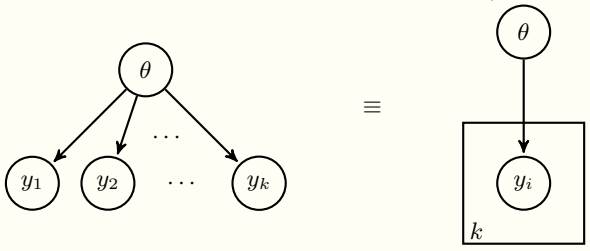
Thus for a given j, $\boldsymbol{\theta}_j$ only appears in the term $\sum_{i=1}^{n} \ln p(x_{ij} | \operatorname{pa}(x_{ij}); \boldsymbol{\theta}_j)$ and no other $\boldsymbol{\theta}_k$ appears in this term. So each $\boldsymbol{\theta}_j$, and thus each conditional probability distribution, can be learned independently of the others, which *significantly* reduces the complexity.

Exercise 10.2 (TABC data and MLE)**.** For the TABC network above, what would our data look like? What is the ML estimate for each parameter based on this data? △

10.2 Bayesian Parameter Estimation for Bayesian Networks

An alternative approach is using Bayesian inference. Since in the Bayesian view, parameters are considered random, we can augment the Bayesian network by adding the parameters as nodes.

Side note 1: the plate notation. Before proceeding, we introduce the plate notation which is helpful for simplifying repeated elements in graphical models, especially a set of iid nodes. Specifically, instead of repeating a node k times, we enclose one instance and indicate how many times that segment of the graph is repeated. Both of the following graphs represent the factorization $p(y_1, \ldots, y_k, \theta) = \prod_{i=1}^{k} p(y_i|\theta)$.



Side note 2: Conditioning for sets of nodes. Consider a BN with nodes $y_1, \ldots, y_m$. Assume that the set of nodes can be partitioned into two sets $S_1 = \{y_1, \ldots, y_r\}$ and $S_2 = \{y_{r+1}, \ldots, y_m\}$ such that there are no edges from S_2 to S_1. Then the following hold

$$p(S_1) = p(y_1, \ldots, y_r) = \prod_{i=1}^{r} p(y_i | \text{ pa}(y_i)), \tag{10.2}$$

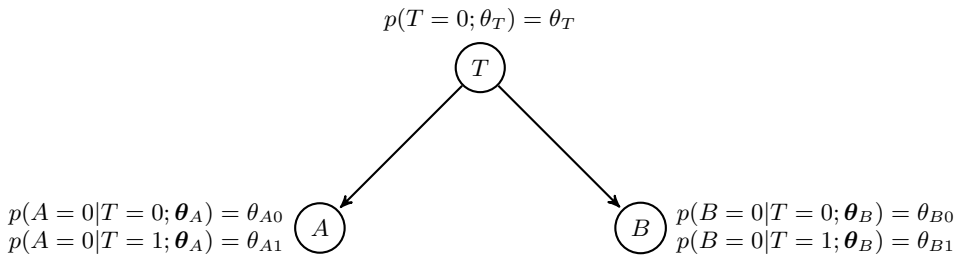
$$p(S_2|S_1) = p(y_{r+1}, \ldots, y_m | y_1, \ldots, y_r) = \prod_{i=r+1}^{m} p(y_i | \text{ pa}(y_i)). \tag{10.3}$$

However, in general,

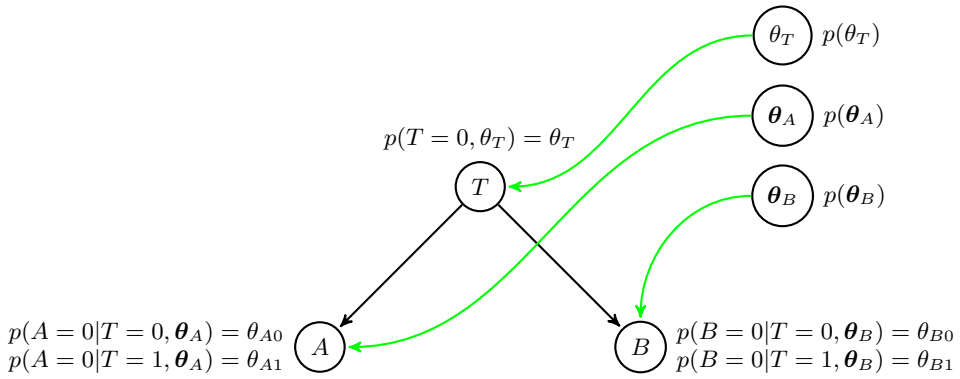
$$p(S_2) = p(y_{r+1}, \ldots, y_m) \neq \prod_{i=r+1}^{m} p(y_i | \text{ pa}(y_i)), \tag{10.4}$$

$$p(S_1|S_2) = p(y_1, \ldots, y_r | y_{r+1}, \ldots, y_m) \neq \prod_{i=1}^{r} p(y_i | \text{ pa}(y_i)) \tag{10.5}$$

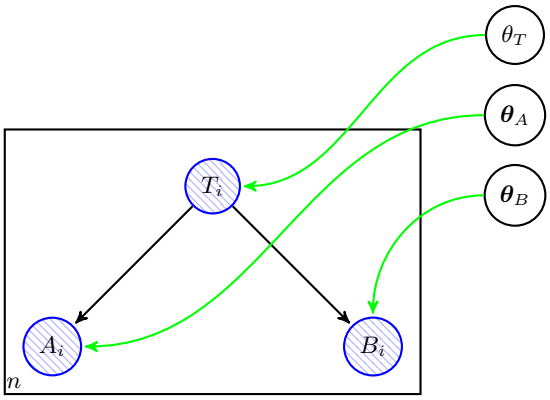
Bayesian networks with explicit representation of the parameters and data. Let us consider a simpler version of the network given in Example 10.1, with unknown parameter vector $\boldsymbol{\theta} = (\theta_T, \boldsymbol{\theta}_A, \boldsymbol{\theta}_B)$:



At this point, we are still viewing the parameters as unknown constants. Now to formulate the Bayesian estimation of these parameters, we need to view them as random and add nodes for them to the graph, estimating the parameters as the network:



Incorporating the n samples $\mathcal{D} = \{(T_1, A_1, B_1), \ldots, (T_n, A_n, B_n)\}$, we represent the problem of estimating the parameters as the network, where the CPDs are omitted for clarity:



Posterior distributions of the parameters. We are interested in finding $p(\boldsymbol{\theta}|\mathcal{D}) = p(\theta_T, \boldsymbol{\theta}_A, \boldsymbol{\theta}_B|\mathcal{D})$. Note that

$$\begin{aligned} p(\boldsymbol{\theta}|\mathcal{D}) &= p(\theta_T|\mathcal{D})p(\boldsymbol{\theta}_A, \boldsymbol{\theta}_B|\mathcal{D}) && (10.6)\\ &= p(\theta_T|\mathcal{D})p(\boldsymbol{\theta}_A|\mathcal{D})p(\boldsymbol{\theta}_B|\mathcal{D}) && (10.7)\\ &= p(\theta_T|T_1^n)p(\boldsymbol{\theta}_A|T_1^n, A_1^n)p(\boldsymbol{\theta}_B|T_1^n, B_1^n), && (10.8) \end{aligned}$$

where the first equality follows from the fact that given $T_1^i \subset \mathcal{D}$, θ_T is independent of all other nodes, including $\boldsymbol{\theta}_A, \boldsymbol{\theta}_B$. In other words, T_1^i is the Markov blanket of θ_T. The second equality also holds because $\mathcal{D}$ contains the Markov blanket for $\boldsymbol{\theta}_A$ and $\boldsymbol{\theta}_B$. Similarly, the last equality follows from a Markov blanket-type argument.

In other words, to estimate each parameter, we need to only consider the part of the data that is in the parameter's Markov blanket. This also makes intuitive sense: For example, to estimate the probability of Alice being late given the state of traffic, only the part of data that deals with Alice's arrival time and traffic is relevant. The fact that the posterior for each parameter can be determined separately significantly reduces the computational complexity.

Example 10.3 (Alice's lateness posterior)**.** Let us find $p(\boldsymbol{\theta}_A|\mathcal{D})$, assuming that the prior satisfies $p(\boldsymbol{\theta}_A) = p(\theta_{A0})p(\theta_{A1})$,

$$\begin{aligned} p(\boldsymbol{\theta}_A|\mathcal{D}) = p(\boldsymbol{\theta}_A|T_1^n, A_1^n) &\propto p(\boldsymbol{\theta}_A)p(T_1^n, A_1^n|\boldsymbol{\theta}_A) \overset{(*)}{\propto} p(\boldsymbol{\theta}_A)\prod_{i=1}^n p(A_i|T_i, \boldsymbol{\theta}_A)\\ &= \left(p(\theta_{A0})\prod_{i:T_i=0} p(A_i|T_i=0, \theta_{A0})\right)\left(p(\theta_{A1})\prod_{i:T_i=1} p(A_i|T_i=1, \theta_{A1})\right). \end{aligned}$$

(Why does the relation shown as $\overset{(*)}{\propto}$ hold?) Since the terms depending on θ_{A0} and θ_{A1} separate, they are conditionally independent and we can estimate them separately: Hence, the estimators of θ_A^0 and θ_A^1 are

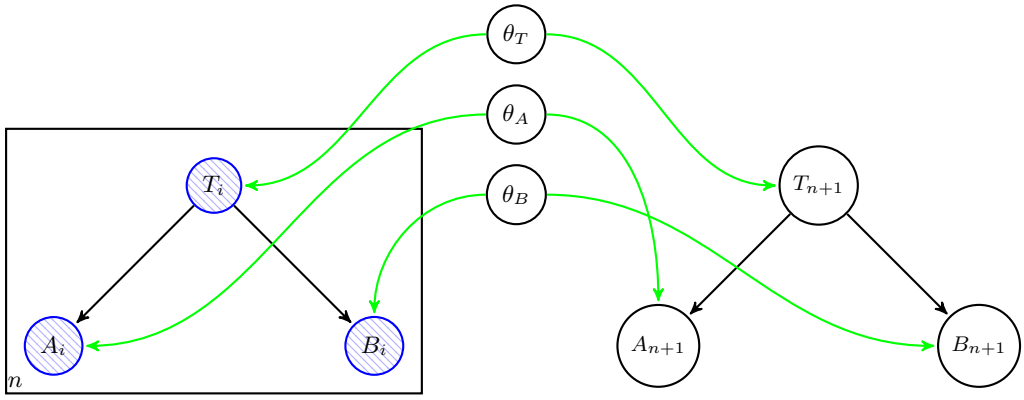
$$\begin{aligned} p(\theta_{A0}|\mathcal{D}) &\propto p(\theta_{A0})\prod_{i:T_i=0} p(A_i|T_i=0, \theta_{A0}),\\ p(\theta_{A1}|\mathcal{D}) &\propto p(\theta_{A1})\prod_{i:T_i=1} p(A_i|T_i=1, \theta_{A1}). \end{aligned}$$

Suppose $p(\theta_A^0) \sim \text{Beta}(1,1)$ and out of 100 days with no traffic, in 40 days Alice was on time. Hence,

$$\theta_{A0}|\mathcal{D} \sim \text{Beta}(41, 61).$$

△

Predicting future outcomes. We can also add future outcomes to predict their value to the network:



Let $\boldsymbol{x}_{n+1} = (T_{n+1}, A_{n+1}, B_{n+1})$. We have

$$p(\boldsymbol{x}_{n+1}, \boldsymbol{\theta}|\mathcal{D}) = p(\boldsymbol{\theta}|\mathcal{D})p(\boldsymbol{x}_{n+1}|\mathcal{D}, \boldsymbol{\theta}) = p(\boldsymbol{\theta}|\mathcal{D})p(\boldsymbol{x}_{n+1}|\boldsymbol{\theta}). \tag{10.9}$$

We have already seen how to find $p(\boldsymbol{\theta}|\mathcal{D})$. We can decompose $p(\boldsymbol{x}_{n+1}|\boldsymbol{\theta})$ as given by the Bayesian network:

$$p(\boldsymbol{x}_{n+1}|\boldsymbol{\theta}) = p(T_{n+1}|\theta_T)p(A_{n+1}|\boldsymbol{\theta}_A, T_{n+1})p(B_{n+1}|\boldsymbol{\theta}_B, T_{n+1}). \tag{10.10}$$

Note that the terms on the right are known probability distributions.

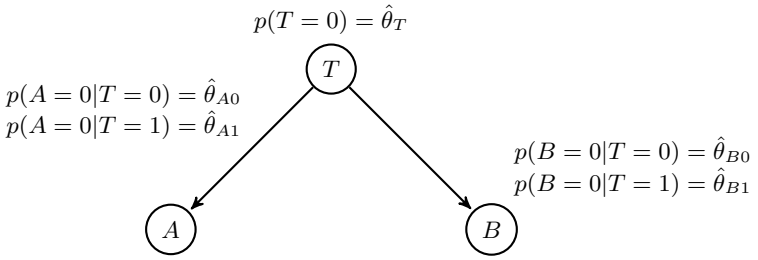
Finally, if we are interested in a specific future outcome, e.g., $p(A_{n+1}|\mathcal{D})$, we can find it through an appropriate integration/summation of $p(\boldsymbol{x}_{n+1}, \boldsymbol{\theta}|\mathcal{D})$.

Example 10.4 (Predicting the next sample)**.** The posterior probability of the next sample (A_{n+1}, B_{n+1}) is

$$p(A_{n+1}, B_{n+1}|\mathcal{D}) = \int_{\boldsymbol{\theta}} \sum_{T_{n+1}} p(A_{n+1}, B_{n+1}, T_{n+1}, \boldsymbol{\theta}|\mathcal{D}) d\boldsymbol{\theta},$$

where we can find the integrand/summand as described in (10.9). In general, such integrals may be difficult to find analytically. In practice, we rely on computational methods such as Markov Chain Monte Carlo (MCMC).

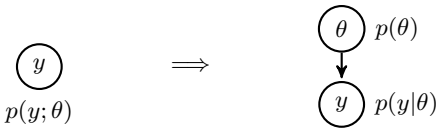
Alternatively, to predict future values, we can use a Bayesian point estimate for $\boldsymbol{\theta}$, and then assume that they are known as shown below.



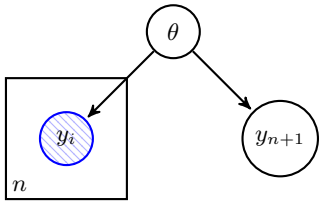
△

We can use graphical models to represent some of the estimation/learning problems we have already discussed in previous chapters.

Example 10.5 (Adding a parameter node)**.** Let y have a distribution p with an unknown parameter θ. Below, on the left a graphical model for y is shown without explicit representation of θ and on the right, θ is added as a node:



We have n independent samples, $\mathcal{D} = \{y_1, y_2, \ldots, y_n\}$, from the distribution and our goal is to predict the next outcome y_{n+1}. We can augment the graph to represent the problem as follows:



with a joind distribution that can be written as $p(\theta, y_1^n, y_{n+1}) = p(\theta)p(y_1^n|\theta)p(y_{n+1}|\theta)$.

We can perform similar analysis as we have done in the Bayesian Estimation chapter, using d-separation to verify independence relations. We have

$$p(y_{n+1}|y_1^n) = \int p(y_{n+1}, \theta|y_1^n)d\theta = \int p(\theta|y_1^n)p(y_{n+1}|\theta, y_1^n)d\theta = \int p(\theta|y_1^n)p(y_{n+1}|\theta)d\theta$$

where in the last step we have used $y_{n+1} \perp\!\!\!\perp y_1^n \mid \theta$, which follows from d-separation. Furthermore,

$$\mathbb{E}[y_{n+1}|y_1^n] = \mathbb{E}[\mathbb{E}[y_{n+1}|\theta, y_1^n]|y_1^n] = \mathbb{E}[\mathbb{E}[y_{n+1}|\theta]|y_1^n]. \tag{10.11}$$

Roughly speaking, to learn about y_{n+1} given y_1^n, we must first learn about θ since this is the node that connects y_1^n and y_{n+1}.

For example, assume $p(\theta) \propto 1$, $y_i|\theta \sim \text{Ber}(\theta)$, and that out of the n samples y_i, there s 1s and f 0s. Then

$$p(y_{n+1} = 1|y_1^n) = \mathbb{E}[y_{n+1}|y_1^n] = \mathbb{E}[\mathbb{E}[y_{n+1}|\theta]|y_1^n] = \mathbb{E}[\theta|y_1^n] = \frac{s+1}{s+f+2}.$$

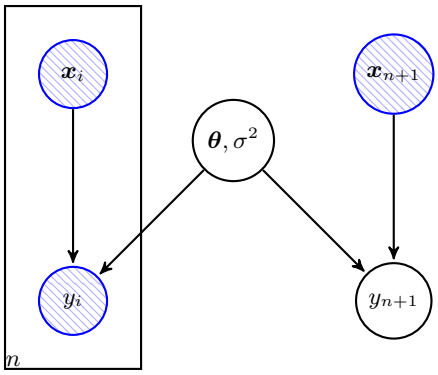
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In more general cases, some of the "future outcomes" may also be known. But the same principles discussed above, still apply.

Example 10.6 (Bayesian Linear Regression)**.** Consider the regression problem

$$p(y_i|\boldsymbol{x}_i, \boldsymbol{\theta}, \sigma^2) \sim \mathcal{N}(\boldsymbol{\theta}^T\boldsymbol{x}_i, \sigma^2),$$

with data $\mathcal{D} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)\}$. We are interested in determining $p(y_{n+1}|\boldsymbol{x}_{n+1}, \mathcal{D})$. The problem can be represented as the graph



We note that

$$p(y_{n+1}, \boldsymbol{\theta}, \sigma^2 | \boldsymbol{x}_{n+1}, \mathcal{D}) = p(\boldsymbol{\theta}, \sigma^2 | \boldsymbol{x}_{n+1}, \mathcal{D}) p(y_{n+1} | \boldsymbol{\theta}, \sigma^2, \boldsymbol{x}_{n+1}, \mathcal{D}) \tag{10.12}$$

$$= p(\boldsymbol{\theta}, \sigma^2 | \mathcal{D}) p(y_{n+1} | \boldsymbol{x}_{n+1}, \boldsymbol{\theta}, \sigma^2), \tag{10.13}$$

where we have used the following facts: $\boldsymbol{\theta}, \sigma^2 \perp\!\!\!\perp \boldsymbol{x}_{n+1} | \mathcal{D}$ and $y_{n+1} \perp\!\!\!\perp \mathcal{D} | \boldsymbol{x}_{n+1}, \boldsymbol{\theta}, \sigma^2$. We know how to find $p(\boldsymbol{\theta}, \sigma^2 | \mathcal{D})$ and $p(y_{n+1} | \boldsymbol{x}_{n+1}, \boldsymbol{\theta}, \sigma^2)$ is given by assumption. While we can find $p(y_{n+1} | \boldsymbol{x}_{n+1})$ through integration analytically, as discussed in the linear regression chapter, we normally produce samples for $\boldsymbol{\theta}, \sigma^2$ and then proceed to produce samples for $p(y_{n+1} | \boldsymbol{x}_{n+1}, \boldsymbol{\theta}, \sigma^2)$.

△

10.3 Parameter Estimation in MRFs

Recall that for an MRF G, the probability distribution is given as

$$p(\boldsymbol{x}; \boldsymbol{\theta}) = \prod_{c \text{ is a clique in } G} \psi_{\boldsymbol{\theta}}(\boldsymbol{x}_c) / Z(\boldsymbol{\theta}),$$

where $Z(\boldsymbol{\theta}) = \sum_{\boldsymbol{x}} \prod_c \psi_{\boldsymbol{\theta}}(\boldsymbol{x}_c)$ is the partition function. Let us consider the frequentist estimation of $\boldsymbol{\theta}$, e.g., maximum likelihood. Unfortunately, the log-likelihood function does not decompose into terms each depending on one component of $\boldsymbol{\theta}$. This is due to the presence of the partition function, which generally depends on all the components of $\boldsymbol{\theta}$, leading to a high-dimensional problem. Furthermore, computing the partition function is a computationally difficult task since it involves computing a sum with possibly exponentially many terms. We will discuss computational approaches to this problem later in the course.

Helpful references: [2, 3, 1]

References

[1] Christopher M. Bishop. *Pattern Recognition And Machine Learning*. New York: Springer, 2006. URL: http://users.isr.ist.utl.pt/~wurmd/Livros/school/Bishop%5C%20-%5C%20Pattern%5C%20Recognition%5C%20And%5C%20Machine%5C%20Learning%5C%20-%5C%20Springer%5C%20%5C%202006.pdf (visited on 02/14/2017).

[2] Michael I Jordan. *An Introduction to Probabilistic Graphical Models (Preprints and Course Notes)*. University of California, Berkeley, 2003.

[3] David JC MacKay. *Information Theory, Inference and Learning Algorithms*. Cambridge university press, 2003.

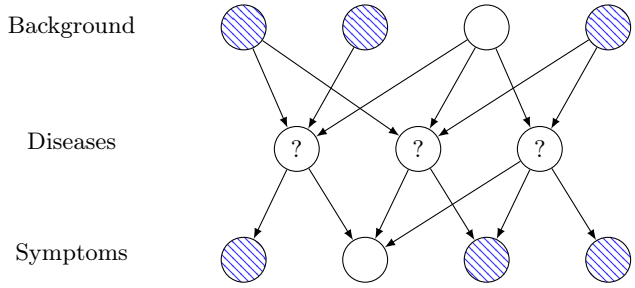
Chapter 11

Inference in Graphical Models

11.1 Introduction

Inference refers to drawing conclusions about unknown quantities based on observations and a model. In the context of graphical models assume, our goal is to learn about a set of query nodes given observed nodes.

For example, consider the following graph with nodes for background information about a patient (e.g., diet, exercise, genetics, etc.), diseases (diabetes, hypertension, etc.), and symptoms/test results (blood pressure, etc). Our goal is assign probabilities to disease based on our observations. Alternatively, we may be interested in identifying the disease that is most likely.



In such a graph, we deal with three types of nodes, evidence (observed) nodes, x_E, query nodes, x_Q, and other nodes, x_O.

Without having made any observations, we can find the probability of the query nodes through *marginalization*:

$$p(x_Q) = \sum_{x_O, x_E} p(x_Q, x_O, x_E),$$

and with observations, through *conditioning*:

$$p(x_Q|x_E) \propto \sum_{x_O} p(x_Q, x_O, x_E).$$

Since we can view the latter case as doing summation over x_E that only consists of a single set of values, from this point on, we will only consider marginalization. Note that we need to compute $\sum_{x_O} p(x_Q, x_O, x_E)$ for all values of x_Q to be able to find $p(x_Q|x_E)$.

11.2 The Elimination Algorithm

Suppose that in a Markov chain $x_1 \to x_2 \to x_3 \to x_4 \to x_5$, we need to find $p(x_4)$,

$$p(x_4) = \sum_{x_1,x_2,x_3,x_5} p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3)p(x_5|x_4).$$

Assume each node can take k different values. In the naive approach, we need to compute and add $O(k^4)$ terms, and we need to do so for each possible value of x_4. So finding the distribution of x_4 has complexity $O(k^5)$.

Alternatively, we could eliminate each variable, which can be done in different orders. The equalities below represent computation performed by an algorithm:

$$\begin{aligned}
p(x_4) &= \sum_{x_1}\sum_{x_2}\sum_{x_3}\sum_{x_5} p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3)p(x_5|x_4) \\
&= \sum_{x_1}\sum_{x_2}\sum_{x_3} p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3)\sum_{x_5} p(x_5|x_4) \\
&= \sum_{x_1}\sum_{x_2}\sum_{x_3} p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3) \\
&= \sum_{x_1}\sum_{x_2} p(x_1)p(x_2|x_1)\sum_{x_3} p(x_3|x_2)p(x_4|x_3) \\
&= \sum_{x_1}\sum_{x_2} p(x_1)p(x_2|x_1)M_1(x_2,x_4) \\
&= \sum_{x_1} p(x_1)\sum_{x_2} p(x_2|x_1)M_1(x_2,x_4) \\
&= \sum_{x_1} p(x_1)M_2(x_1,x_4) \\
&= p(x_4)
\end{aligned}$$

The function $M_1(x_2, x_4)$ is *defined* as the result of the sum $\sum_{x_3} p(x_3|x_2)p(x_4|x_3)$, and a similar statement holds for M_2. We can think of $M_1(x_2, x_4)$ as a table stored in computer memory after it is computed. Computing $M_1(x_2, x_4)$ needs to be done for k different values of x_2 and each of these requires computing and adding k terms, one for each possible value of x_3. The computational complexity for a specific value of x_4 is $O(k^2)$, i.e., we need of the order of k^2 computations. The total computational complexity of finding the distribution $p(x_4)$ is $O(k^3)$ since we need to repeat all operations for the k different values that x_4 can take. More generally, for a Markov chain with n nodes, the complexity is $O(nk^3)$ for computing the distribution $p(x_n)$. But with the naive approach it is $O(k^n)$.

Note that in Bayesian networks, we can ignore downstream nodes since their probability marginalizes to 1 (but not in MRFs).

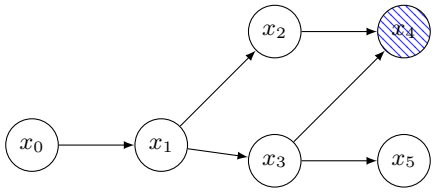
We could also choose the following ordering, which would lead to a different complexity:

$$\begin{aligned}
p(x_4) &= \sum_{x_1}\sum_{x_3}\sum_{x_2} p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3) \\
&= \sum_{x_1}\sum_{x_3} p(x_1)p(x_4|x_3)\sum_{x_2} p(x_2|x_1)p(x_3|x_2) \\
&= \sum_{x_1}\sum_{x_3} p(x_1)p(x_4|x_3)T_1(x_1,x_3) \\
&= \sum_{x_1} p(x_1)\sum_{x_3} p(x_4|x_3)T_1(x_1,x_3) \\
&= \sum_{x_1} p(x_1)T_2(x_1,x_4) \\
&= p(x_4).
\end{aligned}$$

Here, computing $T_1(x_1, x_3)$ has complexity $O(k^3)$, which is also the complexity for one value of x_4. For the distribution, the complexity is $O(k^4)$ for this ordering.

The problem of finding the best ordering for elimination is NP-hard (i.e., computationally difficult) for general graphs.

Now let us find $p(x_0|x_4)$ in the following network:



We have

$$\begin{aligned}
p(x_0|x_4) &\propto \sum_{x_1,x_2,x_3} p(x_0)p(x_1|x_0)p(x_2|x_1)p(x_3|x_1)p(x_4|x_2,x_3) \\
&= \sum_{x_1,x_3} p(x_0)p(x_1|x_0)p(x_3|x_1)\sum_{x_2} p(x_2|x_1)p(x_4|x_2,x_3) \\
&= \sum_{x_1,x_3} p(x_0)p(x_1|x_0)p(x_3|x_1)K_1(x_1,x_3,x_4) \\
&= \sum_{x_1} p(x_0)p(x_1|x_0)\sum_{x_3} p(x_3|x_1)K_1(x_1,x_3,x_4) \\
&= \sum_{x_1} p(x_0)p(x_1|x_0)K_2(x_1,x_4) \\
&= p(x_0)\sum_{x_1} p(x_1|x_0)K_2(x_1,x_4) \\
&= p(x_0)K_3(x_0,x_4).
\end{aligned}$$

The complexity is dominated by $K_1(x_1, x_3, x_4)$, which is $O(k^3)$, assuming each node can take on k values, leading to a total complexity of $O(k^4)$ for the conditional distribution of x_0.

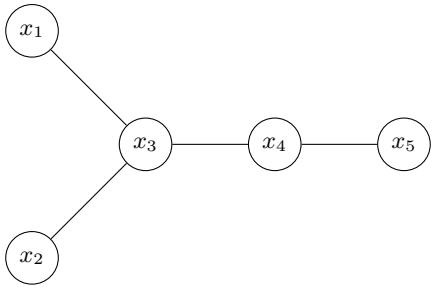
11.3 The Sum-Product Algorithm

The *sum-product algorithm*, also known as *belief propagation* and *sum-product message passing*, provides a simple way of doing exact inference on trees. It is also commonly used on graphs that are not trees since it often provides good approximations.

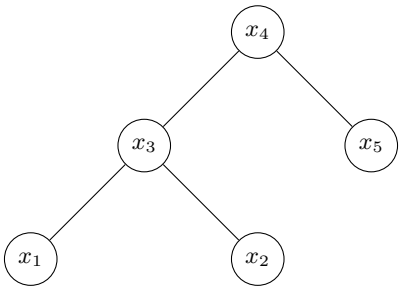
We need to clarify what we mean by trees. For Markov random fields, the algorithm works on trees, in the graph-theoretic sense. But for Bayesian networks, it works for graphs whose equivalent MRF (the moralized graph) is a tree. In particular, no node can have more than one parent. Given the straightforward equivalence between these two categories, we only consider Markov random field trees.

Consider the the following MRF, where we are interested in $p(x_4)$, with

$$p(x_1^4) \propto \psi(x_1, x_3)\psi(x_2, x_3)\psi(x_2)\psi(x_3, x_4)\psi(x_4)\psi(x_4, x_5)$$



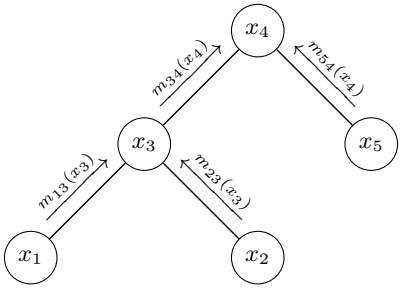
Let's look at this graph as a rooted tree,



and perform elimination starting from the leaves to the roots:

$$\begin{aligned}
p(x_4) &\propto \sum_{x_1,x_2,x_3,x_5} \psi(x_1, x_3)\psi(x_2, x_3)\psi(x_2)\psi(x_3, x_4)\psi(x_4)\psi(x_4, x_5) \\
&= \sum_{x_3} \psi(x_3, x_4)\psi(x_4)\left(\sum_{x_1} \psi(x_1, x_3)\right)\left(\sum_{x_2} \psi(x_2, x_3)\psi(x_2)\right)\left(\sum_{x_5} \psi(x_4, x_5)\right) \\
&= \sum_{x_3} \psi(x_3, x_4)\psi(x_4)\ m_{13}(x_3)\ m_{23}(x_3)\ m_{54}(x_4) \\
&= \psi(x_4)m_{54}(x_4)\sum_{x_3} \psi(x_3, x_4)\ m_{13}(x_3)\ m_{23}(x_3) \\
&= \psi(x_4)m_{54}(x_4)m_{34}(x_4)
\end{aligned} \tag{11.1}$$

We can view this computation as being done on each node and then messages being passed to neighbors:



where

$$m_{13}(x_3) = \sum_{x_1} \psi(x_1, x_3),$$
$$m_{23}(x_4) = \sum_{x_2} \psi(x_2, x_3)\psi(x_2),$$
$$m_{54}(x_4) = \sum_{x_5} \psi(x_4, x_5),$$
$$m_{34}(x_4) = \sum_{x_3} \psi(x_3, x_4)\ m_{13}(x_3)\ m_{23}(x_3).$$

and then at the root, we can find $p(x_4)$ as

$$p(x_4) \propto \psi(x_4)m_{54}(x_4)m_{34}(x_4).$$

Recall that this also works for conditioning. Specifically, if we are interested in the conditional probability $p(x_4|x_3 = a)$, we would compute

$$m_{34}(x_4) = \psi(x_3 = a, x_4)\ m_{13}(a)\ m_{23}(a),$$
$$p(x_4) \propto \psi(x_4)m_{54}(x_4)m_{34}(x_4).$$

We can state the sum-product algorithm for a rooted tree as follows. At each node x_j with parent x_k,

- Product step: After receiving messages $m_{ij}(x_j)$ from all children x_i of x_j, compute the product of all messages and any potential functions containing x_j,

  $$\psi(x_j)\psi(x_j, x_k)\prod_i m_{ij}(x_j).$$

  Note that not all potentials are always present. Do this for each possible pair of values for (x_j, x_k).

- Sum step: Sum over all possible values of x_j to produce the message

  $$m_{jk}(x_k) = \sum_{x_j} \psi(x_j)\psi(x_j, x_k)\prod_i m_{ij}(x_j), \tag{11.2}$$

  and send to x_k. Do this for each possible value for x_k.

A critical point in the correctness of the sum-product algorithm is that the messages received by each node are functions of the value of that node. This is easy to see by induction. After the product step, we get a function of both the current node x_j and its parent x_k. The sum eliminates the current node and so the parent node x_k receives a message that is only a function of x_k.

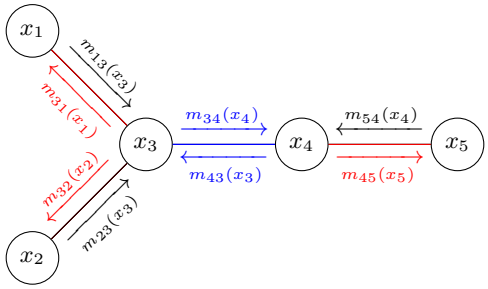
Complexity of computing each message: Suppose each node can take on K different values, namely $\{1, 2, \ldots, K\}$. So the sum in (11.2) contains K terms. Furthermore, $m_{jk}(x_k)$ needs to be computed for $x_k = 1, 2, \ldots, K$. We can imagine a vector

$$\boldsymbol{m}_{jk} = \begin{pmatrix} m_{jk}(1) \\ m_{jk}(2) \\ \vdots \\ m_{jk}(K) \end{pmatrix}$$

being sent to the node x_k. So the complexity at each node is $O(K^2)$ and for n nodes the complexity is $O(nK^2)$.

Computing marginals at all nodes. We can easily extend this algorithm to computing all marginals rather than a single node. We note that the messages sent by the nodes do not depend on the location of the

root. Each node sends a message when it receives messages from all but one of its neighbors. We can extend this by not sending a message only once, but sending a message to each neighbor based on the messages received by the other neighbors:



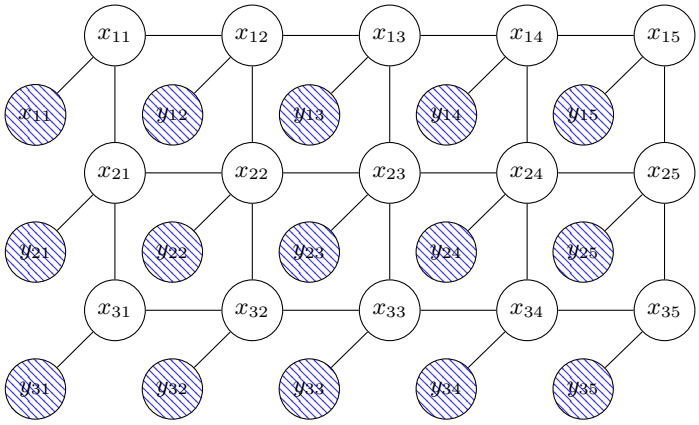
Here the order of messages is color-coded: 1, 2, 3. We can now find the marginal at each node. For example,

$$p(x_2) \propto m_{32}(x_2)\psi(x_2),$$
$$p(x_3) \propto m_{13}(x_3)m_{23}(x_3)m_{43}(x_3).$$

Example 11.1 (Sum-product example)**.** An example for the sum-product algorithm is given at the end of the document. △

11.4 The Max-Product Algorithm

The max-product algorithm is used to identify the configuration that has the maximum probability. Examples include part-of-speech tagging, voice recognition, decoding (communication), and image denoising. The last example is shown below:



where x_{ij} are true image pixels and y_{ij} are observed pixels, e.g., from a camera. Our goal is to find

$$\arg\max_{\boldsymbol{x}} p(\boldsymbol{x}, \boldsymbol{y}).$$

Note that the local maximum-probability configuration does not necessarily agree with the global maximum-probability configuration. As an example, consider

$p(x_1, x_2)$	$x_1 = 0$	$x_1 = 1$
$x_2 = 0$	.3	.4
$x_2 = 1$	.3	0

We have

$$\arg\max_{x_1,x_2} p(x_1,x_2) = (1,0)$$
$$\arg\max_{x_1} p(x_1) = \arg\max_{x_1}(p(x_1,x_2=0)+p(x_1,x_2=1)) = 0.$$

To see the max-product algorithm, suppose we want to find

$$\arg\max_{x_1^5} p(x_1^5)$$

for the tree given in the previous section. To solve this problem, let us start with solving

$$\max_{x_1^5} p(x_1^5)$$

We proceed similar to (11.1). For clarity, we make the partition function Z explicit, but we don't actually need to find it. We replace each summation in the previous derivation with max and write:

$$\begin{aligned}
\max p(x_1^5) &= \max_{x_1,x_2,x_3,x_4,x_5} Z\psi(x_1,x_3)\psi(x_2,x_3)\psi(x_2)\psi(x_3,x_4)\psi(x_4)\psi(x_4,x_5) \\
&= Z \max_{x_4}\max_{x_3} \psi(x_3,x_4)\psi(x_4)\left(\max_{x_1}\psi(x_1,x_3)\right)\left(\max_{x_2}\psi(x_2,x_3)\psi(x_2)\right)\left(\max_{x_5}\psi(x_4,x_5)\right) \\
&= Z \max_{x_4}\max_{x_3} \psi(x_3,x_4)\psi(x_4)\ m_{13}(x_3)\ m_{23}(x_3)\ m_{54}(x_4) \\
&= Z \max_{x_4} \psi(x_4) m_{54}(x_4) \max_{x_3} \psi(x_3,x_4)\ m_{13}(x_3)\ m_{23}(x_3) \\
&= Z \max_{x_4} \psi(x_4) m_{54}(x_4) m_{34}(x_4)
\end{aligned}$$

This is the same as the sum-product algorithm, except that we take the max of product terms. We can again view this as message-passing, but using max instead of sum, with the following messages:

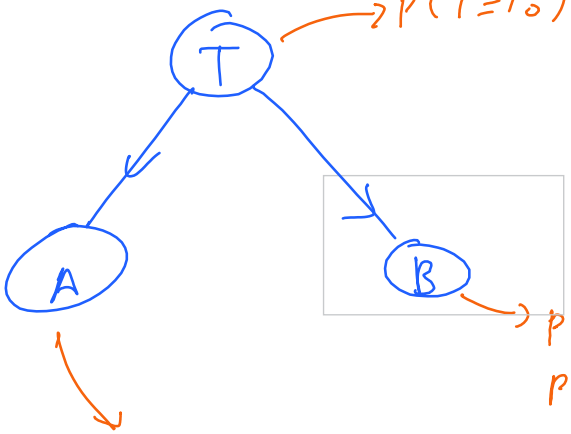
$$\begin{aligned}
m_{13}(x_3) &= \max_{x_1} \psi(x_1,x_3), \\
m_{23}(x_4) &= \max_{x_2} \psi(x_2,x_3)\psi(x_2), \\
m_{54}(x_4) &= \max_{x_5} \psi(x_4,x_5), \\
m_{34}(x_4) &= \max_{x_3} \psi(x_3,x_4)\ m_{13}(x_3)\ m_{23}(x_3).
\end{aligned}$$

If we have Z, we can find the maximum probability. But we are interested in the values x^* of x that achieve this maximum probability (also we don't have Z). To find x^*, we simply need to keep track of which values of x_i maximize the message. Specifically, for a message $m_{ij}(x_j)$, we should know for each value of x_j what value of x_i was used to obtain the maximum value of the message. Then, when we find what value of x_4 maximizes the probability at the last step, we backtrack and find all the other x_is.

11.5 Sum-product Example

In the example below, we are interested in the probability of each node given that $B=0$, i.e., Bob's on time. Specifically, we are after $p(T|B=0), p(A|B=0), p(B|B=0)$.

Example:



$p(T=T_0) = 0.65 = 1-p(T=T_1)$

$p(B=B_0 \mid T=T_0) = 0.82$

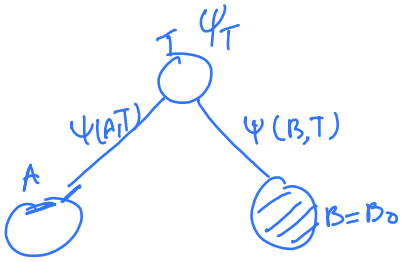
$p(B=B_0 \mid T=T_1) = 0.15$

$p(A=A_0 \mid T=T_0) = 0.9$

$p(A=A_0 \mid T=T_1) = 0.5$

$$p(A,B,T) = p(T)\,p(A \mid T)\,p(B \mid T)$$

We first convert this to an MRF



$\psi_T(T) = p(T) =$

$T=0$	$T=1$
0.65	0.35

$\psi_{BT}(B,T) = p(B \mid T) =$

	$B=0$	$B=1$
$T=0$	0.82	0.18
$T=1$	0.15	0.85

$\psi_{AT}(A,T) = p(A \mid T) =$

	$A=0$	$A=1$
$T=0$	0.9	0.1
$T=1$	0.5	0.5

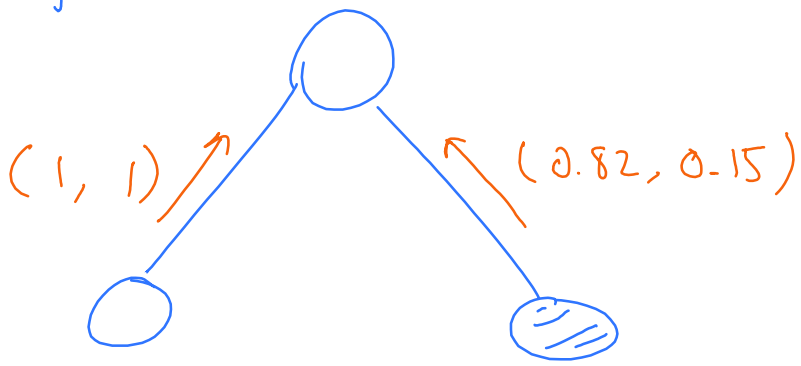
$$\mu_{BT}(T=0) = \sum_{B\in\{0\}} \psi(B, T=0) = 0.82$$

$$\mu_{BT}(T=1) = \sum_{B\in\{0\}} \psi(B, T=1) = 0.15$$

$$\psi(A, T=0) = 0.9 + 0.1 = 1$$

$$\mu_{AT}(T=0) = \sum_{A\in\{0,1\}}$$

$$\mu_{AT}(T=1) = \sum_{A\in\{0,1\}} \psi(A, T=1) = 0.5 + 0.5 = 1$$



$$P(T \mid B=B_0) \propto \mu_{AT}(T)\, \mu_{BT}(T)\, \psi_T(T)$$

$T=0$: $1 \times 0.82 \times 0.65$

$T=1$: $1 \times 0.15 \times 0.35$

Normalization :

$$P(T=0 \mid B=0) = \frac{0.82 \times 0.65}{0.82 \times 0.65 + 0.15 \times 0.35} = 0.91$$

$$\mu_{TA}(A=0) = \sum_{T \in \{0,1\}} \mu_{BT}(T)\,\psi(T, A=0)\,\psi(T)$$

$$= 0.65 \times 0.82 \times 0.9 + 0.35 \times 0.15 \times 0.5$$

$$= 0.506$$

$$\mu_{TA}(A=1) = \sum_{T \in \{0,1\}} \mu_{BT}(T)\,\psi(T, A=1)\,\psi(T)$$

$$= 0.65 \times 0.82 \times 0.1 + 0.35 \times 0.15 \times 0.5$$

$$= 0.08$$

$$P(A \mid B=0) \propto \mu_{TA}(A)$$

$A = A_0$: 0.506

$A = A_1$: 0.08

Normalization

$$P(A=0 \mid B=0) = \frac{0.506}{0.506 + 0.08} = \frac{0.506}{0.586} = 0.863$$

$$\mu_{TB}(B=0) = \sum_{T \in \{0,1\}} \mu_{AT}(T)\,\psi(T, B=0)$$

$$= 1 \times 0.82 + 1 \times 0.15$$

$$= 0.97$$

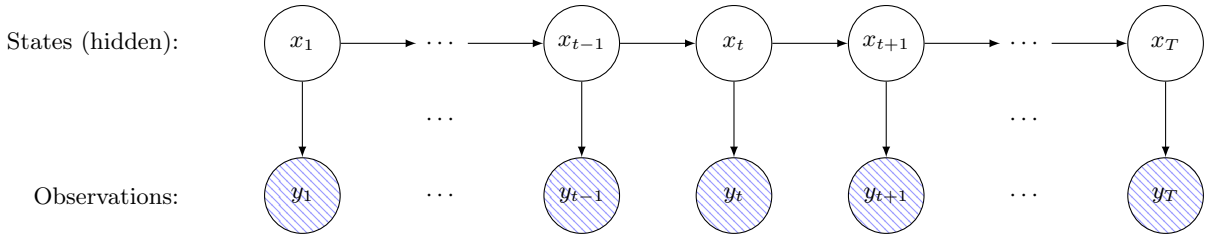
But the only case is $B=0$, so after normalization regardless of the value of $\mu_{TB}(B=0)$, we have

$$P(B=0 \mid B=0) = 1$$

Chapter 12

Inference in Hidden Markov Models

A hidden Markov model (HMM) is a graphical model of the form shown below. The top chain is a Markov chain representing the state of some system. Typically the state cannot be observed directly. However, we can observe some (probabilistic) function of the state. For example, the Markov chain can represent the health status of a patient and the observations are symptoms such as temperature, blood pressure, etc. As another example, the Markov chain can represent the part of speech of words in a text, and the observation is the actual word.



The probability distribution for this model factorizes as

$$p(x_1^T, y_1^T; \theta) = p(x_1) \prod_{t=2}^{T} p(x_t|x_{t-1}) \prod_{t=1}^{T} p(y_t|x_t).$$

Assuming the Markov chain and the observations are both on discrete spaces, we can complete the model by specifying $\theta = (\pi, A, B)$, where:

- The probability distribution π for x_1,
$$\pi_i = p(x_1 = i).$$
- The *transition matrix* A of the Markov chain,
$$A_{ij} = p(x_{t+1} = j|x_t = i).$$
- The *emission matrix* B describing the probabilities of the observations given the state,
$$B_{ij} = p(y_t = j|x_t = i).$$

Below are three common inference problems associated with HMMs and the methods for solving them. We will not derive the solutions but they can be found in [2].

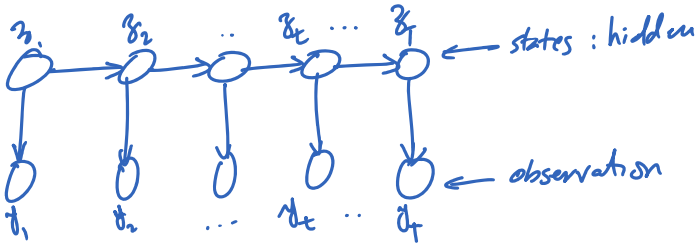
- Evaluation: $p(x_t|y_1^T; \theta) \rightarrow$ *forward-backward algorithm* (sum-product).
- Decoding: $\arg\max_{x_1^T} p(x_1^T|y_1^T; \theta) \rightarrow$ *Viterbi algorithm* (max-product).

- Learning: $\arg\max_\theta p(y_1^T;\theta) \rightarrow$ *Baum-Welch algorithm* (EM).

Below are HMM notes from a previous class. Unless I get a chance to go over these in class, they are not part of the course material and are here for self-study. But note that the methods are sum-product, max-product, and EM algorithms, which are part of the course and so reviewing the material below can be helpful in understanding those.

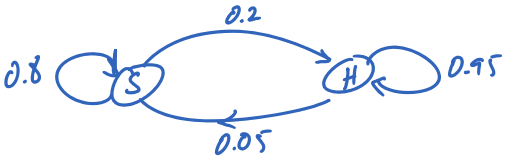
HMM

Tuesday, November 07, 2017 5:06 PM



* A person can be either sick or healthy : hidden state

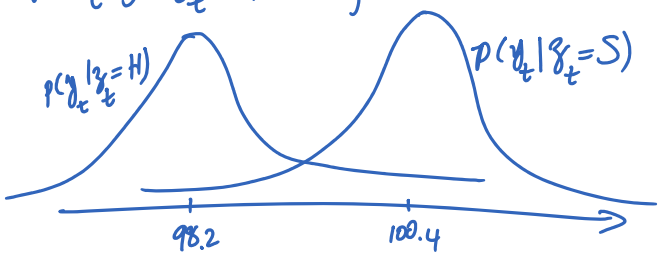
temperature : observation



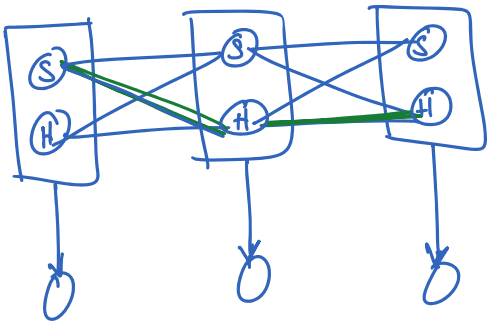
$p(z_{t+1}=j \mid z_t=i) = A_{ij}$: transition probs.

prob. distn over initial state $p(z_1=i) = \pi_i$

$p(y_t=j \mid z_t=i) = B_{ij}$: emission probs.



Trellis :



paths : configurations of hidden states

$$p(z_1^T, y_1^T \mid \theta) = p(z_1)\left(\prod_{t=2}^{T} p(z_t \mid z_{t-1})\right)\left(\prod_{t=1}^{T} p(y_t \mid z_t)\right)$$

$\theta = (\pi, A, B)$ $\quad \pi_i = p(z_1 = i \mid \theta)$

$A_{ij} = p(z_{t+1} = j \mid z_t = i, \theta)$

$B_{ij} = p(y_t = j \mid z_t = i, \theta)$

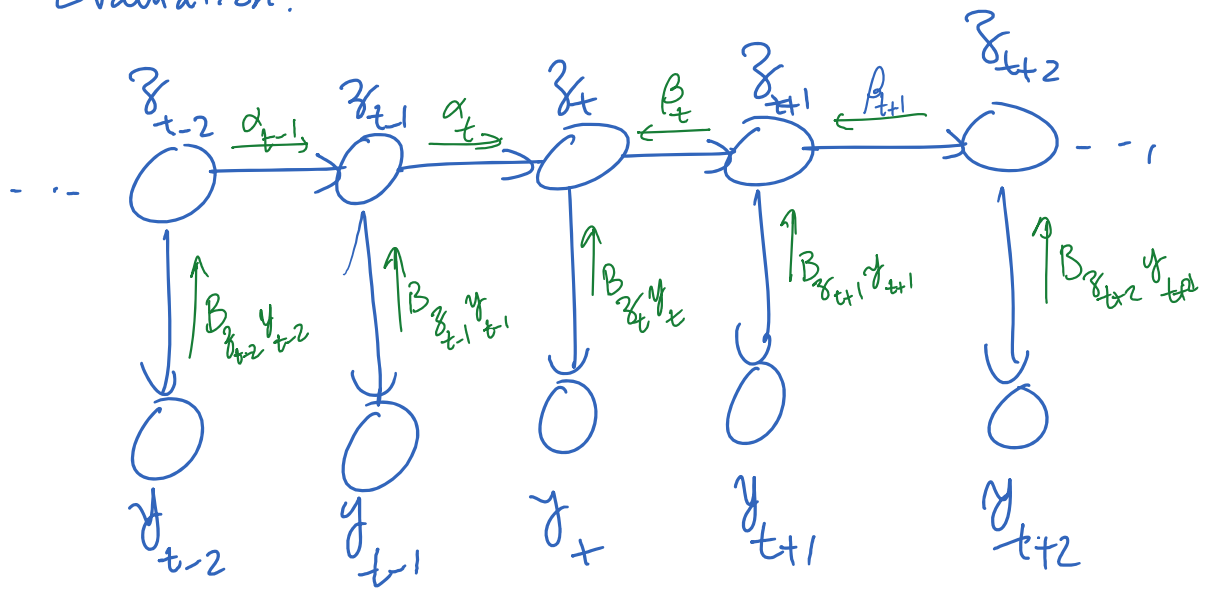
Three HMM problems:

* Evaluation: $p(z_t \mid y_1^T, \theta)$
  - Forward-Backward (Sum-product)
* Decoding: $\arg\max_{z_1^T} p(z_1^T \mid y_1^T, \theta)$
  - Viterbi Alg (Max-product)

Cofounder of Qualcomm

* Learning: $\arg\max_{\theta} p(y_1^T \mid \theta)$
  - Baum-Welch alg (EM)

Evaluation:



Define

... $\alpha_t(i) = p(z_t = i, y_1^{t-1})$ $t \geq 2$ | $\beta_t(i) = p(y_t^T \mid z_t = i)$ $t \leq T-1$

Define

$$\alpha_t(i) = \mu_{z_{t-1} z_t}(z_t = i) \quad t \geq 2 \qquad \alpha_1(i) = \pi_i$$

$$\beta_t(i) = \mu_{z_{t+1} z_t}(z_t = i) \quad t \leq T-1 \qquad \beta_T(i) = 1$$

It can be shown (by induction) that:

$$\alpha_t(i) = p(z_t = i, y_1^{t-1} \mid \theta), \qquad \beta_t(i) = p(y_{t+1}^T \mid z_t = i, \theta)$$

$$\alpha_t(i) = \sum_j \alpha_{t-1}(j) \underbrace{B_{j y_{t-1}}}_{\text{if } z_{t-1}=j} A_{ji} \qquad \beta_t(i) = \sum_j \beta_{t+1}(j) B_{j y_{t+1}} A_{ij}$$

Marginals:

$$p(z_t = i \mid y_1^T, \theta) = \gamma_t(i) \propto \alpha_t(i) \beta_t(i) B_{i y_t}$$

$$p(z_{t-1} = i, z_t = j \mid y_1^T, \theta) = \xi_t(i,j) \propto p(y_1^T, z_{t-1} = i, z_t = j \mid \theta)$$

$$= p(y_1^{t-2}, z_{t-1} = i)\, p(y_{t-1} \mid z_{t-1} = i)\, p(z_t = j \mid z_{t-1} = i)\, p(y_t \mid z_t = j)\, p(y_{t+1}^T \mid z_t = j)$$

$$p(y_{t-1} \mid z_{t-1} = i) = p(y_{t-1} \mid y_1^{t-2}, z_{t-1} = i)$$

$$= \alpha_{t-1}(i) B_{i y_{t-1}} A_{ij} B_{j y_t} \beta_t(j)$$

In traditional form of Forward-Backward, forward msgs included $B_{i y_t}$.

$$\bar{\alpha}_t(i) = p(z_t = i, y_1^t \mid \theta) = \alpha_t(i) B_{i y_t}$$

$$\bar{\alpha}_t(i) = \sum_j \bar{\alpha}_{t-1}(j) A_{ji} B_{iy_t}$$

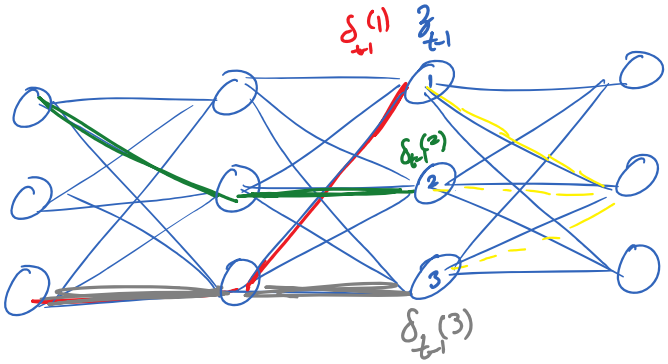
$$\gamma_t(i) \propto \bar{\alpha}_t(i) \beta_t(i)$$

Max-product: Choose z_T as root.

Define $\delta_t(i) = \mu_{z_{t-1} z_t}(z_t = i)$, $t \geq 2$ $\qquad \delta_1(i) = \pi_i$

Can be shown that

$$\delta_t(i) = \max_{z_1^{t-1}} p(z_1^{t-1}, z_t = i, y_1^{t-1} \mid \theta)$$



$$\delta_t(i) = \max_j \delta_{t-1}(j) B_{jy_{t-1}} A_{ji}$$

prob of the max-prob path $= \max_j \delta_T(j) B_{jy_T}$

+ Learning : EM / Baum-Welch

Assume complete data: z_1^T, y_1^T

Estimate $\theta = (\pi, A, B)$

$$\hat{\pi}_i = \begin{cases} 1 & z_1 = i \\ 0 & \text{else} \end{cases}$$

$$\hat{A}_{ij} = \frac{\sum_{t=1}^{T-1} I(z_t = i, z_{t+1} = j)}{\sum T(z = i)}$$

$$\hat{A}_{ij} = \frac{\sum_{t=1}^{T-1} I(z_t = i, z_{t+1} = j)}{\sum_t I(z_t = i)}$$

$$\hat{B}_{ij} = \frac{\sum_{t=1}^{T} I(z_t = i, y_t = j)}{\sum_{t=1}^{T} I(z_t = i)}$$

log likelihood of the complete data

$$\ln p(z_1^T, y_1^T \mid \theta) = \ln \pi_{z_1} + \sum_{t=2}^{T} \ln A_{z_{t-1} z_t} + \sum_{t=1}^{T} \ln B_{z_t y_t}$$

E-step

$$Q(\theta \mid \theta') = E\left[\ln p(z_1^T, y_1^T \mid \theta) \mid y_1^T, \theta'\right]$$

$$E\left[\ln \pi_{z_1} \mid y_1^T, \theta'\right] = \sum_i (\ln \pi_i)\, p(z_1 = i \mid y_1^T, \theta') = \sum_i \gamma_1'(i) \ln \pi_i$$

$\gamma_1'(i)$

$$E\left[\sum_{t=2}^{T} \ln A_{z_{t-1} z_t} \mid y_1^T \theta'\right] = \sum_{t=2}^{T} \sum_i \sum_j (\ln A_{ij})\, p(z_t = j, z_{t-1} = i \mid y_1^T, \theta')$$

$\xi_t'(i,j)$

$$= \sum_i \sum_j \left(\sum_{t=2}^{T} \xi_t'(i,j)\right) \ln A_{ij}$$

$$E\left[\sum_{t=1}^{T} \ln B_{z_t y_t} \mid y_1^T, \theta'\right] = \sum_i \sum_{t=1}^{T} p(z_t = i \mid y_1^T, \theta') \ln B_{i y_t}$$

$\gamma_t'(i)$

ML for p if the LL $= \sum_j n_j \ln p_j \Rightarrow p_j \propto n_j$

$$\pi_i'' \propto \gamma_1'(i) \qquad A_{ij}'' \propto \sum_{t=2}^{T} \xi_t'(i,j)$$

$$= \sum_i \sum_j \left(\sum_{t=1}^{T} \gamma_t'(i)\, I(y_t = j)\right) \ln B_{ij} \Rightarrow B_{ij} \propto \sum_{t=1}^{T} \gamma_t'(i)\, I(y_t = j)$$

Helpful references: [2, 1]

References

[1] Christopher M. Bishop. *Pattern Recognition And Machine Learning*. New York: Springer, 2006. URL: http://users.isr.ist.utl.pt/~wurmd/Livros/school/Bishop%5C%20-%5C%20Pattern%5C%20Recognition%5C%20And%5C%20Machine%5C%20Learning%5C%20-%5C%20Springer%5C%20%5C%202006.pdf (visited on 02/14/2017).

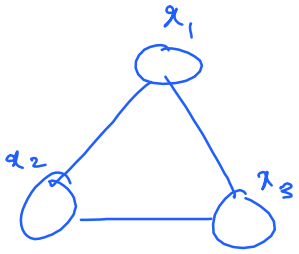
[2] Bruce Hajek. *Random Processes for Engineers*. Illinois, 2014. URL: http://hajek.ece.illinois.edu/Papers/randomprocJuly14.pdf (visited on 01/30/2017).

Chapter 13

Factor Graphs and Sum/Max-product Algorithms **

Factor graphs

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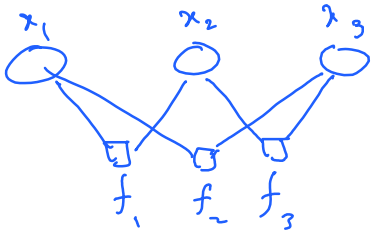
This MRF implies the factorization

$$p(x_1, x_2, x_3) \propto \psi(x_1, x_2, x_3)$$

But suppose we actually want to represent

$$p(x_1, x_2, x_3) \propto f(x_1, x_2)\ f(x_2, x_3)\ f(x_3, x_1)$$

Is there a way to do this with a graph?



Factor graph:

Two types of nodes:

V: variables: x_1, x_2, x_3

F: factors: f_1, f_2, f_3

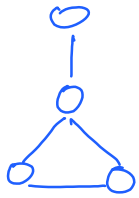
$$p(x_1^m) = \prod_{f_i \in F} f_i(x_{f_i})$$

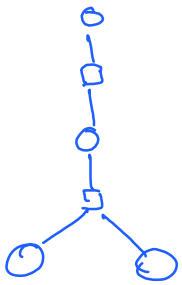
x_{f_i}: variable nodes adjacent to f_i

MRF $\longrightarrow$ FG

nodes $\longrightarrow$ variable nodes

maximal cliques $\longrightarrow$ factor nodes

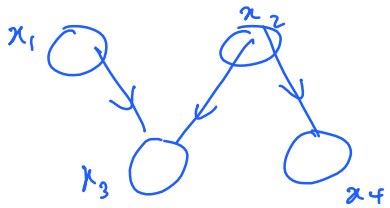


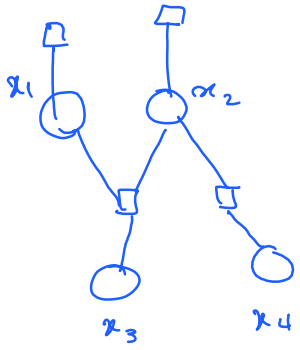


BN $\longrightarrow$ FG

nodes ⟶ variable nodes

(CPDs at) nodes ⟶ factor nodes

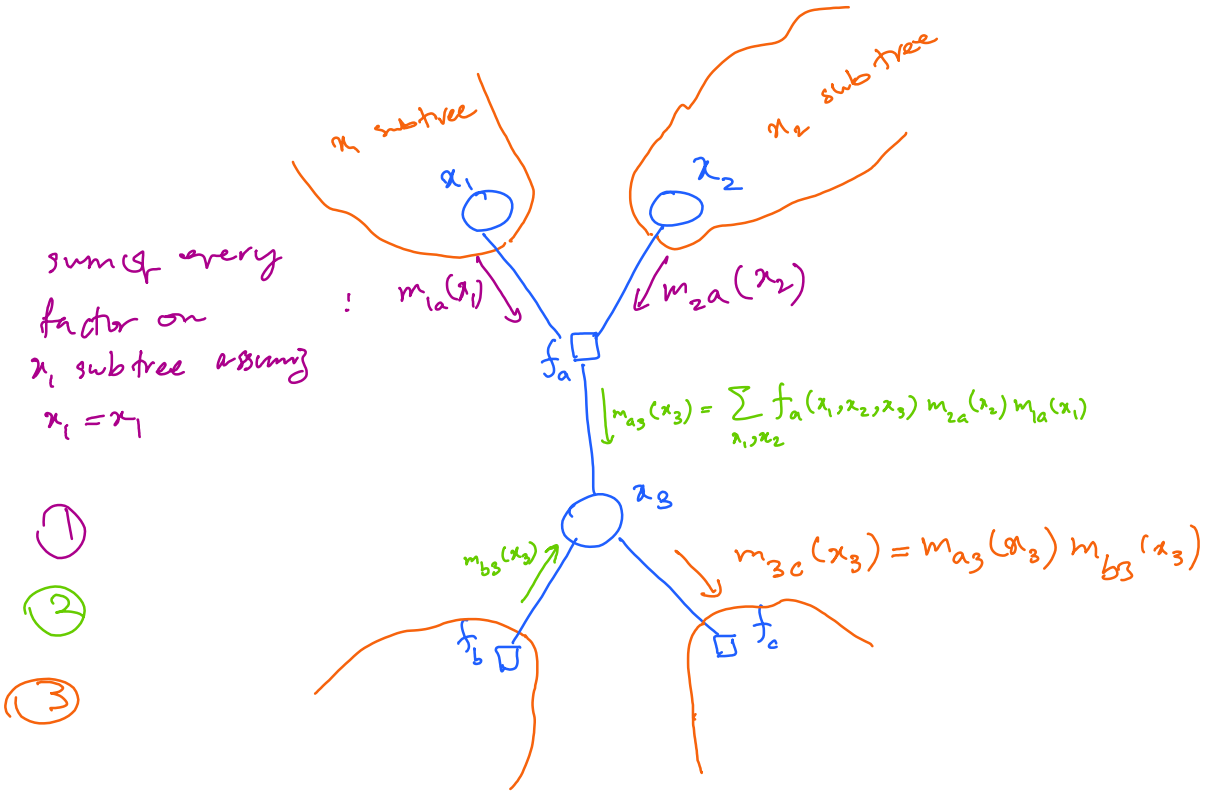




Note: even if the original MRF is not a tree, or the original BN has an MRF which is not a tree, the factor graph may still be a tree: that is discarding the types of the nodes in FG leads to a tree.

This is good because ↓

Sum-product for factor tree graphs:



Starting steps at leaves:



$m_{fx}(x) = f(x)$

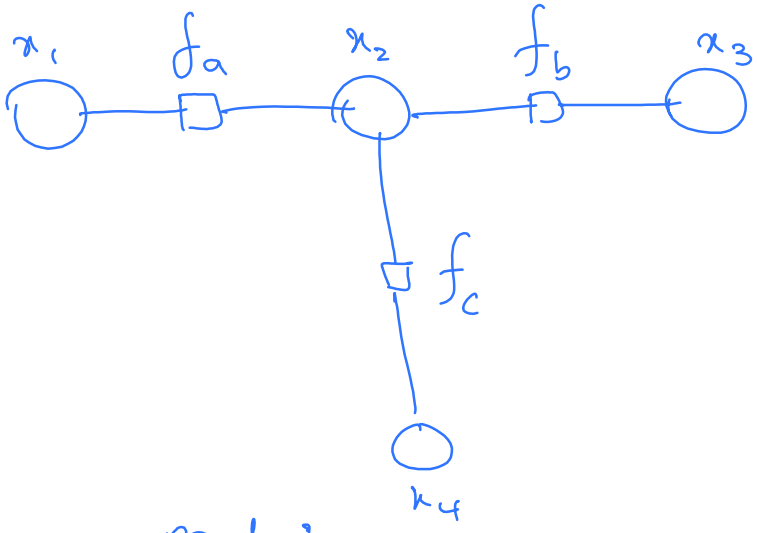


$m_{xf} = 1$

Marginal at each node: product of msgs it receives

marginals at all nodes takes twice as much of that of a single node: two msgs per link as opposed to one.

Example:



round 1

$\mu_{1a}(x_1) = 1$

$\mu_{4c}(x_4) = 1$

$\mu_{3b}(x_3) = 1$

round 2

$$\mu_{a2}(x_2) = \sum_{x_1} f_a(x_1, x_2)\, \mu_{1a}(x_1)$$

$$\mu_{b2}(x_2) = \sum_{x_3} f_b(x_3, x_2)\, \mu_{3b}(x_3)$$

$$\mu_{c2}(x_2) = \sum_{x_4} f_c(x_2, x_4)\, \mu_{4c}(x_4)$$

Max-product

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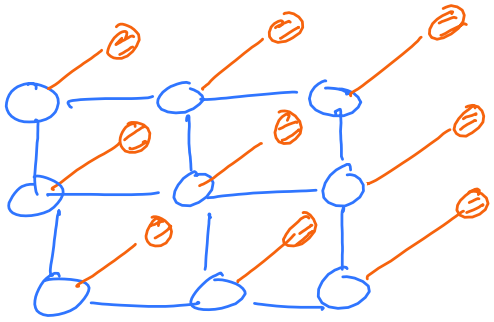
Problem: Identify the most likely configuration:

Find a set of values $x_1^*, \ldots, x_m^*$ s.t.

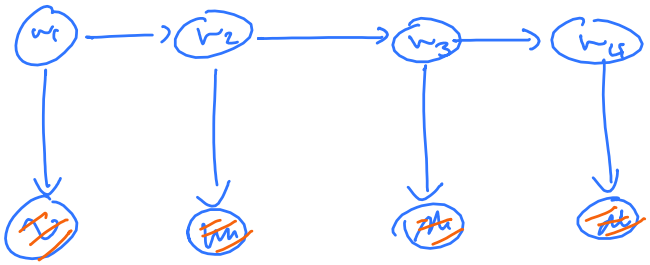
$P(x_1^*, \ldots, x_m^*) \geq P(x_1^m)$ for all x_1^m

Applications:

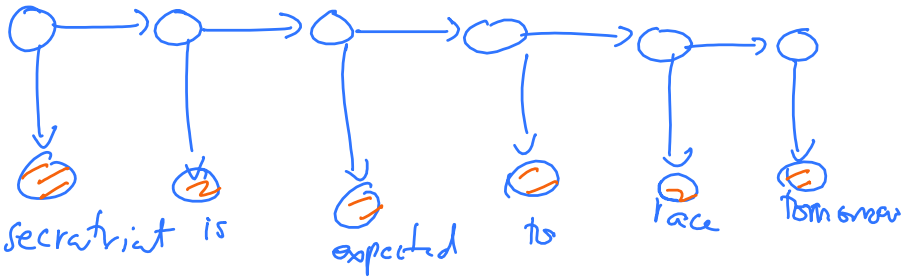
- Image denoising (Lab 1)



- Voice recognition



- Part of speech tagging



Do we already know how to solve this?

Finding most probable state for a node:

run sum-product and find state with max prob.

Finding most probable configuration for graph:

max-product

$$x_{max} = \arg\max_{x} p(x)$$

May not be the same.:

	$x=0$	$x=1$
$y=0$	0.3	0.4
$y=1$	0.3	0

$x=0$ max for x

$y=0$ max for y

$(x,y)=(0,1)$ max for (x,y)

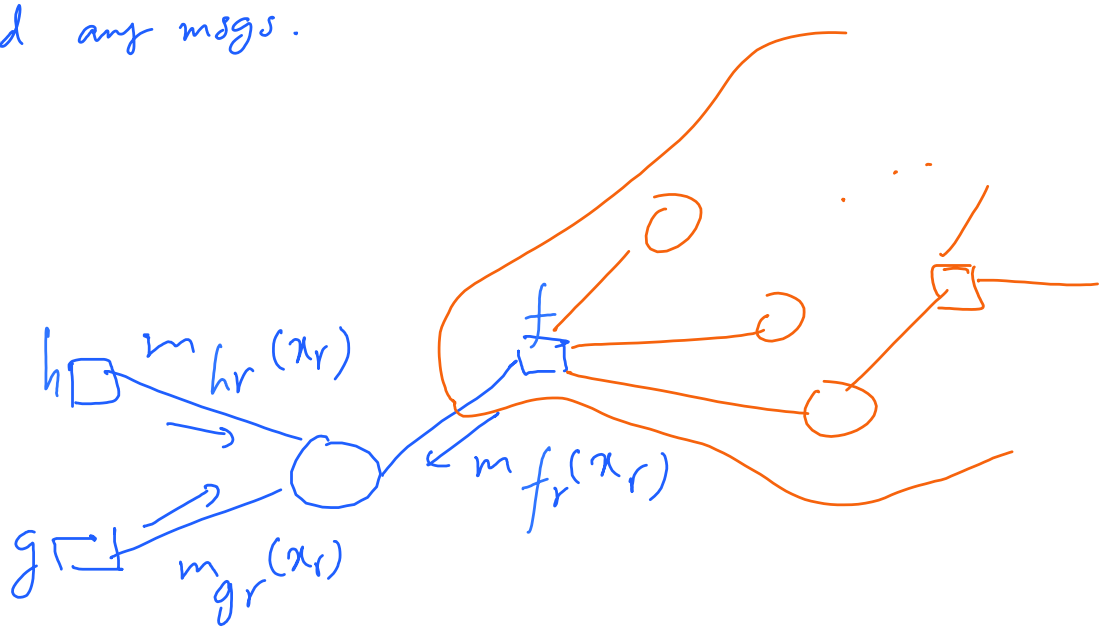
Let's instead try to find $\max_{x_1^m} p(x_1^m)$

$$\max_{x_1^m} p(x_1^m) = \max_{x_1} \max_{x_2} \cdots \max_{x_m} p(x_1^m)$$

* We can use a similar approach to elimination except that Σ is replaced with max.

* Similarly sum-product can be turned to max-product to find $\max p(x)$

* Here, we pick an arbitrary root, which does not send any msgs.



The message from f $\mu_{fr}(x_r)$

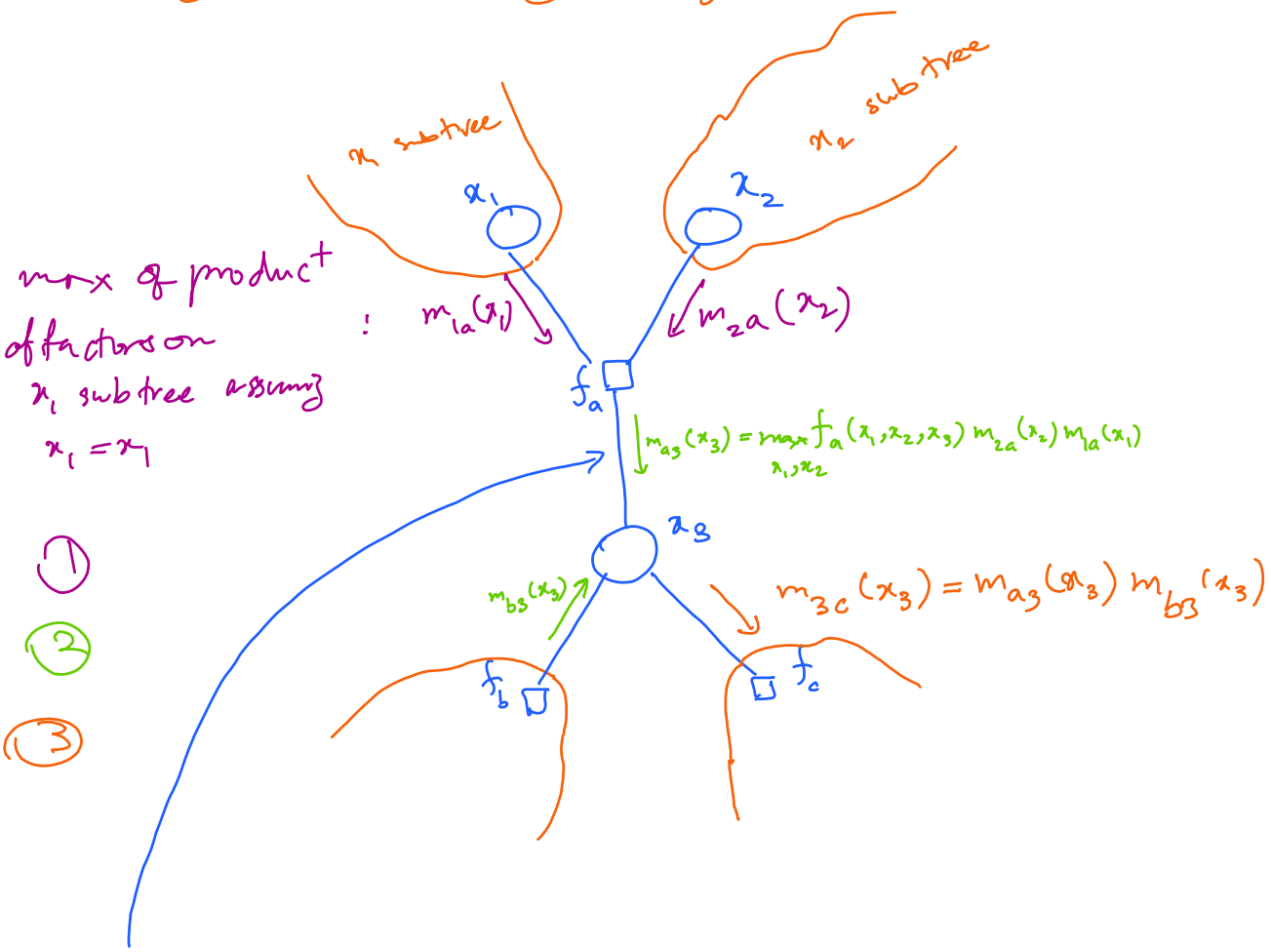
Given the particular value for x_r

* What is the most likely configuration for the subtree of f
* What is the "probability" of this configuration

$$\max p(x_1^m) = \max_{x_r} m_{fr}(x_r)\, m_{gr}(x_r)\, m_{hr}(x_r)$$

↳ from the value of x_r that maximizes this sum and messages we find the most likely configuration

* Finding the maximizing configuration:



For each value of x_3, we also record which values of x_1, x_2 achieved the max.

At the root, we find x_r^* that achieves the max. Then we backtrack and find maximizing values for all nodes.

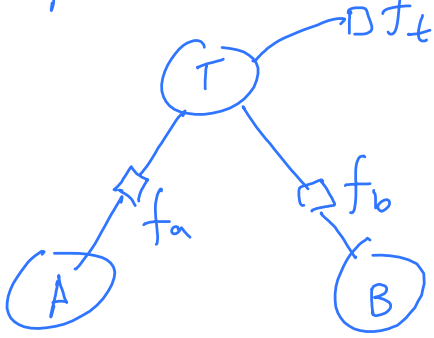
* It may be more convenient to maximize $\ln p(x) \Rightarrow$ max-sum algorithm

$$\sum f_i(x_{f_i})$$

$\max_{x_1 \ldots x_n} p(x) = \max_{x_1 \ldots x_n} \ldots$

Note: Why not continue the alg so that we can find x_i^* for all nodes just like how we found it for x_r? We will find maximizing values, but they may belong do different maximizing configurations.

Example: most-likely configuration



$f_t(T=0) = 0.65$

$f_t(T=1) = 0.35$

$f_a(A=0, T=0) = 0.9$

$f_a(A=0, T=1) = 0.5$

$f_a(A=1, T=0) = 0.1$

$f_a(A=1, T=1) = 0.5$

$f_b(B=0, T=0) = 0.8$

$f_b(B=0, T=1) = 0.15$

$f_b(B=1, T=0) = 0.18$

$f_a(B=1, T=0) = 0.12$

$\mu_{A \to}$: max 1

$A=0 \longrightarrow 1$

$A=1 \longrightarrow 1$

$$\mu_{aT} : \max_{A} f_a(A,T)$$

$T=0 \longrightarrow 0.9$ for $A=0$

$T=1 \longrightarrow 0.5$ for $A=0$ & $A=1$

$$\mu_{tT} : \max f_t(T)$$

$T=0 \longrightarrow 0.65$

$T=1 \longrightarrow 0.35$

$$\mu_{Tb} : \max_{T} \mu_{tT}(T)\, \mu_{aT}(T)\, f_b(B,T)$$

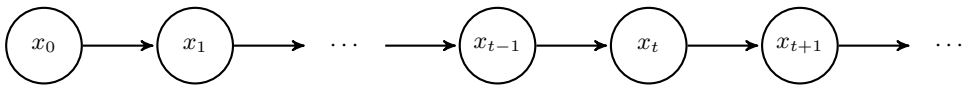
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Chapter 14

Markov Chains

14.1 Introduction

A **Markov chain (MC)** is a stochastic process whose future is independent from its past and can be represented as the following Bayesian network:



The value of x_t is called the **state** of the Markov chain at time t. The set of all possible states is the **state space**. For example,

- We may represent daily weather with the state space: {sunny, cloudy, rainy}
- The state of the disease in a patient may be represented by a MC with two states: {remission, relapse}.
- The number of animals of a certain species can be represented with states $\{0, 1, 2, \dots\}$.

Uncountable state spaces are also possible (e.g., temperature) and we will rely on them for sampling later. But for simplicity, we focus on finite-state MCs. Also, note that a MC is usually an approximation of the world since we like to have a small number of states.

To complete the characterization of a MC, we also need to know the CPDs,

$$p(x_0 = i), \quad p(x_{t+1} = j | x_t = i).$$

We refer to $p(x_0)$ as the **initial distribution** and to the CPD $p(x_{t+1} = j | x_t = i)$ as **transition probabilities**. We are interested in **time-homogeneous** MCs only, in which $p(x_{t+1} = j | x_t = i)$ is independent of t, i.e., the same for all time instances. In such MCs, we can represent the transition probabilities as a transition matrix A with

$$P_{ij} = p(x_{t+1} = j | x_t = i),$$

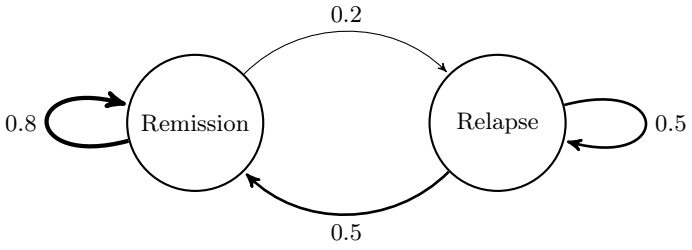
which is particularly useful if the state space is a finite set.

Example 14.1 (The patient's chain)**.** In a Markov chain representing the health of a patient, if we let 1 represent 'remission' and 2 represent 'relapse,' we may have

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \end{pmatrix}.$$

△

Given that the important features of a time-homogeneous MC are its state space and transition probabilities, it is useful to represent the chain as graph, called the **state-transition graph**, whose nodes are the states and edges represent transitions and their probabilities. For example, for a disease, we may have

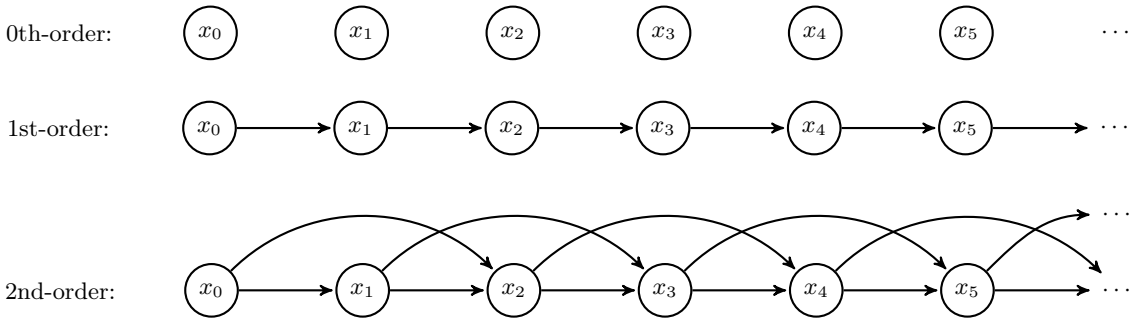


Here are some other examples of common MCs:

- **Random walk** on a grid (1D, 2D, ...). For example, in the 1-dimensional case, we can move left or right at random. This extends to n dimensions. In this context, "a drunk man will find his way home, but a drunk bird may get lost forever."
- **Page-rank**. This is closely related to the previous chain, except that this time the states are webpages, and we click on a link in the current page to transition to another one. This was the main idea behind Google search's ranking of web pages, using stationary probabilities (more on these below).
- **DNA mutations**. There are four states $\{A, C, G, T\}$ and due to mutations, a position in the genome may change from one state to another. Several variations are used in phylogenetics.

As stated before, MCs are usually approximations of real phenomena because we cannot include all relevant information in the state. As an example, consider a MC for weather. Suppose our chain represents a short period where seasonal effects are negligible and so we can assume the chain to be time-homogeneous. Each state of the MC could be the total amount of precipitation. This is already useful since a rainy day is more likely after a rainy day than after a sunny day. But if we add information about temperature, cloud cover, air pressure, etc., the model becomes more accurate and useful.

Another way that MCs can be extended is by allowing dependence on more than previous state, i.e., allowing the **order** to be larger than 1. Graphical examples of zeroth-order, first-order, and second-order MCs are shown below:



Example 14.2 ([2])**.** More accurate models can produce more realistic data, as shown in the following example from Shannon on modeling English text as a MC.

1. Zero-order approximation with uniform distribution (symbols are independent and equally probable).

   XFOML RXKHRJFFJUJ ZLPWCFWKCYJ FFJEYVKCQSGXYD QPAAMKBZAACIB-ZLHJQD

2. Zero-order approximation (symbols independent but their probability is the same as English text).

   OCRO IlLI RGWR NMIELWIS EU LL NBNESEBYA TH EEl ALHENHTTPA OOBTTVA NAH BRL

3. First-order approximation (digram structure; the conditional probability of each symbol given the previous is like English).

   ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TUCOOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE

4. Second-order approximation (trigram structure as in English).

   IN NO 1ST LAT WHEY CRATICT FROURE BIRS GROCID PONDENOME OF DEMONSTURES OF THE REPTAGIN IS REGOACTIONA OF CRE

5. Zero-Order Word approximation; words are chosen independently but with their appropriate frequencies.

   REPRESENTING AND SPEEDILY IS AN GOOD APT OR COME CAN DIFFERENT NATURAL HERE HE THE A IN CAME THE TO OF TO EXPERT GRAY COME TO FURNISHES THE LINE MESSAGE HAD BE THESE.

6. First-Order Word approximation; the word transition probabilities are as in English text.

   THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHARACTER OF THIS POINT IS THEREFORE ANOTHER METHOD FOR THE LETTERS THAT THE TIME OF WHO EVER TOLD THE PROBLEM FOR AN UNEXPECTED

△

14.2 State distribution as a function of time

Consider a MC with m states. Let $\boldsymbol{\pi}_t = (\pi_{t1}, \pi_{t2}, \ldots, \pi_{tm})$ denote the probability distribution over the states at time t, where $\pi_{tj} = p(x_t = j)$. Usually, $\boldsymbol{\pi}_0$, or equivalently, $p(x_0)$ is given. We have the following recursion,

$$\pi_{tj} = \sum_{i=1}^{m} p(x_{t-1} = i)p(x_t = j|x_{t-1} = i) = \sum_{i=1}^{m} \pi_{t-1,i} P_{ij},$$

or more compactly

$$\boldsymbol{\pi}_t = \boldsymbol{\pi}_{t-1} P \qquad \text{and} \qquad \boldsymbol{\pi}_t = \boldsymbol{\pi}_0 P^t.$$

Furthermore, the ijth element of P^t, shown as $(P^t)_{ij}$, is the probability of ending up in state j in t steps if we start from state i.

Example 14.3 (The patient's chain over time)**.** Suppose $\pi_0 = (1, 0)^T$, i.e., the patient starts in remission. Then,

$$\boldsymbol{\pi}_1 = (1,0)\begin{pmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \end{pmatrix} = (0.8, 0.2), \qquad \boldsymbol{\pi}_2 = \boldsymbol{\pi}_1\begin{pmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \end{pmatrix} = (0.74, 0.26)$$

$$\boldsymbol{\pi}_5 = \boldsymbol{\pi}_0 P^5 = (0.71498, 0.28502), \qquad \boldsymbol{\pi}_{10} = \boldsymbol{\pi}_0 P^{10} = (0.71429, 0.28571)$$

So after 10 days, the probability of being in remission is about 71%.

Now suppose the patient starts in relapse. Then

$$\boldsymbol{\pi}_1 = (0.5, 0.5), \qquad \boldsymbol{\pi}_2 = (0.65, 0.35)$$

$$\boldsymbol{\pi}_5 = (0.71255, 0.28745), \qquad \boldsymbol{\pi}_{10} = (0.71428, 0.28572)$$

We can see that, interestingly, $\boldsymbol{\pi}_5$ and $\boldsymbol{\pi}_{10}$ are very close to each other and almost independent of π_0. We will study this further in the next section. △

14.3 Long-term Behavior of Markov Chains

What happens to a MC if we let it run for a long time? This problem is of interest in a variety of contexts, e.g., the Page-rank algorithm above and sampling methods discussed later. We saw in the previous example that as t grows the distribution over the states appears to settle down on a certain distribution, which is called the **limiting distribution**. In the example, the limiting distribution was independent of the initial distribution. In this section, we will study when and why this happens.

A **stationary distribution** of a MC is a distribution $\boldsymbol{\sigma}$ that satisfies

$$\boldsymbol{\sigma} = \boldsymbol{\sigma} P.$$

Any finite-state Markov chain has at least one stationary distribution [1]. The limiting distribution, if it exists, must be a stationary distribution.

Example 14.4 (The patient's stationary distribution)**.** The stationary distribution $\boldsymbol{\sigma} = (\sigma_1, \sigma_2)$ is obtained by solving $(\sigma_1, \sigma_2) = (\sigma_1, \sigma_2)P$ and $\sigma_1 + \sigma_2 = 1$. It can be shown that the unique solution to these equations is

$$\boldsymbol{\sigma} = (5/7, 2/7) = (0.71429, 0.28571),$$

which indeed appears to be the limiting distribution regardless of the initial distribution. △

Graph vs. transition matrix. Whether or not a MC converges to a unique limiting distribution is determined by P. This dependence is only on P_{ij} being zero or nonzero but not how large the values are otherwise. The zero/positive status of each transition probability is given by the MC graph: an edge from states i to state j exists if and only if $P_{ij} > 0$. So the graph is sufficient to decide whether the MC will converge to a unique stationary distribution.

First, let us see some examples when the stationary distribution is not unique:



On the left, the limiting distribution depends on the initial distribution. This arises because of a lack of connectivity between the states. On the right a limiting distribution does not exist because the chain is *periodic* in a certain sense.

We can eliminate both of these possibilities by defining regular Markov chains. A Markov chain is **regular** if there is a positive integer k such that for all i and j it is possible to go from state i to state j in k steps. This is equivalent to $(P^k)_{ij} > 0$ for all i, j and also equivalent to the existence of a path of length k between any two states. In Example 14.1, we have $k = 1$.

Theorem 14.5 (Convergence of regular chains)**.** *If a MC with transition matrix P is regular, then there exists a unique distribution $\boldsymbol{\sigma}$ such that $\boldsymbol{\sigma} = \boldsymbol{\sigma} P$ and for any $\boldsymbol{\pi}_0$, we have $\boldsymbol{\pi}_t = \boldsymbol{\pi}_0 P^t \to \boldsymbol{\sigma}$ as $t \to \infty$.*

The above theorem guarantees that regular MCs converge to their unique stationary distributions. Furthermore, since we can choose $\boldsymbol{\pi}_0$ to have a 1 in any position, the theorem also implies that each row of P^t converges to $\boldsymbol{\sigma}$.

Example 14.6 (Convergence of the patient's chain)**.** Indeed, $\boldsymbol{\sigma} = (5/7, 2/7) = (0.71429, 0.28571)$ is the stationary distribution of

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \end{pmatrix}$$

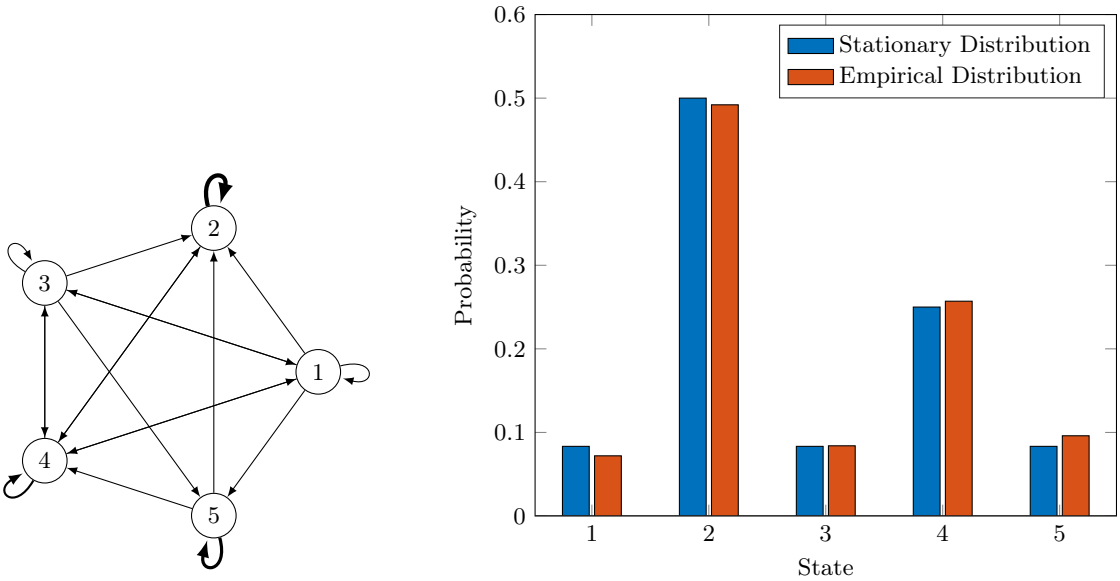


Figure 14.1: In the Markov chain (left), edges between different nodes have probability 1/5 and the probability of self loops is such that the outgoing probabilities sum to 1. The stationary distribution and an empirical (time-averaged) distribution are given on the right.

and $\boldsymbol{\pi}_t \to \boldsymbol{\sigma}$ regardless of $\boldsymbol{\pi}_0$ as we saw in Example 14.1. Furthermore,

$$P^2 = \begin{pmatrix} 0.74 & 0.26 \\ 0.65 & 0.35 \end{pmatrix}, \qquad P^5 = \begin{pmatrix} 0.71498 & 0.28502 \\ 0.71255 & 0.28745 \end{pmatrix}, \qquad P^{10} = \begin{pmatrix} 0.71429 & 0.28571 \\ 0.71428 & 0.28572 \end{pmatrix}$$

△

14.3.1 How often does the Markov Chain visit each state?

For a regular MC with stationary distribution $\boldsymbol{\sigma}$, we know if t is large, at time t, the probability of being in state j is σ_j. But in a time period of length N, how many times state j is visited? The answer is approximately $N\sigma_j$ if N is large. (While this seems natural, similar statements do not necessarily hold for other random processes.)

For example, for a chain with transition matrix,

$$P = \frac{1}{5}\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 4 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 3 \end{pmatrix},$$

whose graph is shown in Figure 14.1 (left), a simulation of length 1000 time units produced an empirical distribution close to the stationary distribution. The first 20 samples are as follows: 32244322242222244122.

14.4 Balance Properties and Finding the Stationary Distribution

14.4.1 Detailed Balance

A distribution $\boldsymbol{\pi}$ satisfies the **Detailed Balance Property (DBP)** if

$$\pi_i P_{ij} = \pi_j P_{ji}.$$

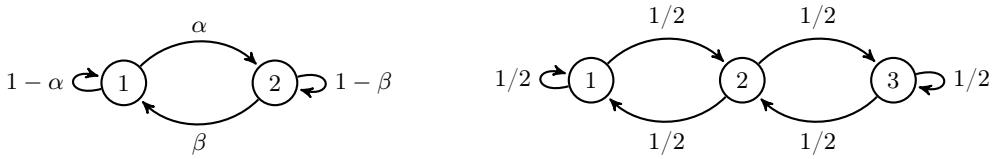
Theorem 14.7 (Stationarity from detailed balance)**.** *For a regular MC, if a vector $\boldsymbol{\pi}$ satisfies the detailed balance property, then $\boldsymbol{\pi}$ is the unique stationary distribution ($\boldsymbol{\pi} = \boldsymbol{\sigma}$).*

Proof. From Theorem 14.5, we know that the stationary distribution is unique, i.e., there is a unique $\boldsymbol{\sigma}$ satisfying $\boldsymbol{\sigma} = \boldsymbol{\sigma} P$. So it suffices to show that $\boldsymbol{\pi}$ satisfies the equation $\boldsymbol{\pi} = \boldsymbol{\pi} P$, where P is the transition matrix. For all j,

$$\pi_j = \pi_j \sum_i P_{ji} = \sum_i \pi_j P_{ji} = \sum_i \pi_i P_{ij}.$$

Hence, $\boldsymbol{\pi} = \boldsymbol{\pi} P$. □

Exercise 14.8 (Stationary distribution via DBP)**.** Using DBP, find the stationary distribution for the following MCs.

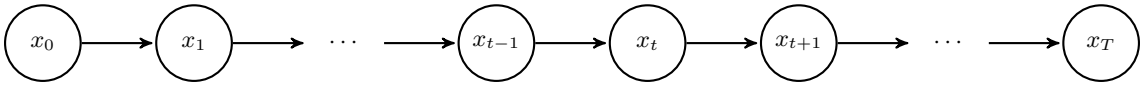


△

If our MC is regular and DBP holds, then we have the stationary distribution. This approach, if possible, is an easy way to find the stationary distribution. For this reason, DBP is commonly used in Markov Chain Monte Carlo (MCMC) methods which we discuss later.

14.4.2 Time-Reversibility **

Consider the Markov chain



and assume that $\boldsymbol{\pi}_t = \boldsymbol{\sigma}$, where $\boldsymbol{\sigma}$ is a stationary distribution. suppose that we run the chain backward in time (or play a movie of it backward). Note that the Markov property still holds as

$$p(x_t | x_{t+1}, \ldots, x_T) = p(x_t | x_{t+1})$$

So what are the transition probabilities P^- for the reversed MC? We have

$$P^-_{ij} = p(x_t = j | x_{t+1} = i) = \frac{p(x_t = j, x_{t+1} = i)}{p(x_{t+1} = i)} = \frac{\pi_j P_{ji}}{\pi_i}.$$

The MC is called **time-reversible** if $P^- = P$, which is equivalent to $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j, which are the detailed balance equations.

14.4.3 Global Balance **

A distribution $\boldsymbol{\pi}$ over the states of the MC satisfies the **Global Balance Property (GBP)** if for any partition[1] $\{R, L\}$ of the states of the MC, we have

$$\sum_{i\in L} \pi_i \sum_{j\in R} P_{ij} = \sum_{j\in R} \pi_j \sum_{i\in L} P_{ji}.$$

In particular, for any node i,

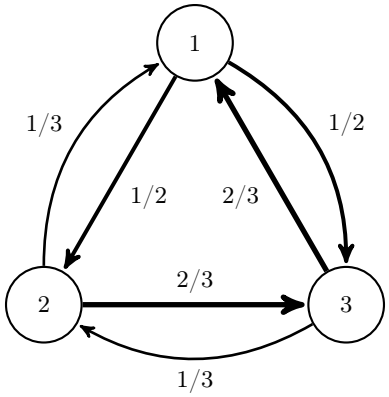
$$\pi_i \sum_{j\neq i} P_{ij} = \sum_{j\neq i} \pi_j P_{ji}.$$

It is not difficult to show mathematically that any stationary distribution $\boldsymbol{\sigma}$ of the Markov chain satisfies the global balance property. To see this intuitively, imagine Alice performs a random walk over the state-transition graph, going from state to state according to the transition probabilities P. Assume that $\boldsymbol{\pi}_0 = \boldsymbol{\sigma}$, i.e., Alice chooses her initial position according to $\boldsymbol{\sigma}$. It follows that $\boldsymbol{\pi}_t = \boldsymbol{\sigma}$. During N steps, where N is large, the number of times that Alice goes from a state in L to a state in R is approximately $N \sum_{i\in L} \pi_i \sum_{j\in R} P_{ij}$. Similarly, the number of times that Alice goes from R to L is about $\sum_{j\in R} \pi_j \sum_{i\in L} P_{ji}$. Since Alice cannot disappear, we must have $\sum_{i\in L} \pi_i \sum_{j\in R} P_{ij} = \sum_{j\in R} \pi_j \sum_{i\in L} P_{ji}$.

We can use the GBP to find the stationary distribution as shown in the next example.

Example 14.9 (Stationary distribution via GBP)**.** Consider a chain with

$$P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}.$$



The DBP equations are

$$\pi_1 \cdot \frac{1}{2} = \pi_2 \cdot \frac{1}{3}$$
$$\pi_1 \cdot \frac{1}{2} = \pi_3 \cdot \frac{2}{3}$$
$$\pi_2 \cdot \frac{2}{3} = \pi_3 \cdot \frac{1}{3},$$

which are not satisfiable. Visually, from the diagram above we may also have guessed that the flow of probability is more counterclockwise than clockwise, and so each pair of states is unbalanced.

[1] A partition of a set S is a collection of disjoint sets whose union is equal to S.

The GBP equations are

$$\pi_1 \cdot (\frac{1}{2} + \frac{1}{2}) = \pi_2 \cdot \frac{1}{3} + \pi_3 \cdot \frac{2}{3}$$
$$\pi_2 \cdot (\frac{1}{3} + \frac{2}{3}) = \pi_1 \cdot \frac{1}{2} + \pi_3 \cdot \frac{1}{3}$$
$$\pi_3 \cdot (\frac{2}{3} + \frac{1}{3}) = \pi_1 \cdot \frac{1}{2} + \pi_2 \cdot \frac{2}{3}$$

which are satisfied for $\pi_2 = \frac{12}{14}\pi_1$ and $\pi_3 = \frac{15}{14}\pi_1$. Taking into account the fact that the probabilities must sum to 1, we find $\pi_1 = \frac{14}{41}, \pi_2 = \frac{12}{41}, \pi_3 = \frac{15}{41}$. △

References

[1] Alex Furman. "WHAT IS . . . a Stationary Measure?" In: *Notices of the AMS* 58.9 (). URL: https://www.ams.org/journals/notices/201109/rtx110901276p.pdf (visited on 04/19/2020).

[2] C.E. Shannon. "A mathematical theory of communication." In: *The Bell System Technical Journal* 27.3 (July 1948), pp. 379–423. ISSN: 0005-8580. DOI: 10.1002/j.1538-7305.1948.tb01338.x.

Chapter 15

Sampling Methods

15.1 Introduction

In Bayesian inference, *distributions* are the ultimate tool for representing knowledge about unknown quantities. This is the reason that we try to find $p(\boldsymbol{\theta}|\mathcal{D})$. If we have the distribution, we can find the *expected value* of various functions of the unknown quantity and in this way find point estimates or the probability of an event,

$$\hat{\boldsymbol{\theta}}_{Bayes} = \mathbb{E}[\boldsymbol{\Theta}|\mathcal{D}],$$
$$p(\boldsymbol{\theta} \in A|\mathcal{D}) = \mathbb{E}[\mathbb{1}(\boldsymbol{\Theta} \in A)|\mathcal{D}],$$

where A is an event and $\mathbb{1}$(condition) equals 1 if the condition holds and is 0 otherwise.

If we find the posterior distribution in closed form and it turns out to be one of the common distributions, e.g., Gaussian, Poisson, etc, then typically, we can easily compute expected values. However, this is not always the case, and we may face two difficulties:

1. Sometimes all we have is a function $q(\boldsymbol{\theta})$ that is proportional to $p(\boldsymbol{\theta}|\mathcal{D})$,

$$p(\boldsymbol{\theta}|\mathcal{D}) \propto p(\boldsymbol{\theta})p(\mathcal{D}|\boldsymbol{\theta}) = q(\boldsymbol{\theta})$$

   and we are not even able to compute $p(\boldsymbol{\theta}|\mathcal{D})$ for a given $\boldsymbol{\theta}$ because the normalization factor is not known.

2. Even if we can compute $p(\boldsymbol{\theta}|\mathcal{D})$, computing expected values requires integration, which may be computationally infeasible.

In such cases, sampling from this distribution will be useful because sampling allows us to find expected values. For a function h, by the law of large numbers we have the following approximation

$$\mathbb{E}[h(X)] \simeq \sum_{i=1}^{N} h(x_i),$$

where x_i are independent samples drawn from the distribution p_X with respect to which the expected value is to be computed.

For example, recall that in Bayesian linear regression, a common likelihood is

$$\boldsymbol{y}|\boldsymbol{\theta}, \sigma^2 \sim \mathcal{N}(X\boldsymbol{\theta}, \sigma^2 I),$$

with prior

$$p(\boldsymbol{\theta}, \sigma^2) \propto 1/\sigma^2.$$

For this model, we found $p(\boldsymbol{\theta}|\mathcal{D}, \sigma^2)$ and $p(\sigma^2|\mathcal{D})$ and stated that while it is possible to obtain $p(\boldsymbol{\theta}|\mathcal{D})$ analytically, doing so is complicated. In practice, we proceed computationally by generating samples from $p(\sigma^2|\boldsymbol{y})$ and then $p(\boldsymbol{\theta}|\boldsymbol{y}, \sigma^2)$. With this sampling approach we can also perform prediction for a given input vector $\boldsymbol{x}_{n+1}$ of by producing samples from $p(y_{n+1}|\boldsymbol{\theta}, \sigma^2) \sim \mathcal{N}(\boldsymbol{x}_{n+1}^T\boldsymbol{\theta}, \sigma^2)$, and answer question such as finding $p(y_{n+1} > a|\boldsymbol{\theta}, \sigma^2)$ for a given constant a.

In this chapter, we will discuss methods for generating samples from a distribution $p(\boldsymbol{\theta})$ which we can only compute up to a multiplicative constant. The approach is identical for conditional distributions such as $p(\boldsymbol{\theta}|\mathcal{D})$. To emphasize the fact that the constant may not be known, we use p to refer to the true distribution and q to the "distribution" without the constant. We will use $\mathbb{E}_p$ to denote expectation with respect to distribution p. For a non-normalized distribution q we define $\mathbb{E}_q = \mathbb{E}_p$.

15.2 Basic Sampling Techniques

In this section, we will review some basic but useful sampling techniques.

15.2.1 Deterministic Integration

This method is not actually a sampling method but rather tries to approximate the expected value by approximating the corresponding integral over a grid,

$$\mathbb{E}_q[h(\boldsymbol{\theta})] = \int h(\boldsymbol{\theta})q(\boldsymbol{\theta})d\boldsymbol{\theta} \simeq \frac{\sum_{i=1}^N h(\boldsymbol{\theta}_i)q(\boldsymbol{\theta}_i)}{\sum_{i=1}^N q(\boldsymbol{\theta}_i)},$$

where $\boldsymbol{\theta}_i$ form a uniform grid covering the support of q. This method becomes computationally prohibitive if the number of dimensions of $\boldsymbol{\theta}$ is large.

15.2.1.1 The Inverse-CDF Method

Suppose θ is one dimensional and that we have the CDF $F(\theta)$. First, assume θ is continuous and $F(\theta)$ is invertible. Inverse-CDF sampling relies on sampling from the uniform distribution to generate samples for potentially more complex distributions. For $i = 1, \ldots, N$,

1. Generate $U_i \sim \text{Uni}[0, 1]$;
2. Let $\theta_i = F^{-1}(U_i)$.

This works by the inverse-CDF transform, proved in the probability review chapter: if $U \sim \text{Uni}[0, 1]$, then $\theta = F^{-1}(U)$ has CDF F.

The algorithm is slightly modified if F has discontinuities or is not invertible. Specifically, we define $F^{-1}(u) = \min\{x : F(x) \geq u\}$.

15.2.2 Rejection Sampling

In rejection sampling, to produce samples for a distribution q, we first produce samples from another distribution g but then only keep some of the samples produced in a way that the resulting distribution is q. The distribution g needs to satisfy

$$\begin{aligned} &g(\boldsymbol{\theta}) > 0, && \text{if } q(\boldsymbol{\theta}) > 0, \\ &q(\boldsymbol{\theta}) \leq Mg(\boldsymbol{\theta}) && \text{for some known } M \text{ and for all } \boldsymbol{\theta}. \end{aligned}$$

We also need to sample $u \sim \text{Uni}(0, 1)$.

Rejection Sampling

1. Sample $\boldsymbol{\theta}' \sim g$.

2. Sample $U \sim \text{Uni}(0,1)$.
3. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}'$ if $U \leq \frac{q(\boldsymbol{\theta}')}{Mg(\boldsymbol{\theta}')}$. (Accept $\boldsymbol{\theta}'$ as a new sample if $U \leq \frac{q(\boldsymbol{\theta}')}{Mg(\boldsymbol{\theta}')}$; else reject the sample.)

We define the normalizing constants for the distribution,

$$Z_q = \int q(\boldsymbol{\theta}) d\boldsymbol{\theta}, \qquad Z_g = \int g(\boldsymbol{\theta}) d\boldsymbol{\theta}.$$

Note that the probability of a sample being accepted is

$$\begin{aligned} p(accepted) &= \int p(\boldsymbol{\theta}', accepted) d\boldsymbol{\theta}' = \int p(\boldsymbol{\theta}') p(accepted|\boldsymbol{\theta}') d\boldsymbol{\theta}' \\ &= \int \frac{g(\boldsymbol{\theta}')}{Z_g} \cdot \frac{q(\boldsymbol{\theta}')}{Mg(\boldsymbol{\theta}')} d\boldsymbol{\theta}' = \frac{Z_q}{MZ_g}. \end{aligned}$$

Let us now find the distribution for an accepted sample,

$$\begin{aligned} p(\boldsymbol{\theta}) &= p(\boldsymbol{\theta}'|accepted) \\ &= \frac{p(\boldsymbol{\theta}')p(accepted|\boldsymbol{\theta}')}{p(accepted)} \\ &= \frac{\frac{g(\boldsymbol{\theta}')}{Z_g} \cdot \frac{q(\boldsymbol{\theta}')}{Mg(\boldsymbol{\theta}')}}{\frac{Z_q}{MZ_g}} \\ &= \frac{q(\boldsymbol{\theta}')}{Z_q}, \end{aligned}$$

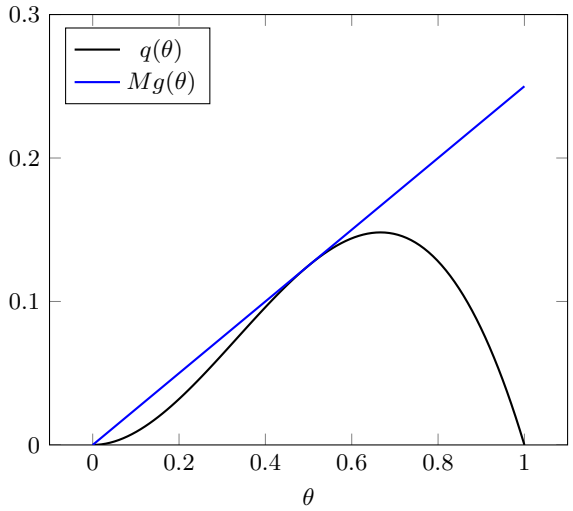
which is the desired distribution.

Rejection sampling does not take advantage of all the samples, unlike importance sampling that we will see next, so in some sense it is inefficient. In particular, if $Z_q = Z_g = 1$, then only a fraction of $\frac{1}{M}$ of the samples will be accepted. If M is large, i.e., g is not a good match for q, then we lose a lot of samples. But rejection sampling has a very important property: it is *self-evaluating*. If we are doing poorly, it is easy to find out by considering the number of samples that are rejected. This is a property that importance sampling lacks.

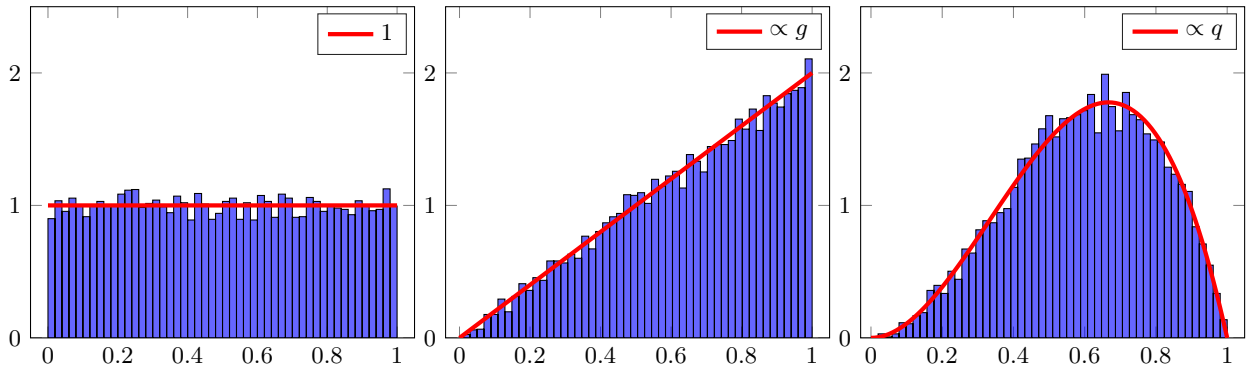
Example 15.1 (Sampling a Beta density)**.** Suppose we need to sample from Beta$(3,2)$ so we let $q(\theta) = \theta^2(1-\theta)$. We would like to do this by sampling from $g(\theta) = \theta$, which we can do using inverse-CDF sampling. First, let us find the required value for M. Observe that

$$Mg(\theta) \geq q(\theta) \iff M\theta \geq \theta^2(1-\theta) \iff M \geq \theta(1-\theta).$$

So the smallest valid value for M is $1/4$, which is what we will choose. Note that in practice, we don't need to find the smallest possible M. For example, here we could argue that the $\theta(1-\theta) \leq 1$ and so it would have been sufficient to let $M = 1$. The plots of q, g are shown below.



To generate samples, first we generate samples from Uni$(0, 1)$, obtaining $S_1 = \{x_1, \ldots, x_N\}$. To generate samples from g, we use the inverse CDF method. The CDF of g is θ^2 and its inverse is $\sqrt{\theta}$. So, our samples become $S_2 = \{\theta'_1, \ldots, \theta'_N\}$, where $\theta'_i = \sqrt{x_i}$. We then accept/reject these based on the rejection sampling rule to obtain S_3, which are samples with distribution q. Specifically, for a sample θ'_i, we accept it with probability $4\theta'_i(1 - \theta'_i)$. Note that this step again requires generating uniform samples, from Uni$(0, 1)$. The graphs below show histograms for x_i, θ'_i and θ_i, as well as the corresponding normalized pdfs. The histograms are normalized so that they are valid pdfs. In this experiment, out of the $N = 1000$ generated samples, 6692 were accepted.



△

15.2.3 Importance Sampling

Again, suppose we are interested in finding

$$\mathbb{E}_q[h(\boldsymbol{\Theta})],$$

where $\mathbb{E}_q$ denotes expectation with respect to distribution q. Now if q is a complicated distribution, we may have a hard time sampling from it. Even if we can sample from q, another issue may arise. The values of $\boldsymbol{\theta}$ such that $h(\boldsymbol{\theta})q(\boldsymbol{\theta})$ are large contribute to the expectation significantly. But $h(\boldsymbol{\theta})q(\boldsymbol{\theta})$ may be large in places where $q(\boldsymbol{\theta})$ is small. So unless we generate a lot of samples, we may not produce one for which $h(\boldsymbol{\theta})q(\boldsymbol{\theta})$ is large, and thus miss significant contributions to the expectation from such points.

Suppose we have a second (possibly unnormalized) distribution $g(\boldsymbol{\theta})$, which is simpler and from which we

can produce samples. Ideally, $g(\boldsymbol{\theta})$ is large if $q(\boldsymbol{\theta})h(\boldsymbol{\theta})$ is large. We have

$$\begin{aligned}\mathbb{E}_q[h(\boldsymbol{\theta})] &= \frac{\int h(\boldsymbol{\theta})q(\boldsymbol{\theta})d\boldsymbol{\theta}}{\int q(\boldsymbol{\theta})d\boldsymbol{\theta}} \\ &= \frac{\int h(\boldsymbol{\theta})[q(\boldsymbol{\theta})/g(\boldsymbol{\theta})]g(\boldsymbol{\theta})d\boldsymbol{\theta}}{\int [q(\boldsymbol{\theta})/g(\boldsymbol{\theta})]g(\boldsymbol{\theta})d\boldsymbol{\theta}} \\ &= \frac{\mathbb{E}_g[h(\boldsymbol{\theta})(q(\boldsymbol{\theta})/g(\boldsymbol{\theta}))]}{\mathbb{E}_g[q(\boldsymbol{\theta})/g(\boldsymbol{\theta})]}.\end{aligned}$$

So we have converted the problem into expectation with respect to g. Define $w(\boldsymbol{\theta}) = \frac{q(\boldsymbol{\theta})}{g(\boldsymbol{\theta})}$ as the importance weight or ratio at $\boldsymbol{\theta}$. Then we can estimate $\mathbb{E}_q[h(\boldsymbol{\theta})]$ as

$$\mathbb{E}_q[h(\boldsymbol{\theta})] = \frac{\mathbb{E}_g[h(\boldsymbol{\theta})w(\boldsymbol{\theta})]}{\mathbb{E}_g[w(\boldsymbol{\theta})]} \simeq \frac{\frac{1}{N}\sum_{i=1}^N h(\boldsymbol{\theta}_i)w(\boldsymbol{\theta}_i)}{\frac{1}{N}\sum_{i=1}^N w(\boldsymbol{\theta}_i)}, \qquad \text{with } \boldsymbol{\theta}_i \sim g(\boldsymbol{\theta}),$$

by producing samples from g rather than q.

Of course, if g is small where $h \times q$ is large, we may miss samples for which $h(\boldsymbol{\theta})g(\boldsymbol{\theta})$ makes significant contributions to the expectation; and this is a drawback of importance sampling.

Example 15.2 (Importance sampling estimate)**.** Let $h(x) = 1 - x$ and $q(x) = x$ for $0 \le x \le 1$. Then

$$\mathbb{E}_q[h(x)] = \int_0^1 (1-x)(2x)dx = \left(x^2 - 2x^3/3\right)_0^1 = 1/3.$$

To estimate this computationally, let $g(x) = 1$. The weights become $w(x) = x$. Generating $N = 100$ samples $x_i \sim \text{Uni}(0, 1)$ using MATLAB, we find

$$\mathbb{E}_q[h(x)] \simeq \frac{\sum_{i=1}^N (1-x_i)x_i}{\sum_{i=1}^N x_i} = 0.34623,$$

which is close to $0.33\cdots$. Of course, for such a simple q we wouldn't resort to importance sampling. △

15.3 Metropolis Monte Carlo

To generate samples from a distribution $p(\boldsymbol{\theta})$, one possible approach is to design a Markov chain whose state space includes all possible values for $\boldsymbol{\theta}$ and its stationary distribution $\boldsymbol{\sigma} = \boldsymbol{\sigma}(\boldsymbol{\theta})$ is equal to the target distribution $p(\boldsymbol{\theta})$. *In the long term, the number of times that the MC spends in a given state is proportional to the probability of that state.* Hence, we can generate samples from the states of the Markov process by letting it run for a long time and record the states that are visited as samples. The distribution of these samples is approximately the same as $\boldsymbol{\sigma}$ and thus the same as $p(\boldsymbol{\theta})$. This is called Markov Chain Monte Carlo (MCMC).

In this section, we present elegant solutions to the challenging problem of finding a MC satisfying $\boldsymbol{\sigma}(\boldsymbol{\theta}) = p(\boldsymbol{\theta})$. In fact, these methods only need $q \propto p$. While MCs can generate samples with the same distribution, we note that the samples are not independent.

We will first discuss the Metropolis algorithm. This algorithm requires a *jump distribution*, $J(\boldsymbol{\theta}'|\boldsymbol{\theta})$, which proposes a new state $\boldsymbol{\theta}'$ given that we are in state $\boldsymbol{\theta}$. We then either move to $\boldsymbol{\theta}'$ or stay at the current state. The jump distribution is chosen in a way that it guarantees $\boldsymbol{\sigma}(\boldsymbol{\theta}) = p(\boldsymbol{\theta})$. We next describe the Metropolis algorithm more formally. We assume θ is one dimensional for simplicity of notation but this is not a requirement.

Metropolis Algorithm

1. Choose θ^0 such that $q(\theta) > 0$.
2. For $t = 1, 2, 3, \cdots$, do
   (a) Generate a proposal θ' based on the jump distribution $J(\theta'|\theta^{t-1})$.
   (b) Move to θ' with probability
   $$r = \frac{q(\theta')}{q(\theta^{t-1})}.$$
   Specifically:
   i. Generate $u \sim \text{Uni}[0, 1]$.
   ii. The next state of the MC, θ^t, is given by
   $$\theta^t = \begin{cases} \theta', & u \leq r \\ \theta^{t-1}, & u > r \end{cases} \tag{15.1}$$

The transition probabilities. The rule (15.1) has interesting implications. Note that if $r > 1$, or equivalently if $q(\theta') > q(\theta^{t-1})$, then we will definitely move to θ'. Otherwise, we move to θ' with probability $r = \frac{q(\theta')}{q(\theta^{t-1})} = \frac{p(\theta')}{p(\theta^{t-1})}$. Define $D = \{\theta : p(\theta) > 0\}$ as the set of all possible values of θ based on target distribution p. If the transition probability of $\theta_a \to \theta_b$ in the MC is denoted by $\Pr(\theta_a \to \theta_b)$, we have

$$\Pr(\theta_a \to \theta_b) = J(\theta_b|\theta_a) \min\left(1, \frac{p(\theta_b)}{p(\theta_a)}\right).$$

The jump distribution. In the Metropolis algorithm, it is not necessary for the jump distribution to have $p(\theta)$ as a stationary distribution. However, the jump distribution $J(\theta'|\theta^{t-1})$ should satisfy certain constraints, discussed below.

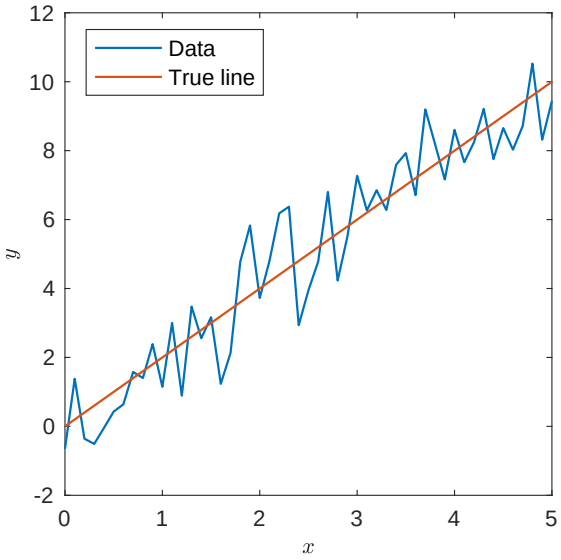
1. *Reachability.* To ensure that the MC is regular, we require that
   $$J(\theta'|\theta) > 0, \qquad \forall \theta, \theta' \in D. \tag{15.2}$$

2. *Symmetry.* For $\theta_a, \theta_b \in D$, the detailed balance property with distribution $\pi(\theta) = p(\theta)$ can be written as
   $$p(\theta_a) \Pr(\theta_a \to \theta_b) = p(\theta_b) \Pr(\theta_b \to \theta_a).$$
   Assume without loss of generality that $p(\theta_a) < p(\theta_b)$. Then, the DBP can be written as
   $$p(\theta_a) J(\theta_b|\theta_a) = p(\theta_b) J(\theta_a|\theta_b) \frac{p(\theta_a)}{p(\theta_b)}.$$
   which is satisfied if the jump distribution is symmetric, i.e.,
   $$J(\theta'|\theta) = J(\theta|\theta'), \qquad \forall \theta, \theta' \in D. \tag{15.3}$$

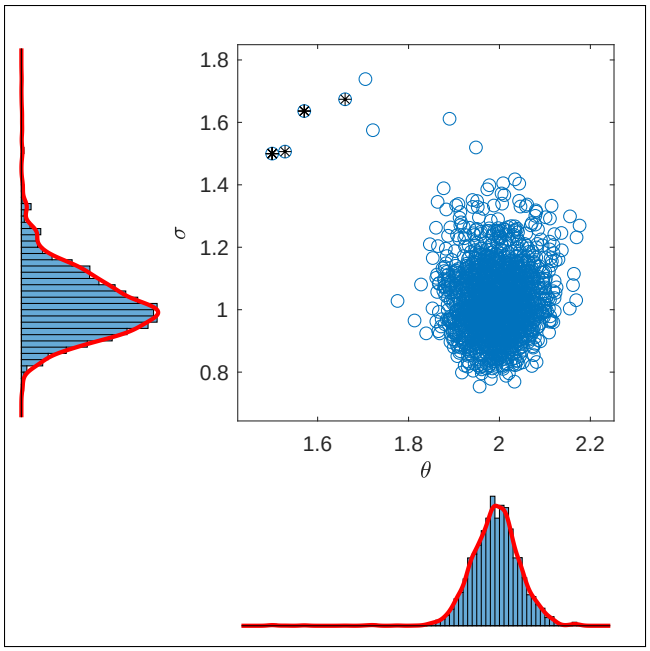
If the jump distribution satisfies (15.2) and (15.3), then the MC is regular and $\boldsymbol{\sigma}(\theta) = p(\theta)$ satisfies the DBP. Hence, $p(\theta)$ is the unique stationary distribution of the Markov chain.

Example 15.3 (Metropolis sampling for regression)**.** Consider a Bayesian regression problem where the data as in Figure 15.1a. The data is generated using the distribution

$$y_i|\theta, \sigma \sim \mathcal{N}(\theta x_i, \sigma^2),$$



(a) Data as well as the true (noiseless) line $y = \theta x$, with $\theta = 2$.



(b) Metropolis sampling for θ and σ. The first 10 samples are marked with $*$.

Figure 15.1: Metropolis sampling for 1-D Bayesian linear regression.

where the true values are $\theta = 2, \sigma = 1$. The figure provides a plot for the samples $\mathcal{D} = \{(x_i, y_i)_{i=1}^N\}$, where $x_i = 0, 0.1, 0.2, \ldots, 5$ as well as the line $y = 2x$.

As we have seen, the Bayesian posteriors for this problem are rather complicated. But it is straightforward to obtain estimates using Metropolis sampling. Assuming the prior $p(\theta, \sigma) \propto 1/\sigma^2$, the posterior is

$$\ln p(\theta, \sigma|\mathcal{D}) \propto -(2+N)\ln\sigma - \frac{1}{2\sigma^2}(\boldsymbol{y} - \theta\boldsymbol{x})'(\boldsymbol{y} - \theta\boldsymbol{x}).$$

We use log-probability because probabilities may be very small and for numerical precision, it is better to work with logs. We can convert these to probabilities if we need to. But in this problem, since we are only interested in the samples, we can keep probabilities in log scale.

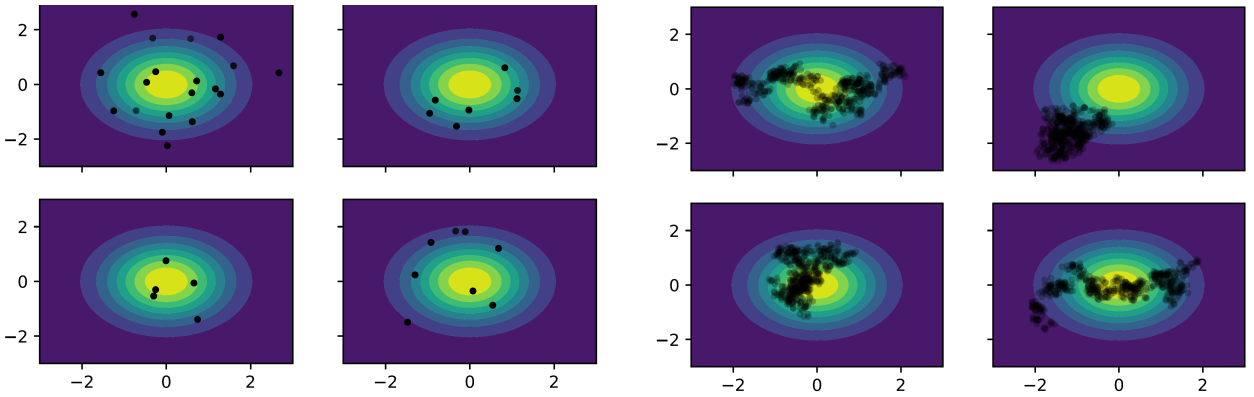
The samples produced by Metropolis are given in Figure 15.1b. As the jump proposal, we use a product of independent Gaussians:

$$\begin{aligned} J(\theta', \sigma'|\theta, \sigma) &= J(\theta'|\theta)J(\sigma'|\sigma), \\ J(\theta'|\theta) &\sim \mathcal{N}(\theta, 0.01), \\ J(\sigma'|\sigma) &\sim \mathcal{N}(\sigma, 0.01). \end{aligned}$$

Based on these samples, the posterior mean for θ is 1.9911 with posterior std 0.055163. The posterior mean for σ is found to be 1.0198. It is also worth noting that the ML estimate for θ is 1.9896. In this example, the estimates are very accurate, which is probably the result of a combination of low noise in the data and chance. △

Metropolis-Hastings algorithm. We can eliminate the symmetry property of the jump distribution if we modify r in the Metropolis algorithm as

$$r = \frac{p(\theta')/J(\theta'|\theta^{t-1})}{p(\theta^{t-1})/J(\theta^{t-1}|\theta')}.$$



(a) Large jumps cause many proposals to be rejected, making the chain stay in its current state.

(b) Small jumps exhibit a random walk behavior which does not necessarily explore the space efficiently.

Figure 15.2: Metropolis sampling with poorly designed jump distributions (4 runs for each case).

Exercise 15.4 (Metropolis-Hastings detailed balance)**.** Prove that with this definition for r, DBP holds even if J is not symmetric. △

Sampling from a MC. Ideally, we should keep only one sample from every m samples for m "large enough" to ensure that the samples are nearly independent. However, there are two issues here:

- It is not easy to determine how large is "large enough."
- If m is too large, the process is inefficient.

However, as long as the empirical distribution (e.g., the histogram) is close to the target distribution, we are not too concerned about independence and the sampling algorithm does not need to throw away any samples, since the order of the samples is not considered. Because the samples at the start states don't satisfy the stationary distribution, it is a good idea to discard the samples produced by the chain at the beginning.

Strong dependence between samples that are close to each other in time could be problematic. For example, suppose we get N samples from a chain whose samples are strongly dependent during intervals of duration not much smaller than N. While each of the N samples may individually have the target distribution, due to strong dependence they all may be from the same area of the probability space and thus the empirical distribution may not look like the target distribution, necessitating obtaining a larger number of samples. This problem can be caused by choosing a poor jump distribution as discussed next.

The Jump distribution. The jumps should be neither too small nor too large!

- When the jumps are large, a large number of proposals will be rejected (we'll stay in the current state) because it is likely that with a large jump, we'll end up with a low probability proposal. In this case, strong dependence manifests as many samples being likely to be equal. An example is shown in Figure 15.2a, where most of the proposals are rejected, resulting in a small number of distinct samples.
- If the jumps are too small, the sampling process is similar to a random walk, because most proposals are accepted but we move only a small step. This means that the MC does not explore the probability space efficiently, again necessitating a large number of samples. An example is given in Figure 15.2b. To see why random walk behavior is not good, consider a random walk with step size ε. How far from

the starting point will we be after T steps? For the random walk, let X_i be the movement in one step:

$$X_i = \begin{cases} \varepsilon, & \text{with } p = \frac{1}{2}; \\ -\varepsilon, & \text{with } p = \frac{1}{2}. \end{cases}$$

After T steps, the expected distance $L = \mathbb{E}\Big[|\sum_{i=1}^{T} X_i|\Big]$ is difficult to find. But we can approximate the distance as

$$L^2 \simeq \mathbb{E}\left[(\sum_{i=1}^{T} X_i)^2\right] = T\varepsilon^2 \text{ (exercise)}.$$

In conclusion, after T steps, we will be approximately at distance $\sqrt{T}\varepsilon$, which is a case of diminishing returns, and not very efficient. In other words, we need $\frac{L^2}{\varepsilon^2}$ steps to move distance L. In the context of MCMC, this means if the probability space has a dimension in which there is a high probability region with length L, we need to run the chain for *at least* $\frac{L^2}{\varepsilon^2}$ steps.

15.4 Gibbs Sampling

At each iteration of the Metropolis algorithm, all the components of $\boldsymbol{\theta}$ are updated at the same time. In Gibbs sampling, for $\boldsymbol{\theta} = (\theta_1, \theta_2, \cdots, \theta_d)$, at each iteration, components are updated one-by-one as

$$\theta_j^t \sim p(\theta_j | \theta_1^t, \cdots, \theta_{(j-1)}^t, \theta_{(j+1)}^{(t-1)}, \cdots, \theta_d^{t-1}), \qquad \text{for } j = 1, \ldots, d.$$

Gibbs sampling may be simpler and more efficient than Metropolis sampling if the joint distribution is too complicated but we can easily sample from the conditional distributions. The components do not need to be one-dimensional necessarily; we can group several dimensions and update each the dimensions in each group simultaneously.

Example 15.5 (Gibbs sampling for bivariate Gaussians)**.** Suppose $\boldsymbol{\theta} = (\theta_1, \theta_2)$ and the observation $\boldsymbol{y} = (y_1, y_2)$ are related by the likelihood

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} | \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right),$$

with the prior $p(\boldsymbol{\theta}) \propto 1$. The posterior distribution $p(\boldsymbol{\theta}|\boldsymbol{y})$ is:

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} | \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right)$$

We can use Gibbs sampling to produce samples for $\boldsymbol{\theta}|\boldsymbol{y}$. The following fact is of use:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right) \Rightarrow x_1 | x_2 \sim \mathcal{N}\left(\mu_1 + \frac{\rho\sigma_1}{\sigma_2}(x_2 - \mu_2), (1-\rho^2)\sigma_1^2 \right)$$

Then, in the t-th iteration, the θ_1^t is sampled by

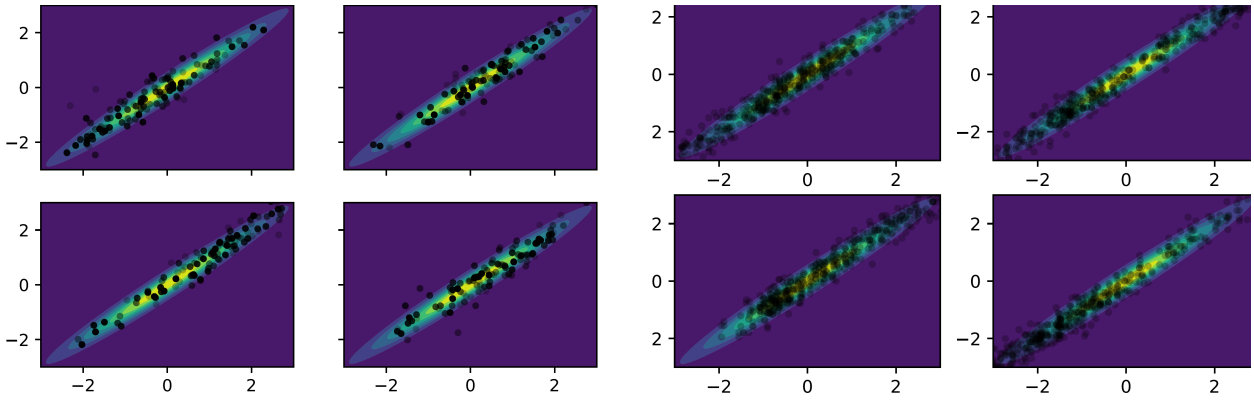
$$\theta_1^t | \theta_2^{t-1} \sim \mathcal{N}\big(y_1 + \rho(\theta_2^{t-1} - y_2), (1-\rho^2)\big).$$

Similarly, the θ_2^t can be updated by

$$\theta_2^t | \theta_1^t \sim \mathcal{N}\big(y_2 + \rho(\theta_1^t - y_1), (1-\rho^2)\big).$$

So we produce a new sample using 1-D distributions. △

Stationary distribution. We prove that Gibbs sampling (with a caveat) satisfies the DBP with distribution $p(\boldsymbol{\theta})$.



(a) Four runs of the Metropolis algorithm.

(b) Four runs of the Gibbs algorithm.

Figure 15.3: Metropolis and Gibbs sampling for highly correlated dimensions. Many proposals for Metropolis are rejected.

Suppose we are in state $\boldsymbol{\theta}$ and we update the jth component to get $\boldsymbol{\theta}'$. We have

$$\theta'_j \sim p(\theta'_j|\boldsymbol{\theta}_{-j}),$$

where $\boldsymbol{\theta}_{-j} = (\theta_1, \ldots, \theta_{j-1}, \theta_{j+1}, \ldots, \theta_d)$. Furthermore, the $\boldsymbol{\theta}'_{-j} = \boldsymbol{\theta}_{-j}$.

To prove DBP for this step, we need to prove $p(\boldsymbol{\theta}) \Pr(\boldsymbol{\theta} \to \boldsymbol{\theta}') = p(\boldsymbol{\theta}') \Pr(\boldsymbol{\theta}' \to \boldsymbol{\theta})$, which holds since

$$\begin{aligned} p(\boldsymbol{\theta}) \Pr(\boldsymbol{\theta} \to \boldsymbol{\theta}') &= p(\boldsymbol{\theta}) p(\theta'_j|\boldsymbol{\theta}_{-j}) = p(\boldsymbol{\theta}_{-j}) p(\theta_j|\boldsymbol{\theta}_{-j}) p(\theta'_j|\boldsymbol{\theta}_{-j}), \\ p(\boldsymbol{\theta}') \Pr(\boldsymbol{\theta}' \to \boldsymbol{\theta}) &= p(\boldsymbol{\theta}') p(\theta_j|\boldsymbol{\theta}'_{-j}) = p(\boldsymbol{\theta}_{-j}) p(\theta'_j|\boldsymbol{\theta}_{-j}) p(\theta_j|\boldsymbol{\theta}_{-j}). \end{aligned}$$

Since the DBP holds for each sub-iteration. We can use this observation to prove that Gibbs sampling works if we choose a component to update at random to produce a new sample or if we update all components in a random order and after a full cycle produce a new sample. Updating the components in a predetermined order works in practice.

Gibbs sampling can be viewed as a special case of Metropolis-Hastings in which the proposal is always accepted and where we don't need to design a jump distribution. Gibbs can use the current state to provide better proposals. An example is shown in Figure 15.3. Here, the dimensions are highly correlated, with most of the probability concentrated in a narrow region. Because of this, many of the Metropolis proposals are rejected. Gibbs, which produces samples based on the conditional distribution given the current state, dose not suffer from this.

Note that θ_j may be independent from some dimensions of $\boldsymbol{\theta}_{-j}$ given others. In particular, if $\boldsymbol{\theta}$ denotes the nodes in a graphical model, given its Markov blanket, θ_j is independent of other elements of $\boldsymbol{\theta}_{-j}$.

15.5 Hamiltonian Monte Carlo **

One problem with the Metropolis algorithm is that, in certain situations, the proposed $\boldsymbol{\theta}'$ by the jump distribution may be rejected too often because $p(\boldsymbol{\theta}')$ is much smaller than $p(\boldsymbol{\theta}^{t-1})$, in which case we will let $\boldsymbol{\theta}^t = \boldsymbol{\theta}^{t-1}$. While the stationary distribution is still $p(\boldsymbol{\theta})$, too many rejection means that it will take a long time to get a sample whose empirical distribution is close to the true distribution.

Let us write our target distribution $p(\boldsymbol{\theta})$ as

$$p(\boldsymbol{\theta}) \propto e^{-E(\boldsymbol{\theta})}$$

and suppose that we can also compute $\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta})$. Note that as $E(\boldsymbol{\theta})$ decreases, the probability increases.

Can we use the fact that we know the gradient to increase the chance of proposals being accepted? At first glance it may seem that we could let $\boldsymbol{\theta}^t = \boldsymbol{\theta}^{t-1} - \epsilon \nabla E(\boldsymbol{\theta})$, similar to gradient descent. But this is a deterministic path rather than a probabilistic MC.

A bit of (questionable) physics. Instead, we use an idea from Hamiltonian Mechanics. We can think of $\boldsymbol{\theta}$ as location and of $E(\boldsymbol{\theta})$ as potential energy. Note that lower potential has a higher probability (a river flows down the valley). Now let us also include momentum (speed) $\boldsymbol{\phi}$, which has the same number of dimensions as $\boldsymbol{\theta}$, in our formulation and define the total energy as

$$H(\boldsymbol{\theta}, \boldsymbol{\phi}) = E(\boldsymbol{\theta}) + K(\boldsymbol{\phi}),$$

where $K(\boldsymbol{\phi})$ is the Kinetic energy

$$K(\boldsymbol{\phi}) = \frac{1}{2}\boldsymbol{\phi}^T \boldsymbol{\phi}.$$

With this physical viewpoint, Hamilton's equations describing the motion of an object with position $\boldsymbol{\theta}$ and momentum $\boldsymbol{\phi}$ are

$$\dot{\boldsymbol{\theta}} = \frac{\partial \boldsymbol{\theta}}{\partial t} = \boldsymbol{\phi}$$
$$\dot{\boldsymbol{\phi}} = \frac{\partial \boldsymbol{\phi}}{\partial t} = -\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta})$$

(A more familiar form of these equations are obtained by representing position with $\boldsymbol{x}$ and speed with $\boldsymbol{v}$. Then, $\dot{\boldsymbol{x}} = \boldsymbol{v}, \dot{\boldsymbol{v}} = -\nabla_{\boldsymbol{x}} E(\boldsymbol{x})$.) It can then be shown that H, the total energy, stays constant in time.

Back to Sampling. Instead of sampling from $p(\boldsymbol{\theta})$, let us define and sample from

$$p(\boldsymbol{\theta}, \boldsymbol{\phi}) \propto e^{-H(\boldsymbol{\theta}, \boldsymbol{\phi})} = e^{-E(\boldsymbol{\theta})} e^{-K(\boldsymbol{\phi})},$$

where $K(\boldsymbol{\phi}) = \frac{1}{2}\boldsymbol{\phi}^T \boldsymbol{\phi}$. We will then discard the $\boldsymbol{\phi}$ component of the samples.

The Hamiltonian Monte Carlo Algorithm is as follows:

1. Randomly choose $\boldsymbol{\theta}^0$ from the domain and choose $\boldsymbol{\phi}^0$ arbitrarily.
2. For $t = 1, 2, ...,$ do
   (a) Pick a random momentum $\boldsymbol{\phi}'$ according to the distribution $p(\boldsymbol{\phi}) \propto e^{-K(\boldsymbol{\phi})}$.
   (b) Starting from $(\boldsymbol{\theta}^{t-1}, \boldsymbol{\phi}')$, simulate the dynamic system for a certain amount of time according to

$$\dot{\boldsymbol{\theta}} = \boldsymbol{\phi},$$
$$\dot{\boldsymbol{\phi}} = -\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta}).$$

   The final values of $(\boldsymbol{\theta}, \boldsymbol{\phi})$ are the new sample, $(\boldsymbol{\theta}^t, \boldsymbol{\phi}^t)$.

It can be shown this process leads to a Markov chain whose stationary distribution is $p(\boldsymbol{\theta}, \boldsymbol{\phi})$. This hinges on step (a) being reversible and step (b) keeping the Hamiltonian and thus the probability constant.

In practice however, we cannot have a perfect simulation. So instead of step (b) above, we perform the following:

(2.b)' For $i = 1, 2, \ldots, L$, perform the following steps, called leapfrog updates:

$$\begin{aligned}\boldsymbol{\phi} &\leftarrow \boldsymbol{\phi} - \frac{1}{2}\epsilon\nabla E(\boldsymbol{\theta}) \\ \boldsymbol{\theta} &\leftarrow \boldsymbol{\theta} + \epsilon\boldsymbol{\phi} \\ \boldsymbol{\phi} &\leftarrow \boldsymbol{\phi} - \frac{1}{2}\epsilon\nabla E(\boldsymbol{\theta})\end{aligned}$$

Let the final $(\boldsymbol{\theta}, \boldsymbol{\phi})$ be denoted by $(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)$. If our simulation is perfect, then this can be accepted as the new state. But because $\epsilon > 0$, we have to perform an accept/reject check similar to Metropolis. That is, we let

$$r = \frac{e^{-H(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)}}{e^{-H(\boldsymbol{\theta}^{t-1}, \boldsymbol{\phi}^{t-1})}}.$$

If $r \geq 1$, we let $(\boldsymbol{\theta}^t, \boldsymbol{\phi}^t) = (\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)$. If $r \leq 1$, then we let $(\boldsymbol{\theta}^t, \boldsymbol{\phi}^t) = (\boldsymbol{\theta}^*, \boldsymbol{\phi}^*)$ with probability r and with probability $1 - r$, we let $(\boldsymbol{\theta}^t, \boldsymbol{\phi}^t) = (\boldsymbol{\theta}^{t-1}, \boldsymbol{\phi}^{t-1})$.

If ϵ is too large, our simulation will be too rough, leading to many rejections. In this case, we decrease ϵ and increase L. On the other hand, if nearly all proposals are accepted, it may be a sign of being too conservative and not exploring the state space as fast as we can, in which case we can be more efficient by increasing ϵ and decreasing L.

Chapter 16

Variational Inference *

Consider the inference problem where our objective is to compute the probability distribution of unknown parameters (Bayesian inference) or latent variables $\boldsymbol{Z}$, conditioned on observations $\boldsymbol{X} = \boldsymbol{x}$. Mathematically, this is represented as

$$p(\boldsymbol{z}|\boldsymbol{x}) = \frac{p(\boldsymbol{x}, \boldsymbol{z})}{\int p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z}} \tag{16.1}$$

Here, the joint distribution $p(\boldsymbol{x}, \boldsymbol{z})$ is typically known, either explicitly defined by the model or through combining $p(\boldsymbol{x}|\boldsymbol{z})$ and $p(\boldsymbol{z})$. However, the integral $\int p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z}$, is challenging to compute, especially in high-dimensional spaces, as it sums over all possible configurations of the latent variables $\boldsymbol{z}$.

While Monte-Carlo methods enable sampling from the target distribution $p(\boldsymbol{z}|\boldsymbol{x})$, they are often computationally intensive. Variational inference, on the other hand, offers a computationally efficient alternative by approximating $p(\boldsymbol{z}|\boldsymbol{x})$ through optimization.

Variational inference simplifies the process by approximating the complex posterior distribution $p(\boldsymbol{z}|\boldsymbol{x})$ with a more tractable distribution $q(\boldsymbol{z})$ from a predefined family Q. The objective is to identify $q^* \in Q$ that is closest to $p(\boldsymbol{z}|\boldsymbol{x})$ as measured by the KL-divergence (reviewed in Chapter 1), $D(q(\boldsymbol{z})||p(\boldsymbol{z}|\boldsymbol{x}))$. The optimization problem thus formulated is

$$q^* = \arg\min_{q \in Q} D(q(\boldsymbol{z})||p(\boldsymbol{z}|\boldsymbol{x})), \tag{16.2}$$

where

$$D(q(\boldsymbol{z})||p(\boldsymbol{z}|\boldsymbol{x})) = \int q(\boldsymbol{z}) \log \frac{q(\boldsymbol{z})}{p(\boldsymbol{z}|\boldsymbol{x})} d\boldsymbol{z} = \mathbb{E}_q \left[\log \frac{q(\boldsymbol{Z})}{p(\boldsymbol{Z}|\boldsymbol{x})} \right], \tag{16.3}$$

with $\mathbb{E}_q$ denoting that the expectation assumes distribution q for $\boldsymbol{Z}$. In variational inference, minimizing $D(q(\boldsymbol{z})||p(\boldsymbol{z}|\boldsymbol{x}))$ is preferred over minimizing $D(p(\boldsymbol{z}|\boldsymbol{x})||q(\boldsymbol{z}))$ because the latter is typically intractable.

Evidence Lower Bound (ELBO): Reformulating the KL-divergence yields:

$$D(q(\boldsymbol{z})||p(\boldsymbol{z}|\boldsymbol{x})) = \int q(\boldsymbol{z}) \log \frac{q(\boldsymbol{z})}{p(\boldsymbol{z}|\boldsymbol{x})} d\boldsymbol{z} \tag{16.4}$$

$$= \int q(\boldsymbol{z}) \log \frac{q(\boldsymbol{z}) p(\boldsymbol{x})}{p(\boldsymbol{z}, \boldsymbol{x})} d\boldsymbol{z} \tag{16.5}$$

$$= \int q(\boldsymbol{z}) \log \frac{q(\boldsymbol{z})}{p(\boldsymbol{z}, \boldsymbol{x})} d\boldsymbol{z} + \log p(\boldsymbol{x}). \tag{16.6}$$

Minimizing the KL-divergence, in this context, is equivalent to maximizing the Evidence Lower Bound (ELBO), defined as:

$$\mathcal{L}(q) = \log p(\boldsymbol{x}) - D(q(\boldsymbol{z})||p(\boldsymbol{z}|\boldsymbol{x})) \tag{16.7}$$

$$= \int q(\boldsymbol{z}) \log \frac{p(\boldsymbol{x}, \boldsymbol{z})}{q(\boldsymbol{z})} d\boldsymbol{z} \tag{16.8}$$

$$= \int q(\boldsymbol{z}) \log p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z} + H(q). \tag{16.9}$$

Note that since $D \geq 0$, ELBO is no greater than $\log p(\boldsymbol{x})$, i.e., $\log p(\boldsymbol{x}) \geq \mathcal{L}(q)$. As $\boldsymbol{x}$ is sometimes referred to as *evidence*, this observation motivates the name "Evidence Lower Bound" or ELBO.

Why does maximizing $\mathcal{L}(q)$ make sense? Let us inspect each term in $\mathcal{L}(q)$. The distribution q that maximizes $\int q(\boldsymbol{z}) \log p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z}$ is the one that puts all the probability mass on $\hat{\boldsymbol{z}} = \arg\max_{\boldsymbol{z}} \log p(\boldsymbol{x}, \boldsymbol{z}) = \arg\max_{\boldsymbol{z}} \log p(\boldsymbol{z}|\boldsymbol{x})$, i.e., the Bayesian mode point estimator. This is a degenerate distribution that tells us that $\boldsymbol{Z}$ is equal to $\hat{\boldsymbol{z}}$ with probability 1. To balance this overconfidence, q that maximizes the second term, $H(q)$, must be high-entropy.

Alternatively, we can rewrite ELBO in the following way to gain more intuition about why maximizing the ELBO gives a reasonable approximation [1]. By (16.8),

$$\mathcal{L}(q) = \int q(\boldsymbol{z}) \log \frac{p(\boldsymbol{z})}{q(\boldsymbol{z})} d\boldsymbol{z} + \int q(\boldsymbol{z}) \log p(\boldsymbol{x}|\boldsymbol{z}) d\boldsymbol{z} \tag{16.10}$$

$$= -D(q(\boldsymbol{z})||p(\boldsymbol{z})) + \int q(\boldsymbol{z}) \log p(\boldsymbol{x}|\boldsymbol{z}) d\boldsymbol{z}. \tag{16.11}$$

The two terms are now the negative KL divergence between $q(\boldsymbol{z})$ and the prior $p(\boldsymbol{z})$, and the expected likelihood assuming $\boldsymbol{Z} \sim q(\boldsymbol{z})$. For the divergence to be small, $q(\boldsymbol{z})$ is encouraged to be close to the prior. On the other hand, for the expected likelihood to be large, $q(\boldsymbol{z})$ should assign more mass to $\boldsymbol{Z}$ that can better explain our observed data $\boldsymbol{x}$, i.e., $\arg\max_{\boldsymbol{z}} p(\boldsymbol{x}|\boldsymbol{z})$. So, the solution balances closeness to the prior with the maximum-likelihood solution, similar to the true posterior.

16.1 Background on Calculus of Variations

Before proceeding further, we review the calculus of variations, a mathematical area that focuses on determining functions that optimize a *functional*. A functional $F : \mathcal{F} \to \mathbb{R}$ is a function that maps the elements of a specified family $\mathcal{F}$ of functions to $\mathbb{R}$. In variational inference, the functional is $D(q(\boldsymbol{z})||p(\boldsymbol{z}|\boldsymbol{x}))$, which assigns a real number to each choice of $q \in \mathcal{Q}$.

A simple class of functionals are those of the form $\int_S J(x, f(x))dz$ for some function J and set S. For the entropy (reviewed in Chapter 1),

$$H[p] = \int_{\mathcal{X}} p(x) \log \frac{1}{p(x)} d(x),$$

we have $J(x, p) = J(p) = p \log \frac{1}{p}$. For KL divergence

$$D(q||p) = \int q(x) \log \frac{q(x)}{p(x)},$$

viewed as a functional of q and for fixed p, we have $J(x, q) = q \log \frac{q}{p(x)}$.

The **functional differential** of F at f in the direction of ϕ is defined as

$$\lim_{\epsilon \to 0} \frac{F[f + \epsilon\phi] - F[f]}{\epsilon} = \left. \frac{dF[f + \epsilon\phi]}{d\epsilon} \right|_{\epsilon=0}$$

This tells us how the functional changes at f if it is perturbed by moving infinitesimally in the "direction"

of ϕ. This quantity is useful for optimizing a functional, that is, for finding a function that maximizes or minimizes the functional. We will explore this concept through an analogy to multivariate calculus.

Analogy from multivariate calculus: Consider $g : \mathbb{R}^n \to \mathbb{R}$, a function that assigns to each vector $\boldsymbol{x} \in \mathbb{R}^n$ a real number $g(\boldsymbol{x})$. If we are at $\boldsymbol{x}$, how does g change if we move in the direction of some vector $\boldsymbol{v}$? The change in g can be quantified by

$$\lim_{\epsilon \to 0} \frac{g(\boldsymbol{x} + \epsilon \boldsymbol{v}) - g(\boldsymbol{x})}{\epsilon}$$

This is useful for optimizing g. For instance, if it is 0 for all $\boldsymbol{v}$, then we are at a local extremum. But for each vector $\boldsymbol{v}$, we would need to compute it from scratch. To address this, we define the gradient

$$\nabla g = \left(\lim_{\epsilon \to 0} \frac{g(\boldsymbol{x} + \epsilon \mathbf{i}_1) - g(\boldsymbol{x})}{\epsilon}, \ldots, \lim_{\epsilon \to 0} \frac{g(x + \epsilon \mathbf{i}_n) - g(\boldsymbol{x})}{\epsilon} \right),$$

where $\mathbf{i}_1, \ldots, \mathbf{i}_n$ are unit vectors in the standard basis. Then we can find the rate of change for any vector $\boldsymbol{v}$ as the inner product of the gradient and the $\boldsymbol{v}$

$$\lim_{\epsilon \to 0} \frac{g(\boldsymbol{x} + \epsilon \boldsymbol{v}) - g(\boldsymbol{x})}{\epsilon} = \langle \nabla g, \boldsymbol{v} \rangle.$$

Furthermore, for small ϵ,

$$g(\boldsymbol{x} + \epsilon \boldsymbol{v}) \simeq g(\boldsymbol{x}) + \epsilon \langle \nabla g, \boldsymbol{v} \rangle.$$

Back to functionals: For functionals of the form $F[f] = \int_S J(x, f(x))dx$, we can find something similar to a gradient. Specifically, there is a function $\frac{\partial F}{\partial f}(x)$, called a **functional derivative** such that

$$\left. \frac{dF[f + \epsilon \phi]}{d\epsilon} \right|_{\epsilon = 0} = \int_{\mathcal{X}} \frac{\partial F}{\partial f}(x) \phi(x) dx$$

This derivative measures how the functional $F[f]$ changes when the function f is perturbed infinitesimally at the point x.

The Defining the inner product in the space of functions as $\langle f(x), g(x) \rangle = \int_{\mathcal{X}} f(x) g(x) dx$ for some predetermined set $\mathcal{X}$, we can write

$$\left. \frac{dF[f + \epsilon \phi]}{d\epsilon} \right|_{\epsilon = 0} \langle \frac{\partial F}{\partial f}(x), \phi(x) \rangle$$

$$F[f + \epsilon \phi] \simeq F[f] + \epsilon \langle \frac{\partial F}{\partial f}(x), \phi(x) \rangle$$

Observe that

$$\begin{aligned} \left. \frac{dF[f + \epsilon \phi]}{d\epsilon} \right|_{\epsilon = 0} &= \frac{d}{d\epsilon} \int J(x, f(x) + \epsilon \phi(x)) dx \Big|_{\epsilon = 0} \\ &= \int \frac{d}{d\epsilon} J(x, f(x) + \epsilon \phi(x)) dx \Big|_{\epsilon = 0} \\ &= \int J_2(x, f(x)) \phi(x) dx, \end{aligned}$$

where J_2 is the prtial dervative of J with respect to its second argument.

Hence,

$$\frac{\partial F}{\partial f}(x) = J_2(x, f(x))$$

Example 16.1 (Functional derivative of entropy)**.** Let us find $\frac{\partial H[p]}{\partial p}(x)$ where H is the entropy function. Here, we have $J(x,p) = p \log \frac{1}{p}$. Hence,

$$\frac{\partial H[p]}{\partial p}(x) = \frac{\partial (p \log \frac{1}{p})}{\partial p}(x) = \log \frac{1}{p(x)} - 1.$$

△

Example 16.2 (Functional derivative of KL divergence)**.** For fixed p, let us find $\frac{\partial D(q||p)}{\partial q}(x)$. Here, we have $J(x,q) = q \log \frac{q}{p(x)}$. Hence,

$$\frac{\partial D}{\partial q}(x) = \frac{\partial (q \log \frac{q}{p(x)})}{\partial q}(x) = 1 + \log \frac{q(x)}{p(x)}.$$

△

Optimization of Functionals: Now that we have functional derivatives, we can optimized functionals by setting the derivative to 0. When we have constrained, we can use Lagrange multipliers.

Example 16.3 (Maximum-entropy distribution)**.** We find the distribution with the highest possible entropy with variance at most σ^2, i.e.,

$$\begin{aligned} &\text{maximize } H[p] \\ &\text{s.t. } S[p] = \int p(x)dx = 1 \\ &\quad V[p] = \int p(x)x^2 dx = 1 \end{aligned}$$

Using Lagrange multipliers:

$$\frac{\partial H}{\partial p}(x) + \lambda_1 \frac{\partial S}{\partial p}(x) + \lambda_2 \frac{\partial V}{\partial p}(x) = 0$$

Hence,

$$\log \frac{1}{p(x)} + \lambda_1 + \lambda_2 x^2 = 0 \Rightarrow p(x) = e^{\lambda_1 + \lambda_2 x^2}.$$

This is a Gaussian distribution. Since we know which Gaussian distribution has variance $sigma^2$, we have

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{x^2/\sigma^2}.$$

We could also find the constants by solving the constraint equations. Note that the mean is arbitrary. △

Example 16.4 (KL divergence minimization)**.** For fixed p, let us find the distribution that minimizes $D(q||p)$, i.e.,

$$\begin{aligned} &\text{maximize } D(q||p) \\ &\text{s.t. } S[q] = \int q(x)dx = 1 \end{aligned}$$

Again, using Lagrange multipliers, we have

$$1 + \log \frac{q(x)}{p(x)} + \lambda_1 = 0 \Rightarrow q(x) = q(x) \propto p(x),$$

which, along with the constraint, leads to

$$q(x) = p(x).$$

△

16.2 Mean-field variational inference

In this chapter, we restrict the "nice" family $\mathcal{Q}$ to be the family of distributions that factorize (being tractable is important!), i.e.,

$$q(\boldsymbol{z}) = \prod_{j=1}^{J} q_j(z_j), \tag{16.12}$$

where $z_1, z_2, \ldots, z_J$ form a partition of all hidden variables in $\boldsymbol{z}$. This is called the **mean-field** approximation and leads to

$$\mathcal{L}(q) = \int \prod_{j=1}^{J} q_j(z_j) \log p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z} + \sum_{j=1}^{J} H(q_j). \tag{16.13}$$

Coordinate ascent variational inference (CAVI) $\mathcal{L}(q)$ is a functional of J functions. The most common way for optimizing (16.13) is coordinate ascent. In other words, we will take turns to optimize $\mathcal{L}(q)$ with respect to one component q_i while fixing the others $q_j, j \neq i$. Now, let us assume that we fix q_j for all $j \neq i$. We can write the ELBO as

$$\mathcal{L}(q) = \int \prod_{j=1}^{J} q_j(z_j) \log p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z} + \sum_{j=1}^{J} H(q_j) \tag{16.14}$$

$$= \sum_{j \neq i} H(q_j) + H(q_i) + \int q_i(z_i) \left(\int \prod_{j \neq i} q_j(z_j) \log p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z}_{-i} \right) dz_i \tag{16.15}$$

$$= \sum_{j \neq i} H(q_j) + H(q_i) + \int q_i(z_i) \tilde{f}_i(z_i) dz_i, \tag{16.16}$$

where $\boldsymbol{z}_{-i} = \{z_j\}_{j \neq i}$ and

$$\tilde{f}_i(z_i) = \int \prod_{j \neq i} q_j(z_j) \log p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z}_{-i} \tag{16.17}$$

$$= \int \boldsymbol{q}_{-i}(\boldsymbol{z}_{-i}) \log p(\boldsymbol{x}, \boldsymbol{z}) d\boldsymbol{z}_{-i} \tag{16.18}$$

$$= \mathbb{E}_{\boldsymbol{z}_{-i} \sim \boldsymbol{q}_{-i}}[\log p(\boldsymbol{x}, \boldsymbol{z})], \tag{16.19}$$

where $\boldsymbol{q}_{-i} = \{q_j\}_{j \neq i}$.

Taking the derivative, we have

$$\frac{\partial \mathcal{L}(q)}{\partial q_i}(z_i) = \log \frac{1}{q_i(z_i)} - 1 + \tilde{f}_i(z_i) = 0 \Rightarrow q_i(z_i) \propto \exp(\tilde{f}_i(z_i)).$$

So we update q_i to

$$q_i^*(z_i) = \frac{\exp\Big(\tilde{f}_i(z_i)\Big)}{\int \exp\Big(\tilde{f}_i(z_i)\Big) dz_i}. \tag{16.20}$$

So $q_i^*(z_i)$ is also a distribution over z_i and since $\tilde{f}_i(z_i)$ is a function of z_i and does not depend on q_i, neither

does $\int \exp\Big(\tilde{f}_i(z_i)\Big) dz_i$.

We summarize the above process in the following algorithm.

Algorithm 1 Coordinate ascent variational inference (CAVI)

Input: visible variables $\boldsymbol{x}$; latent variables $\boldsymbol{z} = (z_1, \ldots, z_J)$; joint distribution $p(\boldsymbol{x}, \boldsymbol{z})$;
Output: an approximation for $p(\boldsymbol{z}|\boldsymbol{x})$;
Initialize distributions $q_1, \ldots, q_J$ over $z_1, \ldots, z_J$, respectively;
while not converged **do**
  for $i = 1$ to J **do**
    $\tilde{f}_i(z_i) = \mathbb{E}_{\boldsymbol{z}_{-i} \sim \boldsymbol{q}_{-i}}[\log p(\boldsymbol{x}, \boldsymbol{z})]$;
    $q_i(z_i) = \frac{\exp\big(\tilde{f}_i(z_i)\big)}{\int \exp\big(\tilde{f}_i(z_i)\big) dz_i}$;
  end for
end while

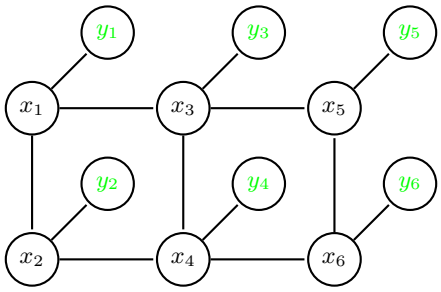
Note that the update rule (16.20) is given in the form of a function involving an integration. In actual implementation, we often derive a parametric form based on $q_i^*(z_i) \propto \exp\Big(\tilde{f}_i(z_i)\Big)$ and perform update over the "variational" parameters. Especially when variables z_i are discrete, we can always represent q_i by $k-1$ parameters, where k is the number of possible values that z_i can take.

16.3 Examples

Next, let us take a look at two examples, one in discrete case and the other in continuous case. The examples are adopted from [3] and [1].

16.3.1 CAVI on a MRF for image denoising

Consider the task of denoising an image using the following MRF



with energy function

$$E(\boldsymbol{x}, \boldsymbol{y}) = -\sum_{i=1}^{m} \alpha_i x_i - \sum_{(i,j)\in\mathcal{E}(G)} \beta_{i,j} x_i x_j - \sum_{i=1}^{m} \zeta_i x_i y_i,$$

where $\mathcal{E}(G)$ is the set of edges between neighboring pixels and $\beta_{i,j}, \zeta_i > 0$. In this task, the visible variables are the noisy pixels y_i and hidden variables are pixels x_i. All variables are discrete and take values in $\{+1, -1\}$.

To recover the original image based on its noisy version, let us apply CAVI to obtain the distribution of $\boldsymbol{x}$ given $\boldsymbol{y}$. The joint distribution $p(\boldsymbol{x}, \boldsymbol{y})$ is

$$p(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{Z} e^{-E(\boldsymbol{x}, \boldsymbol{y})}, \quad Z = \sum_{\boldsymbol{x}} \sum_{\boldsymbol{y}} e^{-E(\boldsymbol{x}, \boldsymbol{y})}. \tag{16.21}$$

We now assume a distribution $q(\boldsymbol{x})$ that factorizes:

$$q(\boldsymbol{x}) = \prod_{i=1}^{m} q_i(x_i). \tag{16.22}$$

Let $\mathbb{E}_{q_i}[x_i] = \mu_i$. Since every x_i takes two values, it suffices to optimize the ELBO over μ_i. We have

$$\begin{aligned}
\log q_i^*(x_i) &= \mathbb{E}_{\boldsymbol{q}_{-i}}[\log p(\boldsymbol{x}, \boldsymbol{y})] + \text{const} && (16.23)\\
&= \mathbb{E}_{\boldsymbol{q}_{-i}}[-E(\boldsymbol{x}, \boldsymbol{y}) - \log Z] + \text{const} && (16.24)\\
&= \mathbb{E}_{\boldsymbol{q}_{-i}}\left[\sum_{i}^{m} \alpha_i x_i + \sum_{(i,j)\in\mathcal{E}(G)} \beta_{i,j} x_i x_j + \sum_{i}^{m} \zeta_i x_i y_i - \log Z\right] + \text{const} && (16.25)\\
&= \mathbb{E}_{\boldsymbol{q}_{-i}}\left[\alpha_i x_i + \sum_{j\in\mathcal{E}(x_i)} \beta_{i,j} x_i x_j + \zeta_i x_i y_i\right] + \text{const} && (16.26)\\
&= \alpha_i x_i + \sum_{j\in\mathcal{E}(x_i)} \beta_{i,j} x_i \mu_j + \zeta_i x_i y_i + \text{const}, && (16.27)
\end{aligned}$$

where $\mathcal{E}(x_i)$ is the set of neighbors of x_i.

It follows that

$$q_i^*(x_i = 1) = \frac{e^{f_i}}{e^{f_i} + e^{-f_i}} = \frac{1}{1 + e^{-2f_i}}, \tag{16.28}$$

where $f_i = \alpha_i + \sum_{j\in\mathcal{E}(x_i)} \beta_{i,j}\mu_j + \zeta_i y_i$. Hence, the updating rules are given by

$$\mu_i^* = +1 \cdot q_i^*(x_i = 1) + (-1) \cdot q_i^*(x_i = -1) = \frac{1}{1 + e^{-2f_i}} - \frac{1}{1 + e^{2f_i}}. \tag{16.29}$$

16.3.2 Bayesian estimation of a univariate Gaussian [3]

Another application where we need to do inference about hidden variables given the visible ones is in Bayesian estimation. For a prior $p(\boldsymbol{\theta})$ and evidence $p(\mathcal{D}|\boldsymbol{\theta})$, we find an approximation for the posterior $p(\boldsymbol{\theta}|\mathcal{D}) \propto p(\boldsymbol{\theta})p(\mathcal{D}|\boldsymbol{\theta})$ by maximizing the ELBO

$$\mathcal{L}(q) = \log p(\mathcal{D}) - KL(q(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathcal{D})) = \int q(\boldsymbol{\theta}) \log p(\mathcal{D}, \boldsymbol{\theta}) d\boldsymbol{\theta} + H(q). \tag{16.30}$$

univariate Guassian Consider Bayesian modeling of a univariate Gaussian. Let our data x follow a Gaussian distribution $\mathcal{N}\left(\mu, \lambda^{-1}\right)$, where λ is the precision. Here we use precision λ as the parameter instead of the variance to simplify our computation.

The likelihood is thus given by

$$p(\mathcal{D}|\mu, \lambda) = \left(\frac{\lambda}{2\pi}\right)^{N/2} \prod_{n=1}^{N} \exp\left(-\frac{\lambda}{2}(x_n - \mu)^2\right). \tag{16.31}$$

We pick the conjugate *Gaussian-Gamma prior* of the form

$$p(\lambda; a_0, b_0) = \text{Gamma}(a_0, b_0) = \frac{\lambda^{a_0-1}\exp(-b_0\lambda)b_0^{a_0}}{\Gamma(a_0)}, \tag{16.32}$$

$$p(\mu|\lambda; \mu_0, \kappa_0) = \mathcal{N}\Big(\mu_0, (\kappa_0\lambda)^{-1}\Big) = \left(\frac{\kappa_0\lambda}{2}\right)^{1/2}\exp\left(-\frac{\kappa_0\lambda}{2}(\mu-\mu_0)^2\right), \tag{16.33}$$

$$p(\mu, \lambda; \mu_0, \kappa_0, a_0, b_0) = \text{GaussGamma}(\mu_0, \kappa_0, a_0, b_0) \tag{16.34}$$

$$\propto \lambda^{a_0-\frac{1}{2}}\exp(-b_0\lambda)\exp\Big(-\frac{\kappa_0}{2}(\mu-\mu_0)^2\lambda\Big). \tag{16.35}$$

Then, $\mathbb{E}\,\lambda = a_0/b_0, \mathbb{E}\,\mu = \mu_0, \text{Var}[\lambda] = a_0/b_0^2, \text{Var}[\mu] = b_0/(\kappa_0(a_0-1))$.

We are interested in the posterior

$$p(\mu, \lambda|\mathcal{D}) \propto p(\mu, \lambda)p(\mathcal{D}|\mu, \lambda). \tag{16.36}$$

Exact posterior ** The exact posterior can be shown to be

$$p(\mu, \lambda|\mathcal{D}) = \text{GaussGamma}\left(\frac{\kappa_0\mu_0 + N\bar{x}}{\kappa_0+N}, \kappa_0+N, a_0+\frac{N}{2}, b_0+\frac{1}{2}\left(Ns + \frac{\kappa_0 N(\bar{x}-\mu_0)^2}{\kappa_0+N}\right)\right), \tag{16.37}$$

where $\bar{x} = \frac{1}{N}\sum_{n=1}^N x_n, s = \frac{1}{N}\sum_{n=1}^N (x_n-\bar{x})^2$.

Approximate posterior Next, we approximate $p(\mu, \lambda|\mathcal{D})$ by

$$q(\mu, \lambda) = q_\mu(\mu)q_\lambda(\lambda). \tag{16.38}$$

Let us derive the updating rules needed by CAVI. Suppose we begin with two guesses $q_\mu(\mu)$ and $q_\lambda(\lambda)$. By (16.20),

$$\log q_\mu^*(\mu) = \mathbb{E}_{q_\lambda}[\log p(\mathcal{D}, \mu, \lambda)] + \text{const} \tag{16.39}$$

$$= \mathbb{E}_{q_\lambda}[\log p(\mathcal{D}|\mu, \lambda) + \log p(\mu|\lambda)] + \text{const} \tag{16.40}$$

$$= \mathbb{E}_{q_\lambda}\left[-\frac{\lambda}{2}\left(\sum_{n=1}^N (x_n-\mu)^2 + \kappa_0(\mu-\mu_0)^2\right)\right] + \text{const} \tag{16.41}$$

$$= -\frac{\mathbb{E}_{q_\lambda}[\lambda]}{2}\left(\sum_{n=1}^N (x_n-\mu)^2 + \kappa_0(\mu-\mu_0)^2\right) + \text{const} \tag{16.42}$$

$$\Rightarrow \quad q_\mu^*(\mu) \sim \mathcal{N}\big(\nu, \tau^{-1}\big), \quad \nu = \frac{\kappa_0\mu_0 + \sum_{n=1}^N x_n}{N+\kappa_0}, \quad \tau = (N+\kappa_0)\mathbb{E}_{q_\lambda}[\lambda]. \tag{16.43}$$

Further,

$$\log q_\lambda^*(\lambda) = \mathbb{E}_{q_\mu}[\log p(\mathcal{D}, \mu, \lambda)] + \text{const} \tag{16.44}$$
$$= \mathbb{E}_{q_\mu}[\log p(\mathcal{D}|\mu, \lambda) + \log p(\mu|\lambda) + \log p(\lambda)] + \text{const} \tag{16.45}$$
$$= \mathbb{E}_{q_\mu}\left[\frac{N}{2}\log\left(\frac{\lambda}{2\pi}\right) + \sum_{n=1}^{N}\left(-\frac{\lambda}{2}(x_n-\mu)^2\right) + \frac{1}{2}\log\left(\frac{\kappa_0\lambda}{2}\right) + \left(-\frac{\kappa_0\lambda}{2}(\mu-\mu_0)^2\right)\right. \tag{16.46}$$
$$\left. + (a_0-1)\log\lambda + (-b_0\lambda)\right] + \text{const} \tag{16.47}$$
$$= \mathbb{E}_{q_\mu}\left[\left(\frac{N+1}{2} + a_0 - 1\right)\log\lambda + \left(-\frac{1}{2}\sum_{n=1}^{N}(x_n-\mu)^2 - \frac{\kappa_0}{2}(\mu-\mu_0)^2 - b_0\right)\lambda\right] + \text{const} \tag{16.48}$$
$$= \left(\frac{N+1}{2} + a_0 - 1\right)\log\lambda - \left(b_0 + \mathbb{E}_{q_\mu}\left[\frac{1}{2}\sum_{n=1}^{N}(x_n-\mu)^2 + \frac{\kappa_0}{2}(\mu-\mu_0)^2\right]\lambda\right) + \text{const} \tag{16.49}$$
$$\tag{16.50}$$
$$\Rightarrow \quad q_\lambda^*(\lambda) \sim \text{Gamma}(a, b), \tag{16.51}$$

where

$$a = \frac{N+1}{2} + a_0, \quad b = b_0 + \frac{1}{2}\mathbb{E}_{q_\mu}\left[\sum_{n=1}^{N}(x_n-\mu)^2 + \kappa_0(\mu-\mu_0)^2\right]. \tag{16.52}$$

As we can see from (16.43) and (16.51), q_μ is a Gaussian and q_λ is a Gamma. Therefore, in practice, we can initialize with these parametric forms and do updating on their parameters. Note that we did not specify the Gaussian and Gamma parametric forms beforehand.

The updating rules for parameters ν, τ, a, b are thus

$$\nu = \frac{\kappa_0\mu_0 + \sum_{n=1}^{N} x_n}{N + \kappa_0}, \tag{16.53}$$
$$\tau = (N+\kappa_0)\mathbb{E}_{q_\lambda}[\lambda] = (N+\kappa_0)\frac{a}{b}, \tag{16.54}$$
$$a = \frac{N+1}{2} + a_0, \tag{16.55}$$
$$b = b_0 + \frac{1}{2}\mathbb{E}_{q_\mu}\left[\sum_{n=1}^{N}(x_n-\mu)^2 + \kappa_0(\mu-\mu_0)^2\right] \tag{16.56}$$
$$= b_0 + \frac{1}{2}\left(\left(\sum_{n=1}^{N} x_n^2\right) + \kappa_0\mu_0^2 - 2\left(\sum_{n=1}^{N} x_n + \kappa_0\mu_0\right)\mathbb{E}_{q_\mu}[\mu] + (N+\kappa_0)\mathbb{E}_{q_\mu}[\mu^2]\right) \tag{16.57}$$
$$= b_0 + \frac{1}{2}\left(\left(\sum_{n=1}^{N} x_n^2\right) + \kappa_0\mu_0^2 - 2\left(\sum_{n=1}^{N} x_n + \kappa_0\mu_0\right)\nu + (N+\kappa_0)\left(\nu^2 + \tau^{-1}\right)\right). \tag{16.58}$$

Figure 16.1 shows the updates when we apply CAVI to approximate the posterior of Gaussian parameters.

16.4 Factorized variational approximations are compact

The variational approximations $q(\boldsymbol{z})$ tend to be more compact than the actual posterior $p(\boldsymbol{z}|\boldsymbol{x})$. This is partly due to the natural asymmetry of KL-divergence. Consider that we approximate $p(x)$ using $q(x)$ by

minimizing

$$D(q(x)||p(x)) = \sum_x q(x) \log \frac{q(x)}{p(x)}. \tag{16.59}$$

It can be seen that when $p(x)$ is close to 0, $q(x)$ being large will contribute a large positive value to the KL. Therefore, to minimize $D(q(x)||p(x))$, wherever $p(x)$ is small, $q(x)$ must also be small. $q(x)$ thus has a tendency of "shrinking" to only regions where $p(x)$ is not close to 0, shown in Figure 16.2a.

On the other hand, if we instead minimize

$$D(p(x)||q(x)) = \sum_x p(x) \log \frac{p(x)}{q(x)}, \tag{16.60}$$

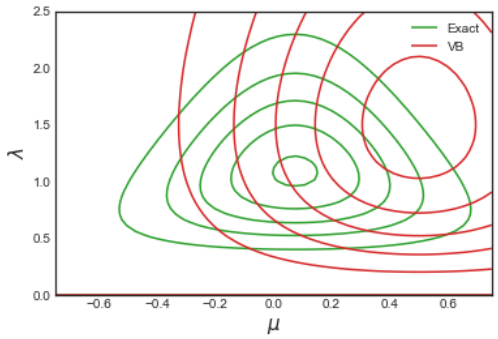
then wherever $p(x)$ is large, $q(x)$ must also be large. Therefore, $q(x)$ will have a tendency of "covering" regions where $p(x)$ is positive, shown in Figure 16.2b.

References

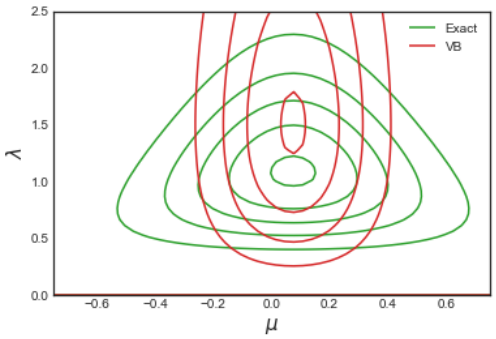
[1] David M Blei, Alp Kucukelbir, and Jon D McAuliffe. "Variational inference: A review for statisticians." In: *Journal of the American statistical Association* 112.518 (2017), pp. 859–877.

[2] Ian Goodfellow. "NIPS 2016 tutorial: Generative adversarial networks." In: *arXiv preprint arXiv:1701.00160* (2016).

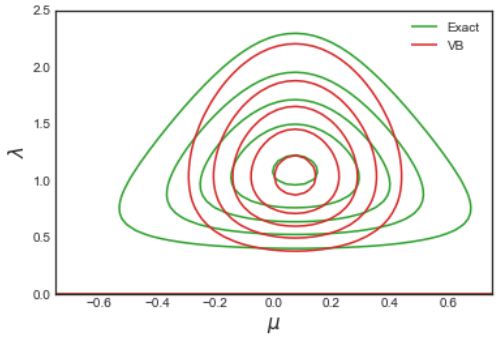
[3] Kevin P Murphy. *Probabilistic machine learning: an introduction*. MIT press, 2022.



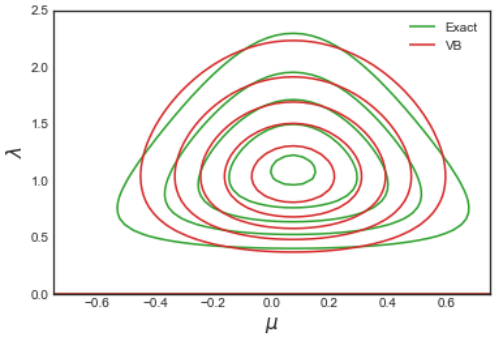
(a) Initial estimates



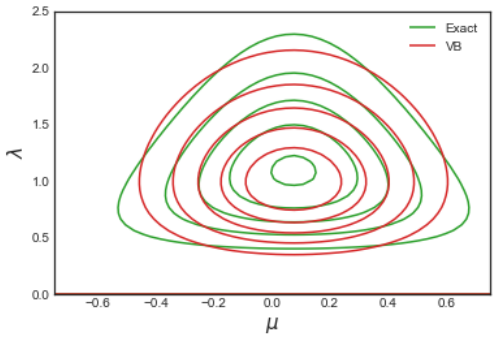
(b) Update q_μ once



(c) Update q_λ once

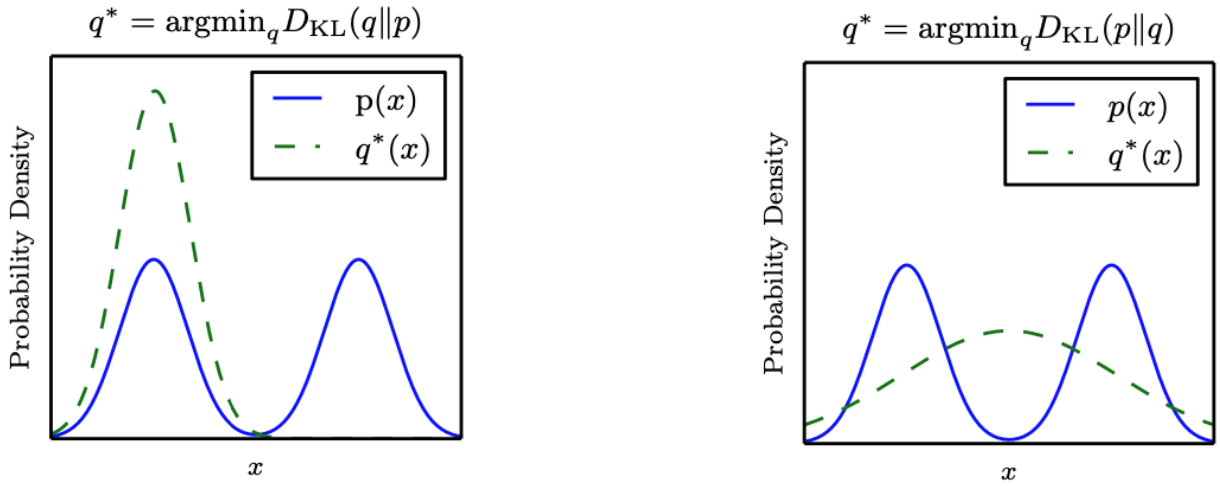


(d) Update q_μ for the second time



(e) Update q_λ for the second time

Figure 16.1: CAVI for the mean μ and precision λ of a univariate Gaussian distribution.



(a) (b)

Figure 16.2: Approximating a bimodal distribution with a uni-modal distribution. Figures are from [2].

Chapter 17

Appendix

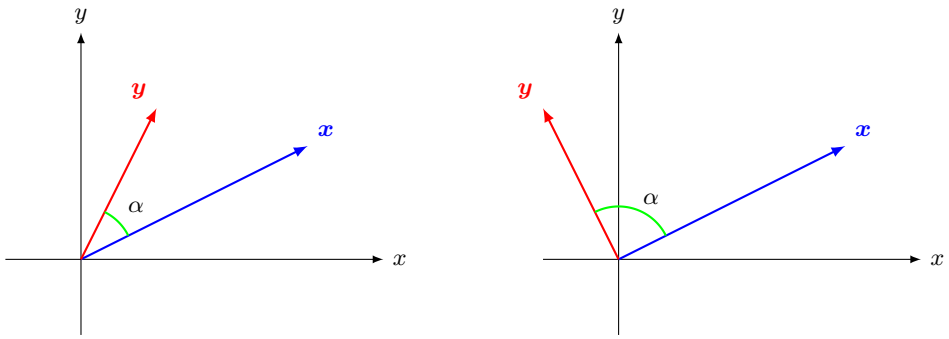
17.1 Review of Linear Algebra

For two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, the **inner product** $\langle \boldsymbol{x}, \boldsymbol{y} \rangle$ of $\boldsymbol{x}$ and $\boldsymbol{y}$ is

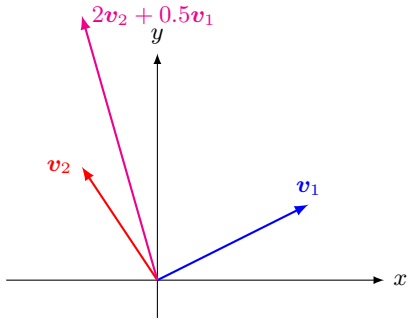
$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \boldsymbol{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^T \boldsymbol{y} = \sum_{i=1}^{n} x_i y_i. \tag{17.1}$$

where $\boldsymbol{x}^T$ is the transpose of $\boldsymbol{x}$.

The **length** or the ℓ_2 norm of a vector $\boldsymbol{x}$ is $\|\boldsymbol{x}\| = \|\boldsymbol{x}\|_2 = \sqrt{\boldsymbol{x}^T \boldsymbol{x}}$ and we have $\|\boldsymbol{x}\|_2^2 = \boldsymbol{x}^T \boldsymbol{x}$. Let α be the angle between $\boldsymbol{x}$ and $\boldsymbol{y}$. Then $\boldsymbol{x}^T \boldsymbol{y} = \|\boldsymbol{x}\| \|\boldsymbol{y}\| \cos \alpha$. If $\boldsymbol{x}^T \boldsymbol{y} = 0$, then the two are called **orthogonal**.



For a collection of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$, a **linear combination** of these is any vector of the form $a_1 \boldsymbol{v}_1 + \cdots + a_m \boldsymbol{v}_m$, $a_i \in \mathbb{R}$. The set of all linear combinations of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ is their **span** and denoted as $\text{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$. This is a **subspace** (think line, plane, or the whole space). For a matrix A, the span of the columns of A is the **column space** of A.

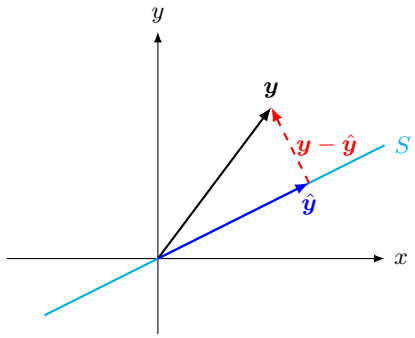


The vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ are **linearly independent** if there is no vector among them that can be written as a linear combination of the others, and linearly dependent otherwise. The vectors are linearly independent if and only if the only values for $a_1, \ldots, a_m$ satisfying $a_1\boldsymbol{v}_1 + \cdots + a_m\boldsymbol{v}_m = \boldsymbol{0}$ are $a_1, \ldots, a_m = 0$. In particular, the columns of a matrix A are linearly independent if and only if the only vector $\boldsymbol{a}$ satisfying $\mathsf{A}\boldsymbol{a} = \boldsymbol{0}$ is $\boldsymbol{a} = \boldsymbol{0}$.

The **inverse** of a square matrix A is a matrix A^{-1} such that $\mathsf{A}\mathsf{A}^{-1} = \mathsf{A}^{-1}\mathsf{A} = I$, where I is the **identity matrix**, which has 1s on the diagonal and 0s elsewhere. A matrix that has an inverse is called **invertible**. For a square matrix A, the following are equivalent:

- It is invertible.
- For all distinct vectors $\boldsymbol{a}$ and $\boldsymbol{b}$, we have $\mathsf{A}\boldsymbol{a} \neq \mathsf{A}\boldsymbol{b}$.
- The only solution to $\mathsf{A}\boldsymbol{x} = \boldsymbol{0}$ is $\boldsymbol{x} = \boldsymbol{0}$.
- Its columns are linearly independent.
- Its determinant $|\mathsf{A}|$ is nonzero. (We also have $|\mathsf{A}^{-1}| = \frac{1}{|\mathsf{A}|}$.)

Given a subspace S (e.g., a plane or the column space of a matrix) and a vector $\boldsymbol{y}$, let $\hat{\boldsymbol{y}}$ be the vector in the subspace that is closest to $\boldsymbol{y}$. That is, we find $\hat{\boldsymbol{y}} \in S$ such that $\|\boldsymbol{y} - \hat{\boldsymbol{y}}\|$ is minimized. Then $\hat{\boldsymbol{y}}$ is called the **projection** of $\boldsymbol{y}$ onto the subspace S.



Lemma 17.1 (Projection Lemma)**.** *Let $\hat{\boldsymbol{y}}$ be the projection of a vector $\boldsymbol{y}$ onto a subspace S. Then $\boldsymbol{y} - \hat{\boldsymbol{y}}$ is orthogonal to every vector in S.*

Proof. Suppose that this is not the case. Then there is a nonzero vector $\boldsymbol{v} \in S$ such that $(\boldsymbol{y} - \hat{\boldsymbol{y}})^T\boldsymbol{v} \neq 0$. We will show that this contradicts the minimality of $\|\boldsymbol{y} - \hat{\boldsymbol{y}}\|$. For any $a \in \mathbb{R}$,

$$\|\boldsymbol{y} - \hat{\boldsymbol{y}} - a\boldsymbol{v}\|_2^2 = (\boldsymbol{y} - \hat{\boldsymbol{y}} - a\boldsymbol{v})^T(\boldsymbol{y} - \hat{\boldsymbol{y}} - a\boldsymbol{v}) \tag{17.2}$$

$$= \|\boldsymbol{y} - \hat{\boldsymbol{y}}\|_2^2 - 2a\boldsymbol{v}^T(\boldsymbol{y} - \hat{\boldsymbol{y}}) + a^2\|\boldsymbol{v}\|_2^2. \tag{17.3}$$

This is a convex function in a. So setting the derivative to 0 gives the value of a that minimizes the error:

$$\frac{\partial}{\partial a}\|\boldsymbol{y} - \hat{\boldsymbol{y}} - a\boldsymbol{v}\|_2^2 = -2\boldsymbol{v}^T(\boldsymbol{y} - \hat{\boldsymbol{y}}) + 2a\|\boldsymbol{v}\|_2^2 = 0 \Rightarrow a = \frac{\boldsymbol{v}^T(\boldsymbol{y} - \hat{\boldsymbol{y}})}{\|\boldsymbol{v}\|_2^2} \neq 0. \tag{17.4}$$

Let

$$\hat{\boldsymbol{y}}' = \hat{\boldsymbol{y}} + \frac{\boldsymbol{v}^T(\boldsymbol{y} - \hat{\boldsymbol{y}})}{\boldsymbol{v}^T\boldsymbol{v}}\boldsymbol{v}, \tag{17.5}$$

and note that $\hat{\boldsymbol{y}}'$ is also in S but it is closer to $\boldsymbol{y}$, contradicting the optimality of $\hat{\boldsymbol{y}}$. □

17.2 Vector and matrix differentiation

Definition 17.2 (The three derivatives)**.** For a matrix A, scalar z, and two vectors $\boldsymbol{x}, \boldsymbol{y}$ (possibly one-dimensional), let

$$\frac{d\mathsf{A}}{dz} = \begin{pmatrix} \frac{\partial \mathsf{A}_{11}}{\partial z} & \cdots & \frac{\partial \mathsf{A}_{1n}}{\partial z} \\ \vdots & \ddots & \vdots \\ \frac{\partial \mathsf{A}_{m1}}{\partial z} & \cdots & \frac{\partial \mathsf{A}_{mn}}{\partial z} \end{pmatrix}, \qquad \frac{dz}{d\mathsf{A}} = \begin{pmatrix} \frac{\partial z}{\partial \mathsf{A}_{11}} & \cdots & \frac{\partial z}{\partial \mathsf{A}_{1n}} \\ \vdots & \ddots & \vdots \\ \frac{\partial z}{\partial \mathsf{A}_{m1}} & \cdots & \frac{\partial z}{\partial \mathsf{A}_{mn}} \end{pmatrix}, \qquad \frac{d\boldsymbol{y}}{d\boldsymbol{x}} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \cdots & \frac{\partial y_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \cdots & \frac{\partial y_n}{\partial x_m} \end{pmatrix}$$

Lemma 17.3 (Vector differentiation rules)**.** *For a scalar a, vectors $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{v}$, and constant matrices A and S,*

$$\begin{aligned} \frac{d\boldsymbol{y}}{d\boldsymbol{v}} &= \frac{d\boldsymbol{y}}{d\boldsymbol{x}}\frac{d\boldsymbol{x}}{d\boldsymbol{v}}, \\ \frac{d}{d\boldsymbol{v}}(a\boldsymbol{x}) &= a\frac{d\boldsymbol{x}}{d\boldsymbol{v}} + \boldsymbol{x}\frac{da}{d\boldsymbol{v}}, \\ \frac{d}{d\boldsymbol{v}}(\boldsymbol{y}^T\mathsf{A}\boldsymbol{x}) &= \boldsymbol{y}^T\mathsf{A}\frac{d\boldsymbol{x}}{d\boldsymbol{v}} + \boldsymbol{x}^T\mathsf{A}^T\frac{d\boldsymbol{y}}{d\boldsymbol{v}}, \\ \frac{d}{d\boldsymbol{v}}(\boldsymbol{y}^T\mathsf{S}\boldsymbol{y}) &= 2\boldsymbol{y}^T\mathsf{S}\frac{d\boldsymbol{y}}{d\boldsymbol{v}}, \qquad (\mathsf{S} \textit{ is symmetric}) \\ \frac{d}{d\boldsymbol{v}}(\mathsf{A}\boldsymbol{x}) &= \mathsf{A}\frac{d\boldsymbol{x}}{d\boldsymbol{v}}. \end{aligned}$$

Lemma 17.4 (Matrix differentiation rules)**.** *For matrix A and constant vector $\boldsymbol{x}$,*

$$\begin{aligned} \frac{d}{d\mathsf{A}}(\boldsymbol{x}^T\mathsf{A}\boldsymbol{x}) &= \boldsymbol{x}\boldsymbol{x}^T \\ \frac{d}{d\mathsf{A}}\ln|\mathsf{A}| &= \mathsf{A}^{-T} \end{aligned}$$

Definition 17.5 (Gradient and Hessian)**.** Let $f : \mathbb{R}^m \to \mathbb{R}$. The gradient of $f(\boldsymbol{x})$ with respect to $\boldsymbol{x}$ is defined as

$$\nabla_{\boldsymbol{x}} f(\boldsymbol{x}) = \left(\frac{df(\boldsymbol{x})}{d\boldsymbol{x}}\right)^T = \begin{pmatrix} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_m} \end{pmatrix}$$

and the Hessian of $f(\boldsymbol{x})$ with respect to $\boldsymbol{x}$ is defined as

$$\mathsf{H}_{\boldsymbol{x}}(f(\boldsymbol{x})) = \frac{d\nabla_{\boldsymbol{x}} f(\boldsymbol{x})}{d\boldsymbol{x}} = \begin{pmatrix} \frac{\partial f(\boldsymbol{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial f(\boldsymbol{x})}{\partial x_m \partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_1 \partial x_m} & \cdots & \frac{\partial f(\boldsymbol{x})}{\partial x_m \partial x_m} \end{pmatrix}$$

Chain rule. Consider $h : \mathbb{R}^m \to \mathbb{R}$, $g : \mathbb{R} \to \mathbb{R}$, and $f(\boldsymbol{x}) = g(h(\boldsymbol{x}))$. From Lemma 17.3,

$$\begin{aligned} \nabla f(\boldsymbol{x}) &= g'(h(\boldsymbol{x}))\nabla h(\boldsymbol{x}), \\ \mathsf{H}f(\boldsymbol{x}) &= g'(h(\boldsymbol{x}))\mathsf{H}h(\boldsymbol{x}) + g''(h(\boldsymbol{x}))\nabla h(\boldsymbol{x})\nabla^T h(\boldsymbol{x}) \end{aligned}$$

since

$$\begin{aligned}\mathsf{H}f(\boldsymbol{x}) &= \frac{d\nabla f}{d\boldsymbol{x}} \\ &= \frac{d(g'(h(\boldsymbol{x}))\nabla h(\boldsymbol{x}))}{d\boldsymbol{x}} \\ &= g'(h(\boldsymbol{x}))\frac{d\nabla h(\boldsymbol{x})}{d\boldsymbol{x}} + \nabla h(\boldsymbol{x})\frac{d(g'(h(\boldsymbol{x})))}{d\boldsymbol{x}} \\ &= g'(h(\boldsymbol{x}))\mathsf{H}h(\boldsymbol{x}) + \nabla h(\boldsymbol{x})\nabla^T h(\boldsymbol{x})g''(h(\boldsymbol{x}))\end{aligned}$$

Example 17.6 (Log-sum-exp derivatives)**.** Let us find the derivatives of $f(\boldsymbol{x}) = \log \sum_{i=1}^m e^{x_i}$. Let $\boldsymbol{z} = (\exp(x_i))_{i=1}^m$ so that $f(\boldsymbol{x}) = \log \mathbf{1}^T\boldsymbol{z}$.

$$\begin{aligned}\nabla f(\boldsymbol{x}) &= \frac{\boldsymbol{z}}{\mathbf{1}^T\boldsymbol{z}}, \\ \mathsf{H}f(\boldsymbol{x}) &= \frac{\operatorname{diag}(\boldsymbol{z})}{\mathbf{1}^T\boldsymbol{z}} - \frac{\boldsymbol{z}\boldsymbol{z}^T}{(\mathbf{1}^T\boldsymbol{z})^2}.\end{aligned}$$

△

Chain rule. Let $\boldsymbol{h} = (h_1, \ldots, h_n) : \mathbb{R}^m \to \mathbb{R}^n$, $g : \mathbb{R}^n \to \mathbb{R}$, and $f(\boldsymbol{x}) = g(\boldsymbol{h}(\boldsymbol{x}))$. Then

$$\begin{aligned}\frac{\partial f}{\partial x_i} &= \sum_{j=1}^n \frac{\partial g}{\partial h_j}\frac{\partial h_j}{\partial x_i} = \frac{dg}{d\boldsymbol{h}} \cdot \frac{d\boldsymbol{h}}{dx_i} = \nabla^T g \cdot \frac{d\boldsymbol{h}}{dx_i}, \\ \frac{df}{d\boldsymbol{x}} &= \frac{dg}{d\boldsymbol{h}}\frac{d\boldsymbol{h}}{d\boldsymbol{x}} = \nabla^T g\frac{d\boldsymbol{h}}{d\boldsymbol{x}}, \qquad \nabla_{\boldsymbol{x}} f = \left(\frac{df}{d\boldsymbol{x}}\right)^T = \left(\frac{d\boldsymbol{h}}{d\boldsymbol{x}}\right)^T \nabla g\end{aligned}$$